



Sensibilités de la croissance foliaire aux contraintes abiotiques chez le blé. Relations entre processus, à différents niveaux, dans une diversité génétique élargie

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THÈSE POUR OBTENIR LE GRADE DE DOCTEUR DE L'INSTITUT AGRO MONTPELLIER ET DE L'UNIVERSITE DE MONTPELLIER

En Ecophysiologie et adaptation des plantes

École doctorale GAIA - Biodiversité, Agriculture, Alimentation, Environnement, Terre, Eau

Portée par l'Unité Mixte de Recherche « Laboratoire d'Ecophysiologie des Plantes sous Stress
Environnementaux » (UMR LEPSE)

**Sensibilités de la croissance foliaire aux contraintes abiotiques chez
le blé.**

**Relations entre processus, à différents niveaux, dans une diversité
génétique élargie.**

**Présentée par Stéphane Leveau
le 17 novembre 2022**

**Sous la direction de Pierre Martre
et Boris Parent**

Devant le jury composé de

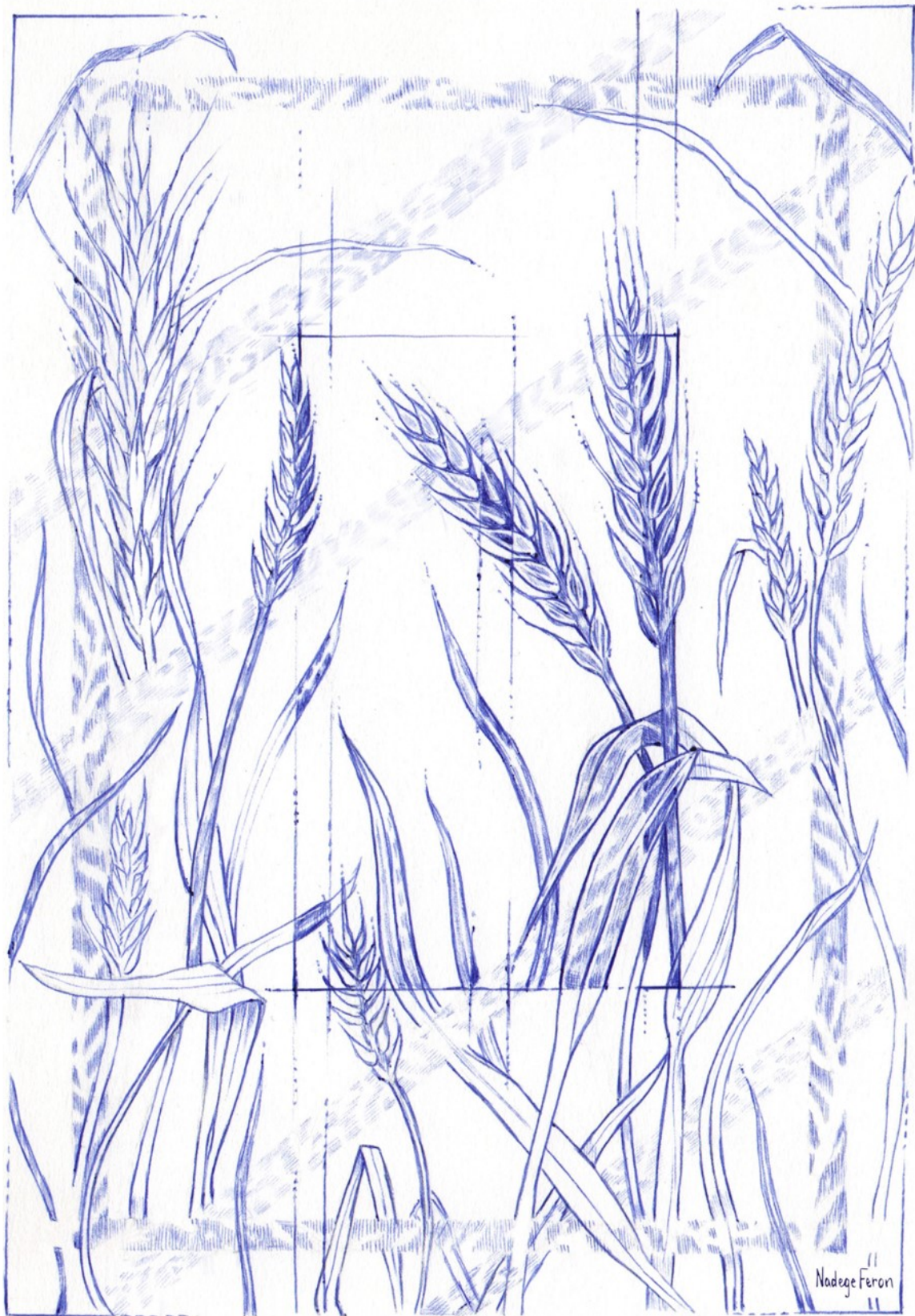
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Preface

Ce mémoire a été rédigé afin d'obtenir le diplôme de docteur de l'Institut Agro Montpellier et de l'Université de Montpellier en écophysiologie et adaptation des plantes. Il est l'aboutissement de mon parcours dans l'enseignement supérieur. Il présente les connaissances accumulées durant ma formation suivie au Laboratoire d'Écophysiologie des Plantes Sous Stress Environnementaux (LEPSE) à Montpellier, en partenariat avec l'entreprise ITK¹ à Clapiers, dans le cadre d'une convention industrielle de formation par la recherche (CIFRE). Ce document réunit une grande partie de mon expérience personnelle et professionnelle dans ce domaine.

Ma première expérience en écophysiologie végétale a commencé lors d'un stage de 3^{ème} année de licence avec Valentina Baldazzi et Michel Génard à l'UMR PHS à l'INRAE d'Avignon. Mon sujet portait sur l'étude de la croissance des fruits en réponse à la température. Ils m'ont donné l'envie de poursuivre dans la recherche et de mieux comprendre les relations plante-environnement.

Les recherches que je présente dans ce mémoire de thèse sont la suite de ma volonté d'évoluer dans ce domaine de connaissance de l'écophysiologie végétale. Les travaux présentés ont été initiés en janvier 2017 lors de mon stage de fin d'étude de Master de Biologie Fonctionnelle des Plantes de l'Université de Montpellier. Ce séjour au LEPSE, en collaboration avec ITK, m'a permis d'approfondir mes compétences et de « mettre les mains à la pâte » en écophysiologie végétale. L'objectif était d'avoir des paramètres de relations liant le fonctionnement de la plante et son environnement pour améliorer les modèles de cultures et les outils d'aides à la décision pour le monde agricole.

Le projet de thèse a commencé en mars 2018. J'ai bénéficié d'une convention CIFRE entre l'ANRT² et ITK. Ce projet, construit par Pierre Martre avec ITK, a permis de financer mon salaire. Durant le doctorat, j'ai profité du projet européen

1. Entreprise d'innovation agronomique qui développe des solutions technologiques
2. Association Nationale Recherche Technologie (convention CIFRE n° 2017/1069)

*Horizon 2020*³ *SoIACE*⁴ pour réaliser une expérimentation en plate-forme de phénotypage et avoir des données en champs. Dans ce cadre, nous avons réalisé un essai en janvier 2019, qui ne sera pas exploité dans ce manuscrit. Nous avons eu de nombreux problèmes techniques et matériels qui ont influencé la croissance et le développement des plantes. Heureusement, un deuxième essai a été effectué en janvier 2020 avec le même panel, financé par le projet *Horizon 2020 EPPN2020*⁵, et est exploité dans ce manuscrit. En mars de la même année, la Covid-19 est passé par là ! Ce qui a modifié la dynamique dans mes activités de recherche. Malgré ces difficultés, j'ai pu orienter mes travaux pour rédiger ce mémoire.

Mon sujet de recherche porte sur la sensibilité de la croissance foliaire aux contraintes abiotiques (eau et azote) chez le blé, avec comme problématique les relations entre la croissance foliaire et le tallage dans une diversité génétique élargie. Il a été établi en collaboration avec Boris Parent et Pierre Martre, que je remercie particulièrement. Ces recherches furent compliquées, mais elles m'ont permis de répondre à une problématique de manière scientifique et je suis heureux de présenter dans ce mémoire et ces résultats. Bien sûr, cette œuvre ne vous serait pas présentée sans l'ensemble de l'équipe du LEPSE, mes camarades et mes directeurs de thèse. Merci à eux.

Je souhaite remercier l'ensemble des personnes de mon entourage pour leurs conseils et leurs soutiens pendant l'ensemble de l'évolution du doctorat. Débattre de mes idées de recherche à vos côtés a été très enrichissant. J'ai également pu bénéficier de ces mêmes débats auprès de mes amis et de ma famille. Vous m'avez aidé à garder ma motivation qui a fluctué durant ce parcours. Je remercie spécialement mes ami.e.s les plus proches pour vos conseils et soutiens. Un remerciement spécial à Adriaan Weetgeest : nous avons réalisé notre thèse dans le même bureau durant 3 ans et nous avons pu évoluer ensemble. Cela a permis de partager nos sentiments, des critiques sur nos activités et de nous soutenir dans nos avancées. Je remercie profondément ma compagne, Nadège Féron, pour m'avoir accompagné et soutenu dans cette aventure. Elle a supporté les décalages successifs de projets personnels et professionnels pour m'accompagner dans cette aventure. Quelle que soit la météo, elle a su être le rayon de soleil de mes journées nuageuses. À vous tous, votre amitié et votre amour m'ont permis d'avancer dans ce projet et de voir les suites possibles celui-ci terminé.

Je vous souhaite une bonne lecture.

Stéphane Leveau

Montpellier, 23 septembre 2022.

3. Programme de financement de la recherche et de l'innovation de l'union européen pour la période 2014-2020

4. Solutions for improving Agroecosystem and Crop Efficiency for water and nutrient use

5. European Plant Phenotyping Network, projet d'infrastructure de recherche

Résumé

La culture des céréales est soumise à des contraintes environnementales dont l'intensité et la fréquence ont augmenté durant deux dernières décennies, notamment sous l'effet du changement climatique. Ces conditions obligent les agriculteurs à mieux raisonner l'utilisation des intrants et le choix des variétés. Pour faire face à ces difficultés, les outils d'aide à la décision, composée de modèle de culture, apportent de l'information utile aux agriculteurs et techniciens agricoles. Ils permettent d'optimiser l'utilisation des intrants et adapter le choix des variétés aux contraintes locales. Ces outils se fondent sur des modèles de simulation de la croissance des plantes, dont la capacité à prédire les interactions entre l'environnement et les variétés reste limitée. La principale limitation des modèles actuels, dans notre cas, réside dans la représentation de la croissance et du développement en réponse à la température, à la fertilisation azotée et au stress hydrique.

Dans ce contexte, l'objectif de la thèse est d'explicitier la plasticité phénotypique de l'expansion foliaire du blé, la céréale la plus cultivée (en surface) au monde, en réponse aux contraintes abiotiques. Nous nous intéressons aux relations entre les différents processus de la croissance foliaire et du rendement, ainsi que leur composante, dans une diversité génétique.

Nous avons bénéficié d'expérimentations en champs dans des conditions de fort et faible intrant hydrique et azotée avec des passages de PhenoMobile. Des expérimentations ont été conduites en plateforme de phénotypage haut-débit, tout d'abord, sur des conditions de température et de demande évaporative dans M3P PhenoDyn et, ensuite, sur des conditions de stress hydrique sur M3P PhenoArch. Nous avons utilisé des ressources génétiques d'espèces sauvages et domestiquées puis des variétés sélectionnées et cultivées en Europe pour étudier la variabilité génétique et leur réponse.

La première étude apporte des formalismes écophysiologiques de la croissance foliaire et de la transpiration en réponse à la température et à la demande évaporative. Dans une seconde partie, nous proposons une méthode de sélection

d'un sous-panel de génotype pour maximiser la diversité du rendement en fonction du nombre de grains chez les blés dur et le nombre d'épis pour le blé tendre. Cette étude a permis également de structurer les génotypes par rapport à leur rang de sensibilité aux contraintes de faible intrant d'irrigation et d'apport en azote. La troisième étude, explore ce sous-panel dans des scénarios de stress hydriques pour apporter des formalismes de réponses au potentiel hydrique du sol et en sortie des caractères génotypique. Ces variables nous permettent d'identifier le tallage comme facteur important de l'expansion foliaire en réponse au potentiel hydrique du sol. Dans notre dernière étude, nous montrons que les paramètres génotypiques estimés en plateforme peuvent être utilisés dans des modèles de cultures pour des projets aux champs.

En sortie de cette thèse, nous avons des formalismes de relation plante environnement, des valeurs de paramètres associées à ces relations et des connaissances sur la structure de la diversité génétique du blé. Ces connaissances, intégrées dans des modèles de cultures, peuvent améliorer la prédiction de la croissance et du développement des végétaux en réponse aux conditions environnementale abiotique, comme la température, la demande évaporative et le potentiel hydrique du sol.

Mots-clés : Blé, écophysiologie, espaces phénotypique, modélisation, formalisme plante-environnement, croissance foliaire, fertilisation azotée, interactions eau-azote, modèle de Culture, outils d'aide à la décision, raisonnement des pratiques agricoles, sécheresse, tallage, transpiration

Abstract

Cereal production is subject to environmental constraints, the intensity and frequency of which have increased over the last two decades, particularly as a result of climate change. These conditions force farmers to reason the use of inputs and the choice of varieties. To face these difficulties, decision support tools, composed of crop models, provide useful information to farmers and agricultural technicians. They allow optimizing the use of inputs and to adapt the choice of varieties to local constraints. These tools are based on plant growth simulation models, whose capacity to predict the interactions between the environment and the varieties remains limited. The main limitation of current models, in our case, is the representation of growth and development in response to temperature, nitrogen fertilization and water stress.

In this context, the objective of this thesis is to explain the phenotypic plasticity of leaf expansion in wheat in response to abiotic constraints. We are interested in the relationships between the different processes of leaf growth and yield, as well as their component, in a genetic diversity.

We have benefited from field experiments under high and low water and nitrogen input conditions with PhenoMobile passages. Experiments were conducted in a high-throughput phenotyping platform, i) under temperature and evaporative demand conditions in M3P PhenoDyn and ii) underwater stress conditions in M3P PhenoArch. We used genetic resources of wild and domesticated species and then selected and cultivated varieties of Europe to study genetic variability and their response.

The first study provides ecophysiological formalism of leaf growth and transpiration in response to temperature and evaporative demand. In a second part, we propose a method for selecting a sub-panel of genotype to maximize yield diversity according to the grains number in durum wheat and the spikes number in bread wheat. This study also structured the genotypes according to their rank of sensitivity to the constraints of low irrigation input and nitrogen supply. The third study explores this sub-panel of underwater stress scenarios to provide formalisms of responses to soil water potential and genotypic trait output. These variables allow us to identify tillering as an important factor of shoot expansion in response to soil water potential. In our last study, we show that the genotypic parameters estimated in the platform can be used in crop models for field projects.

As an output of this thesis, we have formalized plant-environment and variety parameter values that can be used to improve crop models. To improve the prediction of wheat growth and development under abiotic environmental conditions, such as temperature, evaporative demand and soil water potential.

Keywords : Wheat, ecophysiology, phenotypic spaces, modeling, plant environment formalism and interaction, leaf growth, nitrogen fertilization, water-nitrogen interactions, crop model, decision support tools, reasoning of agricultural practices, drought, tillering, transpiration

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Première partie

Introduction

CHAPITRE 1

Contexte Global

1.1 Le changement climatique : une évolution du climat moyen et des évènements extrêmes

Le sixième rapport d'évaluation du groupe d'experts intergouvernemental sur l'évolution du climat (GIEC) publié en 2021 (Masson-Delmotte et al., 2021a) est sans équivoque : l'influence humaine est à l'origine de l'augmentation de la température planétaire d'ampleur inédite depuis au moins 2000 ans (Fig. 1.1).

La température, variable clé du système climatique, s'éloigne d'ores et déjà de la configuration liée à la variabilité naturelle dans laquelle notre agriculture s'est développée et prospère. Selon le rapport du GIEC, la température moyenne du globe a déjà augmenté de 0.99°C [$0.84 - 1.10^{\circ}\text{C}$] entre 1850-1900 et 2001-2020. À partir de cinq scénarios d'évolutions des émissions mondiales de gaz à effet de serre (Masson-Delmotte et al., 2021b) le GIEC estime une augmentation de la température allant de $+2.1^{\circ}\text{C}$ à $+3.5^{\circ}\text{C}$ pour le scénario intermédiaire (SSP2-4.5)

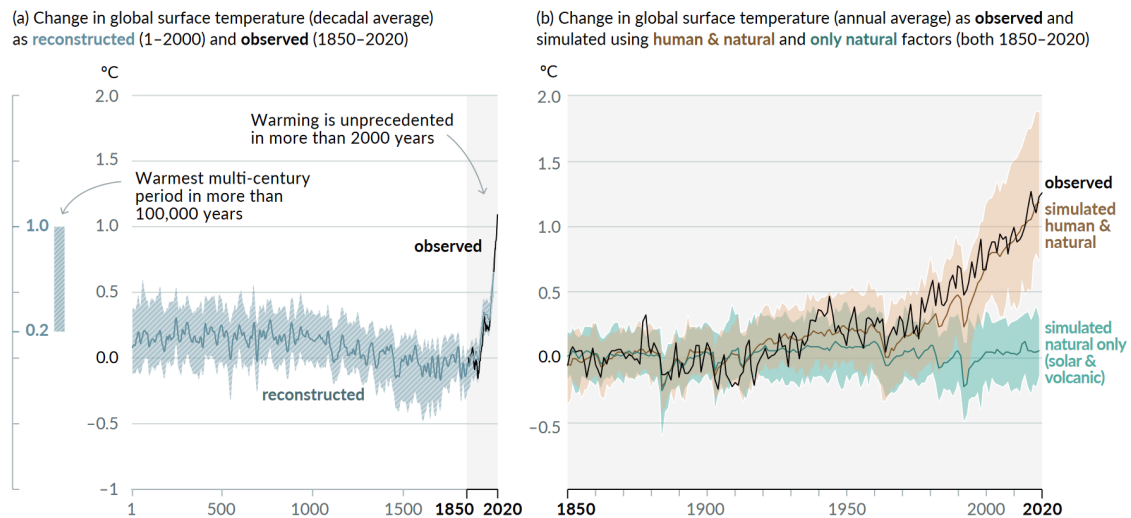


FIGURE 1.1 – Augmentation de la température moyenne à la surface du globe. **a)** changements dans la température de surface du globe reconstitués à partir des archives paléoclimatiques (ligne continue grise, 1-2000) et à partir des observations directes (ligne continue noire, 1850-2020) rapportés à la période 1850-1900 (le 0 correspond à cette période) et moyennés par décennie. La barre verticale à gauche montre la température estimée (plage très probable) durant la période multiséculaire la plus chaude de ces 100 000 dernières années, qui est survenue depuis environ 6 500 ans durant la période interglaciaire actuelle (Holocène). L'ombrage gris avec les diagonales blanches montre la plage très probable pour les reconstructions de température. **b)** Changement de la température globale de surface sur les 170 années passées (ligne noire) rapportés à la période 1850-1900 et moyennés annuellement, comparés aux simulations de l'ensemble de modèles climatiques CMIP6 (Coupled Model Intercomparison Project Phase 6) de la réponse de la température aux moteurs humains et naturels (en brun), et aux moteurs naturels seuls (activité solaire et volcanique, en vert). Les lignes colorées solides montrent la moyenne de l'ensemble de modèles, et les ombres colorées montrent la plage très probable des simulations. Source (Masson-Delmotte et al., 2021b).

et jusqu'à +3.3°C à +5.7°C pour le scénario le plus pessimiste (SSP5-8.5) (Fig. 1.2).

Le changement climatique n'est pas aussi marqué par une hausse de la concentration en CO₂ de l'atmosphère et un réchauffement moyen ; on observe aussi une augmentation de la récurrence et de l'intensité des phénomènes intenses tels que les vagues de chaleur, les épisodes de sécheresse intense et les pluies diluviennes (Masson-Delmotte et al., 2021b). Les dernières décennies en fut un parfait exemple avec une augmentation des vagues de chaleur plus longue et plus fréquentes depuis les années 1950 visible dans les régions du monde (Perkins-Kirkpatrick and Lewis, 2020). Les modèles utilisés dans le rapport du GIEC estiment que les épisodes de chaleur, sur une période de 10 ans future, pourront se produire 9.4 (8.3-9.6) fois en moyenne dans un climat avec une augmentation de 4°C par rapport à la période de référence (Fig. 1.3), encadré « Hot temperature extremes over land ». Durant ces événements de température extrême, nous aurons une augmentation de température en moyenne de plus, 5.1° C. Cela dans les différents scénarios de réchauffement global mondial. Nous aurons en moyenne 9 occasion sur 10 ans d'avoir un pic de chaleurs de + 5°C, soit presque une fois par an. En parallèle, pour la même période de 10 ans future, les événements de sécheresses agricole et écologique pourront être de 4.1 (1.7-7.2) fois plus présent en moyenne par rapport à la référence avec des conditions de changement climatique future comprennent +4°C (Fig. 1.3), encadré « Agricultural & ecological droughts in drying regions ». La période de référence est comprise de 1850 à 1900, elle représente la plus ancienne période d'observations suffisamment complètes à l'échelle mondiale pour estimer la température de la surface du globe. Elle est utilisée comme une approximation des conditions pré-industrielles.

Les preuves de changements ont été renforcés depuis le précédent rapport du GIEC concernant des extrêmes climatiques tels que des vagues de chaleur, de fortes précipitations, des sécheresses, des cyclones tropicaux, ainsi que leur attribution à l'influence humaine. Le World Weather Attribution a ainsi attribué au changement climatique de nombreux événements extrêmes récents ayant impacté l'agriculture : les canicules européennes de juin et juillet 2019, celle de Sibé-

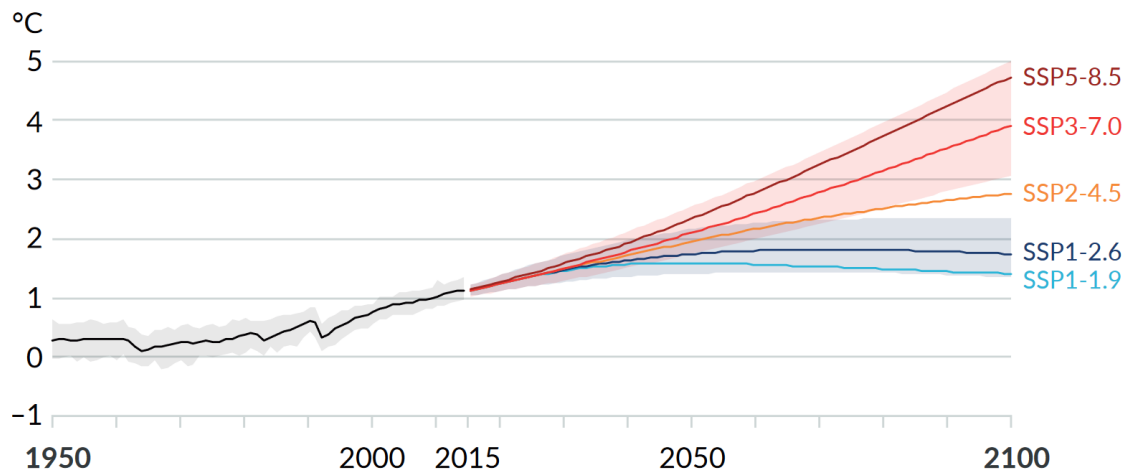


FIGURE 1.2 – Changement de la température de surface du globe par rapport à 1850-1900. Les projections pour chacun des cinq scénarios sont indiquées en couleur. Les nuances représentent les fourchettes d'incertitude. Les nuances représentent les fourchettes d'incertitude. La courbe noire représente les simulations historiques. Les cinq scénarios illustratifs sont désignés par l'abréviation SSPx-y, où "SSPx" désigne la trajectoire socio-économique commune (SSP, Shared Socio-economic Pathway) décrivant les tendances socio-économiques qui sous-tendent le scénario, et "y" désigne le niveau approximatif du forçage radiatif (en $W m^{-2}$) résultant du scénario en 2100. Reproduit depuis (Masson-Delmotte et al., 2021b).

rie en 2020, ou du Canada en juin 2021 ($49,6^{\circ}C$ au Canada ; Philip et al., 2021), les impacts des gels tardifs sur l'arboriculture en Europe d'avril 2021 ou encore les inondations en Allemagne de juillet 2021 (Kreienkamp et al., 2021 ; Vautard et al., 2021).

Dans cette thèse, j'ai étudié les effets des principaux stress dérivés du changement climatique subis par les plantes, la température, la sécheresse du sol et de l'air.

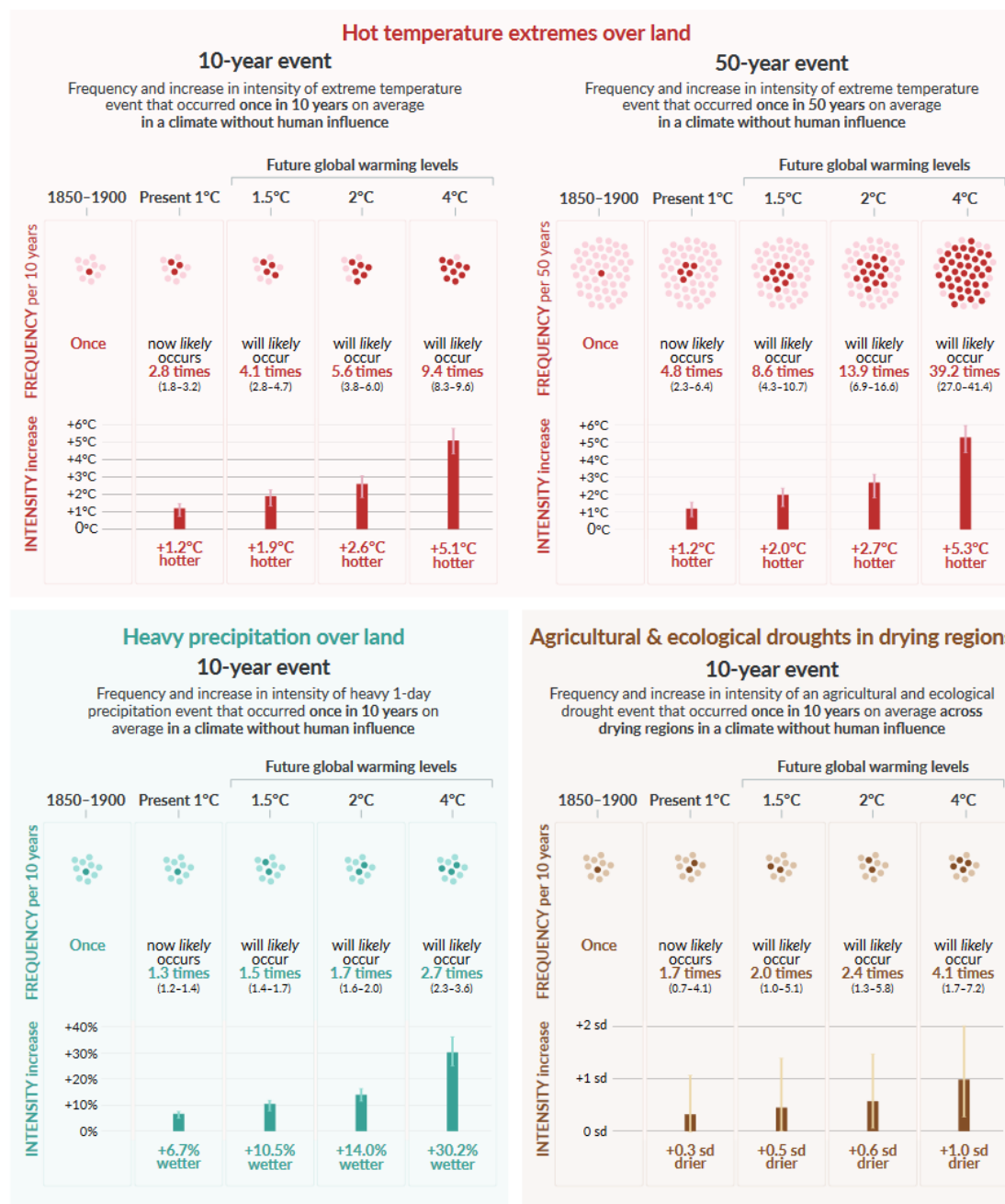


FIGURE 1.3 – Les changements projetés sont indiqués pour des niveaux de réchauffement planétaire de 1°C, 1,5°C, 2°C et 4°C et sont relatifs à la période 1850-1900 représentant un climat sans influence humaine. La figure illustre la fréquence et l'augmentation de l'intensité des événements extrêmes décennaux ou cinquantennaux par rapport à la période de référence (1850-1900) pour différents niveaux de réchauffement planétaire. Les extrêmes de température chaude (« hot temperature extremes over land » dans la figure) sont définis comme les températures maximales quotidiennes sur Terre dépassées en moyenne une fois par décennie (événement décennal) ou une fois par 50 ans (événement cinquantennal) pendant la période de référence 1850-1900. Les précipitations extrêmes sont définies comme la quantité de précipitations quotidiennes sur les terres dépassée en moyenne une fois par décennie durant la période de référence 1850-1900. Les événements de sécheresse agricole et écologique sont définis comme la moyenne annuelle de l'humidité totale de la colonne de sol inférieure au 10e percentile de la période de référence 1850-1900. Ces extrêmes sont définis à l'échelle des mailles du modèle. Pour les extrêmes de température chaude et les précipitations extrêmes, les résultats sont montrés pour la terre globale.

FIGURE 1.3 – Pour la sécheresse agricole et écologique, les résultats sont présentés pour les régions sèches uniquement, qui correspondent aux régions AR6 dans lesquelles il existe une confiance au moins moyenne dans une augmentation projetée de la sécheresse agricole/écologique au niveau de réchauffement de 2°C par rapport à la période de base 1850-1900 dans Coupled Model Intercomparison Project Phase 6 (CMIP6). Ces régions comprennent l'Amérique du Nord occidentale, l'Amérique du Nord centrale, l'Amérique centrale septentrionale, l'Amérique centrale méridionale, les Caraïbes, l'Amérique du Sud septentrionale, l'Amérique du Sud orientale, la mousson sud-américaine, l'Amérique du Sud occidentale, l'Amérique du Sud méridionale, l'Europe occidentale et centrale, la Méditerranée, l'Afrique australe occidentale, l'Afrique australe orientale, Madagascar, l'Australie orientale et l'Australie méridionale (les Caraïbes ne sont pas incluses dans le calcul de la figure en raison du nombre trop faible de cellules de grille terrestres complètes). Les régions non sèches ne présentent pas d'augmentation ou de diminution globale de la gravité des sécheresses. Les projections de l'évolution des sécheresses agricoles et écologiques dans l'ensemble multimodèle CMIP5 diffèrent de celles du CMIP6 dans certaines régions, notamment dans une partie de l'Afrique et de l'Asie. Les évaluations des changements projetés dans les sécheresses météorologiques et hydrologiques sont fournies au chapitre 11 [section 11.6, 11.9] du rapport du GIEC 2021. Dans la section "fréquence", chaque année est représentée par un point. Les points sombres indiquent les années où le seuil extrême est dépassé, tandis que les points clairs sont les années où le seuil n'est pas dépassé. Les valeurs correspondent aux médianes (en gras) et à leur fourchette respective de 5 à 95 % sur la base de l'ensemble multimodèle issu des simulations du CMIP6 sous différents scénarios SSP. Par cohérence, le nombre de points sombres est établi sur la médiane arrondie. Dans la section "intensité", les médianes et leur fourchette de 5 à 95 %, également établies sur l'ensemble multimodèle issu des simulations de CMIP6, sont affichées sous forme de barres sombres et claires, respectivement. Les changements dans l'intensité des températures extrêmes chaudes et des précipitations extrêmes sont exprimés en degrés Celsius et en pourcentage. Concernant la sécheresse agricole et écologique, les changements d'intensité sont exprimés en fractions de l'écart type de l'humidité annuelle du sol. Reproduit depuis (Masson-Delmotte et al., 2021a).

1.2 Impact du réchauffement climatique sur les rendements

L'évolution du climat et son cortège associé d'épisodes de sécheresse et de vagues de chaleur menace la capacité de l'agriculture à nourrir d'une manière durable la population mondiale. En effet, dans de nombreux pays, et en particulier les pays européens, on observe une stagnation, voire une baisse des rendements depuis les dernières décennies, chez les céréales et en particulier chez le blé (Fig. 1.4) (Brisson et al., 2010). Pourtant, le progrès génétique n'a pas diminué (de 1 à 2 % par an selon l'espèce), et la cause du plafond des rendements en blé serait donc en premier lieu d'origine climatique (Gate et al., 2010). Deux indicateurs pédo-climatiques affectent le rendement. En particulier, le déficit hydrique cumulé à la phase de montaison et pendant le remplissage des grains et le nombre de jours avec des températures excessives (température maximale journalière supérieure à 25°C) pendant le remplissage, expliquent très bien les variations inter-annuelles de rendement observés dans la plupart des régions de France (Gate et al., 2010).

Lors d'évènements extrêmes, ces impacts sur le rendement peuvent menacer la sécurité alimentaire globale. En 2010, la sécheresse historique en Russie a provoqué des incendies importants, fortement affecté les rendements, modifié les stocks à l'échelle globale et fortement affecté les courts des denrées alimentaires à l'échelle globale, avec des conséquences graves dans certains pays en voie de développement (Jégo, 2010). Dans une moindre mesure, d'après les données de l'Agreste, la sécheresse historique en France en 2022 a affecté les rendements sur l'ensemble du territoire français et une grande partie de l'Europe (Charrière, 2022).

Sous des latitudes moins clémentes, ces effets sont même décuplés. Par exemple, les variations observées des températures moyennes pendant la saison de culture du blé de ± 2 °C dans les principales régions de culture du blé en Australie peuvent entraîner des réductions de la production de grains allant jusqu'à 50 %. (Asseng

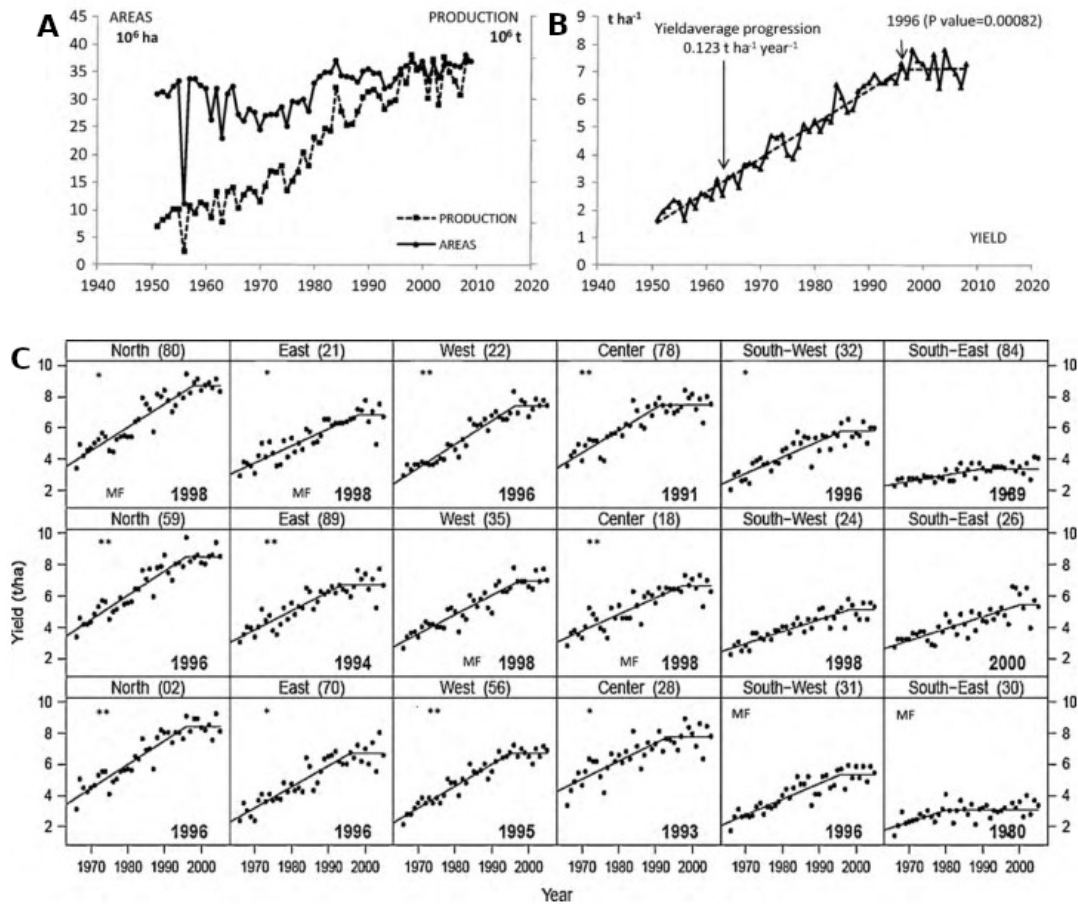


FIGURE 1.4 – Évolution annuelle des surfaces et de la production de blé panifiable en France (A) et du rendement ultérieur (B). L'optimisation du modèle statistique à plateaux ascendants donne 1996 comme l'année d'inflexion. La valeur P est le résultat du test de Fisher comparant le modèle à plateau ascendant avec la régression linéaire. C, Rendements départementaux répartis dans les 6 secteurs géographiques (3 départements par secteur) avec la régression en plateau ascendant, l'année de stagnation et la significativité de cette évolution par rapport à la droite à pente unique (** très significatif $P < 0,01$, * significatif $P < 0,05$, pas d'étoile $P > 0,05$). "MF" indique les départements où sont localisées les stations météorologiques utilisées dans l'étude sont situées. Source des données : AGRESTE. Source des figures (Brisson et al., 2010).

et al., 2011). En France, l'augmentation du déficit hydrique sera plus importante en moyenne dans le sud de la France (Fig. 1.5). Les prédictions montrent une aggravation du déficit hydrique de 30 % à 50 % dans toute la France et de plus de 50 % dans le sud (Kapsambelis et al., 2022).

Cette thèse vise à mieux comprendre la réponse du rendement à des environnements limitants affectés par le réchauffement climatique, par l'analyse des réponses de ses composantes sous-jacentes et de leur variabilité génétique.

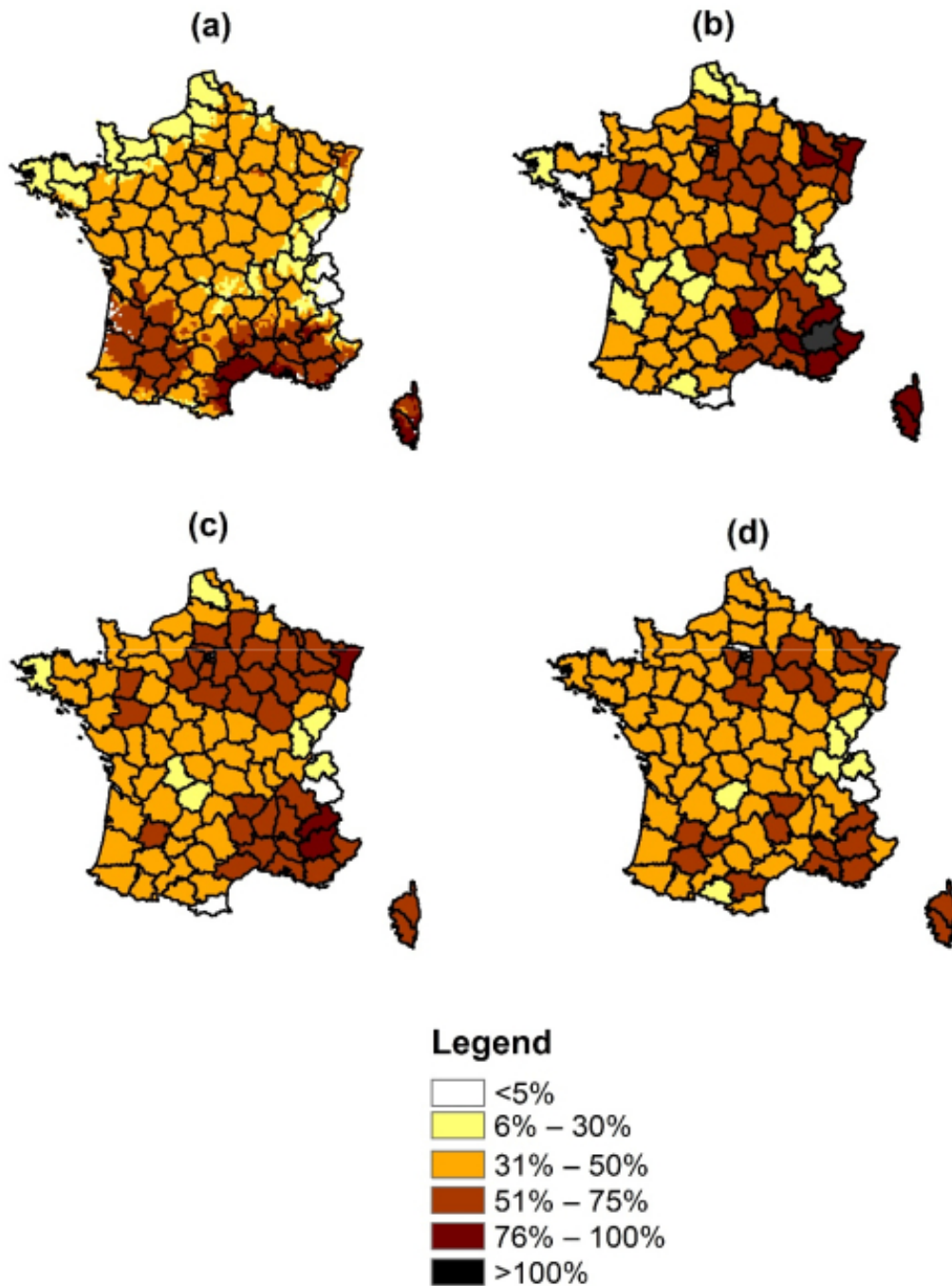


FIGURE 1.5 – Évolution annuelle moyenne entre le climat en 2000 et en 2050 des (a) valeurs de l'indicateur DOWKI (Drought and Overwhelmed Water Key Indicator, indice météorologique basé sur une anomalie hydrique cumulée, permettant d'expliquer les sécheresses et leurs conséquences sur les pertes de rendement agricole à l'échelle départementale), (b) des pertes de rendement du blé tendre d'hiver, (c) des pertes de rendement de l'orge d'hiver, et (d) des prairies. Source Kapsambelis et al. (2022)

1.3 Vers une nouvelle révolution agricole ?

Dans ce contexte, l'agriculture est à la fois un fardeau et un rempart pour le changement climatique. Un fardeau, puisque, d'après le rapport du CITEPA¹ (Citepa, 2022), en 2021, l'agriculture représente 19% des émissions totales de GES ; dernière le transport avec 30% (29% pour le routier seul) ; au même niveau que l'industrie avec 19% et le bâtiment avec 18% (11% pour le résidentiel seul) ; et devant la production d'énergie à 10 %. En 2020, l'agriculture (Élevage, Cultures et Engins) était à l'origine de 20.6 % des gaz à effet de serre en France. L'essentiel des émissions est constitué de méthane (CH₄, 46%), principalement liées à l'élevage et de protoxyde d'azote (N₂O, 40%), principalement liées à la fertilisation des cultures. Les émissions liées à la consommation d'énergie du secteur représentent 13 % du total. Ils présentent dans leur rapport de 1990 à 2020 une diminution de 12 % des émissions de CO₂e dans ce secteur, due principalement à la diminution du cheptel bovin et à la baisse de la fertilisation azotée. En 2020, ils observent une diminution montrée par un recul de la fertilisation azotée expliqué par des conditions climatiques défavorable. L'agriculture est également un rempart pour le changement climatique grâce au stockage du carbone dans le végétal (via la photosynthèse) puis dans les sols. Dans l'étude de l'INRAE « Quelle contribution de l'agriculture française à la réduction des émissions de gaz à effet de serre ? Potentiel d'atténuation et coût de dix actions techniques » (Pellerin et al., 2013), les chercheurs s'accordent sur les marges de progrès importantes de l'agriculture pour participer à la diminution des émissions de GES. Ils identifient trois leviers majeurs : la réduction des émissions de N₂O et de CH₄, le stockage de carbone dans les sols et dans la biomasse (couverture des sols entre deux cultures, reconstitution des haies entre les champs, non-retournement des prairies par exemple), et la production d'énergie à partir de biomasse (biocarburants, biogaz) réduisant les émissions du territoire, par effet, de substitution à des énergies fossiles.

Renverser cet équilibre (entre fardeaux et rempart) constituerait une nouvelle transformation agricole. L'histoire de l'humanité a connu plusieurs périodes

1. Centre interprofessionnel technique d'études de la pollution atmosphérique

d'évolutions profondes des systèmes agricoles ayant pour objectif d'en augmenter la productivité. Celles-ci sont entretenues par des liens de causes à effet, réciproque et étroit, avec des progrès industriels, techniques ou scientifiques et des transitions démographiques : ces transformations sont désignées par les révolutions agricoles (Regnault et al., 2012).

La première révolution agricole (18^e siècle) est caractérisée techniquement par l'abandon de la jachère, des rotations de cultures et assolements, des amendements et augmentation des cheptels. La deuxième révolution agricole, aussi appelé "révolution verte", est désigné par l'utilisation d'engrais chimiques, pesticides, semences hybrides, mécanisation et motorisation généralisée (milieu du 20^e siècle). Cette révolution des pays développés, apparaît dès la fin de la Deuxième Guerre mondiale, profitant des avancées scientifiques de la chimie et de la génétique. En parallèle de cette assimilation des progrès scientifiques, ces changements du monde agricole ont pu apparaître comme une réponse efficace et suffisante face à l'augmentation de la population mondiale et à la persistance d'une insuffisance de l'offre alimentaire dans de nombreuses régions du monde. Mais, cette dernière révolution agricole a mis en place un modèle productiviste à l'origine de graves inconvénients bien connus pour l'environnement. Notamment la pollution de la biosphère (sol, eau, air) par les pesticides et les fertilisants avec des effets non intentionnels sur les écosystèmes et les organismes. Cela a dégradé les sols, diminué la biodiversité, affecté le climat et modifier le paysage en réalisent une spécialisation géographique (avec un remembrement et une augmentation des tailles des parcelles). La troisième révolution agricole, la révolution biotechnologique, qui a commencé à la fin du 20^{ème} siècle et qui est en cours, est caractérisé par les techniques de la transgénèse (organisme génétiquement modifier) et des techniques culturales simplifiées (travailler limitant la modification du sol, exemple : sans labour et semis direct). Cette révolution à un effet non intentionnel sur les écosystèmes et les organismes non-cibles. Elle est également accompagnée des mêmes effets sur l'environnement que la seconde révolution en ajoutant le bilan carbone, le changement climatique, les puits de carbones et la préservation

d'espaces naturels terrestre (Regnault et al., 2012).

Les défis auxquels doit faire face l'agriculture mondiale en ce début du 21^{ème} siècle sont plus divers et plus complexes. Il s'agit de garantir une production agricole durable, suffisante pour nourrir une population mondiale en expansion rapide et qui demande plus de calories par jour (Hickey et al., 2019 ; Miranda et al., 2019 ; Rockström et al., 2017). Tout en réduisant la pression qu'exerce l'agriculture sur l'environnement et en s'adaptant à un contexte de changement global, exemple réduction des contributions aux rejets de gaz à effet de serre et adaptations des pratiques (Pellerin et al., 2013 ; Rial-Lovera et al., 2017). Cela dans le sens de maintenir les moyens de subsistance des personnes travaillant dans les 570 millions d'exploitations agricoles dans le monde, la plupart se trouvent être des entreprises familiales (Lowder et al., 2016).

Pour répondre à ces récents enjeux sociétaux, environnementaux et climatiques, l'agriculture doit évoluer vers une nouvelle révolution agricole ! Elle serait basée sur l'assimilation des connaissances apportées par des systèmes agricoles minoritaires regroupées dans une palette de termes techniques : agriculture biologique, agriculture de conservation, agriculture intégrée, agriculture écologiquement intensive et agriculture de précision pour en citer quelques exemples. Tout comme dans les évolutions de l'agriculture connues, cette révolution en cours s'accompagne d'une insécurité alimentaire liée à l'évolution démographique. *"Si l'on veut améliorer la nutrition et faire reculer l'insécurité alimentaire et la sous-alimentation, la production future devra progresser plus vite que la croissance démographique et les régimes de consommation devront évoluer"*, affirme le communiqué de presse du rapport de la FAO en 2011 (FAO, 2011 ; George Kourous, 2011). L'alimentation est d'ailleurs l'un des premiers objectifs de l'agenda 2030 du développement durable adopté le 25 septembre 2015 par les Nations Unies (Gouvernement de la République française, 2018 ; Nation Unies, 2015). L'analyse des médias, politiciens, conseillers et agriculteurs, révéler que la quatrième révolution agricole serait associée à des technologies émergentes comme l'intelligence artificielles, l'édition des gènes et la robotique (Barrett and Rose, 2022 ; Zaka, 2022). De plus, les connais-

sances techniques, les progrès dans les domaines de l'informatique (modélisation), des technologies embarquées (drones, GPS, stations connectées) et de la génétique semblent être une voie pour continuer l'évolution agricole.

Par l'utilisation des nouvelles technologies innovantes (phénotypage haut débit) appliquées à l'analyse d'une diversité génétique élargie, cette thèse s'inscrit dans cette nouvelle révolution agricole qui se dessine. De plus, en contribuant à l'obtention de variétés plus économes en eau et en intrants, ma thèse peut avoir un impact positif direct sur la charge qu'à l'agriculture sur l'environnement. Pour finir, les travaux de ma thèse peuvent être repris pour améliorer la prédiction des modèles cultureaux. Elle s'inscrit donc également dans l'innovation numérique.

Approches mises en œuvre dans la thèse

2.1 L'Ecophysiologie, comprendre les interactions Géno- type x Environnement

Il existe une grande variabilité de scénarios climatiques (exacerbés par le réchauffement climatique) qui peuvent conduire à un fonctionnement différent de la plante, et à *plus forte raison* à un impact sur sa production. Il ne suffit donc pas de comprendre comment fonctionne un organisme (ici les plantes cultivées) dans un environnement artificiellement optimisé, il faut également comprendre comment il répond à des scénarios climatiques.

Dans ce contexte, l'*Ecophysiologie* est l'étude du fonctionnement des organismes biologiques en interaction avec leur environnement. Elle permet de mieux comprendre les modifications des processus biologiques lors de contraintes environnementales, et de déterminer les capacités des individus à s'adapter lorsque ces contraintes changent. L'écophysiologie est un positionnement de la physio-

logie et en ce sens étudie des variables quantitatives mesurables. Pour cela, elle réunit des mesures des caractéristiques quantitatives de l'environnement proche des plantes, le phylloclimat (Chelle, 2005), et des mesures indicatrices de l'état physiologique des plantes étudiées. Ces relations sont formalisées pour définir la plasticité des plantes à l'environnement, à différentes échelles spatiales, de la cellule à la canopée et de façon temporelle, de réponses instantanées aux effets intégrés sur la durée du cycle de culture. Ces formalismes entre organisme et environnement débouchent sur des relations mathématiques (normes de réaction ou lois de réponse) et, par conséquent, des paramètres et des variables caractérisant la sensibilité de l'individu à chaque variable quantitative environnementale, seules ou en interactions.

Il n'existe pas une réponse unique d'une « espèce » à une modification de l'environnement, mais il existe une variabilité génétique de ces réponses dans chaque espèce (Bertin et al., 2010). Le rendement en grain résulte de l'intégration tout au long du cycle de multiples processus sous-jacents, qui répondent différemment à l'environnement, qui ont chacun leur propre variabilité génétique, et qui interagissent entre eux directement ou au travers de rétroaction multiples (Tardieu & Parent, 2017).

Cela a comme conséquence que chaque variable d'état biologique mesurée dépend du génotype étudié (G), l'environnement (E) et les interactions Génotype x Environnement ($G \times E$; Bertin et al., 2010). Il est bon de noter ici que la conduite de culture (souvent notée « M » pour Management) est considérée en écophysiologie comme une modification de l'environnement et est donc comprise dans le « E » (par exemple, une date de semis différente conduira à des températures différentes lors d'un stade précis du développement d'une plante cultivée). Ces interactions $G \times E$ impliquent que la performance relative d'une variété par rapport aux autres dépend des conditions environnementales. Par exemple, en cas de sécheresse importante, chaque millimètre d'eau économisé pendant la phase végétative permet d'obtenir un état hydrique plus favorable pendant les stades plus tardifs que sont la floraison et le remplissage du grain (Sinclair, 2017). À

l'opposé, en cas de sécheresse modérée, une forte réduction d'accumulation de biomasse accompagnant une réduction de la transpiration ne sera pas compensée pendant le cycle, aboutissant à des biomasses finales plus faibles (Tardieu, 2012). Ainsi, la meilleure variété observée dans un environnement peut être un très mauvais choix dans un scénario de sécheresse différent. Plus généralement, il est difficile de préjuger quelle combinaison de traits physiologiques sera la plus bénéfique dans tel ou tel environnement, tant ces traits interagissent (Drewry *et al.*, 2014).

Durant ma thèse, j'ai utilisé l'ensemble des concepts de l'écophysiologie, via (i) l'analyse des interactions Génotype x Environnement, (ii) sur différents processus à différentes échelles, de la vitesse de croissance des organes en conditions contrôlées au rendement au champ, (iii) pour une diversité génétique élargie (panels, espèces apparentées), pour (iv) mieux comprendre et prédire l'impact de la variabilité génétique des sensibilités aux contraintes environnementales dans des scénarios contrastés et bien définis.

2.2 La croissance foliaire et le tallage comme levier génétique pour l'optimisation d'utilisation de l'eau chez le blé

L'agriculture fait donc face au triple défi de (i) l'augmentation de la population mondiale, qui exacerbe les tensions sur l'usage de l'eau, (ii) l'attente de la société et des pouvoirs publics pour une révolution rapide de l'agriculture vers une agriculture plus durable (Pe'er *et al.*, 2019), moins gourmande en intrant et plus économe en eau (Alamanos *et al.*, 2019), et (iii) un changement climatique, dont les effets s'accroissent chaque année.

Dans ce contexte, le maintien des rendements ne sera assuré que par l'optimisation de stratégies d'utilisation de l'eau (Berger *et al.*, 2016; Tardieu, 2012). Il est donc urgent de développer de nouvelles variétés plus économes en eau (Davies

and Bennett, 2015) adaptées aux spécificités agronomiques et climatiques locaux (Jeuffroy et al., 2014). Or, une vision classique de l'amélioration variétale pour la tolérance à la sécheresse est établie sur le compromis entre disponibilité en eau pendant le cycle végétatif et pendant le remplissage du grain (Sinclair, 2017). Dans ce cadre, les leviers génétiques possibles pour développer de nouvelles variétés adaptées à des épisodes de sécheresse de plus en plus fréquents sont : (i) l'évitement, en jouant sur la précocité des variétés et leur vitesse de développement, afin que la floraison ait lieu avant les épisodes de sécheresse sévères (Shavrukov *et al.*, 2017), (ii) la croissance racinaire (Lynch, 2013); et (iii) la modération de la transpiration par le contrôle hydraulique de la transpiration et la croissance foliaire (Basu *et al.*, 2016).

C'est ce troisième levier que j'analyserai dans cette thèse. Le contrôle de la transpiration en utilisant le levier de la surface foliaire permet de s'affranchir des effets pléiotropiques¹ du contrôle stomatique, et en particulier l'augmentation de la température foliaire. Dans le contexte du changement climatique, il est important de pouvoir jouer sur les deux types de contrôles de la transpiration, tant température et déficit hydrique sont liés, et tant ces processus sont au cœur de l'augmentation des interactions G x E dus aux changements climatiques. En effet, le contrôle de la transpiration affecte autant les réponses des plantes à la température qu'au déficit hydrique. Au champ, des épisodes de sécheresse et de stress thermique sont très susceptibles de se produire simultanément (Barnabas *et al.*, 2008), en raison d'un temps ensoleillé, chaud et sec, mais également en raison du contrôle de la transpiration et de ses effets sur la température et l'état hydrique de la plante (Rizhsky *et al.*, 2002). Les plantes réagissent à la sécheresse en fermant leurs stomates, ce qui augmente la température du couvert. L'augmentation de la température de l'air augmente la demande évaporative imposée aux feuilles par l'atmosphère, entraînant une augmentation de la transpiration qui peut à son tour provoquer une déshydratation des tissus, et celle du sol à plus long terme.

1. En biologie génétique, la **pléiotropie**, ou polyphénie, qualifie les effets secondaires qui se manifestent lorsque l'animal ou la plante présente un phénotype qualitatif particulier

Ma thèse vise donc à apporter des connaissances sur la variabilité génétique disponible pour permettre l'optimisation de l'utilisation de l'eau, par l'étude d'un processus adaptatif, la croissance foliaire, en réponse à des contraintes environnementales que sont (1) la sécheresse du sol, (2) la sécheresse de l'air et (3) et la température.

2.3 Phénotyper la croissance

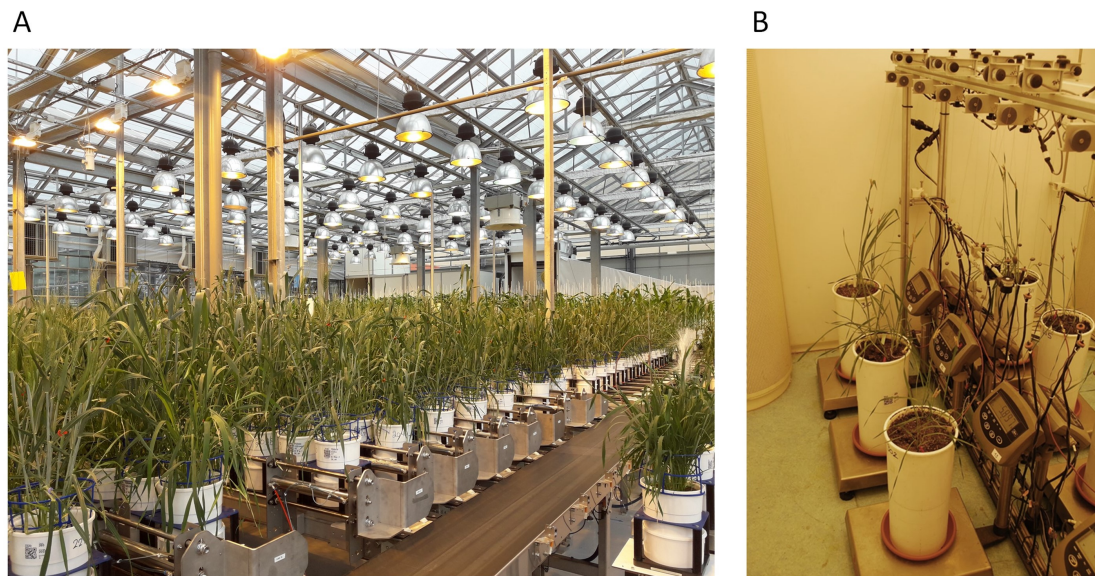
Pendant des années, le phénotypage (la caractérisation du phénotype, soit d'un/des caractère-s observable-s chez un individu), est resté un obstacle majeur en génétique quantitative (Berger *et al.*, 2010; Furbank & Tester, 2011; Yang *et al.*, 2020), mais également pour la sélection. Jusqu'à récemment, obtenir des jeux de paramètres génotypiques, en particulier les paramètres représentant des processus dynamiques en réponse à des contraintes environnementales, pour quelques dizaines de variétés était une gageure, tant l'effort expérimental pouvait être important pour mesurer plusieurs dizaines de valeurs de paramètres.

Ces 15 dernières années ont été marquées par l'explosion du nombre de plateformes de phénotypage haut-débit (Fiorani & Schurr, 2013), en vue de pouvoir caractériser des phénotypes dynamiques, à différentes échelles, de métabolites (HitMe, Bordeaux, HiTMe), jusqu'à la croissance de la plante entière en conditions contrôlées (e.g. parties aériennes, PhenoArch, M3P; ou racines, Dijon, 4PMI), dans des champs semi-contrôlés (DiaPHEN; Pheno3C); et à l'exploitation agricole (Pieruschka & Schurr, 2019). Cependant, dans un premier temps, cette démonstration technologique ne s'est pas transformée automatiquement en données biologiques d'intérêt (Parent *et al.*, 2015; Yang *et al.*, 2020), du fait du manque des pipelines d'analyses de ces données hétérogènes à différents pas de temps. Aujourd'hui, la « Phénomique » végétale est devenue une science à part entière (Houle *et al.*, 2010; Furbank & Tester, 2011; Yang *et al.*, 2020), développant de nouvelles méthodes de traitement et de modélisation des données (Tardieu *et al.*, 2017), pour tirer parti de l'imagerie à haut débit, des technologies de capteurs,

et procurer aux modélisateurs et aux généticiens des ensembles de caractères sur des centaines de variétés.

Au LEPSE, les plateformes de phénotypes regroupées dans M3P (Montpellier Plant Phenotyping Platforms) permettent de caractériser la transpiration, la croissance et l'architecture des plantes à différents par de temps et échelles d'organisation dans des scénarios de déficit hydrique et de température définis (Fig. 2.1). La plate-forme PhenoDyn permet d'analyser la réponse de la croissance de feuilles individuelles et la transpiration de centaines de plantes à la température, au déficit hydrique du sol et au déficit de saturation de pression d'eau de l'air (VPD) au pendant de 15 minutes (Sadok et al., 2007). La Plate-forme PhenoArch permet de mesurer au pas de temps journalier la transpiration et la croissance de 2400 plantes. Dans PhenoArch, les plants peuvent être imagées tous les jours dans des cabinets d'imageries automatisées, ce qui permet de caractériser leur croissance et architecture en 3-dimensions (Cabrera-Bosquet *et al.*, 2016; Perez *et al.*, 2019).

Dans ma thèse, j'ai utilisé les plateformes de phénotypage haut débit de M3P pour analyser les variations dynamiques de vitesse de transpiration et des différents processus sous-jacents de la croissance foliaire du blé (vitesse d'apparition des feuilles, croissance des feuilles individuelles, vitesse d'apparition et de régression des talles) dans des scénarios contrastés de déficits hydriques chez des panels de diversités de blé tendre et de blé dur, et leurs ancêtres.



A, PhenoArch est une plateforme de phénotypage dédiée à l'analyse des déterminismes génétiques de la réponse des plantes (croissance, transpiration, développement) aux conditions environnementales en particulier la sécheresse, la température et la lumière. Elle est établie sur un convoyeur qui transporte 2500 plantes en pots entre des espaces d'irrigation et une chambre d'imagerie. Les caractéristiques mesurées sont la biomasse (via les biovolumes), la transpiration et l'architecture. Les images recueillies et le suivi du poids des pots permettent d'estimer la croissance et la transpiration et d'estimer l'efficacité d'utilisation de l'eau. Des pipelines d'analyse permettent d'estimer le rayonnement incident ainsi que le rayonnement intercepté par chaque plante et en déduire l'efficacité d'utilisation du rayonnement. Elle permet de réaliser des études d'association multi-trait sous un état contrasté de l'eau du sol et des scénarios climatiques.

B, Phenodyn est une plateforme de phénotypage dédiée à l'enregistrement dynamique de la croissance des organes d'une plante. Elle est basée sur l'enregistrement dynamique de la croissance des organes à l'aide de transducteurs de déplacement, tandis que les pots sont placés individuellement sur des balances permettant l'enregistrement et la manipulation automatiques (à travers des pompes péristaltiques) d'eau du sol dans la serre. La même configuration est installée avec un nombre limité de balances dans trois chambres de croissance permettant d'imposer des changements de pas ou des conditions climatiques extrêmes. Il permet de caractériser la croissance et la transpiration de 500 plantes dans des conditions environnementales fluctuantes. Les populations typiquement testées sont la population multipartite, les bibliothèques d'introggression et la collecte de mutants et des GM spécifiques. Les plateformes sont équipées de capteurs environnementaux permettant d'estimer les conditions perçues par les plantes à chaque instant.

2.4 Des réservoirs d'allèles cachés dans la diversité génétique élargie ?

Chez un grand nombre d'espèces cultivées, la sélection variétale a fortement réduit la diversité phénotypique dans les groupes de sélections, ainsi que les groupes allélique correspondant (Charmet, 2011 ; Fu et al., 2006 ; Hyten et al., 2006) . C'est particulièrement vrai chez le blé, chez qui par exemple la diversité des réponses aux stress a chuté dans la plupart des pays européens après 2002-2009 (Kahiluoto et al., 2019). Chez ces espèces, les sélectionneurs se sont récemment emparés de la problématique et essayent d'élargir leur groupe génétique, mais, très souvent, en restant dans des gammes génétiques limitées (par exemple des variétés déjà adaptées aux conditions européennes).

Pourtant, en raison de la disponibilité d'espèces sauvages liées au blé, il existe potentiellement un grand réservoir d'allèles et de gènes qui pourraient être introduits dans les programmes de sélection du blé par des croisements interspécifiques. Par exemple, le blé tendre (*Triticum aestivum*) est issu de l'hybridation naturelle entre un blé sauvage diploïde *Aegilops tauschii* et un blé tétraploïde (*T. turgidum* subsp. *dicoccum*), mais toute la diversité de *Ae. Tauschii* est loin d'être représentée dans le blé moderne. Des croisements synthétiques entre blés tétraploïdes et *Ae. tauschii* (Rosyara et al., 2019) pourraient donc potentiellement permettre d'augmenter fortement la diversité génétique dans les groupes de sélection (Gorafi et al., 2018).

L'étude comparative des sous-espèces de blé peut amener des éclairages en écophysiologie et en modélisation :

- 1) La structure de la variabilité génétique peut permettre d'approcher les mécanismes explicatifs sous-jacents, ainsi que les facteurs externes structurant la diversité (ex : climat d'origine, pression de sélection, ploïdie, allométrie, ontogénie, ...)
- 2) L'échelle interspécifique permet d'exploser la diversité génétique et aug-

menter la puissance pour tester des hypothèses fonctionnelles. Un test d'hypothèse à cette échelle et une validité trans-espèces amène une robustesse qui n'aurait pas été possible dans une gamme de diversité génétique limitée.

- 3) L'analyse d'une large diversité génétique peut permettre de déterminer de possibles sources de diversité intéressante pour la réponse de la croissance foliaire aux changements climatiques et ses conséquences dans des scénarios variés.

Dans ma thèse, outre l'analyse de panels de diversité chez le blé tendre et le blé dur, j'ai analysé une diversité génétique élargie en considérant 12 sous-espèces apparentées au blé.

2.5 Définir l'espace Phénotypique : pour des idéotypes réalistes

Aussi indépendants que possible du point de vue mécaniste, des caractères génotypiques peuvent être corrélés entre eux dans des panels de diversité en raison de contraintes communes sur ces caractères (Donovan et al., 2011). Ainsi, toutes les combinaisons de valeurs de n traits ne représentent pas un hypercube de n dimensions et ne sont pas disponibles dans la variabilité naturelle. L'ensemble des contraintes et des corrélations génétiques entre les traits définit l'ensemble des possibilités, appelé *espace phénotypique*, ne représentant qu'une partie de l'hyper-cube. L'espace phénotypique est un concept très utile en écologie, en recherche perpétuelle de lois simples et validés chez un grand nombre d'espèces et de genres. Parmi, les cas concrets les plus connus d'espaces phénotypiques très contraints, on peut citer le compromis foliaire entre allocation et acquisition des ressources (Leaf Economics Spectrum, Wright et al., 2004; Shipley et al., 2006)

Le phénotypage haut-débit et l'analyse de dizaines de caractères sur des cen-

taines de variétés (Tardieu *et al.*, 2017) peut permettre de mettre en évidence les compromis potentiels entre les traits, et de déterminer la limite multidimensionnelle des combinaisons de paramètres possibles (Townsend *et al.*, 2018). *C'est un point commun et un fil rouge entre les différents chapitres cette thèse.*

2.6 Stratégie Générale et plan de la thèse

Cette thèse s'inscrit pleinement dans les grandes thématiques liées au maintien des rendements dans le contexte des crises climatiques et environnementales et des attentes sociétales. Tout d'abord, en (i) abordant **les effets de stress environnementaux** liées à la crise climatique (sécheresse du sol, Chapitres 4, 5, et 6 ; sécheresse de l'air, Chapitre 3 ; et température, Chapitre 3) et environnementale (azote, Chapitres 4 et 6) sur (ii) **un processus adaptatif, la croissance foliaire** étudiée dans tous ses composantes et leurs interrelations (Chapitres 4, 5, 6), dans (iii) **une diversité génétique élargie** (espèces apparentées, Chapitre 3 ou panels de blé tendre et de blé dur, Chapitres 4, 5, et 6), pour analyser leur **impact sur le rendement** et ses composantes dans des environnements multi-contraints (Chapitres 4 et 6). En détail :

- Le Chapitre 3 analyse la variabilité des réponses à des variables environnementales liées au changement climatique (température et demande évaporative) chez 12 espèces apparentées au blé. Je conclus que ces réponses sont très conservées, mais qu'il existe potentiellement des ressources génétiques d'intérêt pour ces traits chez *Ae. tauschii*, accessibles par hybridations interspécifiques.
- Le Chapitre 4 analyse l'espace phénotypique des composantes du rendement et sa plasticité sous contraintes hydriques et azotées chez le blé tendre et le blé dur. Outre son intérêt académique pour la compréhension des interrelations entre ces composantes, ce chapitre a permis de sélectionner un sous panel de blé tendre (~40 génotypes) et un sous panel de blé dur (~40 génotypes) qui respectent l'espace phénotypique observé dans

- les panels initiaux. Ces deux panels sont utilisés dans les chapitres 5 et 6.
- Le Chapitre 5 analyse les réponses des différentes composantes de la croissance foliaire au déficit hydrique dans la plateforme de phénotypage à haut débit PhenoArch. Ce chapitre démontre que le tallage est non seulement le premier pourvoyeur de variabilité génétique de croissance potentielle (comparé à la vitesse d'apparition des feuilles et à la croissance des feuilles individuelles), mais surtout le premier « fusible » expliquant la variabilité de la sensibilité de la croissance foliaire observé chez le blé tendre, comme chez le blé dur.
 - Le Chapitre 6 analyse les relations entre les paramètres de vitesse et les sensibilités des composantes de la croissance mesurées ou calculées au Chapitre 5 avec des paramètres de croissance mesurés au champ et les composantes du rendement. En observant (i) que les vitesses de croissance et leurs sensibilités mesurées en plateforme sont fortement liées à celles observées au champ, et (ii) qu'ainsi elles sont corrélées aux composantes du rendement mesurées au champ, ce Chapitre (iii) démontre l'intérêt des plateformes de phénotypage en conditions contrôlées pour l'analyse de scénarios hydriques spécifiques, très difficiles à maîtriser au champ.

Chaque chapitre a pour projet d'être publié dans des revues scientifiques. Le chapitre 3 a été publié dans la revue Journal of Experimental Botany (Leveau et al., 2021), les chapitres 5 et 6 ont été combinées dans un article soumis pour publication, et le chapitre 4 sera également prochainement soumis pour publication.

Références

Alamanos, A., Latinopoulos, D., Papaioannou, G., Mylopoulos, N., 2019. Integrated Hydro-Economic Modeling for Sustainable Water Resources Management in Data-Scarce Areas : The Case of Lake Karla Watershed in Greece. *Water Resour Manage* 33, 2775–2790. <https://doi.org/10.1007/s11269-019-02241-8>

Asseng, S., Foster, I., Turner, N.C., 2011. The impact of temperature variability on wheat yields. *Global Change Biology* 17, 997–1012. <https://doi.org/10.1111/j.1365-2486.2010.02262.x>

Barrett, H., Rose, D.C., 2022. Perceptions of the Fourth Agricultural Revolution : What's In, What's Out, and What Consequences are Anticipated? *Sociologia Ruralis* 62, 162–189. <https://doi.org/10.1111/soru.12324>

Basu, S., Ramegowda, V., Kumar, A., Pereira, A., 2016. Plant adaptation to drought stress [version 1 ; peer review : 3 approved]. *F1000Research*, 5(F1000 Faculty Rev) :1554. <https://doi.org/10.12688/f1000research.7678.1>

Berger, J., Palta, J., Vadez, V., 2016. Review : An integrated framework for crop adaptation to dry environments : Responses to transient and terminal drought. *Plant Science* 253, 58–67. <https://doi.org/10.1016/j.plantsci.2016.09.007>

Bertin, N., Martre, P., Genard, M., Quilot, B., Salon, C., 2010. Under what circumstances can process-based simulation models link genotype to phenotype for complex traits? Case-study of fruit and grain quality traits. *Journal of Experimental Botany* 61, 955–967. <https://doi.org/10.1093/jxb/erp377>

Brisson, N., Gate, P., Gouache, D., Charmet, G., Oury, F.-X., Huard, F., 2010. Why are wheat yields stagnating in Europe? A comprehensive data analysis for France. *Field Crops Research* 119, 201–212. <https://doi.org/10.1016/j.fcr.2010.07.012>

Charmet, G., 2011. Wheat domestication : Lessons for the future. *Comptes*

Rendus Biologies, On the trail of domestications, migrations and invasions in agriculture 334, 212–220. <https://doi.org/10.1016/j.crv.2010.12.013>

Charrière, P., 2022. Moisson 2022 : baisse générale des rendements pour les céréales à paille. Agreste Infos rapides Grandes cultures, 2022–87.

Chelle, M., 2005. Phylloclimate or the climate perceived by individual plant organs : what is it? How to model it? What for? *New Phytol* 166, 781–790. <https://doi.org/10.1111/j.1469-8137.2005.01350.x>

Citepa, 2022. Inventaire des émissions de polluants atmosphériques et de gaz à effet de serre en France – Format Secten. (No. 2070sec / 2022).

Davies, W.J., Bennett, M.J., 2015. Achieving more crop per drop. *Nature Plants* 1, 1–2. <https://doi.org/10.1038/nplants.2015.118>

Donovan, L.A., Maherali, H., Caruso, C.M., Huber, H., de Kroon, H., 2011. The evolution of the worldwide leaf economics spectrum. *Trends in Ecology & Evolution* 26, 88–95. <https://doi.org/10.1016/j.tree.2010.11.011>

FAO, 2011. The state of the world's land and water resources for food and agriculture (SOLAW) – Managing systems at risk. Food and Agriculture Organization of the United Nations, Rome and Earthscan, London. <https://doi.org/10.4324/9780203142837>

Fu, Y.-B., Peterson, G.W., Yu, J.-K., Gao, L., Jia, J., Richards, K.W., 2006. Impact of plant breeding on genetic diversity of the Canadian hard red spring wheat germplasm as revealed by EST-derived SSR markers. *Theor Appl Genet* 112, 1239–1247. <https://doi.org/10.1007/s00122-006-0225-2>

Gate, P., Brisson, N., Gouache, D., 2010. Les causes du plafonnement du rendement du blé en France : d'abord une origine climatique. *Comptes-Rendus de l'Académie d'Agriculture de France* 96, 17.

George Kourous, 2011. FAO - Nouvelles : Appauvrissement et dégradation des terres et des eaux : menace grandissante pour la sécurité alimentaire [WWW Document]. URL <https://www.fao.org/news/story/fr/item/95178/icode/> (accessed 9.7.22).

Gorafi, Y.S.A., Kim, J.-S., Elbashir, A.A.E., Tsujimoto, H., 2018. A population of wheat multiple synthetic derivatives : an effective platform to explore, harness and utilize genetic diversity of *Aegilops tauschii* for wheat improvement. *Theor Appl Genet* 131, 1615–1626. <https://doi.org/10.1007/s00122-018-3102-x>

Gouvernement de la République française, 2018. ODD2 - Éliminer la faim, assurer la sécurité alimentaire, améliorer la nutrition et promouvoir une agriculture durable [WWW Document]. L'Agenda 2030 en France. URL (accessed 9.7.22).

Hickey, L.T., N. Hafeez, A., Robinson, H., Jackson, S.A., Leal-Bertioli, S.C.M., Tester, M., Gao, C., Godwin, I.D., Hayes, B.J., Wulff, B.B.H., 2019. Breeding crops to feed 10 billion. *Nat Biotechnol* 37, 744–754. <https://doi.org/10.1038/s41587-019-0152-9>

Hyten, D.L., Song, Q., Zhu, Y., Choi, I.-Y., Nelson, R.L., Costa, J.M., Specht,

J.E., Shoemaker, R.C., Cregan, P.B., 2006. Impacts of genetic bottlenecks on soybean genome diversity. *Proceedings of the National Academy of Sciences* 103, 16666–16671. <https://doi.org/10.1073/pnas.0604379103>

Jégo, M., 2010. La sécheresse en Russie provoque une envolée inédite des prix alimentaires. *Le Monde.fr*.

Jeuffroy, M.-H., Casadebaig, P., Debaeke, P., Loyce, C., Meynard, J.-M., 2014. Agronomic model uses to predict cultivar performance in various environments and cropping systems. A review. *Agronomy for Sustainable Development* 34, 121–137. <https://doi.org/10.1007/s13593-013-0170-9>

Kahiluoto, H., Kaseva, J., Balek, J., Olesen, J.E., Ruiz-Ramos, M., Gobin, A., Kersebaum, K.C., Takac, J., Ruget, F., Ferrise, R., Bezak, P., Capellades, G., Dibari, C., Makinen, H., Nendel, C., Ventrella, D., Rodriguez, A., Bindi, M., Trnka, M., 2019. Decline in climate resilience of European wheat. *Proc. Natl. Acad. Sci. U. S. A.* 116, 123–128. <https://doi.org/10.1073/pnas.1804387115>

Kapsambelis, D., Moncoulon, D., Veyssiere, M., Soubeyroux, J.-M., Cordier, J., 2022. Modeling the Impact of Extreme Droughts on Agriculture under Current and Future Climate Conditions Using a Spatialized Climatic Index. *Applied Sciences* 12, 2481. <https://doi.org/10.3390/app12052481>

Kreienkamp, F., Philip, S.Y., Tradowsky, J.S., F. Kew, S., Lorenz, P., 2021. Heavy rainfall which led to severe flooding in Western Europe made more likely by climate change. *World Weather Attribution*. (accessed 9.14.22).

Leveau, S., Parent, B., Zaka, S., Martre, P., 2021. Differential sensitivity to temperature and evaporative demand in wheat relatives. *Journal of Experimental Botany*. <https://doi.org/10.1093/jxb/erab431>

Lowder, S.K., Skoet, J., Raney, T., 2016. The Number, Size, and Distribution of Farms, Smallholder Farms, and Family Farms Worldwide. *World Development* 87, 16–29. <https://doi.org/10.1016/j.worlddev.2015.10.041>

Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S.L., Péan, C., Berger, S., Caud, N., Chen, Y., Goldfarb, L., Gomis, M.I., Huang, M., Leitzell, K., Lonnoy, E., Matthews, J.B.R., Maycock, T.K., Waterfield, T., Yelekçi, Ö., Yu, R., Zhou, B. (Eds.), 2021a. IPCC, 2021 : Climate Change 2021 : The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press. ed. Cambridge, United Kingdom and New York, NY, USA.

Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S.L., Péan, C., Berger, S., Caud, N., Chen, Y., Goldfarb, L., Gomis, M.I., Huang, M., Leitzell, K., Lonnoy, E., Matthews, J.B.R., Maycock, T.K., Waterfield, T., Yelekçi, Ö., Yu, R., Zhou, B. (Eds.), 2021b. IPCC, 2021 : Summary for Policymakers, in : Climate Change 2021 : The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, Cambridge, United Kingdom and New York. NY, USA, pp. 3–32.

Miranda, J., Ponce, P., Molina, A., Wright, P., 2019. Sensing, smart and sustainable technologies for Agri-Food 4.0. *Computers in Industry* 108, 21–36. URL

Nation Unies, 2015. Les Etats membres de l'ONU adoptent un nouveau programme de développement audacieux [WWW Document]. ONU Info. URL (accessed 9.7.22).

Pe'er, G., Zinngrebe, Y., Moreira, F., Sirami, C., Schindler, S., Müller, R., Bontzorlos, V., Clough, D., Bezák, P., Bonn, A., Hansjürgens, B., Lomba, Â., Möckel, S., Passoni, G., Schleyer, C., Schmidt, J., Lakner, S., 2019. A greener path for the EU Common Agricultural Policy. *Science* 365, 449–451. URL

Pellerin, S., Bamière, L., Angers, D., Béline, F., Benoit, M., Butault, J.-P., Chenu, C., Colnenne-David, C., de Cara, S., Delame, N., Doreau, M., Dupraz, P., Faverdin, P., Garcia-Launay, F., Hassouna, M., Hénault, C., Jeuffroy, M.-H., Klumpp, K., Metay, A., Moran, D., Recous, S., Samson, E., Savini, I., Pardon, L., 2013. Quelle contribution de l'agriculture française à la réduction des émissions de gaz à effet de serre? Potentiel d'atténuation et coût de dix actions techniques (Other No. convention n° 11-60-C0021, convention n° 11-60-C0021.). Inra - DEPE. <https://doi.org/10.15454/rgfm-wh23>

Perkins-Kirkpatrick, S.E., Lewis, S.C., 2020. Increasing trends in regional heatwaves. *Nat Commun* 11, 3357. <https://doi.org/10.1038/s41467-020-16970-7>

Philip, S.Y., Kew, S.F., van Oldenborgh, G.J., Yang, W., Vecchi, G.A., Anslow, F.S., Li, S., Seneviratne, S.I., Luu, L.N., Arrighi, J., Singh, R., van Aalst, Hauser, M., Schumacher, D.L., Marghidan, C.P., Ebi, K.L., Bonnet, R., Vautard, R., Tradosky, J., Coumou, D., Lehner, F., Rodell, C., Stull, R., Howard, R., Gillett, N., Otto, F.E.L., 2021. Rapid attribution analysis of the extraordinary heatwave on the Pacific Coast of the US and Canada June 2021. 37.

Pieruschka, R., Schurr, U., 2019. Plant Phenotyping : Past, Present, and Future. *Plant Phenomics* 2019, 1–6. <https://doi.org/10.34133/2019/7507131>

Regnault, H., Sartre, X.A. de, Regnault-Roger, C., 2012. Les révolutions agricoles en perspective - Introduction 204.

Rial-Lovera, K., Davies, W.P., Cannon, N.D., 2017. Implications of climate change predictions for UK cropping and prospects for possible mitigation : a review of challenges and potential responses. *Journal of the Science of Food and Agriculture* 97, 17–32. <https://doi.org/10.1002/jsfa.7767>

Rockström, J., Williams, J., Daily, G., Noble, A., Matthews, N., Gordon, L., Wetterstrand, H., DeClerck, F., Shah, M., Steduto, P., de Fraiture, C., Hatibu, N., Unver, O., Bird, J., Sibanda, L., Smith, J., 2017. Sustainable intensification of agriculture for human prosperity and global sustainability. *Ambio* 46, 4–17. URL

Rosyara, U., Kishii, M., Payne, T., Sansaloni, C.P., Singh, R.P., Braun, H.-J., Dreisigacker, S., 2019. Genetic Contribution of Synthetic Hexaploid Wheat to CIM-MYT's Spring Bread Wheat Breeding Germplasm. *Sci Rep* 9, 12355–12355. URL

Sadok, W., Naudin, P., Boussuge, B., Muller, B., Welcker, C., Tardieu, F., 2007. Leaf growth rate per unit thermal time follows QTL-dependent daily patterns in hundreds of maize lines under naturally fluctuating conditions. *Plant Cell Environ.* 30, 135–146. <https://doi.org/10.1111/j.1365-3040.2006.01611.x>

Shipley, B., Lechowicz, M.J., Wright, I., Reich, P.B., 2006. Fundamental Trade-

Offs Generating the Worldwide Leaf Economics Spectrum. *Ecology* 87, 535–541. <https://doi.org/10.1890/05-1051>

Tardieu, F., 2012. Any trait or trait-related allele can confer drought tolerance : just design the right drought scenario. *J Exp Bot* 63, 25–31. URL

Townsend, A.J., Retkute, R., Chinnathambi, K., Randall, J.W.P., Foulkes, J., Carmo-Silva, E., Murchie, E.H., 2018. Suboptimal Acclimation of Photosynthesis to Light in Wheat Canopies. *Plant Physiology* 176, 1233–1246. URL

Vautard, R., Van Oldenborgh, G.J., Bonnet, R., Robin, Y., Kew, S., Soubeyroux, J.-M., Dubuisson, B., Viovy, N., Reichstein, M., Otto, F., 2021. Human-caused climate change increased the likelihood of early growing period frost in France. *World Weather Attribution*. (accessed 9.14.22).

Wright, I.J., Reich, P.B., Westoby, M., Ackerly, D.D., Baruch, Z., Bongers, F., Cavender-Bares, J., Chapin, T., Cornelissen, J.H.C., Diemer, M., Flexas, J., Garnier, E., Groom, P.K., Gulias, J., Hikosaka, K., Lamont, B.B., Lee, T., Lee, W., Lusk, C., Midgley, J.J., Navas, M.-L., Niinemets, Ü., Oleksyn, J., Osada, N., Poorter, H., Poot, P., Prior, L., Pyankov, V.I., Roumet, C., Thomas, S.C., Tjoelker, M.G., Veneklaas, E.J., Villar, R., 2004. The worldwide leaf economics spectrum. *Nature* 428, 821–827. <https://doi.org/10.1038/nature02403>

Zaka, S., 2022. Comment l’agriculture française doit s’adapter aux sécheresses. Brut Media.

Deuxième partie

Recherches

Differential sensitivity to temperature and evaporative demand in wheat relatives

Résumé

Il existe des sources potentielles d'allèles et de gènes actuellement présents chez les espèces apparentées au blé qui pourraient être introduites dans les programmes de sélection du blé visant les climats chauds et secs actuels et futurs. Cependant, à ce jour, ni la diversité intra- ou interspécifique des réponses de la croissance et de la transpiration des feuilles à la température et à la demande d'évaporation n'a été étudiée dans une gamme significative d'espèces liées au blé. En analysant 12 groupes d'espèces et de sous-espèces liées au blé, nous avons pu examiner la structure multidimensionnelle de la diversité génétique pour les traits liés aux structures végétatives des plantes et à leur développement, ainsi qu'à l'expansion et à la transpiration des feuilles, de même que leurs réponses à des plages de température et de demande d'évaporation " non stressantes ". En plus de fournir de nouvelles informations sur la façon dont le type de génome,

le niveau de ploïdie, la phylogénie et la pression de sélection agissent ensemble pour structurer cette diversité génétique, notre étude fournit également de nouveaux formalismes mathématiques et des paramètres associés pour les réponses aux traits dans une large gamme de diversité génétique chez les espèces liées au blé. Cela permettra potentiellement aux modèles de culture de prédire l'impact de cette diversité sur le rendement, et donc d'indiquer les sources potentielles d'amélioration variétale pour les germplasm¹ de blé modernes par le biais de croisements interspécifiques.

Mots-clés : Aegilops, demande évaporative, diversité génétique, allongement foliaire, croissance des feuilles, température, transpiration, Triticum, déficit de pression de vapeur, demande évaporative, blé.

Abstract

There are potential sources of alleles and genes currently present in wheat-related species that have the potential to be introduced into wheat breeding programs targeting current and future hot and dry climates. However, to date neither the intra- nor the interspecific diversity of the responses of leaf growth and transpiration to temperature and evaporative demand have been investigated in across a significant range of wheat-related species. By analysing 12 groups of wheat-related species and subspecies, we were able to examine the multi-dimensional structure of the genetic diversity for traits linked to plant vegetative structures and their development, and to leaf expansion and transpiration, together with their responses to 'non-stressing' ranges of temperature and evaporative demand. In addition to providing new insights on how genome type, ploidy level, phylogeny, and breeding pressure act together to structure this genetic diversity, our study also provides new mathematical formalisms and associated parameters for trait responses across a wide range of genetic diversity in

1. Le germoplasme est le matériel héréditaire transmis à la progéniture au moyen de cellules germinales capables de permettre de préserver directement la biodiversité au niveau génétique et spécifique

wheat-related species. This will potentially allow crop models to predict the impact of this diversity on yield, and thus to indicate potential sources of varietal improvement for modern wheat germplasms through interspecific crosses.

Keywords : *Aegilops*, evaporative demand, genetic diversity, leaf expansion, leaf growth, temperature, transpiration, *Triticum*, vapour pressure deficit (VPD), wheat.

3.1 Introduction

Cereal production worldwide is affected by high temperature, high evaporative demand, and drought (Lobell et al., 2011). The frequency of such events is increasing as a result of climate change (Handmer et al., 2012; Ben-Ari et al., 2018; Rojas et al., 2019), while at the same time agriculture is being challenged by societal and policy requirements for more sustainable and water-sparing production methods. Developing new cultivars that are adapted to meet the needs of this dual context is not straightforward, given the wide range of climates and management practices in wheat-growing regions and the fact that an individual genotypic trait can have positive effects on crop performance in one type of environment while being deleterious in another (Tardieu, 2012).

A purely experimental approach cannot explore the effects on yield, plant resilience, and environmental balances of each combination of traits under all possible environmental scenarios. A combination of experimental and modelling approaches is therefore required to define the best combination of traits in a given target environment (Reyer et al., 2013; Jeuffroy et al., 2014; Perego et al., 2014; Rotter et al., 2015), using models that predict genotypic effects under various environmental scenarios (Parent and Tardieu, 2014). These models require formalisms of specific responses to environmental constraints together with associated trait/parameter values that cover the wide range of genetic variability that breeders can encounter within enlarged breeding pools (Hammer et al., 2019). This has not yet been achieved for the responses of leaf expansion and transpira-

tion to temperature and evaporative demand in crop growth models for wheat. This is despite the fact that, for example, the control of transpiration under high evaporative demand can be a key factor in designing future cultivars that can produce ‘more crop per drop’ (Davies et al., 2001). In wheat models, the availability of efficient formalisms and parameter values differ greatly between processes, as follows.

The response of leaf expansion to temperature has been found to be similar to that of plant development (Parent et al., 2019), at least within a ‘non-stressing’ range in which processes are not durably affected, and are reversible (Parent et al., 2010). Within such ranges, which are not considered as ‘heat’ or ‘cold’ stresses, development responses to temperature are well-formalized and parameter values have been determined in most cultivated species. Indeed, these responses are the basis of the calculation of thermal time, a central concept of most process-based models. Parent and Tardieu, (2012) reported a very low variability of this response within each species they studied, while a wider variability was present between species. However, to date, the variability of temperature responses—and the possibility of seeking new alleles controlling these responses—between closely-related species such as within different wheats has not been documented.

While evaporative demand has been found as mostly affecting leaf expansion rate in several species (e.g. Parent et al., 2010; Tardieu et al., 2014), very little is known about how growth is affected in wheat and whether significant genetic variability exists in it and related species. This is probably due to the fact that temperature and vapour-pressure deficit (VPD) are difficult to dissociate in natural environments. There is an urgent need to characterize these responses, both for the accuracy of model predictions and to inform wheat breeders on the availability (or not) of associated genetic variability.

The response of transpiration to VPD has been studied in several crops including wheat (Schoppach and Sadok, 2012; Schoppach et al., 2017). In wheat,

breeders have indirectly selected genotypes that present a limitation in hydraulic conductance (Schoppach et al., 2017), allowing decreased transpiration at high VPD. However, we do not know if the reported range of responses represents a large part of the genetic variability in wheat-related species, or if some genetic variability has been lost during wheat domestication and breeding.

Wheat-related species form a large family (>300 species and subspecies), including wild and cultivated species domesticated over 10 000 years, three subgenomes (A, B, and D), and three levels of ploidy (diploids : Au, Am, B, and D genome; tetraploids : AB subgenomes; hexaploids : ABD subgenomes). Polyploid Triticum species originated by natural hybridization events between Triticum and Aegilops species (Fig. 3.1) (Petersen et al., 2006). Bread wheat (*T. aestivum*) represents 95% of global wheat production and is grown across very wide ranges of latitude and altitude (Curtis et al., 2002). Durum wheat (*T. turgidum* subsp. *durum*) is mainly grown under Mediterranean-type climates. Other wheat species are grown more marginally, such as spelt (*T. aestivum* subsp. *spelta*), einkorn (*T. monococcum* subsp. *monococcum*), and emmer wheat (*T. turgidum* subsp. *dicoccum*).

Because of the availability of wild wheat-related species, there is a large reservoir of alleles and genes that can potentially be introduced into wheat breeding programs through interspecific crosses. For example, bread wheat emerged from the natural hybridization between wild goatgrass (*Ae. tauschii*) and a tetraploid wheat (*T. turgidum* subsp. *dicoccum*, a progenitor of modern durum wheat). However, the genetic diversities of *T. turgidum* and *Ae. tauschii* are not fully represented in *T. aestivum*, probably because it has evolved from only a few hybridization events. Introducing genes from wheat-relative species can therefore increase the genetic diversity of modern germplasms (Gorafi et al., 2018), for example as seen in synthetic crosses between tetraploid wheats and *Ae. tauschii* (Rosyara et al., 2019) and between *T. aestivum* and *T. turgidum* (Martin et al., 2011).

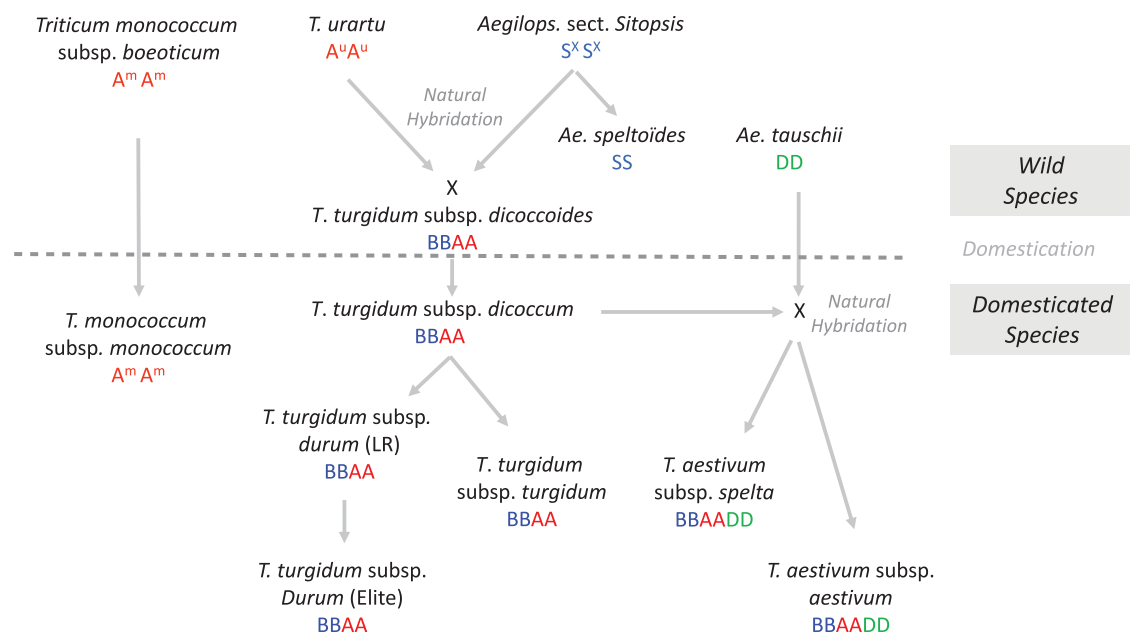


FIGURE 3.1 – The evolutionary and genome relationships between the cultivated wheat species and related wild species used in this study. The origin of the b genome is still unclear but it comes from a species of *ae. Sect. Sitopsis*, represented here by *ae. Speltoïdes*. *Triticum turgidum* subsp. *Durum* was separated in two groups : landrace populations (lr) grown before the green revolution and elite cultivars released afterwards.

The higher tolerance of durum wheat to heat and drought stress compared with bread wheat is still debated (Marti and Slafer, 2014; Giunta et al., 2019), and comparisons have generally been performed based only on yield or yield components, and our knowledge regarding the differences in plant responses to abiotic constraints is still limited (Allahverdiyev, 2015). Physiological responses at fine scales can be very different from macroscopic responses, which emerge as properties of numerous underlying processes, integrated over the whole growth cycle (Tardieu and Parent, 2017).

Such interspecific genetic variability can originate not only from genes or alleles currently residing in wild species but also from ploidy itself. Traits expressed in wild diploids might not predict the traits in progeny of higher ploidy after synthetic hybridizations. Indeed, ploidy itself can provide agronomic advantages or disadvantages (Comai, 2005), for example by conferring resistance to biotic and abiotic stresses (Tu et al., 2014; Schoenfelder and Fox, 2015; Hias et al., 2018).

In this study, we analysed 12 groups of wheat-related species and subspecies (five accessions each) for traits linked to vegetative structure and development, and to leaf expansion and transpiration, together with their responses to ranges of ‘non-stressing’ temperatures and evaporative demand during the vegetative phase. We used an original experimental protocol and selected model formalisms adapted to this broad genetic diversity in order to compare the intra- and interspecific diversities. We determined that the genome type, ploidy level, and phylogeny act together to structure the genetic diversity, and that breeding pressure may have indirectly pushed these traits in specific directions. Overall, in addition to providing new formalisms of trait responses together with model parameters to be included in process-based models for large-scale simulations, this study indicates potential sources of varietal improvement for modern wheat germplasms that currently reside in related species.

3.2 Materials and methods

3.2.1 Plant material

We analysed 12 wheat-related species and subspecies (Fig. 3.1) and selected five accessions per species in order to maximize the genetic diversity within each (Supplementary Tables S1, S2). This set included wild and cultivated species with different levels of ploidy, as follows. Wild diploid species : *Triticum monococcum* subsp. *boeoticum* (Am genome), *T. urartu* (Au genome), *Aegilops tauschii* (D genome), and *Ae. speltoides* (S genome, related to species belonging to *Ae. sect. sitopsis* from which the B genome originated). Cultivated diploid species : einkorn wheat *T. monococcum* subsp. *monococcum* (Am genome), domesticated from *T. monococcum* subsp. *Boeoticum*. Wild tetraploid species : wild emmer wheat *T. turgidum* subsp. *dicoccoides* (AB genome), originating from a natural hybridization event between *T. urartu* and an unknown species belonging to *Aegilops sect. sitopsis*. Cultivated tetraploid species : cultivated emmer wheat *T. turgidum* subsp. *dicoccum* (AB genome), domesticated from wild emmer wheat, rivet wheat (or poulard wheat, *T. turgidum* subsp. *turgidum*; AB genome), which evolved from *T. turgidum* subsp. *dicoccum*, and durum wheat (*T. turgidum* subsp. *durum*; AB genome), which was domesticated from *T. turgidum* subsp. *dicoccum*. We separated these into two groups : landraces from before and elite cultivars released after the ‘Green Revolution’. Cultivated hexaploid wheats : spelt wheat (*T. aestivum* subsp. *spelta*, ABD genome) and bread wheat (*T. aestivum* subsp. *aestivum*, ABD genome) originating from natural hybridization events between *T. turgidum* subsp. *dicoccum* and *Ae. tauschii*.

3.2.2 Plant growth conditions

All experiments were conducted in the plant phenotyping platform Pheno-dyn, part of the plant phenotyping platforms of M3P (Montpellier, France, https://www6.montpellier.inrae.fr/lepse_eng/M3P). Briefly, this platform is set up with balances to monitor plant transpiration, displacement transducers to moni-

tor leaf expansion, a set of climatic sensors, and automatic drip-feed systems. Air temperature, relative humidity, and photosynthetic photon flux density (PPFD) at the plant level were measured and stored every 15 min in a datalogger (CR10X, Campbell Scientific Ltd, Leicestershire, UK). The temperature of the leaf meristematic (growth) zone was measured using thermocouples inserted vertically inside the sheath of leaf 5 of 3–6 plants, and used to calculate meristem-to-air VPD. VPD was controlled based on measurements of relative humidity and temperature.

Seeds were imbibed for 24 h at 4 °C on water-saturated filter papers in Petri dishes, then placed at room temperature for 24 h before being returned to 4 °C for 48 h. Six germinated seeds each were planted in 9-l plastic pots (20 cm diameter) filled with a 40 :60 (v :v) mixture of clay and organic compost, and then thinned to three uniform plants per pot when the third leaf emerged. A total of three pots (nine plants) per accession were used, together with four pots containing non-transpiring artificial plastic plants mimicking real ones that were used to calculate evaporation from the soil surface. Sowing was spread over 11 weeks in order to stagger measurements in the growth chambers. A fully randomized design was used. Pots with fewer than three emerged plants were replaced with newly sown ones during the 12th week.

Plants were grown for the first 3 weeks in a growth chamber where temperature and VPD were controlled at 20.6 ± 2.6 / 16.3 ± 1.9 °C and 1.4 ± 0.4 / 0.9 ± 0.3 kPa (day/night), respectively. PPFD at plant level averaged $300 \mu\text{mol s}^{-1} \text{m}^{-2}$ during the 12-h photoperiod. Plants were watered daily to maintain the soil water potential above -0.05 MPa. At 3 weeks after plant emergence, two pots (six plants) of each accession were transferred to another growth chamber and were subjected to varying temperature and VPD conditions (scenario 1) over a 4-d period, whilst one pot (three plants) was subjected to a second scenario (scenario 2) with more extreme temperatures (Supplementary Fig. S1). This period of treatment was selected because it corresponded to the appearance of leaf 5 in all the genotypes, and it allowed the comparison of absolute values of leaf elongation rate

because internode elongation had not started yet. At later plant stages, the automatic measurement of elongation with displacement transducers did not make it possible to discriminate between the elongations of internodes and leaves. For both scenarios, the plants were harvested at the end of the 4-d period. We used the two scenarios applied in different growth chambers and on different plants to compare the response of leaf elongation rate (LER) to temperature on plants at similar development stages over the full range of target temperatures.

In scenario 1, each night the plants were subjected to two air temperatures : a reference temperature of 20 °C for 6 h, and a target temperature of either 12, 16, 24, or 28 °C for the other 6 h. The lowest temperature between the reference and the target was applied first. For the first 3 d, during the light period the plants were subjected to two VPD values : a 6-h period with a reference VPD of 1.3 kPa and a 6-h period with a target VPD value of either 0.5, 2.5, or 3.0 kPa. Again, the lowest VPD value was applied first. The daytime air temperature was set in such a way that the target air VPD values were attainable (20 °C for 0.5 kPa, 26 °C at 2.5 kPa, and 28 °C at 3.0 kPa), but in all cases the temperature was similar for both the reference and the tested VPD. On the fourth day, the air temperature and VPD were maintained at 20 °C and 1.3 kPa, respectively. In scenario 2, a protocol similar to that of scenario 1 was applied but with more extreme temperatures and VPD values. The target night-time air temperatures were 4, 8, 32, and 36 °C, and the target day-time VPD values were 0.3, 2.5, and 4 kPa. However, we found that VPD was not sufficiently stable under these conditions, and hence in later analyses we used the data for plant responses to night temperature not the data obtained during the light period. As in scenario 1, on the fourth day the temperature was maintained at 20 °C and VPD at 1.3 kPa.

3.2.3 Response of leaf elongation rate to temperature

The youngest growing leaf (which was leaf 4 or 5 during the first night, and leaf 5 or 6 for the last day of measurements) was attached to a rotary displacement transducer (601–1045 Full 360 Smart Position Sensor; Spectrol Electronics,

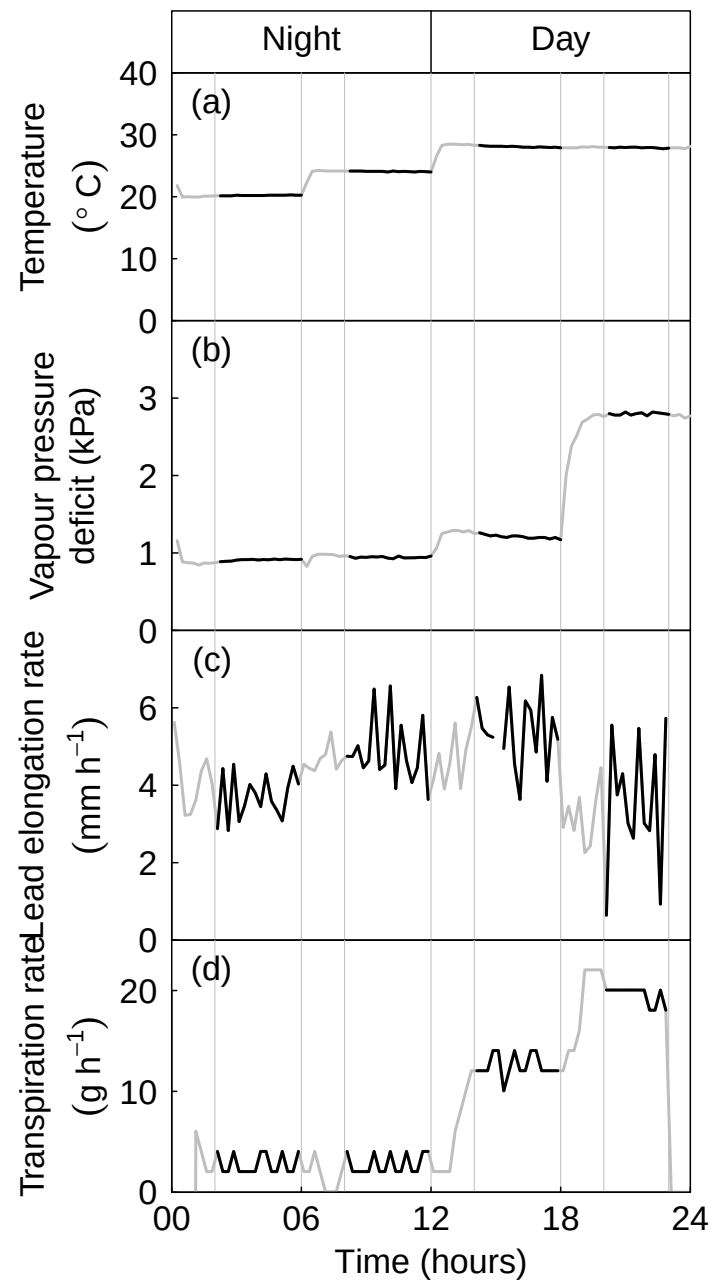


FIGURE 3.2 – Illustration of time-courses of air temperature and vapour-pressure deficit (*vpd*) applied to plants, and example responses. (a) air temperature and (b) *vpd* regimes, (c) leaf elongation rate (*ler*), and (d) transpiration rate (*tr*). Measurements were taken during day 3 of scenario 1 (see supplementary Fig. S1). The vertical grey lines indicate the 4-h periods during which the environmental conditions were considered to be stable, and the mean values of *ler* and *tr* were determined during these periods (black lines). The data in (c, d) are from *t. Turgidum* subsp. *Durum*, cultivar *brumaire*. All measurement were performed on the same plant.

Ltd, Wiltshire, UK). Leaf elongation was transmitted to the sensor via a pulley attached to it, which carried a thread attached to the leaf tip and to a 10-g counterweight. Leaf elongation was recorded on the datalogger every 15 min. The response of LER to the temperature of the leaf growth zone was calculated from both the temperature scenarios during the night period in order to avoid confounding effects of temperature and evaporative demand. In all cases, LER stabilized within 2 h after the beginning of the night period or after the change in temperature (Fig. 3.2; checked on all plants). LER was therefore averaged over the last 4 h of each period. In wheat, the LER of a given leaf is not steady and depends on leaf rank. Therefore, in order to compare leaves of different lengths (relative to their final length) and rank, the LER at the target temperature was normalized by LER at the reference temperature (20 °C) measured during the same night. To determine the cardinal temperatures of normalized LER, the data were fitted with a 3-parameter segmented linear-function equation constrained to be equal to 1 at the reference temperature :

$$f(T) = \begin{cases} \frac{(T - T_{min})}{(T_{ref} - T_{min})}, & T < T_{opt} \\ \frac{(T - T_{min})(T_{opt} - T_{min})}{(T_{ref} - T_{min})(T_{opt} - T_{max})}, & T \leq T_{opt} \end{cases} \quad (3.1)$$

where T is the temperature of the leaf meristematic zone, T_{ref} is the reference temperature at which f(T) is equal to 1, T_{min} is the base temperature at which the normalized LER equals zero, T_{opt} is the temperature at which the normalized LER is maximal, and T_{max} is the supra-optimal temperature at which the normalized LER equals zero (all temperatures in °C; see Table 3.1 for a full list of definitions of traits). Equation (3.1) was compared with a beta function (Wang and Engel, 1998) and an Arrhenius-type function (Parent and Tardieu, 2012); Supplementary Fig. S2). Regressions were performed using the BFGS method of the “optim” function in the R software (www.r-project.org). The goodness of fit of the three equations was evaluated using the root mean-squared error and models

were compared using the Akaike information criterion (AIC; Akaike, 1974) and the Bayesian information criterion (BIC; Schwarz, 1978).

3.2.4 Responses of leaf elongation rate and transpiration to vapour-pressure deficit

The responses of LER and transpiration to leaf–air VPD were calculated during the day in scenario 1 only, with LER being measured as described above. Transpiration rate (TR) was calculated from the weight loss of each pot divided by the total leaf (sheath + blade) area. Direct evaporation from the soil was estimated by measuring the weight loss of pots containing artificial plants, which were watered to maintain a soil water potential similar to that of the pots containing real live plants. The weight of each pot was measured every minute, and then the average was calculated and stored every 15 min. The LER of the youngest growing leaf measured at the 15 min intervals was first expressed with units of thermal time in order to remove the effects of temperature. It was then averaged over the period of stabilized VPD (i.e. the last 4 h of the 6-h period) and then normalized by the mean TR during those 4 h to remove the effects of plant size and leaf area. The normalized LER and TR data were then fitted with a linear model constrained to 1 at the reference VPD, so that only one parameter described each response, the sensitivities to VPD of LER (s_{LER} , kPa^{-1}) and TR (s_{TR} , kPa^{-1}).

3.2.5 Traits linked to rates of development, leaf expansion, and transpiration

The progress of plant development on the main stem was determined from the leaf stage (LS), adapted from Haun (1973) :

$$LS = n + \frac{l}{L} \quad (3.2)$$

where n is the number of ligulate leaves, l is the exposed length of leaf $n+1$

TABLE 3.1 – Summary of the phenotypic structural, rate, and response traits measured in this study

Trait type	Abbreviation	Units	Definition
Structural	LA2	cm ²	Leaf area of leaf 2
	rlw2	–	Length/width ratio for leaf 2
	TL	Leaves plant ⁻¹	Total number of leaves per plant (main stem + tillers)
	LAT	cm ² plant ⁻¹	Leaf area per plant (sheath + leaf blade)
	MLA	g m ²	Specific leaf mass
	rLA	–	Ratio of sheath to blade leaf area
	sLAK	–	Slope of the relationship between individual leaf area and leaf rank, normalized by the area of leaf 2
	PN	g N plant ⁻¹	Shoot nitrogen mass per plant
	SLN	g N m ²	Specific leaf (sheath + blade) nitrogen
Rates	LERm	mm h ⁻¹	Maximal elongation rate of leaf 5 at 20 °C and 1.0 kPa vapour pressure deficit
	TR	g H ₂ O h ⁻¹ m ⁻²	Plant transpiration rate per area at 20 °C and 1.3 kPa vapour pressure deficit
	LAR	leaves °Cd ⁻¹	Average leaf appearance rate (thermal time, base 0 °C)
Responses	Tmin	°C	Minimum temperature for leaf elongation
	Topt	°C	Optimum temperature for leaf elongation
	Tmax	°C	Maximum temperature for leaf elongation
	sLER	kPa ⁻¹	Response of leaf elongation rate to vapour pressure deficit
	sTR	kPa ⁻¹	Response of transpiration rate to vapour pressure deficit

at the time of measurement, and L is the final length of the blade of leaf n+1. The exposed length of a leaf was measured with a ruler as the distance from leaf tip to the upper collar of the sheath tube. LS was determined on all plants in scenario 1 from the day before to the last of the LER measurements. The rate of leaf appearance (LAR, leaves °Cd⁻¹; inverse of the phyllochron) was calculated as the slope of the relationship between LS and thermal time (base 0 °C).

Because the elongation of leaf 5 was measured during at least one night for all the accessions, the absolute LER of all accessions was compared based on the LER of leaf 5 (LERm, mmh⁻¹) measured at 20 °C, 2 - 3 d after leaf appearance (i.e. when LER was at its maximum value). The transpiration rate at 20 °C and VPD=1.3 kPa was calculated as the mean of absolute transpiration rate as measured on each of the four nights during the 4-h period of stabilised VPD. At the end of the 4-d period, the whole-plant leaf area was measured (see below). The transpiration rate per unit leaf area (TR, gcm⁻²h⁻¹) was then calculated by first deducting the evaporation recorded in pots with artificial plants, and then dividing by the whole-plant leaf area.

3.2.6 Traits linked to plant structure

All plant structural traits were measured after the fourth day of LER and TR measurements. The total number of leaves per plant ($TL, leaf\ plant^{-1}$) was counted. The leaf blades and sheaths of the main stem and tillers were then separated and their lengths, widths, and surface areas were measured using an electronic planimeter (LI-3100C, LI-COR) and used to calculate various traits linked to plant structure. The total leaf area per plant was calculated ($LAT, cm^2\ plant^{-1}$). The length-to-width ratio ($rlw2$, dimensionless) and surface area ($LA2, cm^2\ plant^{-1}$) of the blade of main-stem leaf 2 were calculated as proxies of early plant vigour (Rebetzke et al., 2004). It was correlated to seed size (Supplementary Fig. S3). We also determined the slope of the relationship between main-stem individual leaf-blade surface area and leaf rank normalized by the surface area of the blade of leaf 2 ($sLAK$, dimensionless), and the ratio of leaf-blade to total leaf (sheath + blade) surface area (rLA , dimensionless).

The main-stem leaf blades and sheaths of each plant were pooled and oven-dried at 85 °C until they reached constant mass. The dry mass of the samples was measured and specific leaf mass (mass per leaf area, $MLA; gDMcm^{-2}leaf$) was calculated by dividing total plant dry mass by the total leaf area LAT . The samples were then milled and their total N concentration (N mass per unit dry mass) was determined using the Dumas combustion method (Horwitz, 1980) using a FlashEA 1112 N/Protein Analyzer (ThermoFisher Scientific) and the mass of N per plant was calculated ($PN, gN\ plant^{-1}$). Specific leaf N ($SLN, gNm^{-2}leaf$) was calculated by dividing PN by the total leaf area of the plant.

subsectionData analyses All data analyses were carried out using the R software (www.r-project.org). ANOVAs on each variable were performed with the function `aov`, considering species as factors and genotypes as replicates. Confidence intervals of differences between means of pairs of species were calculated using a Tukey test (R function `TukeyHSD`) and the significance of these differences were calculated with the function `HSD.test` with $\alpha = 0.05$. The procedure

for building correlograms consisted of the following steps : (i) normality Shapiro–Wilk test ; (ii) calculation of Spearman correlations ; and (iii) calculation of q-values (P-value corrected for false-discovery rate). Significant correlations were considered at $q < 0.01$. To compare the intra- versus interspecific variability of each trait, we calculated the ratio of the corresponding coefficient of variation of standard deviation. Principal component analyses (PCAs) were performed using the function PCA of the FactorMineR R package (Lê et al., 2008). We considered several sets of traits (e.g. all traits, or traits linked to sensitivities to temperature and VPD only), with data at the accession level. Values of each species on the first five axes of the PCA were determined by calculating the barycentre of all accessions in each species. A hierarchical clustering on principal components (HCPC) analysis of species was then performed based on these first five axes of the PCA using the HCPC function of the FactorMineR package. In the case of the analysis of the set of traits comprising only the five traits linked to sensitivities, and because the phenotypic space was then automatically reduced to five dimensions (five variables), the hierarchical clustering analysis was carried out based on these five traits rather than on principal components of the PCA. Approximately unbiased (AU) P-values (%) were calculated using the pvclust function of the pv.clust R package, with 1000 bootstrap replications.

3.3 Results

3.3.1 Interspecific variability of traits related to plant structure and rates of development, leaf expansion, and transpiration, and their response to temperature and VPD

We measured nine traits related to plant structure and three traits related to the rate of plant development and growth in 60 accessions covering a wide genetic diversity in 12 groups (five accessions in each group) of wheat-related species and subspecies (Fig. 3.1). Traits related to the rates of plant development and growth (LAR, LERm) and plant size (LA2, LAT, and PN) showed a high variability

lity and significant differences between species (Fig. 3.3, see Supplementary Table S3 for results of ANOVA and Supplementary Table S4 for comparing intra- vs interspecific variations). Among these traits, LAR increased with ploidy level (Fig. 3.3L, $P < 0.001$), while LA2, LAT, and PN were higher for the diploid species than for the polyploid species ($2n > 4n > 6n$; Fig. 3.3A, D, H; all $P < 0.001$). Traits linked to plant structure but not expected to be linked to plant size (i.e. rlw2, rLA, and sLAK) showed no clear patterns with ploidy (Fig. 3.3B, F, G), and the other traits (TL, SLN, and TR) also showed no clear pattern. MLA and SLN both showed low overall variability compared to the other traits that we measured.

We examined the sensitivities of LER to temperature and VPD in the 60 accessions. Each night, a temperature between 4 °C and 36 °C was applied for 6 h together with a period of 6 h at the reference temperature of 20 °C (Supplementary Fig. S1). An example for *T. turgidum* subsp. *durum* cultivar Brumaire is shown in Fig. 3.2, where temperature was first maintained at 20 °C for 6 h and then increased to 24 °C. The LER oscillated around 4 mm h⁻¹ during the first period and then increased rapidly to $\approx 5 \text{ mm h}^{-1}$ when temperature was increased (Fig. 3.2C). We normalized the LER measured at the tested temperature by that measured at the reference temperature on the same night in order to examine the effect of the temperature change regardless of differences in LER due to differences in leaf rank or time since leaf emergence (Fig. 3.4A). We fitted the data to three different function equations to estimate the cardinal temperatures of LER, namely a beta function (Wang and Engel, 1998), an Arrhenius function (Parent and Tardieu, 2012), and a segmented linear equation (Eqn. 3.1). For all accessions, the segmented linear equation provided the best fit (Supplementary Fig. S2, Supplementary Table S5), and all data were therefore analysed using this model (Fig. 3.4A provides an example of the type of response we obtained).

The three cardinal temperatures showed moderate but significant variability between the 12 groups of species and subspecies (Fig. 3.4D–F), with Tmin ranging from -1.7 °C to 3.9 °C, Topt from 24.0 °C to 31.6 °C, and Tmax from 36.2 °C to 53.4 °C (Supplementary Table S6 for values at the accession level and Sup-

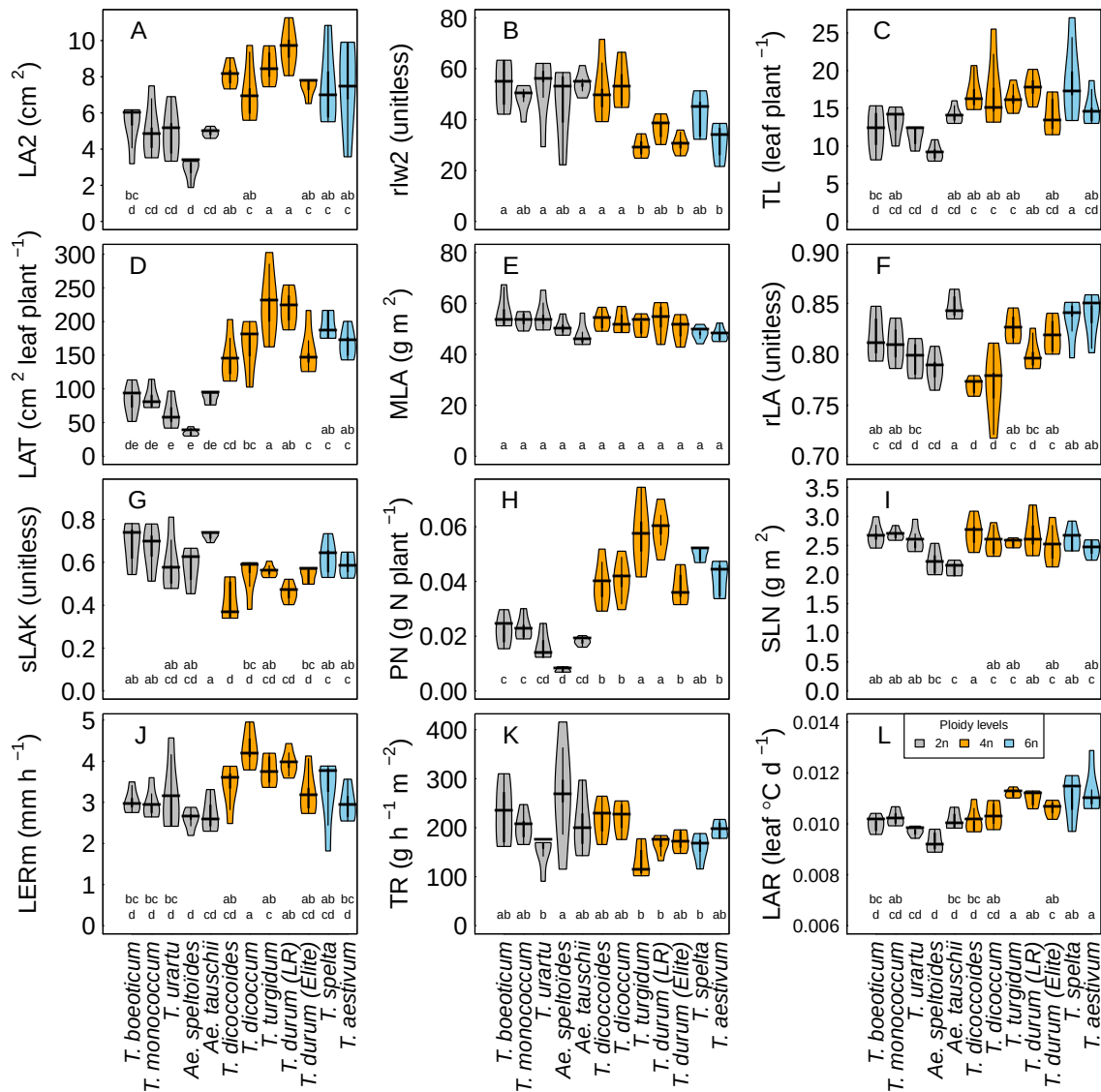


FIGURE 3.3 – Traits related to plant structure and rate of development, leaf expansion, and transpiration for the 12 groups of wheat-related species and subspecies. (a) surface area of leaf 2 (la2), (b) length-to-width ratio of leaf 2 (rlw2), (c) total number of leaves per plant (tl), (d) total leaf surface area per plant (lat), (e) specific leaf mass (mass per leaf area, mla), (f) ratio of leaf blade to total leaf surface area (rla), (g) slope of leaf blade surface area versus leaf rank, normalized by the surface area of leaf 2 (slak), (h) total n mass per plant (pn), (i) specific leaf n (sln), (j) leaf elongation rate of leaf 5 at 20 °c and vapour-pressure deficit (vpd) of 1.0 kpa (lerm), (k) transpiration rate at 20 °c and 1.3 kpa vpd (tr), and (l) leaf appearance rate (lar; thermal time, base 0 °c). Data are means ($\pm$ sd) for $n=5$ accessions. *Triticum turgidum* subsp. *Durum* was separated in two groups : landrace (lr) populations grown before the green revolution and elite cultivars released afterwards.

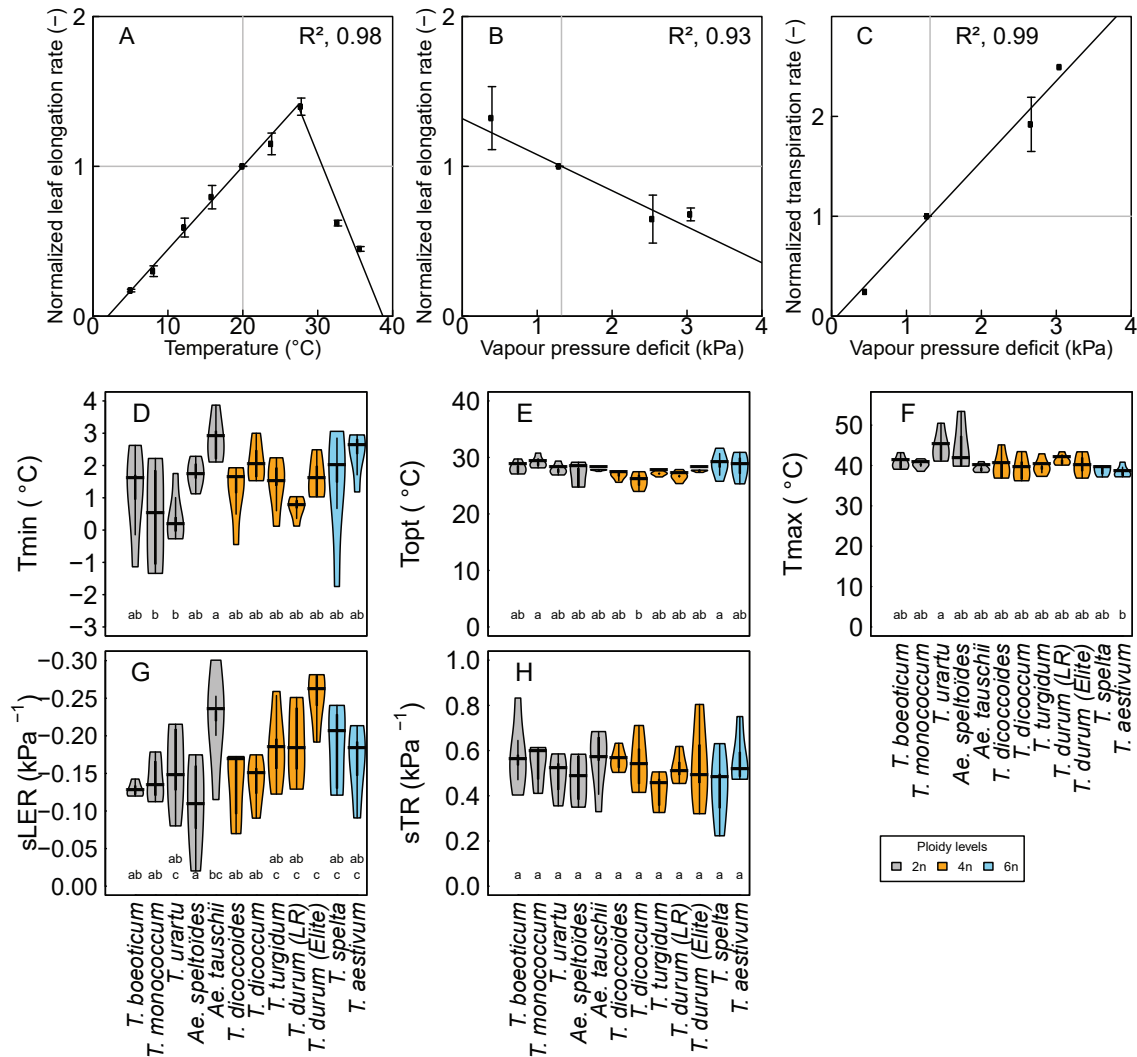


FIGURE 3.4 – Typical responses of leaf elongation rate (*ler*) and transpiration rate (*tr*) to temperature and vapour-pressure deficit (*vpd*), and variation in cardinal temperatures and sensitivities of *ler* and *tr* to *vpd* across the 12 groups of wheat-related species and subspecies. (a, b) normalized *ler* versus (a) temperature of the leaf growth zone and (b) *vpd*, and (c) normalized *tr* versus *vpd* for *triticum turgidum* subsp. *Durum* cultivar *neodur*. Data are means ($\pm$ sd) for $n=6$ independent replicates, except for temperatures $<12^\circ\text{C}$ and $>28^\circ\text{C}$ where $n=3$. The black lines are linear (a, c) or segmented-linear (c) regressions, and the grey horizontal and vertical lines indicate the normalization point constraining the fitted function equations. (d) minimum (*tmin*), (e) optimum (*topt*), and (f) maximum (*tmax*) temperatures for *ler* of the leaf meristematic zone for the 12 groups of wheat-related species and subspecies. (g, h) variability of the sensitivity of (g) *ler* (*sler*) and (h) *tr* (*str*) to *vpd*. Data in (d–h) are means ($\pm$ sd) for $n=5$ accessions. *Triticum turgidum* subsp. *Durum* was separated in two groups : landrace (*lr*) populations grown before the green revolution and elite cultivars released afterwards.

plementary Table S7 for values at the species level). There was no clear pattern between ploidy levels, genomes, or between wild versus domesticated species.

We quantified the responses of LER and TR to VPD during the day using an approach similar to that described above for the response of LER to temperature during the night. Each day, a VPD between 0.5 kPa and 3.0 kPa was applied for 6 h together with a period of 6 h at the reference VPD of 1.3 kPa. Examples of time courses we obtained for LER and Transpiration are shown in Fig. 3.2C, D. LER oscillated around 5.5 mmh^{-1} at the reference VPD and decreased to 4 mmh^{-1} when VPD was increased to 2.7 kPa. At the same time, transpiration rate, which stabilized at 13 gh^{-1} at the reference VPD, increased to 19 gh^{-1} . LER and TR (plant transpiration rate per area) at the tested values of VPD were normalized by their corresponding value at the reference VPD to remove the effects of leaf rank, time since leaf emergence, and plant leaf area. Both responses to VPD were fitted with a linear model, the slope of which is the sensitivity of LER (sLER) or TR (sTR) to VPD (Fig. 3.4B, C). sLER showed significant variability between species and subspecies (Fig. 3.4G), ranging from $\sim 0.268 \text{ kPa}^{-1}$ to $\sim 0.037 \text{ kPa}^{-1}$, while sTR (Fig. 3.4H) showed lower variability (Supplementary Table S6 for values at the accession level and Supplementary Table S7 for values at the species level). However, at the level of single traits, we did not observe any clear patterns between the 12 groups of species.

3.3.2 Ploidy level, phylogeny, and breeding together explain the phenotypic space of all traits

An understanding of the structure of the interspecific diversity and how the intraspecific diversity is nested inside is made possible by considering the multi-dimensional space constituted by all traits together (the ‘phenotypic space’). We found significant correlations between most traits (Fig. 3.5, Supplementary Table S8). The strongest positive correlations were between traits related to plant structure and plant size, but we also found significant correlations between traits of different types, for example between response traits and traits related to plant

structure.

We used principal component analyses (PCAs) to reduce the number of dimensions of the phenotypic space to five. PCA based on all traits at the accession level (Fig. 3.6A–D) showed that within each group of species, the diversity of values for the first three components was low compared to the overall diversity observed at the interspecific level (Fig. 3.6B, D). The first principal component (PC1) explained 30% of the observed variance and was mainly explained by traits related to development rate (e.g. LAR and LERm) and traits linked to plant size (e.g. LAT and TL; Fig. 3.6A), and it discriminated diploid species from tetraploid and hexaploid species (fig. 3.6B). Diploid species had a smaller leaf 2 area (LA2), lower leaf appearance rate (LAR), fewer leaves (TL), and a lower total leaf area per plant (LAT) compared with tetraploid and hexaploid species (Fig. 3.3). However, because most of the wild species used in this study were diploids, and because *T. turgidum* subsp. *dicoccoides* was close to the origin of PC1, we are not able to conclude if this axis discriminated species based on domestication or on ploidy level.

The hierarchical clustering on principal components (HCPC) analysis based on the first five components of the PCA classified the 12 groups into six clusters (fig. 3.6E). The first branching of the dendrogram (corresponding to the highest relative distance) separated the diploid and polyploid species. For the diploids, the second order of the dendrogram separated genome A from the other two diploid genomes, whilst for the polyploids it separated *T. turgidum* subsp. *dicoccoides* and subsp. *dicoccum* from the other polyploid species. The third order of clustering grouped together *T. turgidum* subsp. *durum* elite cultivars and the two hexaploid species (*T. aestivum* subsp. *aestivum* and subsp. *spelta*), while *T. turgidum* subsp. *durum* landraces were grouped with *T. turgidum* subsp. *turgidum*. The wild and cultivated emmer wheat species (*T. turgidum* subsp. *dicoccoides* and subsp. *dicoccum*) were grouped together. Thus, the phenotypic space based on the 17 traits that we studied discriminated wheat-related species and subspecies based on their ploidy level, domestication, and breeding, and the resulting

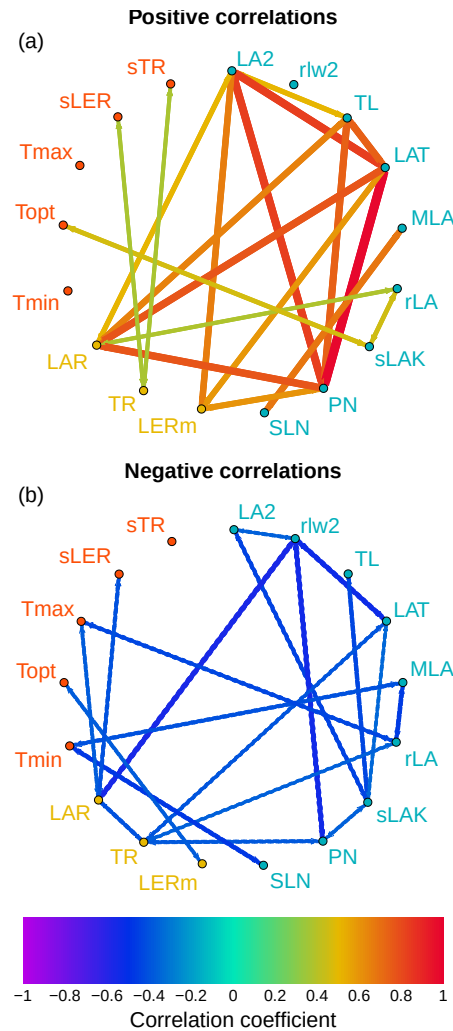


FIGURE 3.5 – Correlograms of spearman's correlation coefficients between traits at the accession level across the 12 groups of wheat-related species and subspecies. See table 3.1 for definitions of variables. (a) positive correlations and (b) negative correlations. Traits in blue are related to plant structure, those in yellow are related to absolute rates of development, expansion, or transpiration, and those in red are related to the responses of ler to temperature and vpd, and of tr to vpd. The colour and thickness of the lines indicate the value of the correlation coefficient. Only significant correlations are shown (q -value <0.01).

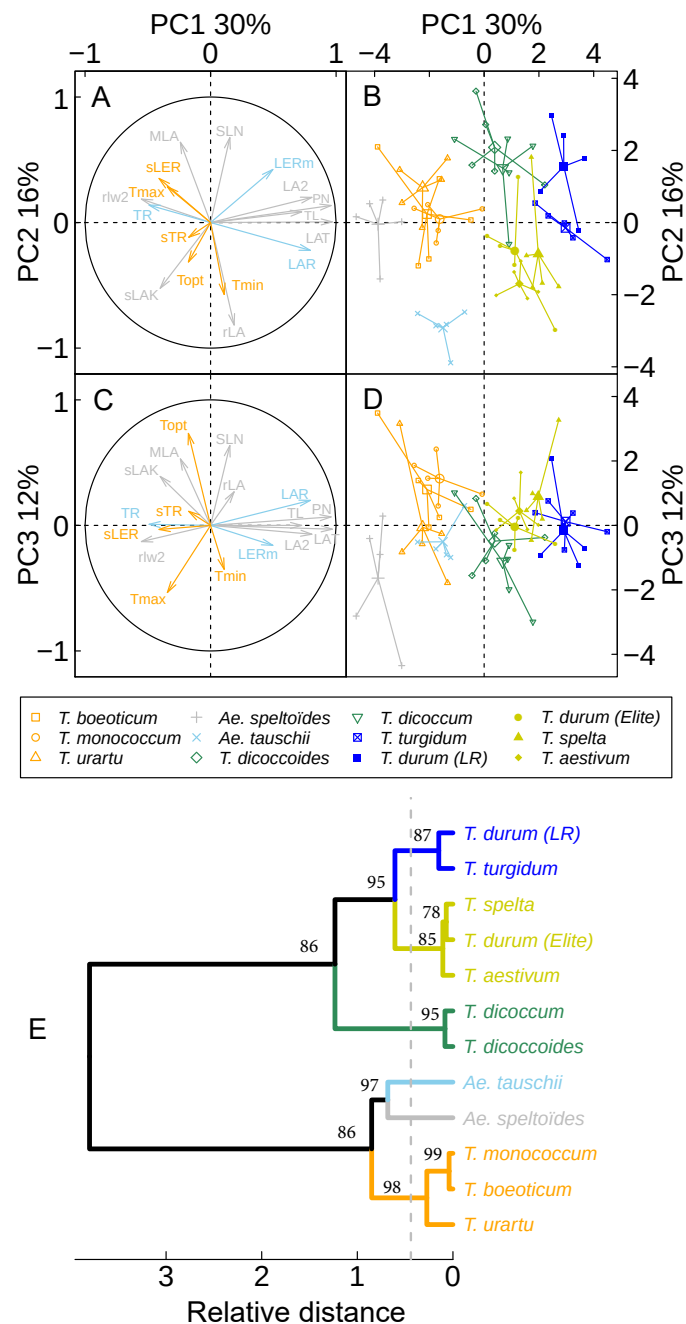


FIGURE 3.6 – Principal component analysis (pca) and hierarchical clustering on principal components (hcpc) analysis for the 17 traits measured for the 60 accessions across the 12 groups of wheat-related species and subspecies. See table 3.1 for definitions of variables. (a, c) representation of structural traits (grey), absolute rates of development, expansion, and transpiration (blue), and responses of leaf elongation rate to temperature and vapour-pressure deficit (vpd), and of transpiration rate to vpd (orange) for (a) pc1 versus pc2 and (c) pc1 versus pc3. (b, d) factor maps of the accessions and the barycentre of each cluster identified by the hcpc analysis corresponding to (a) and (b), respectively. (e) dendrogram of the hcpc analysis performed on the barycentre of each species in the first five axes of the pca. The dashed vertical line shows the threshold of relative distance for clustering species. The numbers indicate the approximately unbiased (au) p-value obtained from 1000 bootstrap replications.

clustering of species fitted well with the phylogeny of the 12 groups of species and subspecies.

3.3.3 Phylogeny structures the phenotypic space of traits related to the rate of plant development and growth, and their responses to temperature and VPD

We conducted a similar analysis combining PCA and HCPC analysis but based on traits linked to plant structure only (Supplementary Fig. S4) and obtained similar results to when all 17 traits were used (with marginal differences), suggesting that the structure traits drove the clustering of the 12 groups of species. We therefore performed a similar analysis on traits linked to rates and sensitivities whilst excluding structure traits (Fig. 3.7). The first axis discriminated species with high LAR from those with low TR, sLER, and Tmax, while the second axis discriminated species with high Tmax from those with low Tmin, Topt, and sTR. Topt was negatively correlated with LERm (Supplementary Table S8) and these were the main contributing traits to the third axis. The resulting HCPC analysis first segregated two groups (Fig. 3.7E). The first consisted of the cultivated tetraploid and hexaploid species, as well as *Ae. tauschii*, while the second consisted of the wild and cultivated diploids and the wild tetraploid species *T. turgidum* subsp. *dicoccoides*. Within these two groups, the HCPC analysis produced a few unexpected results, with *T. turgidum* subsp. *dicoccoides* clustered with *T. monococcum* (subsp. *monococcum* and subsp. *beoticum*), *T. aestivum* subsp. *spelta* clustered with a group of three *T. turgidum* subspecies, and *Ae. tauschii* in the same cluster as *T. aestivum* subsp. *aestivum*. Overall, these results indicated that the rate of development of growth and the sensitivities to temperature and VPD are well-conserved, with low variability within species compared to the phenotypic space of wheat-related species, and that their interspecific variability is mainly structured by phylogeny.

Finally, we analysed the structure of the phenotypic space of response traits

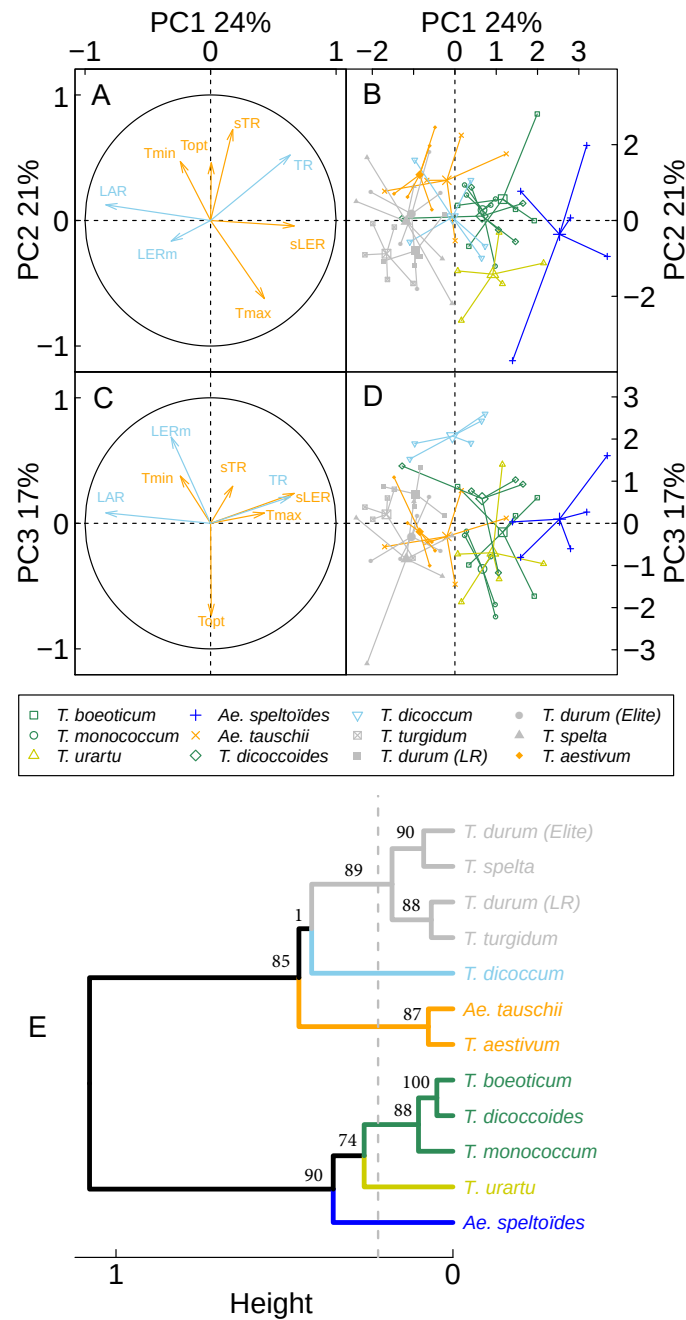


FIGURE 3.7 – Principal component analysis (pca) and hierarchical clustering on principal components (hcpc) analysis for the traits related to the absolute rates of development, expansion and transpiration, and response to temperature and vapour-pressure deficit (vpd) for the 60 accessions across the 12 groups of wheat-related species and subspecies. See table 3.1 for definitions of variables. (a, c) representation of absolute rates of development, expansion, and transpiration rate (blue) and responses of leaf elongation rate to temperature and vpd, and transpiration rate to vpd (orange) for (a) pc1 versus pc2 and (c) pc1 versus pc3. (b, d) factor maps of the accessions and the barycentre of each cluster identified by the hcpc analysis corresponding to (a) and (b), respectively. (e) dendrogram of the hcpc analysis performed on the barycentre of each species in the first five axes of the pca. The dashed vertical line shows the threshold of relative distance for clustering species. The numbers indicate the approximately unbiased (au) p-value obtained from 1000 bootstrap replications.

only, knowing that the values of individual traits did not show any clear trends (Fig. 3.4). Because we only measured five response traits, the phenotypic space was automatically reduced to five dimensions (five variables), which appeared as independent in the PCA (Fig. 3.8A), we then performed a hierarchical clustering analysis based on these five traits. The resulting dendrogram again showed clear patterns (Fig. 3.8E). The clustering first segregated two wild diploid species, *Ae. speltoïdes* and *T. uratu*, from the other species, indicating that they had significantly different responses to temperature and evaporative demand. Apart from this, the analysis clustered three other groups, the first of which comprised all tetraploid species, regardless of wild/domesticated subgroups, the second group contained the two wild and domesticated *T. monococcum* species (subsp. *monococcum* and *boeiticumm*, respectively), and the third group contained the two hexaploid species (*T. aestivum* subsp. *spelta* and subsp. *aestivum*) and *Ae. tauschii*, that is, all species with a D genome. This analysis therefore indicated that the interspecific variability in temperature and VPD responses was structured by both the ploidy level and the phylogenetic relationships. However, domestication and/or breeding pressure does not appear to have played a role in modifying these responses..

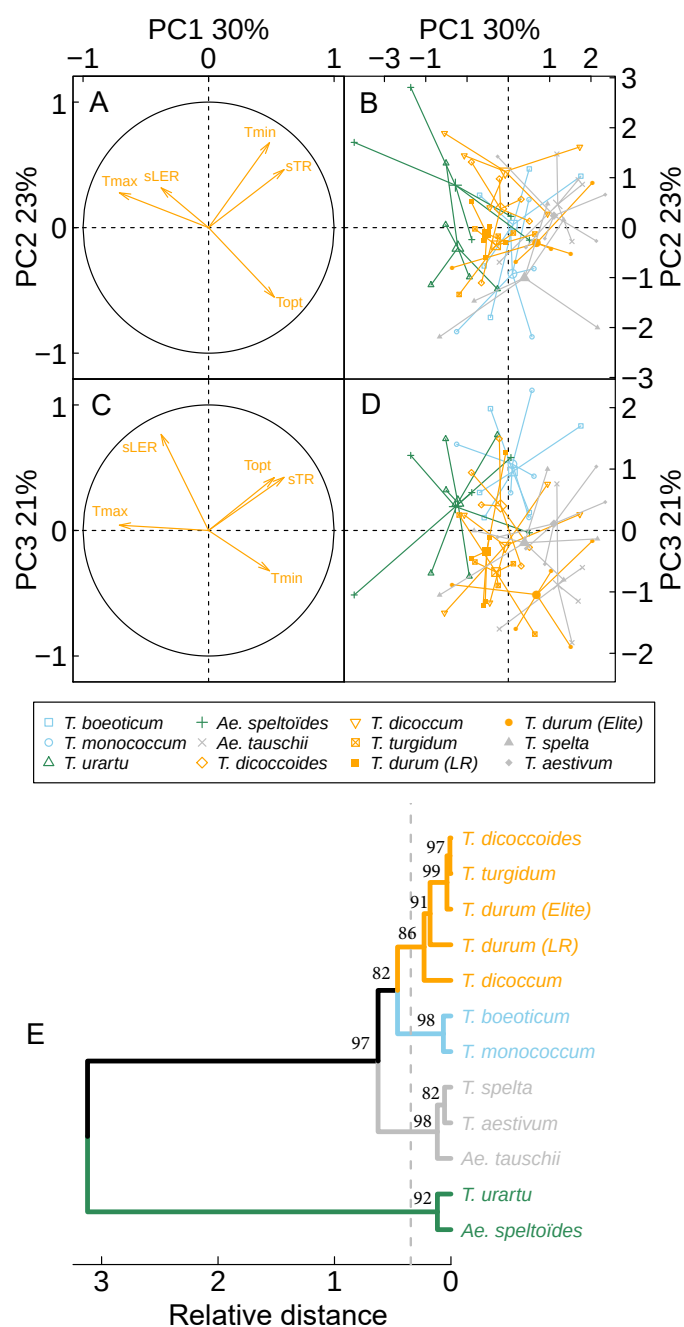


FIGURE 3.8 – Principal component analysis (pca) and hierarchical clustering on principal components (hcpc) analysis based on trait values related to response to temperature and vapour-pressure deficit (vpd) for the 60 accessions across the 12 groups of wheat related species and subspecies. See table 3.1 for definitions of variables. (a, c) representation of the response of leaf elongation rate to temperature and vpd, and transpiration rate to vpd for (a) pc1 versus pc2 and (c) pc1 versus pc3. (b, d) factor maps of the accessions and the barycentre of each cluster identified by the hcpc analysis corresponding to (a) and (b), respectively. (e) dendrogram of the hcpc analysis performed on the barycentre of each species in the five variables. The dashed vertical line shows the threshold of relative distance for clustering species. The numbers indicate the approximately unbiased (au) p-value obtained from 1000 bootstrap replications.

3.4 Discussion

3.4.1 Modelling the sensitivities of leaf elongation and transpiration rates to temperature and vapour pressure deficit for a large genetic diversity

The responses of expansion processes of plant organs to temperature have been studied extensively (reviewed in Parent et al., 2012). Most of these studies have concluded that the response is similar to that of plant development, and the few studies that have assessed intraspecific variability have suggested that it is low (Parent and Tardieu, 2012). The result is that most crop models use similar formalisms and parameter values for both the response of development and expansion processes without including genetic variability in these parameters (Parent and Tardieu, 2014). A range of formalisms are used in crop models, from the simplest linear model to more complex, curvilinear models (Kumudini et al., 2014; Parent and Tardieu, 2014; Wang et al., 2017). In this study, we used a segmented linear formalism, which provided an adequate fit to the observed sensitivities in 60 accessions of wheat-related species (Supplementary Fig. S2, Supplementary Table S6). As expected, the observed intraspecific variability was low (Supplementary Tables S4 and S6), and so was the interspecific variability (albeit not zero; see below, Fig. 3.4; Supplementary Table S4). This indicates that a modelling strategy based on parsimony with a single set of parameter values for all wheat-related species is probably a good strategy (Parent et al., 2016). Such a model with no genotypic variation would be probably very close to one that assumes organized genetic variation, but would have the advantage of avoiding time-consuming experiments on each individual wheat accession.

To our knowledge, there have been no studies to date that have assessed the response of leaf elongation to VPD in wheat. This is despite the fact that most cereal species react strongly to changes in evaporative demand, and large genetic variability in the response of LER to VPD has been found in the species in which

it has been studied (e.g. Parent et al., 2010b; Welcker et al., 2011). Here, we have demonstrated that a linear model can fit this response across a large diversity of wheat-related species. The range of sensitivities at the interspecific level (from -10% to -25% kPa^{-1} ; Fig. 3.4G) was very close to that observed in maize (Welcker et al., 2011; Lacube et al., 2017). Given that consideration of this response can greatly improve prediction of leaf area in contrasting field sites for maize, we would also expect it to be the case in wheat growth predictions.

Although the response of transpiration to VPD has been investigated in many species including bread wheat (e.g. Schoppach and Sadok, 2012; Schoppach et al., 2012), and very high variability has been found across generations of selections of Australian wheat (Schoppach et al., 2016), there has been no information about its variability in wheat-relative species. Unexpectedly, the variability of responses observed in this study was very low at both the intra- and interspecific level compared to that found in 23 cultivars released from 1890 to 2010 in Australia (Fig. 3.4H; Supplementary Tables S4 and S6; Schoppach et al., 2016). For some genotypes, Schoppach et al. (2012) observed a break-point in the TR–VPD relationship between 2.4 kPa and 3.9 kPa, whereas in the range of VPDs that we tested a linear model fitted the response of all accessions without any break-point (Fig. 3.4C). This could be explained by the limited range of VPD used here (0.5–3.0 kPa compared with 0.8–4.5 kPa in Schoppach et al., 2012), which corresponded to the environmental range observed in the spring in most temperate regions.

3.4.2 Most traits have probably evolved slowly, following polyploidization, phylogeny, and breeding pressure

A striking result of this study was the fact that a hierarchical clustering of species based on traits that have not been directly selected was able to reproduce almost perfectly the known phylogeny of wheat-related species and subspecies. Our hierarchical clustering based on traits linked to plant structure discriminated diploid from polyploid species, regardless of domestication (e.g. *T. monococcum* subsp. *monococcum* and subsp. *boeoticum* clustered together, and *T. turgidum*

subsp. *dicoccoides* and subsp. *dicoccum* clustered together; fig. 3.6 and Supplementary Fig. S4). This suggests a specific effect of ploidy on plant physiology, such as that observed in trees (De Baerdemaeker et al., 2018). In addition, we found variations depending on the genome, with different sets of trait values between diploid species with genome A, D, or S (Fig. 3.1, fig. 3.6 and Supplementary Fig. S4). Finally, selection pressure seems to have affected these traits (but much less than phylogeny and ploidy), with different clusters of trait values for the earliest tetraploid species (*T. turgidum* subsp. *dicoccoides* and subsp. *dicoccum*) and landraces of *T. turgidum* subsp. *durum* selected before the Green Revolution, compared with durum wheat cultivars released later (Fig. 6 and Supplementary Fig. S4). At the level of individual traits, the values of the five traits related to sensitivities to temperature and VPD did not indicate any clear patterns, with no obvious impact of selection pressure, domestication, genome or sub-genome, ploidy, or phylogenetic links between species (Fig. 3.4). In contrast, a clustering based on these traits did show clear patterns, with an effect of both ploidy and phylogenetic relationships (Fig. 3.8). Overall, species that were phylogenetically close showed similar phenotypic spaces, indicating that these traits are well-conserved and have probably evolved slowly, following phylogeny, as has been proposed previously for the response to temperature in a much wider genetic diversity of cultivated plants (Parent and Tardieu, 2012).

3.4.3 What sources of genetic improvement are available within species and between related species?

In addition to providing formalisms and parameter values for process-based models, we also hypothesized that our study could identify new sources of genetic diversity for the responses of LER and TR to VPD and temperature within wheat-related species, in particular in wild or cultivated *T. turgidum* or diploid species, which could then potentially enrich the genetic diversity of cultivated wheat via the use of a synthetic-wheat strategy.

Most synthetic-derived wheat cultivars have been developed by crossing *Ae.*

tauschii with elite durum wheat cultivars, and more recently with *T. dicoccoides*, *T. dicoccum*, or *T. turgidum* (Singh and Trethowan, 2008; Li et al., 2018; Rosyara et al., 2019). Several studies have reported significant improvements in heat and drought tolerance of bread wheat cultivars derived from *Ae. tauschii* × *T. durum* crosses compared with locally adapted elite cultivars (e.g. Lopes and Reynolds, 2011; Mariano Cossani and Reynolds, 2015). However, the interspecific variability of wild and cultivated wheat-related species for traits related to temperature and evaporative demand has not yet been explored.

As expected, our results showed that the genetic variability of structural traits observed within wheat ancestors or the secondary gene pool was very different from that observed in modern durum or bread wheat cultivars, and mostly presented lower values (Fig. 3.3), which would not be advantageous in an agronomic context. A source of enrichment of genetic diversity for these traits therefore needs to be sought in other wild species. In contrast, for response traits, some wild relative species or wheat ancestors might provide adaptive value if introduced into cultivated wheat species (Figs 3.4 and 3.4).

The responses of LER and TR to temperature or VPD in *Ae. tauschii* and *T. dicoccum* were similar to those of *T. aestivum* (Figs 3.4 and 3.4). In particular, accessions of *Ae. tauschii* presented very similar trait values compared to modern wheat lines and they delimited a phenotypic space that was almost nested in that of *T. aestivum* (Fig. 3.8E). In contrast, *T. uratu* and *Ae. speltoïdes* showed very distinct responses and had higher T_{max} and lower sLER than elite bread and durum wheat cultivars (Fig. 3.4). Synthetic tetraploid and hexaploid lines have been created with *T. urartu* and *Ae. speltoïdes* but, to our knowledge, they have not been evaluated for their responses to temperature or evaporative demand (Valkoun, 2001; Kishii, 2019). Although introgressions from these species within wheat is practically and methodically challenging, recent developments of genomic resources and biotechnology have good potential to accelerate the use of diploid wheat-related species other than *Ae. tauschii* (Brunazzi et al., 2018). Several studies have focused on the responses of vegetative and/or reproductive

traits to high temperature in *Aegilops* species, and genetic variability at both the intra- and interspecific levels has been observed (Pradhan et al., 2012). Although we considered only a very limited number of accessions in each species, our results suggest that *T. urartu* and *Ae. speltaoides* could be interesting sources of genes to develop wheat cultivars that are able to maintain high LER under high temperature and evaporative demand.

3.4.4 Conclusion

In addition to providing new formalisms for trait responses together with model parameters for 12 wheat-related species and subspecies, this study indicates that traits linked to the responses of LER and TR to temperature and VPD are well-conserved, with an unexpected low genetic variability within species. Our results suggest that vegetative traits have probably evolved slowly, following phylogenetic links between accessions, and that both domestication and phylogenetic relationships have driven the slow evolution of temperature and VPD responses. Finally, the study has identified potential sources for varietal improvement that currently reside in wheat-related species.

3.5 References

Akaike, H., 1998. Information Theory and an Extension of the Maximum Likelihood Principle, in : Parzen, E., Tanabe, K., Kitagawa, G. (Eds.), *Selected Papers of Hirotugu Akaike*, Springer Series in Statistics. Springer New York, pp. 199–213.

Allahverdiyev, T.I., 2015. Physiological Traits of Durum Wheat (*Triticum durum* Desf.) and Bread Wheat (*Triticum aestivum* L.) Genotypes under Drought Stress. *Agricultural Sciences* 06, 848–859. <https://doi.org/10.4236/as.2015.68082>

Ben-Ari, T., Boe, J., Ciais, P., Lecerf, R., Van der Velde, M., Makowski, D., 2018. Causes and implications of the unforeseen 2016 extreme yield loss in the bread-basket of France. *Nat. Commun.* 9, 1627. <https://doi.org/10.1038/s41467-018-04087-x>

Brunazzi, A., Scaglione, D., Talini, R.F., Miculan, M., Magni, F., Poland, J., Pè, M.E., Brandolini, A., Dell'Acqua, M., 2018. Molecular diversity and landscape genomics of the crop wild relative *Triticum urartu* across the Fertile Crescent. *The Plant Journal* 94, 670–684. <https://doi.org/10.1111/tpj.13888>

- Comai, L., 2005. The advantages and disadvantages of being polyploid. *Nature Reviews Genetics* 6, 836–846. <https://doi.org/10.1038/nrg1711>
- Curtis, B., Gómez Macpherson H, Rajaram S, 2002. Bread Wheat. Improvement and Production, Vol 30 [WWW Document]. URL <http://www.fao.org/docrep/006/Y4011E/Y4011E00.HTM> (accessed 6.5.17).
- Davies, W.J., Wilkinson, S., Loveys, B., 2001. Stomatal control by chemical signalling and the exploitation of this mechanism to increase water use efficiency in agriculture. *New Phytologist* 153, 449–460. <https://doi.org/10.1046/j.0028-646X.2001.00345.x>
- De Baerdemaeker, N.J.F., Hias, N., Van den Bulcke, J., Keulemans, W., Steppe, K., 2018. The effect of polyploidization on tree hydraulic functioning. *Am. J. Bot.* 105, 161–171. <https://doi.org/10.1002/ajb2.1032>
- Giunta, F., Pruneddu, G., Zuddas, M., Motzo, R., 2019. Bread and durum wheat : Intra- and inter-specific variation in grain yield and protein concentration of modern Italian cultivars. *Eur. J. Agron.* 105, 119–128. <https://doi.org/10.1016/j.eja.2019.02.011>
- Gorafi, Y.S.A., Kim, J.-S., Elbashir, A.A.E., Tsujimoto, H., 2018. A population of wheat multiple synthetic derivatives : an effective platform to explore, harness and utilize genetic diversity of *Aegilops tauschii* for wheat improvement. *Theor Appl Genet* 131, 1615–1626. <https://doi.org/10.1007/s00122-018-3102-x>
- Hammer, G., Messina, C., Wu, A., Cooper, M., 2019. Biological reality and parsimony in crop models—why we need both in crop improvement! *in silico Plants* 1, diz010. <https://doi.org/10.1093/insilicoplants/diz010>
- Handmer, J., Honda, Y., Kundzewicz, Z.W., Arnell, N., Benito, G., Hatfield, J., Mohamed, I.F., Peduzzi, P., Wu, S., Sherstyukov, B., Takahashi, K., Yan, Z., Vicuna, S., Suarez, A., Abdulla, A., Bouwer, L.M., Campbell, J., Hashizume, M., Hattermann, F., Heilmayr, R., Keating, A., Ladds, M., Mach, K.J., Mastrandrea, M.D., Mechler, R., Nobre, C., Sanghi, A., Screen, J., Smith, J., Velegrakis, A., Vergara, W., Waite, A.M., Westrich, J., Whittaker, J., Yunhe, Y., Yamano, H., 2012. Changes in Impacts of Climate Extremes : Human Systems and Ecosystems, Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation. Cambridge Univ Press, Cambridge.
- Haun, J.R., 1973. Visual Quantification of Wheat Development1. *Agronomy Journal* 65, 116. <https://doi.org/10.2134/agronj1973.00021962006500010035x>
- Hias, N., Svara, A., Keulemans, J.W., 2018. Effect of polyploidisation on the response of apple (*Malus × domestica* Borkh.) to *Venturia inaequalis* infection. *European Journal of Plant Pathology* 151, 515–526. <https://doi.org/10.1007/s10658-017-1395-2>
- Horwitz, W. (Ed.), 1980. AOAC : Official Methods of Analysis, 1980, AOAC International. ed, Thirteenth edition.
- Jeuffroy, M.-H., Casadebaig, P., Debaeke, P., Loyce, C., Meynard, J.-M., 2014. Agronomic model uses to predict cultivar performance in various environments and cropping systems. A review. *Agronomy for Sustainable Development* 34,

121–137. <https://doi.org/10.1007/s13593-013-0170-9>

Kishii, M., 2019. An Update of Recent Use of *Aegilops* Species in Wheat Breeding. *Front. Plant Sci.* 10. <https://doi.org/10.3389/fpls.2019.00585>

Kumudini, S., Andrade, F.H., Boote, K.J., Brown, G.A., Dzotsi, K.A., Edmeades, G.O., Gocken, T., Goodwin, M., Halter, A.L., Hammer, G.L., Hatfield, J.L., Jones, J.W., Kemanian, A.R., Kim, S.-H., Kiniry, J., Lizaso, J.I., Nendel, C., Nielsen, R.L., Parent, B., Stoeckle, C.O., Tardieu, F., Thomison, P.R., Timlin, D.J., Vyn, T.J., Wallach, D., Yang, H.S., Tollenaar, M., 2014. Predicting Maize Phenology : Intercomparison of Functions for Developmental Response to Temperature. *Agron. J.* 106, 2087–2097. <https://doi.org/10.2134/agronj14.0200>

Lacube, S., Fournier, C., Palaffre, C., Millet, E.J., Tardieu, F., Parent, B., 2017. Distinct controls of leaf widening and elongation by light and evaporative demand in maize. *Plant Cell Environ.* 40, 2017–2028. <https://doi.org/10.1111/pce.13005>

Leveau, S., Parent, B., Zaka, S., Martre, P., 2021. Data : Sensitivities to temperature and evaporative demande in wheat relatives. <https://doi.org/10.5281/zenodo.5511805>

Li, A., Liu, D., Yang, W., Kishii, M., Mao, L., 2018. Synthetic Hexaploid Wheat : Yesterday, Today, and Tomorrow. *Engineering* 4, 552–558. <https://doi.org/10.1016/j.eng.2018.07.001>

Lobell, D.B., Schlenker, W., Costa-Roberts, J., 2011. Climate Trends and Global Crop Production Since 1980. *Science* 333, 616–620. <https://doi.org/10.1126/science.1204531>

Lopes, M.S., Reynolds, M.P., 2011. Drought Adaptive Traits and Wide Adaptation in Elite Lines Derived from Resynthesized Hexaploid Wheat. *Crop Sci.* 51, 1617–1626. <https://doi.org/10.2135/cropsci2010.07.0445>

Mariano Cossani, C., Reynolds, M.P., 2015. Heat Stress Adaptation in Elite Lines Derived from Synthetic Hexaploid Wheat. *Crop Sci.* 55, 2719–2735. <https://doi.org/10.2135/cropsci2015.02.0092>

Marti, J., Slafer, G.A., 2014. Bread and durum wheat yields under a wide range of environmental conditions. *Field Crops Research* 156, 258–271. <https://doi.org/10.1016/j.fcr.2013.10.008>

Martin, A., Simpfendorfer, S., Hare, R.A., Eberhard, F.S., Sutherland, M.W., 2011. Retention of D genome chromosomes in pentaploid wheat crosses. *Heredity* 107, 315–319. <https://doi.org/10.1038/hdy.2011.17>

Parent, B., Millet, E.J., Tardieu, F., 2019. The use of thermal time in plant studies has a sound theoretical basis provided that confounding effects are avoided. *J. Exp. Bot.* 70, 2359–2370. <https://doi.org/10.1093/jxb/ery402>

Parent, B., Suard, B., Serraj, R., Tardieu, F., 2010. Rice leaf growth and water potential are resilient to evaporative demand and soil water deficit once the effects of root system are neutralized. *Plant, Cell & Environment* 33, 1256–1267. <https://doi.org/10.1111/j.1365-3040.2010.02145.x>

- Parent, B., Tardieu, F., 2014. Can current crop models be used in the phenotyping era for predicting the genetic variability of yield of plants subjected to drought or high temperature? *J Exp Bot* 65, 6179–6189. <https://doi.org/10.1093/jxb/eru223>
- Parent, B., Tardieu, F., 2012. Temperature responses of developmental processes have not been affected by breeding in different ecological areas for 17 crop species. *New Phytologist* 194, 760–774. <https://doi.org/10.1111/j.1469-8137.2012.04086.x>
- Parent, B., Turc, O., Gibon, Y., Stitt, M., Tardieu, F., 2010. Modelling temperature-compensated physiological rates, based on the coordination of responses to temperature of developmental processes. *J. Exp. Bot.* erq003. <https://doi.org/10.1093/jxb/erq003>
- Parent, B., Vile, D., Violle, C., Tardieu, F., 2016. Towards parsimonious eco-physiological models that bridge ecology and agronomy. *New Phytologist* 210, 380–382. <https://doi.org/10.1111/nph.13811>
- Perego, A., Sanna, M., Giussani, A., Chiodini, M.E., Fumagalli, M., Pilu, S.R., Bindi, M., Moriondo, M., Acutis, M., 2014. Designing a high-yielding maize ideotype for a changing climate in Lombardy plain (northern Italy). *Sci. Total Environ.* 499, 497–509. <https://doi.org/10.1016/j.scitotenv.2014.05.092>
- Petersen, G., Seberg, O., Yde, M., Berthelsen, K., 2006. Phylogenetic relationships of *Triticum* and *Aegilops* and evidence for the origin of the A, B, and D genomes of common wheat (*Triticum aestivum*). *Molecular Phylogenetics and Evolution* 39, 70–82. <https://doi.org/10.1016/j.ympev.2006.01.023>
- Pradhan, G.P., Prasad, P.V.V., Fritz, A.K., Kirkham, M.B., Gill, B.S., 2012. High Temperature Tolerance in *Aegilops* Species and Its Potential Transfer to Wheat. *Crop Sci.* 52, 292–304. <https://doi.org/10.2135/cropsci2011.04.0186>
- Rebetzke, G.J., Botwright, T.L., Moore, C.S., Richards, R.A., Condon, A.G., 2004. Genotypic variation in specific leaf area for genetic improvement of early vigour in wheat. *Field Crops Research* 88, 179–189. <https://doi.org/10.1016/j.fcr.2004.01.007>
- Reyer, C.P.O., Leuzinger, S., Rammig, A., Wolf, A., Bartholomeus, R.P., Bonfante, A., de Lorenzi, F., Dury, M., Gloning, P., Abou Jaoude, R., Klein, T., Kuster, T.M., Martins, M., Niedrist, G., Riccardi, M., Wohlfahrt, G., de Angelis, P., de Dato, G., Francois, L., Menzel, A., Pereira, M., 2013. A plant's perspective of extremes: terrestrial plant responses to changing climatic variability. *Glob. Change Biol.* 19, 75–89. <https://doi.org/10.1111/gcb.12023>
- Rojas, M., Lambert, F., Ramirez-Villegas, J., Challinor, A.J., 2019. Emergence of robust precipitation changes across crop production areas in the 21st century. *Proc. Natl. Acad. Sci. U. S. A.* 116, 6673–6678. <https://doi.org/10.1073/pnas.1811463116>
- Rosyara, U., Kishii, M., Payne, T., Sansaloni, C.P., Singh, R.P., Braun, H.-J., Dreisigacker, S., 2019. Genetic Contribution of Synthetic Hexaploid Wheat to CIM-MYT's Spring Bread Wheat Breeding Germplasm. *Sci Rep* 9, 12355–12355. <https://doi.org/10.1038/s41598-019-47936-5>
- Rötter, R.P., Tao, F., Hohn, J.G., Palosuo, T., 2015. Use of crop simulation modelling to aid ideotype design of future cereal cultivars. *J. Exp. Bot.* 66, 3463–3476. <https://doi.org/10.1093/jxb/erv098>

Schoenfelder, K.P., Fox, D.T., 2015. The expanding implications of polyploidy. *The Journal of Cell Biology* 209, 485–491. <https://doi.org/10.1083/jcb.201502016>

Schoppach, R., Fleury, D., Sinclair, T.R., Sadok, W., 2017. Transpiration Sensitivity to Evaporative Demand Across 120 Years of Breeding of Australian Wheat Cultivars. *Journal of Agronomy and Crop Science* 203, 219–226. <https://doi.org/10.1111/jac.12193>

Schoppach, R., Sadok, W., 2012. Differential sensitivities of transpiration to evaporative demand and soil water deficit among wheat elite cultivars indicate different strategies for drought tolerance. *Environmental and Experimental Botany* 84, 1–10. <https://doi.org/10.1016/j.envexpbot.2012.04.016>

Schoppach, R., Taylor, J.D., Majerus, E., Claverie, E., Baumann, U., Suchecki, R., Fleury, D., Sadok, W., 2016. High resolution mapping of traits related to whole-plant transpiration under increasing evaporative demand in wheat. *J Exp Bot* 67, 2847–2860. <https://doi.org/10.1093/jxb/erw125>

Schwarz, G., 1978. Estimating the Dimension of a Model. *Ann. Statist.* 6, 461–464. <https://doi.org/10.1214/aos/1176344136>

Singh, R.P., Trethowan, R., 2008. Breeding Spring Bread Wheat for Irrigated and Rainfed Production Systems of the Developing World, in : *Breeding Major Food Staples*. John Wiley & Sons, Ltd, pp. 107–140. <https://doi.org/10.1002/9780470376447.ch5>

Tardieu, F., 2012. Any trait or trait-related allele can confer drought tolerance : just design the right drought scenario. *J Exp Bot* 63, 25–31. <https://doi.org/10.1093/jxb/err269>

Tardieu, F., Parent, B., 2017. Predictable ‘meta-mechanisms’ emerge from feedbacks between transpiration and plant growth and cannot be simply deduced from short-term mechanisms. *Plant, Cell & Environment* 40, 846–857. <https://doi.org/10.1111/pce.12822>

Tardieu, F., Parent, B., Caldeira, C.F., Welcker, C., 2014. Genetic and physiological controls of growth under water deficit. *Plant Physiology* pp.113.233353. <https://doi.org/10.1104/pp.113.233353>

Tu, Y., Jiang, A., Gan, L., Hossain, M., Zhang, Jinming, Peng, B., Xiong, Y., Song, Z., Cai, D., Xu, W., Zhang, Jianhua, He, Y., 2014. Genome duplication improves rice root resistance to salt stress. *Rice* 7. <https://doi.org/10.1186/s12284-014-0015-4>

Valkoun, J.J., 2001. Wheat pre-breeding using wild progenitors. *Euphytica* 119, 17–23. <https://doi.org/10.1023/A:1017562909881>

Wang, E., Engel, T., 1998. Simulation of phenological development of wheat crops. *Agricultural Systems* 58, 1–24. [https://doi.org/10.1016/S0308-521X\(98\)00028-6](https://doi.org/10.1016/S0308-521X(98)00028-6)

Wang, E., Martre, P., Zhao, Z., Ewert, F., Maiorano, A., Rötter, R.P., Kimball, B.A., Ottman, M.J., Wall, G.W., White, J.W., Reynolds, M.P., Alderman, P.D., Aggarwal, P.K., Anothai, J., Basso, B., Biernath, C., Cammarano, D., Challinor, A.J.,

De Sanctis, G., Doltra, J., Dumont, B., Fereres, E., Garcia-Vila, M., Gayler, S., Hoogenboom, G., Hunt, L.A., Izaurralde, R.C., Jabloun, M., Jones, C.D., Kersebaum, K.C., Koehler, A.-K., Liu, L., Müller, C., Kumar, S.N., Nendel, C., O’Leary, G., Olesen, J.E., Palosuo, T., Priesack, E., Rezaei, E.E., Ripoche, D., Ruane, A.C., Semenov, M.A., Shcherbak, I., Stöckle, C., Stratonovitch, P., Streck, T., Supit, I., Tao, F., Thorburn, P., Waha, K., Wallach, D., Wang, Z., Wolf, J., Zhu, Y., Asseng, S., 2017. Author Correction : The uncertainty of crop yield projections is reduced by improved temperature response functions. *Nature Plants* 3, 833–833. <https://doi.org/10.1038/s41477-017-0032-6>

Welcker, C., Sadok, W., Dignat, G., Renault, M., Salvi, S., Charcosset, A., Tardieu, F., 2011. A Common Genetic Determinism for Sensitivities to Soil Water Deficit and Evaporative Demand : Meta-Analysis of Quantitative Trait Loci and Introgression Lines of Maize. *Plant Physiology* 157, 718–729. <https://doi.org/10.1104/pp.111.176479>

Analysis of the phenotypic space of sensitivities of grain yield and its components to water and nitrogen limitations for durum wheat and bread wheat : constitution of sub-panels for ecophysiological studies

Résumé

Dans le contexte du changement climatique et de politiques publiques, pour une agriculture plus durable utilisant moins d'intrants, le rendement risque d'être de plus en plus affecté par des contraintes abiotiques telles que les limitations en eau et en azote. En fonction de chaque scénario environnemental spécifique, les composantes du rendement peuvent être affectées différemment par ces contraintes, car leurs déterminismes physiologiques ont leur propre diversité génétique. De plus, il existe de nombreux compromis entre ces variables, souvent liés à la compétition pour les ressources, ce qui limite l'espace phénotypique des réponses aux stress abiotiques. Dans cette étude, nous proposons une méthode visant à sélec-

tionner des géotypes sensibles et tolérants pour le rendement, couvrant l'espace phénotypique original des réponses de ses composantes. Nous avons analysé la sensibilité aux déficits hydrique et en azote, et l'espace phénotypique qui en résulte, de deux panels de diversité de 250 géotypes de blé tendre et 200 géotypes de blé dur, cultivés respectivement dans sept et trois essais au champ comprenant des conditions optimales et des conditions limitantes en azote et/ou eau. Nous avons d'abord étudié comment les réponses du rendement et de ses composantes variaient, observé que les réponses de ces composantes aux limitations hydriques et azotées étaient elles-mêmes fortement dépendantes de l'interaction avec les autres covariables expérimentales (site x année). Nous avons observé que l'espace phénotypique de la sensibilité du rendement et de ses composantes aux limitations hydriques et azotées était fortement contraint, indiquant la nécessité de le prendre en compte pour la sélection de sous-panels de diversité. Nous décrivons une méthode de sélection de sous-panels permettant d'explorer l'espace phénotypique initial dans les analyses écophysiologiques. Enfin, en utilisant une sélection aléatoire du nombre de sites, nous avons questionné l'importance du nombre de sites et leur impact respectif dans la sélection des sous-panels. Outre l'amélioration de nos connaissances sur les relations entre la sensibilité des composantes du rendement aux multiples contraintes, cette étude présente également un intérêt opérationnel pour la sélection de sous-panels respectant l'espace phénotypique de la sensibilité du rendement aux contraintes abiotiques des panels génétiques de départ, utilisables pour des études d'écophysiologie.

Mots-clés : blé, *triticum durum*, *triticum aestivum*, blé tendre, blé dur, rendement en grains, limitation de l'eau et de l'azote, espace phénotypique, composants de rendement, nombre de grains.

Abstract

In the context of climate change and policies making, for a more sustainable agriculture using less inputs, grain yield is likely to be increasingly affected by

abiotic constraints such as water and nitrogen (N) deficits. Depending on each specific environmental scenario, grain yield components may be differently affected by these constraints because their underlying determinisms have their own genetic diversity. In addition, there are many tradeoffs between grain yield components, often related to the competition for resources, constraining the phenotypic space of responses to abiotic stresses. In this study, we propose a method aiming at selecting sensitive and tolerant genotypes for grain yield covering the initial phenotypic space of the responses of grain yield components. We analyzed the sensitivity to water and N deficits, and the resulting phenotypic space of two diversity panels of 250 bread wheat and 200 durum wheat genotypes grown in seven and three field trials, respectively, with optimal conditions and conditions limited by N and/or water. We first studied how the responses of grain yield and its components varied under different environmental scenarios, observed that the response to stresses was strongly dependent on the interaction with experimental co-variables (site x year). We found that the phenotypic space of the sensitivity of grain yield and its components to water and N deficits was strongly constrained, which should be considered for the selection of sub-panels of diversity. Finally, by using random selection of the number of sites, we questioned the importance of the number of sites and their respective impact in the selection of subpanels. In addition to improving our knowledge of the relationships between the sensitivity of grain yield components to multiple constraints, this study is also of practical interest for the selection of subpanels covering the phenotypic space of grain yield sensitivity to abiotic constraints of the initial genetic panels, which can be used for ecophysiology studies.

Keywords : wheat, *triticum durum*, *triticum aestivum*, bread wheat, durum wheat, grain yield, water and nitrogen limitation, phenotypic space, yield components, grain number.

4.1 Introduction

Agriculture production must at the same time, (i) feed a rapidly growing global population, which exacerbates tensions over water use and increases the risk of food insecurity; (ii). adapt to climate change, which increases the frequency and intensity of drought events (Lobell et al., 2011); and reduce its environmental footprint (Pe'er et al., 2019), by using less fertilizers and water (Alamanos et al., 2019). However, water and nitrogen (N) can be considered as the main limiting environmental factors affecting grain yield (Jafarikouhini et al., 2021). In this context, maintaining grain yields will only be ensured by optimizing water and N use (Berger et al., 2016; Tardieu, 2012), though varietal improvement (Sinclair, 2017) and crop management practices.

Addressing this challenge is not straightforward. Indeed, climate change tends to increase the multiplicity of environmental scenarios, even within a given site, leading to increased interannual variations in agricultural production (Urban et al., 2015), mainly due to the multiplicity of drought scenarios (Fischers and Knutti, 2013). In addition, abiotic constraints tend to increase Genotype $\times$ Environment ($G \times E$) interactions :). Thus, the best variety observed in one environment may be a very poor choice in a different drought scenario.

Grain yield components may be differently affected by N and water deficits. Not only because the periods of sensitivity during the crop growth cycle of grain yield related physiological processes are not the same (Parent et al., 2017), but also because the underlying mechanisms have their own sensitivities and genetic determinisms. In each given scenario, a given "strategy" will therefore be beneficial, or not, to better maintain final grain yield, and grain quality (Tardieu, 2012).

In addition, there are many tradeoffs between grain yield components. As independent as mechanistically possible, grain yield components and their sensitivities can be correlated with each other in germplasms or diversity panels, due to the many tradeoffs, and constraints (Donovan et al., 2011) including evolution and selection, genetics, biophysics, and plants' structure and architecture. Such

constraints are strongly related to the competition for resources, the same pool of water, N and carbon serving to the different sink organs. As a result, a variety that tends to strongly reduce the number of spike per square meter under drought will tend to reduce less the number of grains per spike (Vahamidis et al., 2019).

To take these trade-offs into account, it is necessary to consider the different responses of grain yield components as a whole. The set of constraints and genetic correlations between traits defines the set of possibilities, called phenotypic space, representing only a part of the n-dimensional hypercube (set of biological variables considered). Among the best known concrete cases of highly constrained phenotypic spaces, is the Leaf Economics Spectrum (Wright et al., 2004; Shipley et al., 2006), which defines trade-off between leaf resource allocation and acquisition.

For the breeder, it is therefore necessary to consider (i) the diversity of temporal scenarios of abiotic constraints; (ii) the potential non-additive interactions between the tolerance to multiple-abiotic constraints; and (iii) the diversity of responses of the underlying grain yield components.

In this study, we present a method to select sensitive and tolerant genotypes for grain yield, the number of spikes per square meter, the number of grains per square meter, and the average grain dry mass in response to LI conditions (water and N). This was performed in order to get contrasting sensitivities while maximizing the diversity of the selected genotypes. The selected genotypes were then analyzed in the PHENOARCH platform (Chapter 5) and in the field (Chapter 6). . We analyzed the sensitivity to water and N deficits, and the resulting phenotypic space of two diversity panels of 250 bread wheat and 200 durum wheat genotypes grown in seven and three field trials, respectively, with optimal conditions and conditions limited by N and/or water. We found that the response to water and N stresses strongly depended on the interactions between experimental co-variables (site x year). We then described a method to select subsets of genotypes

in genetic panels that cover phenotypic space. Finally, by using random selection of the number of sites, we questioned the importance of the number and diversity of environmental scenarios and their respective impact in the selection of subpanels. In addition to improving our knowledge of the relationships between the sensitivity of grain yield components to multiple constraints, this study is also of practical interest for the selection of subpanels covering the phenotypic space of grain yield sensitivity to abiotic constraints that can be used for ecophysiology analyses.

4.2 Material and methods

4.2.1 Genetic material

A pane of 235 bread wheat (*Triticum aestivum* L.) genotypes were selected from a diversity panel of 450 genotypes sampled from 4,506 accessions in a world-wide collection (Balfourier et al., 2019) by minimizing linkage disequilibrium and by imposing agronomic constraints on phenology and plant height (Paux et al., 2022). Among these 235 genotypes, 194 were grown in all field trials.

For durum wheat, a panel of 257 durum wheat genotypes was constituted from 126 elite lines of worldwide origin registered between 1973 and 2015, 69 elite lines developed by the Center Research for Agricultural and Economics and the University of Foggia, Italy, and 61 lines randomly selected from 480 evolutionary pre-breeding lines derived from an open-pollinated population based on the undercrossing of about 650 accessions from wild subspecies and elite lines (David et al., 2014). For technical reasons, only 232 durum wheat genotypes were common to all trials.

4.2.2 Field trials and experimental conditions

Ten field trials conducted in the frame of the European project H2020 Solace and the French ANR project BreedWheat were analyzed in this study (seven sites

over two years for *T. aestivum*, and three sites over one year for *T. durum*; Table 4.1 and Fig. S7; note that an additional site with *T. aestivum* has not been analysed because of sever logging). In each trial, two treatments were tested, a control condition with optimal water and N supplies (named high-input, HI hereafter) and another condition with either low N input or rainfed conditions (named low-input, LI hereafter; Table 4.2 and Fig. S8). Plots were 2.9 to 11.9 m² depending on the site. All crops were sown at the locally recommended date, with sowing density varying from 196 to 300 seeds m⁻² for the bread wheat trials depending on the site, and of 350 seeds m⁻² for durum wheat in all trial (Table 4.2). Each trial used a split-plot design in which the LI and HI treatments were randomized on main plots, cultivar was randomized on the sub-plots and each treatment was replicated two to three times.

In all trials, soil water content at 30-cm and 60-cm below the soil surface was monitored throughout the growing season using tensiometric probes. All other crop inputs including weed, disease and pest control, and potassium, phosphate, and sulfur fertilizers, were applied at levels to prevent non-N nutrients or weeds, diseases and pests from limiting grain yield.

Among the seven bread wheat trials, the LN treatments consisted in low N supply under rainfed conditions in four trials (trials DEScap2017, INRcle2017, INRmon2017, and SYNmoi2017; Table 4.2), in rainfed conditions with high N supply similar to the HI treatment in one trial (BAYmil2017), and rainfed conditions and low N supply in two trials (ARVgre2018 and SYNlev2018; Table 4.2). In all durum wheat trials, the LI treatments were under rainfed conditions and low N supply compared to the corresponding HI treatment (Table 4.2).

In all trials, N fertilizer applications in the HI treatments were intended to replicate commercial practice. The growth stages and application rates were calculated based on provisional N balance associated with expected grain yields using the N sheet balance method described in Rémy and Hébert (1977). For the LI treatments with low N fertilizer rate, N fertilizer was reduced by 37% to 100% compared with the HI treatments. In the HI treatments with low water supply, irrigation was applied to ensure that water supply did not limit plant growth. In all trials, the LI treatments were rainfed

4.2.3 Crop measurements

Crop m Several plant variables were measured and analyzed jointly in this study. Crop emergence and heading dates were scored at least twice weekly during two weeks around the scored growth stage. Growth stages were determined as the date when 50% of the plants in plot had reached the considered growth stage. In the bread wheat trials, the number of spikes was counted about one week after anthesis on two to four non-adjacent rows over 0.75 to 1 m (0.25 to 0.7 m²). At harvest maturity, all plots were harvested with a combiner. One to two

TABLE 4.2 – *Environmental conditions in the 10 field trials analysed in this study. Temperature, solar radiation, and water supply were calculated between sowing to final harvest.*

Species	Trial name	Average daily minimum temperature (°C)	Average daily maximal temperature (°C)	Thermal time (°C)	Cumulative solar radiation (MJ m-2)	Water supply ^{a,b} (mm)		Fertilizer application rate ^a (kg N ha-1)	
						HI	LI	HI	LI
Bread wheat	BAYmil2017	4.6	13.3	2251	2489	563 (85)	478 (0)	120	120
	DEScap2017	9.7	10.6	2968	2997	719 (0)	719 (0)	100	0
	INRcle2017	5.1	15.1	2463	2907	678 (20)	658(0)	123	78
	INRmon2017	5.2	13.6	2398	2888	710 (0)	695 (0)	50	0
	SYNmoi2017	5.7	14.9	2923	3295	447 (0)	447 (0)	80	0
	ARVgre2018	3.9	17.8	2779	3532	716 (70)	646 (0)	180	50
Durum wheat	SYNlev2018	6.2	14.9	2812	3264	707 (125)	612 (30)	201	59
	INRdia2018	7.3	18.6	3221	3477	710 (15)	695 (0)	240	60
	CREFog2018	7.8	18.7	2883	3133	623 (66)	556(0)	240	60
	CREFog2019	7.3	18.6	2705	3082	914 (323)	592 (0)	240	60

^a LI, Low-input conditions; HI, high input conditions.

^b Water supply is the amount of precipitations plus irrigation, number of parenthesis are irrigation amounts.

rows on each side and 30 to 50 cm on the two extremities of each plot were excluded. Samples of about 1,200 seeds were oven dry until constant mass to determined the average grain dry mass. Grain number per square meter was calculated by dividing grain yield by the average grain dry mass. For all bread wheat trials, grain number per spike was calculated by dividing grain number per square meter by the spike number per square meter. Spike number per square meter, and thus grain number per spike, were not determined in the durum wheat trials. All grain yields and average grain dry masses were expressed at 0% moisture.

4.2.4 Ranks of sensitivity to low-input conditions and genotype selection

The method for sub-sampling genotypes presented in the Results section was based on the calculation of ranks of sensitivity to LI conditions . In each trial, the residuals ($Yield_{res}$) of the linear relationship for grain yield under LI conditions ($Yield_{LI}^{obs}$) and HI conditions ($Yield_{HI}^{obs}$) were calculated as :

$$Yield_{LI}^{est} = \hat{\beta}_0 + \hat{\beta}_1 \times Yield_{HI}^{obs} + \varepsilon \quad (4.1)$$

$$Yield_{res} = Yield_{LI}^{obs} - Yield_{LI}^{est} \quad (4.2)$$

where $Yield_{HI}^{obs}$ and $Yield_{LI}^{obs}$ are the observed grain yield in HI and LI conditions, respectively; $Yield_{LI}^{est}$ is the estimated grain yield in LI condition; $Yield_{res}$ the residuals of observed versus estimated grain yield in LI condition; $\hat{\beta}_0$, Intercept parameter of linear model; $\hat{\beta}_1$, slope parameter of linear model.

The standardized residuals ($Yield_{rstandard}$) were then calculated as :

$$Yield_{rstandard} = \frac{(Yield_{res} - Yield_{res}^{mean})}{Yield_{res}^{sd}} \quad (4.3)$$

where $Yield_{res}^{mean}$ is the mean of the grain yield residuals, and $Yield_{res}^{sd}$ is the standard deviation of the grain yield residuals.

In each trial, a stress index ($Stress.index^{trial}$) was calculated as the average grain yield reduction normalized by grain yield under HI condition. It was calculated in each trial based on the average grain yield of all genotypes.

$$Stress.Index^{trial} = \left(\frac{\widehat{Yield}_{HI}^{obs} - \widehat{Yield}_{LI}^{obs}}{\widehat{Yield}_{HI}^{obs}} \right) \quad (4.4)$$

where $\widehat{Yield}_{HI}^{obs}$ and $\widehat{Yield}_{LI}^{obs}$ are the average (across genotypes) observed grain yield in HI and LI conditions, respectively.

This index was then normalized by the sum of indexes given by :

$$Stress.Index_{normalized}^{trial} = \frac{Stress.Index^{trial}}{\sum_{itrial=1}^n (Stress.Index^{trial})} \quad (4.5)$$

Finally, the rank of each genotype ($rank_{genotype}$) was calculated as its average across all trials, weighted by the stress index of the trial :

$$rank_{genotype} = \sum_{itrial=1}^n \left(rank_{genotype}^{trial} \times Stress.Index_{normalized}^{trial} \right) \quad (4.6)$$

where $rank_{genotype}^{trial}$ is the rank of each genotype based on the standardized residuals in each trial.

The metrics $rank_{genotype}$ was used for genotype selection. In this study, we also applied specific conditions adapted to our dataset, which we do not consider as part of the method itself. We considered here only the genotypes that were present in at least two trials. In bread wheat, we have chosen only high-yielding genotypes with average grain yield $> 5 \text{ t ha}^{-1}$ in HI conditions, and within a flowering window of 13 days (average day-of-year of heading across all trials from 132 to 145). We selected the 20 genotypes with the lowest average rank of sensitivity and the 20 genotypes with the highest rank, considered respectively as the most “sensitive” and “tolerant” genotypes, respectively.

4.2.5 Data analysis

All data analyses were carried out using the R software (R Core Team, 2018). The correlation matrix of the heatmap, we used the function "heatmap" and "hclust" of the package R "stats" (correlation matrix with the method "euclidean" and the hierarchical clustering with the method of "ward.D"). For the analysis of variances, we used the "anova" function of the "stats" package in R. For the correlations between the variables, we used a linear regression and for the correlation tables, we used the method of "pearson" with the function "rcorr" of the package R "Hmisc" and the function "cor.test" of the package R "stats".

4.3 Results

4.3.1 Genotype x Environment interactions on grain yield are driven by site/trials contrasts rather than targeted water and nitrogen limitations

Differences in grain yield between the HI and LI conditions in the seven bread wheat trials were analysed for the 273 genotypes (Fig. 4.1). Depending

on the considered trial, the observed range of grain yield differed, even under non-limiting conditions (HI), indicating an effect of the trial on grain yield in HI conditions. Under more limiting LI conditions, observed grain yield were globally lower compared to that in HI but the difference depended on the trial. For example, the largest differences in grain yield between HI and LI was observed in the trial SYNlev2018 and the lowest differences were observed in SYNmoi2017 (Fig. 1). Note that because these trials were not repeated over several years, the respective effects of specific site and year cannot be separated and are therefore summarized into the effect of trial (defined here as E_{trial}). By contrast we define the specific effect of the treatment within a trial as $E_{\text{treatment}}$.

In each trial, the correlation between LI and HI conditions was high and significant, even in trials in which the effect of the treatment was severe (SYNlev2018, INRcle2017 and DEScap2017). It indicated that $G \times E$ interactions for grain yield were low compared to the effect of the treatment ($E_{\text{treatment}}$). While this was expected in trials in which the effect of the LI conditions was low, this was observed even in trials for which the LI treatment was severe, and regardless of the origin of the stress, N and/or water.

To analyze the genetic correlations between trials and treatments, we performed a hierarchical clustering of environments (trial $\times$ treatments), based on the ranking of each genotype for grain yield (Fig. 4.2). It appeared clearly that the environments clustered based on their the site $\times$ year, rather than the treatments. In other words, a genotype with high grain yield in one environment, was also one with the highest grain yield in the other environment of the same trial (low $G \times E_{\text{treatment}}$ compared to $G \times E_{\text{trial}}$). Overall, it indicated that one environment was more similar to the other environment of the same trial, even if they differed in terms of abiotic conditions (e.g. HI and LI conditions of the same trial) than to another environment, in another trial with similar abiotic conditions (e.g. HI conditions of two different trials).

Overall, it appeared that the effect of the trial E_{trial} , and so the effects of en-

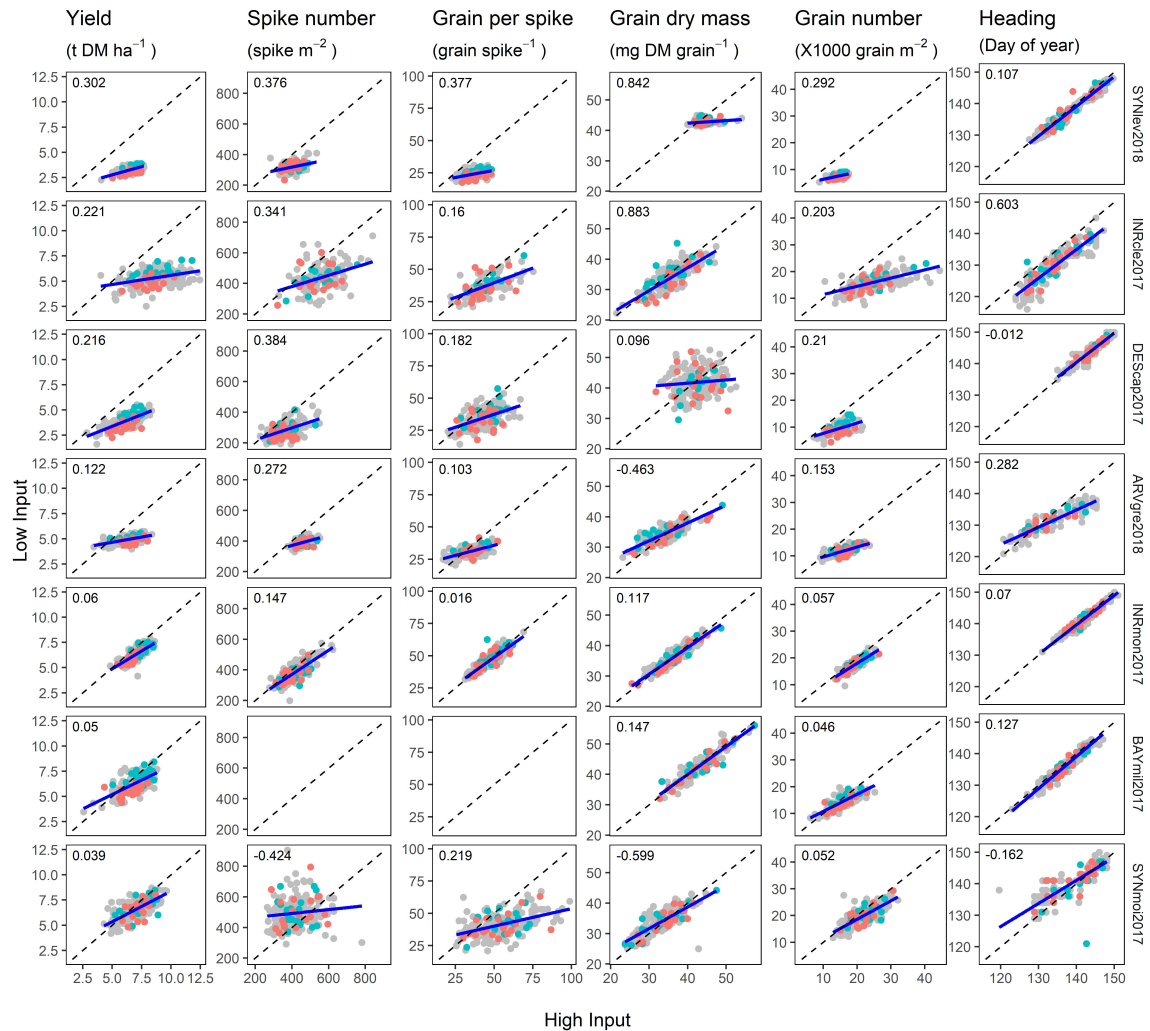


FIGURE 4.1 – Relationships between grain yield and its components for 273 bread wheat genotypes grown in low-input conditions with their values in high-input conditions. Dashed lines are 1 :1 relationship and solid blue lines are linear regressions (Pearson's correlation coefficients and P-values are given in the Supplementary Information Table S11). Blue and red circles are the 20 "tolerant" and "sensitive" genotypes selected in this study, respectively. In each panel, the value at the top left corner are the stress-index of the trial (see the Material and Method section), calculated for each variable (note that only the value for grain yield were used for genotype selection).

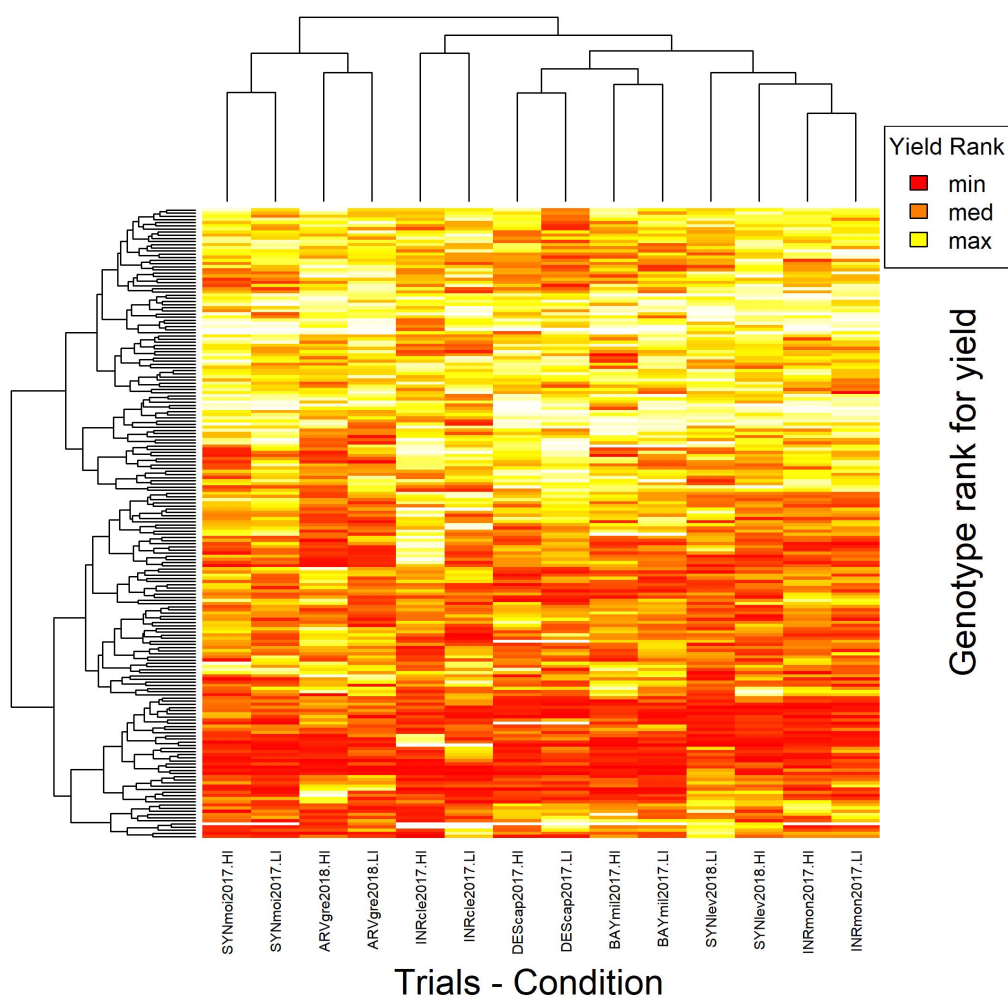


FIGURE 4.2 – Heat map of grain yield rank for 273 bread wheat genotype grown in the field in 14 treatment $\times$ trial combinations. The treatments $\times$ trial combinations and genotypes were group using a Hierarchical clustering analysis.

vironment x management (other than water and N) provided stronger G x E interactions than those driven by the N and water treatments, and that, even in the trials in which the overall effect of the treatment was high.

Taken together, these results show that G x E interactions due to N and water treatments can be highly dependent on the trial, highlighting the need of several trials and a proper method, if one aim at selecting genotypes differing in their response to LI conditions ($G \times E_{\text{treatment}}$ interactions).

4.3.2 Trials and treatments both participate to E and G x E effects on grain yield components under nitrogen and drought constraints

The analysis of heading date for each trial and each treatment showed (Fig. 4.1, Table S11) (i) that earliness was genotype-dependent (± 20 days depending on the trial; G effect), (ii) that heading dates were highly correlated between the two treatments of each trial (no G x E interaction), and this, on the 1 :1 line (no treatment-specific effect). These results indicated that treatment-specific effects on grain yield components could not be explained by any change of development rate between treatments (no G x E effect observed on heading date). On the other hand, the observed G effect (different heading dates between genotypes) may lead to differences in the timing of stresses between genotypes. This difference may differ between trials and may therefore result in different values of sensitivities calculated for the same genotype in different trials. Overall, in order to increase the robustness of the classification of genotypes according to their sensitivity to N or drought stresses, it is necessary to consider several trials that are different in their temporal scenarios of abiotic stresses.

Differences between trials were observed in the four grain yield components analyzed here, both for the median value of one component in one trial (E_{trial} effect on one component) and for the range of values covered by the genotypes ($G \times E_{\text{trial}}$). In addition, for most grain yield components and most trials, the observed values in LI conditions were significantly correlated with those in HI conditions

(Table S11, $P < 0.01$ in 24 out of the 26 tested correlations). This indicated, as for grain yield, that the G effect was high, compared to the $G \times E_{\text{treatment}}$ effect.

However, the correlation was not total for each component in each trial indicating some $G \times E_{\text{treatment}}$ effects. Overall, the data showed the importance of considering the trial effect and $G \times E_{\text{trial}}$ effect in order to properly analyze the diversity of responses to water and N deficits ($G \times E_{\text{treatment}}$).

4.3.3 Which component drives the diversity of the sensitivity of yield to abiotic constraints

In order to understand how grain yield components could drive the sensitivity of grain yield to LI conditions, we first analyzed the relationships between the sensitivity of grain yield to LI conditions in each trial and that of its components, the number of grain per square meter and the individual grain dry mass (Fig. 4.3 and Table 4.3). The sensitivity of the number of grain per square meter was then itself analyzed as a function of its two underlying components, the sensitivity of the grain number per spike and the number of spike per meter square (Fig. 4.3 and Table 4.4). The sensitivity of grain number was highly correlated with that of grain yield in all trials (average $r = 0.84$, all $P < 0.001$), and overall largely more than that of individual grain dry mass (r ranged from -0.04 to 0.45, non-significant in two trials). The analysis of variance of the sensitivity of grain yield considering these two components as predictors resulted in similar conclusions, grain number explaining 10-times more variance than individual grain dry mass, and that, largely independently of the trial (Table 4.3). Overall, both analyses indicate that the sensitivity of grain yield to LI condition ($G \times E_{\text{treatment}}$ on grain yield) was largely related to that of grain number per square meter rather than individual grain dry mass.

It was less clear which components of grain number (either grain number per spike or spike number per square meter) were most related to the sensitivity of grain yield to LI conditions (Fig. 4.3). For example, the sensitivity of the number

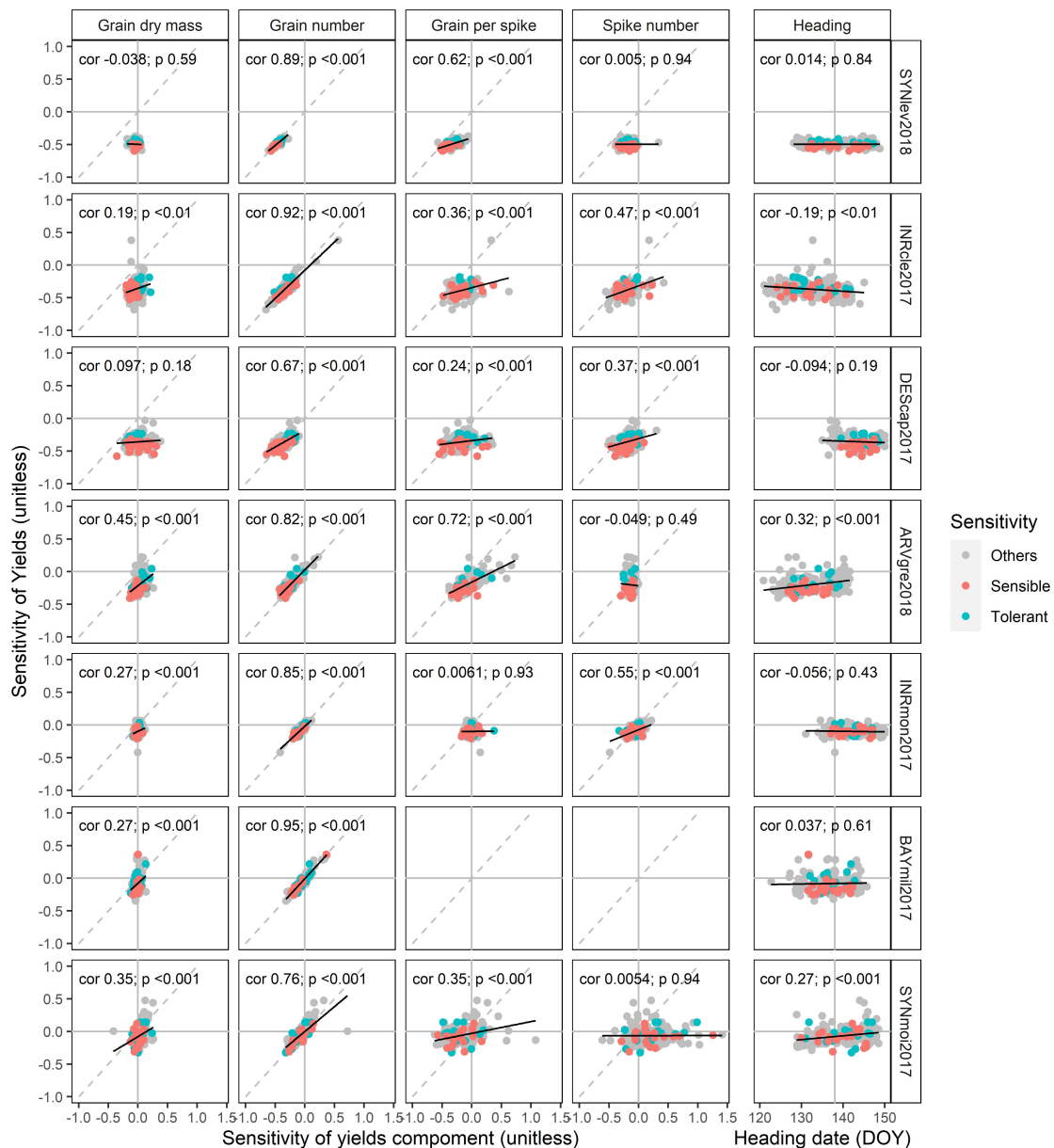


FIGURE 4.3 – Relationships between the sensitivity of grain yield and the sensitivities of grain number per square meter, grain per spike, spike number per square meter, and heading date for 273 bread wheat genotypes grown in field in seven trials. Grey dashed line are 1 :1 relationships and solid lines are linear regressions (Pearson correlation coefficients and the associated P-values are shown). Blue and red circles are the 20 “tolerant” and “sensitive” genotypes selected in this study, respectively.

TABLE 4.3 – Analysis of variance of the sensitivity of grain yield to water and nitrogen deficit, considering the sensitivities of its components (grain number per square meter and individual grain dry mass) as predictors together with the interactions with the trial.

Factors	Degree of freedom	Sum of squares	Mean squares	F-value	Pr(> F)	P-value
Grain number	1	42.54	42.54	90991	0	< 0.001
Grain dry mass	1	4.32	4.32	9236	0	< 0.001
Environment : Grain number	6	0.28	0.05	98	0	< 0.001
Environement : Grain dry mass	8	0.16	0.03	57	0	< 0.001
Residuals	1366	0.64	0	NA	NA	NA

NA : not applicable

of ears per square meter correlated with that of grain yield in the INRmon2017 trial, while this was not the case for the sensitivity of the number of grains per ear. The opposite was observed in ARVgre2018 and SYNmol2017. (Fig. 4.3).

We then tested the relative contributions of the sensitivities of spike number per square meter and grain number per spike to that of grain number (and therefore grain yield ; Fig. 4.3 and Table 4.4). Both underlying components explained the variance observed on the sensitivity of grain number (Table 4.4), with a higher effect for the number of grain per spike. This was also observed on the sensitivity of grain yield (Fig. 3), with most correlations (but not all, see above) between the sensitivity of grain yield and that of these two components being significant.

Overall, this analysis indicated that grain number per square meter drove most of the sensitivity of grain yield to LI conditions, but with different underlying components, either the number of spike per square meter (as in INRmon2017) or the grain number per spike (as in SYNev2018).

4.3.4 A method for subsetting genotypes and maximizing variability in the sensitivity of grain yield and its components, and their causal relationships

The results presented above demonstrate that (i) the effect of the trial E_{trial} could be very strong compared to that of the LI condition, (ii) that it was the same for the $G \times E_{\text{trial}}$, indicating the need to consider different trials robustly classify genotypes based on their grain yield sensitivity to LI conditions, (iii) depending on the trial, the sensitivity of grain yield was explained by the diversity of the

TABLE 4.4 – Analysis of variance of the sensitivity of grain number to water and nitrogen deficit, considering the sensitivities of its components (spike number per square meter and grain number per spike) as predictors together with the interactions with the trial.

Factors	Degree of freedom	Sum of squares	Mean squares	F-value	Pr(> F)	P-value
Spike number per square meter	1	8.79	8.79	4664	0	< 0.001
Grain number per spike	1	21.40	21.40	11358	0	< 0.001
Environment : Spike number per square meter	5	1.06	0.21	122	0	< 0.001
Environment : Grain number per spike	5	0.31	0.06	32	0	< 0.001
Residuals	1114	2.10	0.00	NA	NA	NA height

sensitivity of spike number per square meter or that of the number of grain per spike, indicating that an advantage on one of these components could have no effect on grain yield depending on the trial and therefore the genetic diversity of such traits could be lost when selecting genotype in such trials.

Taken together, these results indicated that a method considering several trials and the diversity of sensitivities to all grain yield components is necessary to select genotypes representing the diversity of grain yield responses to LI conditions. Therefore, we proposed a method to select a subset of genotypes that maximizes diversity of grain yield sensitivity to water and N constraints (i) based on sensitivities calculated in different trials (ii) by giving more weight to trials in which constraints had stronger effects on grain yield, (iii) with a particular attention to the relationships between grain yield components and grain yield, i.e. covering the whole phenotypic space of grain yield sensitivities and its components.

The metrics used for genotype selection was the average rank of grain yield sensitivity calculated here for grain yield of each genotype, the ranks in each trial being weighted by an index depending on the average effect of the LI treatment of the considered trial (see the Materials and Method section). Genotypes with the lowest rank were therefore considered as the most “sensitive” and genotypes with the highest rank the most “tolerant”. We selected the 20 genotypes with the lowest and higher average rank of sensitivity and considered them as the most “sensitive” and “tolerant” genotypes, respectively.

In bread wheat, contributions of each trial to the average sensitivity of grain yield to Li conditions are indicated in the top-left corner of each panel of Fig.1. For example, the trial with the highest weight (higher contribution to the calculation of average rank) was SYNlev2018 followed by INRcle2017 and DEScap2017 which contributed 30.2, 22.1, 21.6 % to the rank of genotype sensitivity to LI conditions (Fig. 4.1).

We then checked if the distribution of the 40 genotypes (sub-panel) covered the phenotypic space of grain yield and its components in LI and HI conditions

of the different trials (Fig. 4.1), together with the sensitivities of grain yield and its components in each trial (Fig. 4.3) and the average sensitivities (Fig. 4.4). Overall, we discriminated genotypes between those maintaining grain yield in LI conditions (“tolerant” genotypes) from those being more “sensitive” (Fig. 4.4D). This sensitivity was linked to the sensitivity of grain number in both the whole panel (see above) and our sub-panel (Fig. 4.4D). It was not the case for individual grain dry mass but the subset of genotypes covered the range of sensitivity observed in the whole panel (Fig. 4.4C). The sensitivity of grain number in our panel was due either to the sensitivity of the number of grains per spike (Fig. 4.4B) or the sensitivity of the number of spike per square meter (Fig. 4.4 A), as observed in the whole panel.

Overall, this selected sub-panel covered the phenotypic space of average sensitivities of grain yield components (Fig. 4.4), and we did not observe any kind of artefact which may have appeared in single trials (Fig. 4.1 and Fig. 4.3). The average ranks of sensitivity for all grain yield components of the selected bread wheat genotypes are given in Supplementary Information Table S9.

In durum wheat, the results were generally similar to those for bread wheat. As for bread wheat, we discriminated well “tolerant” and “sensitive genotypes” for grain yield. As in the whole panel, their grain yield sensitivity to LI conditions was related to their grain number sensitivity (Fig. 4.5B), and not to that of individual grain dry mass (Fig. 4.5A). The subset of genotypes covered the range of sensitivity observed in the whole panel for both underlying components of grain yield. As for grain yield we also check the effect of selection at the trial level (Fig. S9 and Fig. S10).). The average ranks of sensitivity for all grain yield components of the selected durum wheat genotypes are given in Supplementary Information Table S10.

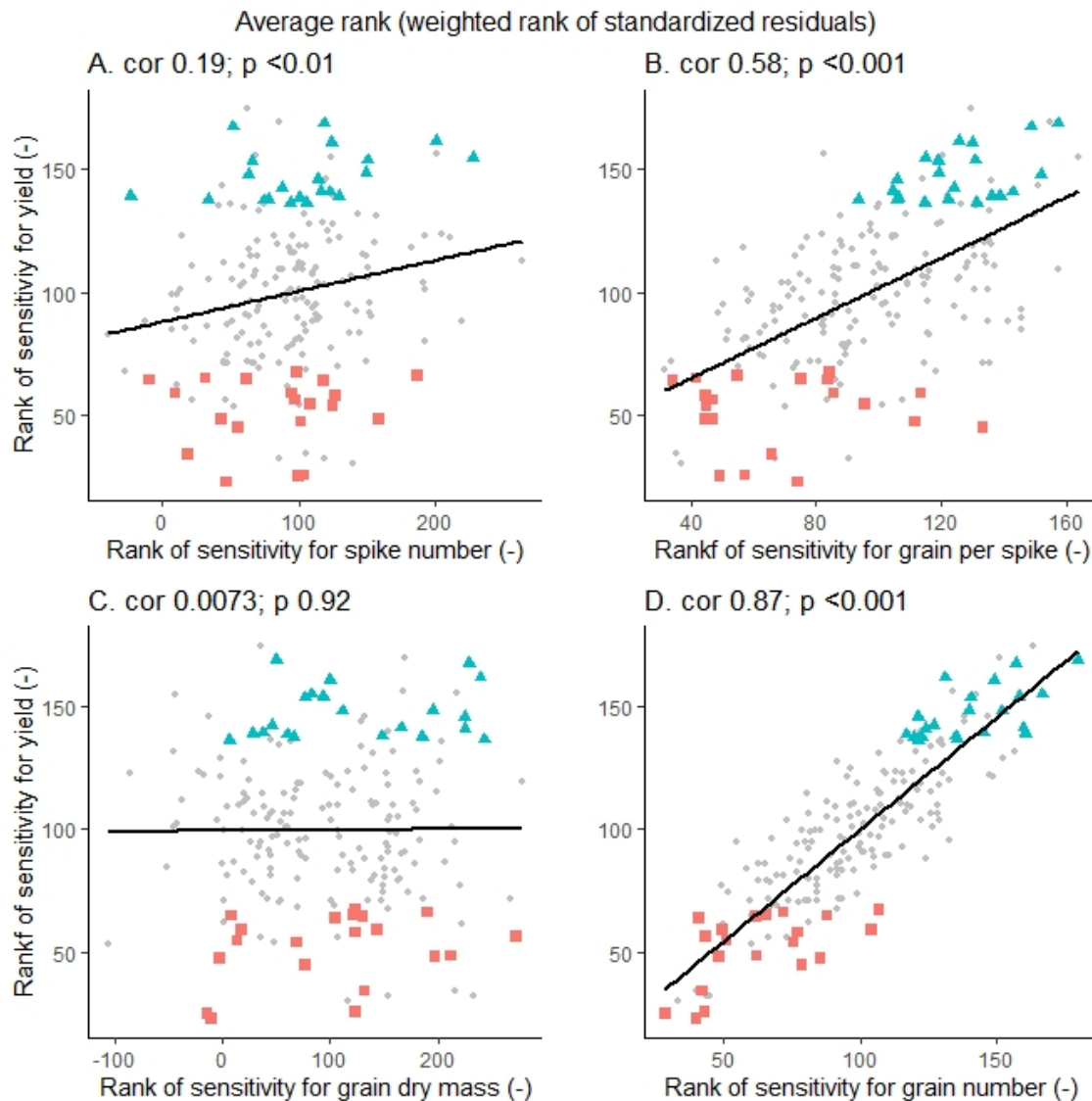


FIGURE 4.4 – Relationship between the weighted rank of the 273 bread wheat genotype ($rank_{genotype}$), based on all trials, for grain yield and for spike number per square meter (A), grain per spike (B), individual grain dry mass (C), and grain number per spike (D). Solid lines are linear regressions (Pearson's correlation coefficients and the associated P-values are shown). Blue and red circles are the 20 "tolerant" and "sensitive" genotypes selected in this study, respectively.

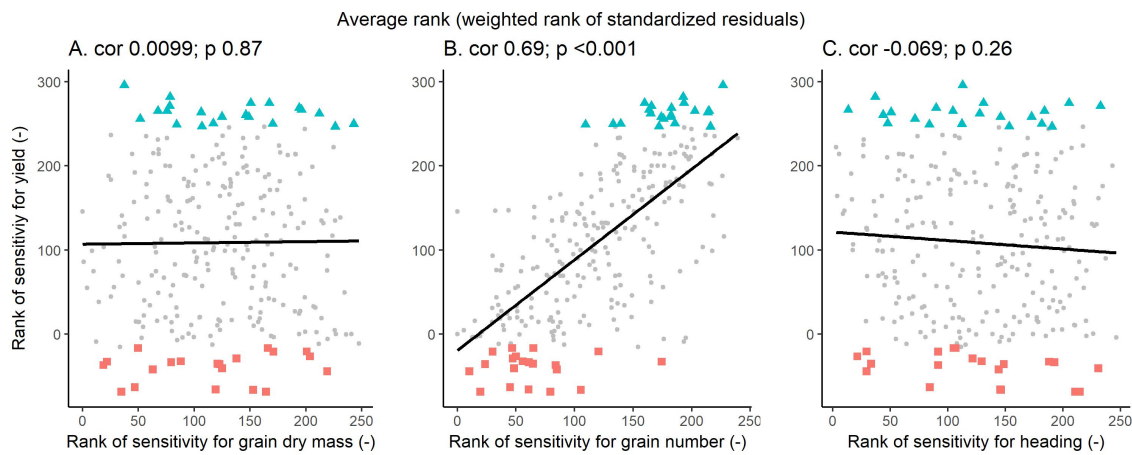


FIGURE 4.5 – Relationship between the weighted rank of the 232 durum wheat genotype ($rank_{genotype}$), based on all trials, for grain yield and for individual grain dry mass (A), and grain number per square meter (B). Solid lines are linear regressions (Pearson's correlation coefficients and the associated P-values are shown). Blue and red circles are the 20 “tolerant” and “sensitive” genotypes selected in this study, respectively.

4.3.5 Robustness of genotype selection as a function of the number of trials

The selection method for bread wheat genotypes, based on seven contrasting trials, respected the relationships between the sensitivities of grain yield components. Therefore, we investigated how the number of trials considered for selection of sensitive and tolerant genotypes might affect the results by analysing the effect of the number of trials on selection similarity when all trials are considered (and considered here as the ground-truth). We performed two types of analysis (Fig. 4.6). For both analysis, the selection of genotypes was performed using the method described in the Material and Methods section. These analyses were not performed for durum wheat because of the low number trials.

The first analysis considers the relative effect of each trial on genotype selection, starting with the trial with the highest stress-index and adding trials step by step, from the trial with the highest stress index to the trial with the lowest value. For example, we found that trial SYNlev2018 had the strongest contribution (30%) to the final rank (Fig. 4.1). Selection of genotypes based on this trial only resulted in about 55% of similarity with the selection that considered all trials (Fig. 4.6). The addition of trials improved the similarity with the final selection, even for

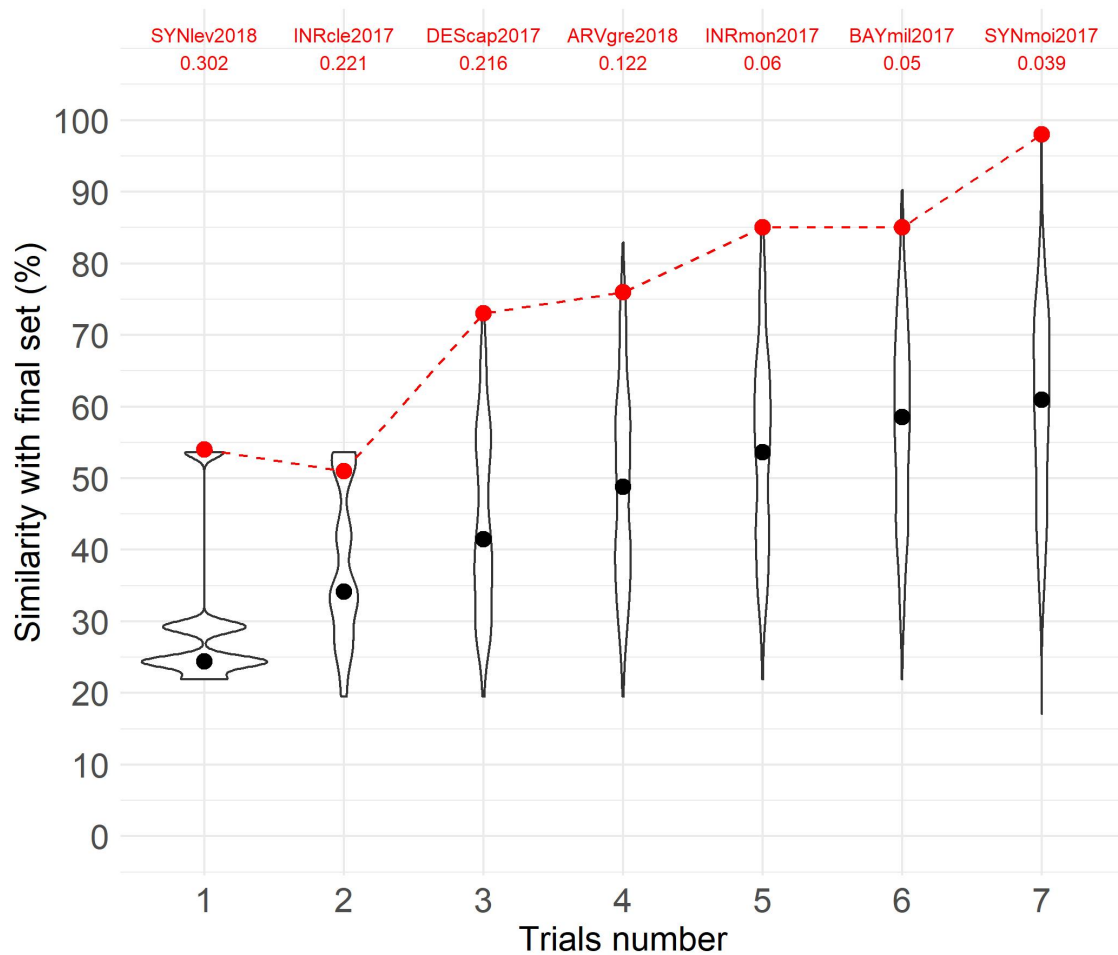


FIGURE 4.6 – Percentage of genotypes similar to the final selection for genotype selection based on one to seven trials. Red circles and dashed line are the selection of genotypes based on one to seven trials, starting with the one with the highest stress index to the one with the lowest stress index. The names of the trials and their average stress index are shown above the figure. Black circles and violin plots are random sampling (with replacement) of one to seven trials. Genotypes were selected using the method described in the Material and Methods section.

trials with very little weight in the final rank calculation.

The second analysis considered random sampling (with replacement) of a number of trials between 1 and 7 (Fig. 4.6). This analysis showed that, on average, a selection based on a single trial resulted in about 20-25% similarity with our final selection. The number of genotypes selected was 40 out of 232 genotypes at the beginning, i.e. 17%, so this selection was close to a random selection of genotypes. The more trials were analyzed, the closer the selection was to our final selection.

Overall, both analyses showed that increasing the number of trials increased the similarity with the ground-truth selection, and that, even with trials in which the effect of the LI condition was low.

4.4 Discussion

4.4.1 Phenotypic structure of grain yield sensitivity to water and nitrogen limitations

In this study, we have clearly observed that grain yield sensitivity to water and N constraints was related to that of grain number per spike rather than individual grain dry mass. This was observed in all trials, in bread wheat and durum wheat. This result was expected in well-watered conditions, but not necessarily under water and N limiting conditions. We considered a large diversity of environmental scenarios, and we would have expected that final stresses in rainfed environments would have affected individual grain dry mass. Indeed, the sensitivity to water shortage is higher at around anthesis than at the vegetative and grain filling stages (Fischer, 1993; Reynolds et al., 2004). On the contrary, grain yield is more sensitive to N uptake during the vegetative stage than during the reproductive and maturation stages (Sharma et al., 1992), with an overlap between these two periods around the stage of stem elongation. In addition, the grain yield sensitivity to water stress would increase at higher N application rates.

The correlation between grain yield and grain number has long been a matter of debate. Indeed, grain number and grain yield are correlated in many species (Sadras, 2021). The strong correlation between grain yield and grain number had been largely interpreted as a cause and effect relationship until Sinclair & Jamieson challenged this view (Sinclair and Jamieson, 2008). Variation in grain number and grain yield would likely be caused by variations in resource accumulation, particularly N, by the crop (Sinclair and Jamieson, 2008). In our study, this relationship was valid under all tested conditions, in bread wheat, and in durum wheat, regardless of the environmental scenario, with a weak effect of individual grain dry mass, especially for bread wheat. Overall, these results rather validate the causal theory, according to which the number of grains (carbon sinks) would set a large part of the final grain yield very early in the cycle, with very little room for the sensitivity of individual grain dry mass in explaining the variance of final grain yield.

The sensitivity of the number of grains per square meter was linked to the sensitivity of the number of spike per square meter in some trials and sometimes to the number of grains per spike, depending on the trial. This is consistent with the fact that there are different sensitivity periods for the number of spike per square meter and the number of grains per spike (Parent et al., 2017). Indeed, the traits related to spike development (length of spike and number of spikelets) are rather sensitive during the early vegetative stages, i.e. the period of spike development (Vahamidis et al., 2014). In Parent et al. (2017), the number of grains per spikelet was very sensitive during a short period around flowering, possibly related to male or female sterility during gametocyte development (Saini et al., 1984; Saini and Aspinall, 1982), pollination (Ferris et al., 1998), or competition for carbon at an early stage of grain filling. The number of grains per spikelet is less sensitive to early environmental conditions, which could have affected spikelet development (Saini and Aspinall, 1982). In our study, the diversity of scenarios has therefore led to different sensitivities between components of grain number, and to make the best use of the genetic diversity of sensitivities of these two

underlying grain number variables.

4.4.2 Subset genotypes cover the phenotypic space of the whole panel

In our study, the set of correlations between grain yield, its components, and their sensitivities, defined a constrained phenotypic space. Our genotype selection respected this phenotypic space, in the sense that it kept all its components, the relationships between variables and the genetic variability leading to these relationships.

Several types of constraints can lead to strong correlations between *a priori* independent biological variables, leading to a bounded phenotypic space : evolution and selection, genetics, biophysics, and plant structure (Donovan et al., 2011). In particular, here, varieties/species have been able to evolve in the same direction for several traits because they were subjected to the same environmental or agronomic constraints (Mayr and Provine, 1998). Evolution and selection have therefore been able to constrain this space, in particular selection for grain quality, by selecting varieties with fewer grains under limiting conditions but maintaining quality grains. As these relationships are similar in durum wheat (tetraploid) and bread wheat (hexaploid), the hypothesis of strong genetic constraints (proximity, link between genomic regions) is less likely than the hypothesis of common evolutionary and agronomic constraints.

4.4.3 Possible improvements of the genotype selection method

In this study we described a method to sample genotypes in durum and bread wheat genetic panels that respected the phenotypic space of the whole panels. However, these relationships were only verified afterwards, the method being based on grain yield only. Although a given number of safeguards have been determined (contrasted trials, weightings, validation afterwards), it could be interesting to build a multi-criteria selection on the rank of the sensitivity of the grain yield,

as well as on those of its components, to guarantee, from the start, that the phenotypic space is fully covered in the final selection. Currently we have not finalized and/or use such method, mainly because of the lack of time (the following activities for my thesis and in SolAce depended on it), and because of a needed consensus between different actors in the SolACE project.

4.4.4 Conclusions

The set of correlations between yield, its components and their sensitivities defined a very constrained phenotypic space. Our genotypic selection respected this phenotypic space, in the sense that it preserved all its components, the relationships between variables and the genetic variability leading to these relationships. Several types of constraints can lead to strong correlations between independent biological variables "a priori", leading to a bounded phenotypic space : evolution and selection, genetics, biophysics, plant structure. In particular, here, varieties/species were able to evolve in the same direction for several traits, because they were subject to the same environmental or agronomic constraints (Mayr and Provine, 1998). Evolution and breeding were thus able to constrain this space, particularly breeding for grain quality, by selecting varieties with fewer grains under limiting conditions, but maintaining grain quality. As these relationships are similar in durum wheat (tetraploid) and bread wheat (hexaploid), the hypothesis of strong genetic constraints (proximity, linkage between genomic regions) is less likely than the hypothesis of common evolutionary and agronomic constraints.

The genotypes selected in this study, where used in several phenotyping experiments under semi-field conditions at Clermont-Ferrand, France (for bread wheat), and Groux-les-Bains (for durum wheat), and in platforms to phenotype the response of shoot growth (in the PhenoArch platform at UMR LEPSE, see chapter 6 of this thesis), and root growth and architecture (in the RooPhAir platform at UCL, Louvain, Belgium, and 4PMI, UMR Agrecolgy, Dijon) to water and N limitations. The results of the PhenoArch experiment are presented in the follo-

wing chapter 5, and those of the semi-field experiments are presented in Chapter 6.

4.5 References

Alamanos, A., Latinopoulos, D., Papaioannou, G., Mylopoulos, N., 2019. Integrated Hydro-Economic Modeling for Sustainable Water Resources Management in Data-Scarce Areas : The Case of Lake Karla Watershed in Greece. *Water Resour Manage* 33, 2775–2790. <https://doi.org/10.1007/s11269-019-02241-8>

Balfourier, F., Bouchet, S., Robert, S., De Oliveira, R., Rimbart, H., Kitt, J., Choulet, F., Appels, R., Feuillet, C., Keller, B., Praud, S., Baumann, U., Budak, H., Rogers, J., Eversole, K., Alaux, M., Bejar, B., Lafarge, S., Lagendijk, E., Derory, J., Le Gouis, J., Paux, E., 2019. Worldwide phylogeography and history of wheat genetic diversity. *Sci. Adv.* 5, eaav0536. <https://doi.org/10.1126/sciadv.aav0536>

Berger, J., Palta, J., Vadez, V., 2016. Review : An integrated framework for crop adaptation to dry environments : Responses to transient and terminal drought. *Plant Science* 253, 58–67. <https://doi.org/10.1016/j.plantsci.2016.09.007>

David, J., Holtz, Y., Ranwez, V., Santoni, S., Sarah, G., Ardisson, M., Poux, G., Choulet, F., Genthon, C., Roumet, P., Tavaud-Pirra, M., 2014. Genotyping by sequencing transcriptomes in an evolutionary pre-breeding durum wheat population. *Mol Breeding* 34, 1531–1548. <https://doi.org/10.1007/s11032-014-0179-z>

Donovan, L.A., Maherali, H., Caruso, C.M., Huber, H., de Kroon, H., 2011. The evolution of the worldwide leaf economics spectrum. *Trends in Ecology & Evolution* 26, 88–95. <https://doi.org/10.1016/j.tree.2010.11.011>

Ferris, R., Ellis, R.H., Wheeler, T.R., Hadley, P., 1998. Effect of High Temperature Stress at Anthesis on Grain Yield and Biomass of Field-grown Crops of Wheat. *Annals of Botany* 82, 631–639. <https://doi.org/10.1006/anbo.1998.0740>

Fischer, R.A., 1993. Irrigated spring wheat and timing and amount of nitrogen fertilizer. II. Physiology of grain yield response. *Field Crops Research* 33, 57–80. [https://doi.org/10.1016/0378-4290\(93\)90094-4](https://doi.org/10.1016/0378-4290(93)90094-4)

Fischers, E., Knutti, R., 2013. Robust projections of combined humidity and temperature extremes | Nature Climate Change. *Nature Climate Change* 3, 126–130.

Jafarikouhini, N., Kazemeini, S.A., Sinclair, T.R., 2021. Fresh sweet corn yield sensitivity to deficit nitrogen and water conditions. *Journal of Crop Improvement* 0, 1–11. <https://doi.org/10.1080/15427528.2021.1995560>

Lobell, D.B., Schlenker, W., Costa-Roberts, J., 2011. Climate Trends and Global Crop Production Since 1980. *Science* 333, 616–620. URL

Mayr, E., Provine, W.B., 1998. *The Evolutionary Synthesis : Perspectives on the Unification of Biology*. Harvard University Press.

Parent, B., Bonneau, J., Maphosa, L., Kovalchuk, A., Langridge, P., Fleury, D., 2017. Quantifying Wheat Sensitivities to Environmental Constraints to Dissect Genotype x Environment Interactions in the Field. *Plant Physiol.* 174, 1669–1682. <https://doi.org/10.1104/pp.17.00372>

Paux, E., Lafarge, S., Balfourier, F., Derory, J., Charmet, G., Alaux, M., Perchet, G., Bondoux, M., Baret, F., Barillot, R., Ravel, C., Sourdille, P., Le Gouis, J., on behalf of the BREEDWHEAT Consortium, 2022. Breeding for Economically and Environmentally Sustainable Wheat Varieties : An Integrated Approach from Genomics to Selection. *Biology* 11, 149. <https://doi.org/10.3390/biology11010149>

Pe'er, G., Zinngrebe, Y., Moreira, F., Sirami, C., Schindler, S., Müller, R., Bontzorlos, V., Clough, D., Bezák, P., Bonn, A., Hansjürgens, B., Lomba, Â., Möckel, S., Passoni, G., Schleyer, C., Schmidt, J., Lakner, S., 2019. A greener path for the EU Common Agricultural Policy. *Science* 365, 449–451. URL

R Core Team, 2018. R : A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

Rémy, J.C., Hébert, J., 1977. Le devenir des engrais dans le sol. *Compte Rendu de l'Académie d'Agriculture de France, français, biblio* 63, 700–710.

Reynolds, M.P., Trethowan, R., Crossa, J., Vargas, M., Sayre, K.D., 2004. Erratum to “Physiological factors associated with genotype by environment interaction in wheat.” *Field Crops Research* 85, 253–274. [https://doi.org/10.1016/S0378-4290\(03\)00057-1](https://doi.org/10.1016/S0378-4290(03)00057-1)

Sadras, V.O., 2021. Evolutionary and ecological perspectives on the wheat phenotype. *Proceedings of the Royal Society B : Biological Sciences* 288, 20211259. <https://doi.org/10.1098/rspb.2021.1259>

Saini, H.S., Aspinall, D., 1982. Abnormal Sporogenesis in Wheat (*Triticum aestivum* L.) Induced by Short Periods of High Temperature. *Annals of Botany* 49, 835–846. <https://doi.org/10.1093/oxfordjournals.aob.a086310>

Saini, H.S., Sedgley, M., Aspinall, D., 1984. Development Anatomy in Wheat of Male Sterility Induced by Heat Stress, Water Deficit or Abscissic Acid. *Functional Plant Biol.* 11, 243–253. <https://doi.org/10.1071/pp9840243>

Sharma, B.D., Jalota, S.K., Kar, S., Singh, C.B., 1992. Effect of nitrogen and water uptake on yield of wheat. *Fertilizer Research* 31, 5–8. URL

Shipley, B., Lechowicz, M.J., Wright, I., Reich, P.B., 2006. Fundamental Trade-Offs Generating the Worldwide Leaf Economics Spectrum. *Ecology* 87, 535–541. <https://doi.org/10.1890/05-1051>

Sinclair, T.R. (Ed.), 2017. Water-Conservation Traits to Increase Crop Yields in Water-deficit Environments, *SpringerBriefs in Environmental Science*. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-319-56321-3>

Sinclair, T.R., Jamieson, P.D., 2008. Yield and grain number of wheat : A correlation or causal relationship ? : Authors' response to “The importance of grain or kernel number in wheat : A reply to Sinclair and Jamieson” by R.A. Fischer. *Field Crops Research* 105, 22–26. <https://doi.org/10.1016/j.fcr.2007.07.003>

Tardieu, F., 2012. Any trait or trait-related allele can confer drought tolerance : just design the right drought scenario. *J Exp Bot* 63, 25–31. URL

Urban, D.W., Sheffield, J., Lobell, D.B., 2015. The impacts of future climate and carbon dioxide changes on the average and variability of US maize yields under two emission scenarios. *Environ. Res. Lett.* 10, 045003. <https://doi.org/10.1088/1748-9326/10/4/045003>

Vahamidis, P., Karamanos, A., Economou, G., Fasseas, C., 2014. A new scale for the assessment of wheat spike morphogenesis. *Annals of Applied Biology* 164, 220–231. <https://doi.org/10.1111/aab.12097>

Vahamidis, P., Karamanos, A.J., Economou, G., 2019. Grain number determination in durum wheat as affected by drought stress : An analysis at spike and spikelet level. *Annals of Applied Biology* 174, 190–208. URL

Wright, I.J., Reich, P.B., Westoby, M., Ackerly, D.D., Baruch, Z., Bongers, F., Cavender-Bares, J., Chapin, T., Cornelissen, J.H.C., Diemer, M., Flexas, J., Garnier, E., Groom, P.K., Gulias, J., Hikosaka, K., Lamont, B.B., Lee, T., Lee, W., Lusk, C., Midgley, J.J., Navas, M.-L., Niinemets, Ü., Oleksyn, J., Osada, N., Poorter, H., Poot, P., Prior, L., Pyankov, V.I., Roumet, C., Thomas, S.C., Tjoelker, M.G., Veneklaas, E.J., Villar, R., 2004. The worldwide leaf economics spectrum. *Nature* 428, 821–827. <https://doi.org/10.1038/nature02403>

Tillering structures the genetic variability of wheat vegetative growth and its plasticity under water deficit

Abstract

Leaf growth in response to water stress in wheat plants is determined by several factors, such as tillering and leaf number as well as their time-dependent rates. In this chapter, we present the trade-offs between the components of leaf expansion at the whole plant level and their sensitivities to water deficit. We explain that the observed diversity of leaf growth under different water scenarios is a result of the diversity of tillering, at least more than that of the rate of leaf emission or elongation of individual leaves. The plasticity of leaf area was mainly determined by that of tillering under our growing conditions. In addition, we show the dynamics of leaf growth, as well as the respective shares of its subcomponents, their plasticities and their mutual relationships. We used simple formalisms for the response of the leaf area components, as well as the associated genetic parameter structure that can be used for modelling. As an output, we identify tillering

as a trait that drives leaf area underwater deficit and that the genetic variability associated with this trait explains the genetic variability of leaf area in wheat.

Key words : bread wheat, drought sensitivity, durum wheat, genetic variability, leaf expansion, phenotypic plasticity, tillering.

5.1 Introduction

Agriculture is facing both the increasing global demand for food and climate change. Increasing frequency of drought events, and other extreme weather events (Lobell et al., 2011), increases interannual variability in crop yields in different regions of the globe (Urban et al., 2015), mainly due to multiple drought scenario (Fischer and Knutti, 2013).

In this context, plants constantly adapt to changes in their microclimate and soil water status, particularly through the control of transpiration (Faralli et al., 2019a, 2019b; Hatfield and Dold, 2019). Transpiration is controlled in the short term by stomatal conductance and in the long term by leaf area dynamics (Sinclair, 2017). Control of stomatal conductance and leaf expansion are thus key players in the genetic and environmental variability of transpiration (Tardieu and Parent, 2017). However, climate change appears to increase the frequency of combined high temperature and drought events (Barnabás et al., 2008; Lobell et al., 2011; Webber et al., 2018). In this case, the decrease in stomatal conductance can lead to over-heating (Rizhsky et al., 2002), with significant and sometimes irreversible consequences.

Reduction of leaf growth under water deficit allows to adjust transpiration without negative heating. The use of traits associated with leaf expansion as genetic levers in plant breeding may seem sensible but is not straightforward. Indeed, the control of leaf area and its dual consequences on photosynthesis and transpiration is a major player in Genotype x Environment (G x E) interactions. For example, under adverse conditions, every millimeter of water saved during

the vegetative phase allows for a more favorable water status during the later stages of flowering and grain filling (Sinclair, 2017). In contrast, under moderate drought, the reduction in biomass accumulation accompanying lower leaf area and transpiration is not compensated for during the growth cycle, resulting in lower biomass and grain production (Tardieu, 2012). In addition, while in the short-term expansive growth can be view as largely uncoupled from structural growth (Hilty et al., 2021), in the longer term feedbacks can occur between the “container” and the “content” (e.g. carbon sink/source relationships), so that a greater leaf area can lead to a greater final biomass. Thus, the best variety observed in one environment may be a very poor choice in a different drought scenario (Tardieu, 2012).

"Tuning growth to the environmental demand" (Rymen and Sugimoto, 2012) requires focusing not only on the plants constitutive growth potential but also on its ability to modulate its growth under drought (plasticity, Bradshaw, 1965). Plasticity of plant architectural traits allows plants to modify their architecture when environment changes (Pierik et al., 2021), increasing the range of adaptation of a species (Bacher et al., 2022). It is therefore at the basis of plant performance in both natural and crop production environments (Pierik et al., 2021). Phenotypic plasticity occurs at all levels of organization within plants (Funk et al., 2007). This is particularly true for expansive growth (Belaygue et al., 1996). Indeed, the whole-plant leaf area is the consequence of three underlying processes (branching or tillering, individual-leaf expansion rate, and leaf apperance rate). The fact that these processes have their own timing within the crop-cycle and potentially their own genetic variability and response to water deficit makes it complex to analyze the genetic variability of whole plant growth under fluctuating environments. There is no single response to water deficit, but potentially as many as there are water deficit scenarios (duration, intensity, timing for example) (Koch et al., 2019).

There is potentially a large variability of stress-adaptive plasticity in wheat, as found in wheat relatives (Leveau et al., 2021 ; Suneja et al., 2019).The rate of leaf

appearance is often considered to be stable and independent of the environment (except for photoperiod and temperature). In the only paper to our knowledge that considered the relative sensitivities of the underlying components of shoot expansion (in white clover), Belaygue et al. (1996) found that leaf appearance rate was the least sensitive process. However, in both durum and bread wheat, water deficit has been shown to decrease the rate of leaf appearance. In addition, Baumont et al. (2019) showed that the rate of leaf appearance in wheat plants is sensitive to the carbon status of the plant, and thus to the environmental conditions driving this status (intercepted radiation, photoperiod, CO₂, ...), and that genetic variability exists for this trait. Overall, the relative contribution of leaf appearance rate to the genetic variability of whole-plant expansive growth and its plasticity under drought remains largely unknown.

In most studied species, there is a large variability in the response of individual leaf expansion to soil water potential (SWP; e.g. Parent et al., 2009 in rice; Welcker et al., 2011 in maize), as is its response to other environmental conditions in wheat (VPD; Leveau et al., 2021). In non-tillering maize hybrids, the response of individual leaves to SWP correlated to that of whole plant (Lacube et al., 2020). However, its relative influence in tillering species such as wheat species has been far less studied.

Tillering traits under drought have long been a breeding target, especially for drought-prone environment. However, most studies have focused on the intrinsic value of tillering potential, with the aim of reducing tillering. Indeed, the advantage of reduced-tillering traits is close to a “drought avoidance strategy”, reducing leaf area in order to shift soil water availability from pre- to post-anthesis when periods of drought stress are frequent (Berry et al., 2003). A reduced-tillering *tin* (tiller inhibition) gene (Atsmon and Jacobs, 1977) increases grain yield in dryland environments but not under wetter conditions (Hendriks et al., 2016; Reynolds et al., 2009). While it appears that tillering is a key process for phenotypic plasticity (Sanad et al., 2016), we could find only few results on the plasticity of free-tillering genotypes and the associated genetic diversity, scatters

into more general studies.

A purely experimental approach cannot explore the consequences on yield of each combination of plasticity-related traits across the full range of drought scenarios. Model-based ideotyping (Martre et al., 2015; Parent and Tardieu, 2014; Rötter et al., 2015) uses a crop model with genetic parameters, capable of simulating the Genotype $\times$ Environment $\times$ Management interactions under a wide variety of present and future local scenarios (Parent et al., 2018), combined with phenotypic measurements of the genetic variability of its parameters. Such a model therefore needs to be built according to the possibilities of large-scale phenotyping (genetic diversity for hundreds of genotypes) (Lacube et al., 2017), and defining the constraints between parameters. It allows the model to not be over-parameterized and to define idéotypes that are achievable by breeding. The set of constraints and genetic correlations between traits defines the set of possibilities, called phenotypic space (Shipley et al., 2006; Wright et al., 2004), representing only a part of the hyper-cube of parameters (Raup and Michelson, 1965).

In the case of whole plant shoot expansion, if its underlying components and their sensitivities to environmental constraints were fully independent, with different genetic controls, this would imply many genetic parameters that could be very difficult to measure in hundreds of genotypes. In one sense, these components could all have their own response to environmental constraints, at different time scales, and could integrate differently the “history” of the plant micro-environment. However, the independency of these traits can be questioned because (i) some common underlying processes can drive several components, such as water potential and turgor, and (ii) the potential of one process (e.g., Individual leaf expansion rate) under well-watered conditions is linked to its sensitivity to water deficit, the biggest plants being the most sensitive to water stress (e.g. Welcker et al., 2011).

In this study, we used a non-destructive imaging platform in order to analyze the temporal shoot growth dynamics under well-defined water deficit scenarios

(Jin et al., 2012) and determine the links and trade-off between the components of shoot expansion and their sensitivities to water deficit. We conducted a large experiment in a high throughput platform (PHENOARCH; Cabrera-Bosquet et al., 2016; Prado et al., 2018) on 84 bread wheat (*Triticum turgidum* L. subsp. *durum* (Desf.) Husn.) and durum wheat (*T. aestivum*) genotypes in four drought scenarios. We combined automatic measurements of shoot area over time (> 25,000 measurements) with manual measurements of individual leaf length, tiller number and leaf number over time (> 10,000 measurements for each trait), to analyze the time courses of shoot area expansion and its underlying components under drought scenarios. We calculated the rates of each process and their sensitivity to SWP for each genotypes (Table 5.1). Our results demonstrate that the observed diversity of plasticity for shoot area expansion is mostly driven by that of tillering. In addition, the most plastic component under drought was tillering, explaining more than two third of the phenotypic plasticity observed at whole-plant scale. This work can have large consequences in (i) ecophysiology, by describing the response of specific processes in a large genetic diversity of bread wheat and durum wheat, (ii) in breeding, by determining the genetic variability for target traits under drought in these two species, and (iii) in modelling, by simplifying the parameterization of the plasticity of leaf area under drought and by describing the phenotypic space of the shoot expansion traits that can be used in ideotyping studies.

TABLE 5.1 – *Processes and variables analyzed in this study.*

Process	Variables		Rates		Sensitivity to soil water potential
	Absolute values		Metrics		
	Dynamic variable (observed)	Dynamic variable	Dynamic variable	Metrics	
Shoot expansion	Shoot area (SA, cm ² plant-1)	Maximum shoot area per plant (SAm _{ax} , cm ² plant-1)	Shoot expansion rate (SER, cm ² °Cd-1)	Maximum shoot expansion rate (SER _{max} , cm ² °Cd-1)	Sensitivity of shoot expansion rate (SSER, Mpa-1)
Leaf appearance	Main-stem leaf number (LN, leaf main stem-1)	Main-stem final leaf number (FLN, leaf main stem-1)	Main-stem leaf appearance rate (LAR, leaf °Cd-1)	Average main-stem leaf appearance rate (LAR _{mean} , leaf °Cd-1)	Sensitivity of main-stem leaf appearance rate (SLAR, Mpa -1)
Tiller appearance	Tiller number per plant (TN, tiller plant-1)	Maximum tiller number per plant (TN _{max} , tiller plant-1)	Tiller appearance rate (TiR, tiller °Cd-1)	Maximum tiller appearance rate (TiR _{max} , tiller °Cd-1)	Sensitivity of tiller appearance rate (STiR, Mpa -1)
Individual leaf elongation	Cumulative leaf length per plant (CLL _{max} , cm plant-1)	Maximum cumulative leaf length per plant (CLL _{max} , cm plant-1)	Main-stem leaf elongation rate (LER, cm °Cd-1)	Maximum main-stem leaf elongation rate (LER _{max} , cm °Cd-1)	Sensitivity of main-stem leaf elongation rate (SLER, Mpa-1)

5.2 Results

5.2.1 Dynamic responses of leaf area expansion and underlying processes to water deficit

The dynamics of whole-plant leaf area (shoot area, SA) expansion was analyzed for 49 genotypes of bread wheat and 37 genotypes of durum wheat under a well-watered scenario (S1) and three water-deficit scenarios (S2 to S4) in the high-throughput plant phenotyping platform PhenoArch (Cabrera-Bosquet et al., 2016). The time course of SA versus thermal time after leaf-3 emergence were fitted to a cubic smoothing spline function (> 36 time points per plant, see Material and Methods for details; Fig. 5.1A). For all plants under well-watered conditions (scenario S1), time courses of SA showed a typical sigmoid curve with a period of exponential increase, followed by an almost-linear increase, a period of slowing-down, a maximum value, and a decrease due to leaf senescence after anthesis (Fig. 5.1A).

Whole-plant leaf expansion rate (shoot expansion rate, SER), calculated as the first derivation of SA curves had a bell shape (Fig. 5.1B), with maximum values between 400 and 800°Cd after leaf-3 appearance depending on the genotype. The three water deficit scenarios affected the time course of SER. In scenario S4 (a 31-d period of water deficit starting at the beginning of tillering), as illustrated in Figure 1B for the durum wheat cultivar Balsamo, SER increased similarly to that of well-watered plants during a first period, and then increased more slowly from around 250°Cd after leaf-3 appearance, when SWP decreased below -0.2 MPa (Fig. 5.1D). It resulted that the relative value of SER of droughted plants compared to that of well-watered plants was close to one until around 20°Cd after L-3 appearance (Fig. 5.1C), when the soil water potential (SWP) was about 0.2 MPa (Fig. 5.1D), and then decreased almost linear as SWP continued to decrease. When plotted against SWP, during a period combining maximum values of SER in well-watered conditions (S1) and between the time at which SWP decreased below -0.2 MPa and the day of first rewatering, the relative SER showed a linear

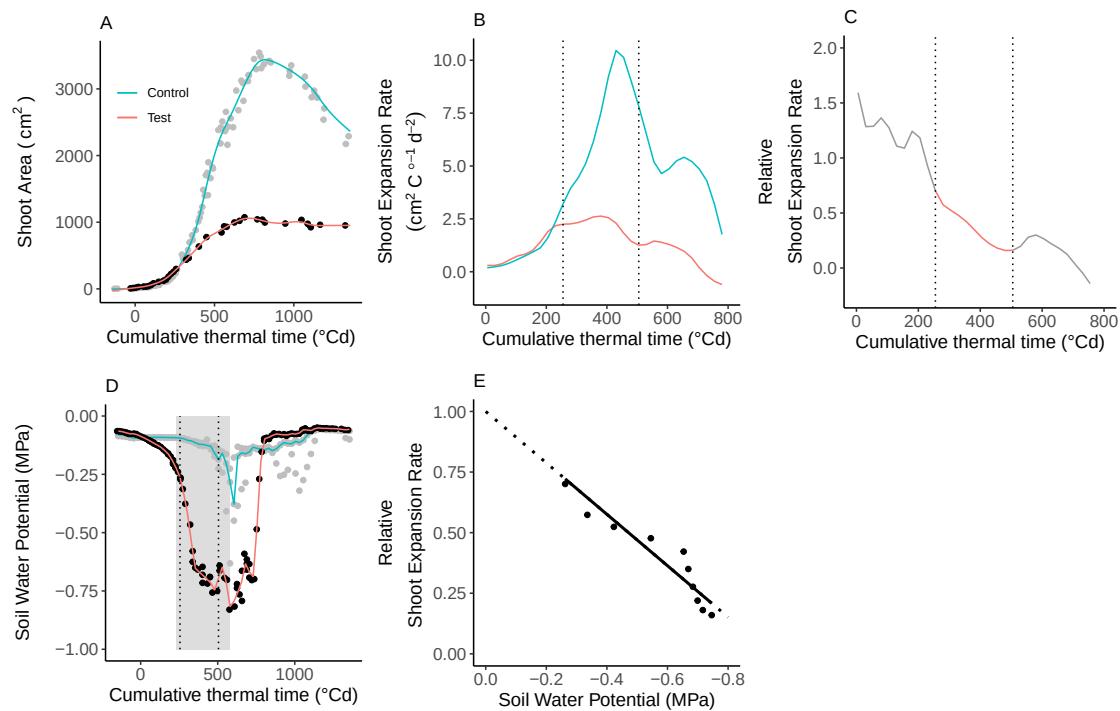


FIGURE 5.1 – Example of calculation of dynamic variables and their sensitivity to soil water potential. Data are for whole-plant leaf area (shoot area, SA) for the durum wheat cultivar Balsamo in the well-watered (control, scenario 1) and the water deficit scenario S4 (long water deficit). (A) Time courses of SA versus thermal time after leaf-3 appearance. A cubic smoothing spline (lines) was fitted to experimental data (circles). For the control treatment, the spline function was fitted to the data of all the plants ($n = 3$ to 6 replicates depending on the group), while for the water deficit scenario, the data for each plant were fitted separately. (B) Whole-plant leaf area expansion rate (shoot expansion rate, SER) versus thermal time after leaf-3 appearance. SER was calculated as the first derivative of the spline function equation fitted to SA versus thermal time after leaf-3 appearance. (C) Relative SER versus thermal time after leaf-3 appearance. Relative SER was calculated by dividing SER of each plant in the water-deficit scenario (red curve in B) by that of the plants in the well-watered scenario (blue curve in B). (D) Soil water potential (SWP) versus thermal time after leaf-3 appearance for the well-watered (blue circles and curves) and water-deficit (red symbols and lines). As in (A), for the control treatment, data of all the plants ($n = 3$ to 6 replicates) were fitted together, while for the water deficit scenario data, the data of each plant were fitted separately. In (B), (C), and (D), the vertical dashed lines indicate the period beginning when SER in the control treatment was at 30% of its maximum value and ending when it had decreased by 20% of its maximum value. In (D) the gray band indicates the period beginning when SWP in water deficit scenario reached to -0.2 MPa and ending when at the plants were first rewatered. (E) relative SER versus SWP during the consensus period defined by the vertical dashed lines and the gray band. Solid line is linear regression forced to 1 when SWP is null. The slope of this regression corresponds to the sensitivity of SER to SWP.

response to soil water deficit (Fig. 5.1E). All genotypes showed similar SER time course and response to SWP, but with different absolute values (Fig. 5.2).

Other drought scenarios affected SER in a similar way but with different timings (Fig. 5.2). For example, in scenarios S3, for which plant watering was reduced 10 d later compared to S2 and S4, for most genotypes SER started to diverge from that of well-watered plants slightly later compared to S2 and S4 plants (Fig. 5.2C). SER of early-rewatered plants (S2) diverged from that of non-rewatered plants (S4) rapidly after rewatering but with values lower than those of well-watered plants for 450 to 550°Cd after rewatering depending on genotype, and exceeded those of well-watered plants after anthesis (Fig. 5.2C). The time period defined above to calculate the sensitivity of SER to SWP resulted, for all the genotypes, in a unique response in the three tested water-deficit scenarios (Fig. 5.2D).

Similar analyses of the time course of absolute values, rates, relative rates and sensitivities to SWP were performed for the three underlying components of shoot area (i.e. number of appeared tillers, number of appeared leaves, and leaf length; Table 5.1). The tiller appearance rate (TiR) of well-watered plants (S1), calculated from the number of appeared tillers over time, had a time course very-similar to that of SER (Fig. 5.3 and Supplementary Information Fig. S15), but with maximum values occurring slightly sooner compared to SER. TiR of droughted plants diverged from that well-watered plants at around -0.2 MPa, similarly to SER. This resulted in a unique linear response of relative TiR and SWP in the three water-deficit scenarios for each genotype (Fig. 5.3 and Supplementary Information Fig. S15) but with lower sensitivity than for SER.

Because water-deficit can affect the growth duration of individual organs more than their initiation and appearance rate (Verbraeken et al., 2021), we analyzed the rate of elongation of all growing leaves on the main stem together, to avoid the a compensating effect by duration. Cumulated main-stem leaf length over time for droughted plants was closer to that of well-watered plants than

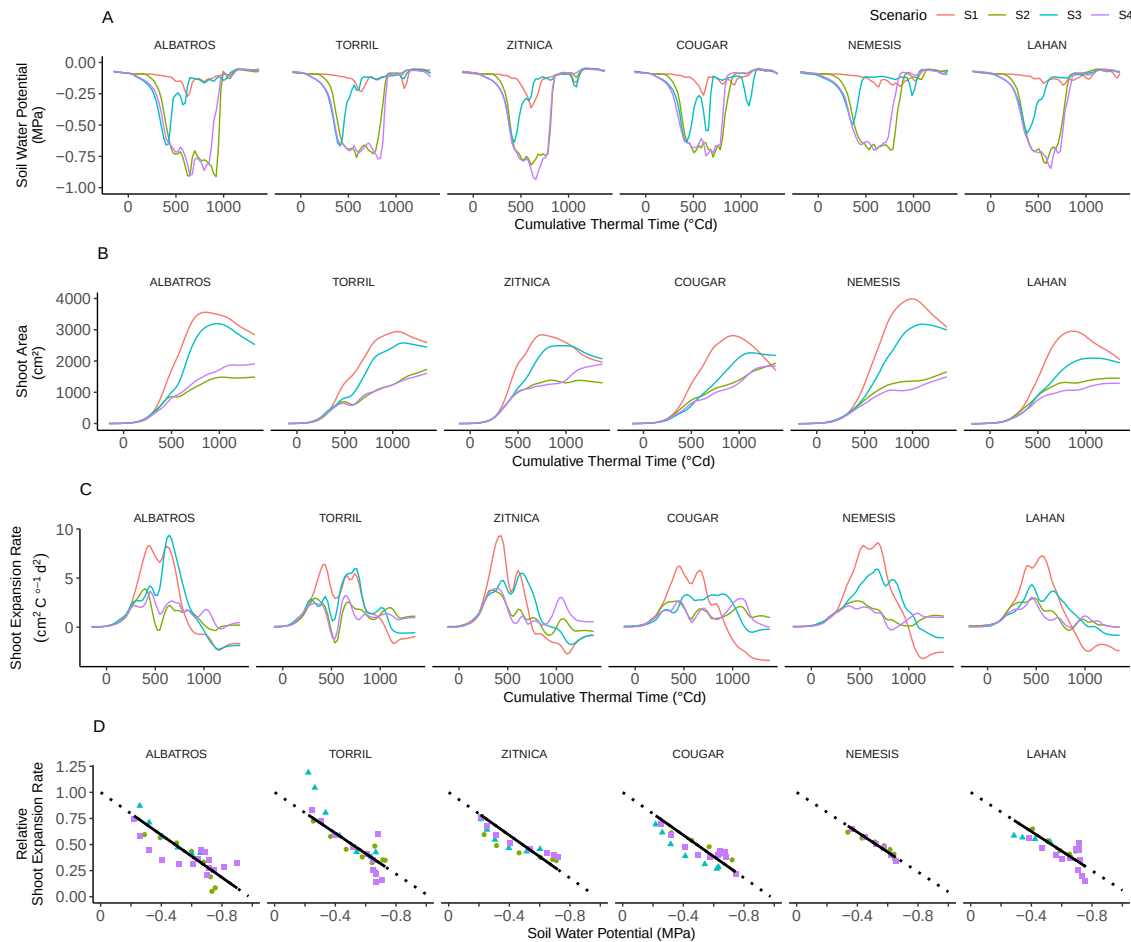


FIGURE 5.2 – Time courses of whole-plant leaf area (shoot area) and its sensitivity to soil water potential for the six cultivars of group A (four bread cultivars : Albatros, Torril, Zitnica, and Cougar, and two durum wheat cultivars : Nemesis and Lahan) under the four watering scenarios. (A) to (C) soil water potential, shoot area, and shoot expansion rate versus thermal time after leaf-3 appearance, respectively. (D) relative shoot expansion rate versus soil water potential. In (A) and (B) lines are cubic smoothing splines fitted to the experimental data. In (C) lines are the first derivatives of the fitted splines shown in (B). In (C) symbols are the calculated daily values during the period determined as illustrated in Figure 1, and lines are linear regressions forced to unit when soil water potential is null. Red lines, well-watered scenario (S1); blue triangles and lines, early drought scenario (S3); green circles and lines, late drought scenario (S2), and purple square and lines, long-lasting drought scenario (S4).

was observed for SA and the number of appeared tillers (Fig. 5.3). This resulted in small differences in the elongation rate of individual main-stem leaves (LER) of plants in the four scenarios (Fig. 5.3 and Supplementary Information Fig. S16), and a lower sensitivity of LER to SWP than that of TiR and SER.

In all scenarios and for all genotypes, leaf appearance rate (LAR) on the main stem calculated between leaf-3 and flag-leaf appearance showed a slight decrease with time (Fig. 5.3 and Supplementary Information Fig. S17). It resulted that the sensitivity of LAR to SWP was very small, and even positive in some genotypes (Supplementary Tables S3 and S4).

The dynamic responses of leaf area expansion and underlying processes to water deficit were analyzed for 86 genotypes, allowing analysis of the relative impact of genetic diversity and phenotypic plasticity of the three processes underlying the variability of whole plant leaf expansion under drought.

5.2.2 Under well-watered conditions, the genetic variability of tiller appearance rate is the main contributor to that of whole plant leaf expansion

To understand the relative contribution of the genetic variability of the three underlying processes to the genetic variability of shoot expansion in well-watered conditions, we first compared the coefficients of variation of the maximum rates for shoot expansion (SER_{max}), tiller appearance (TiR_{max}), leaf appearance (LAR_{mean}) and individual main-stem leaf elongation (LER_{max}) in well-watered conditions (Fig. 5.4). For both species, the genetic variability of LAR_{mean} and LER_{max} was small compared to that of SER_{max} , indicated by lower coefficients of variation. On the contrary, that of TiR_{max} was higher than that of SER_{max} , especially in bread wheat.

The first dimension of a principal component analysis (Supplementary Information Fig. S18) indicated covariations of TiR_{max} and SER_{max} , while the two other

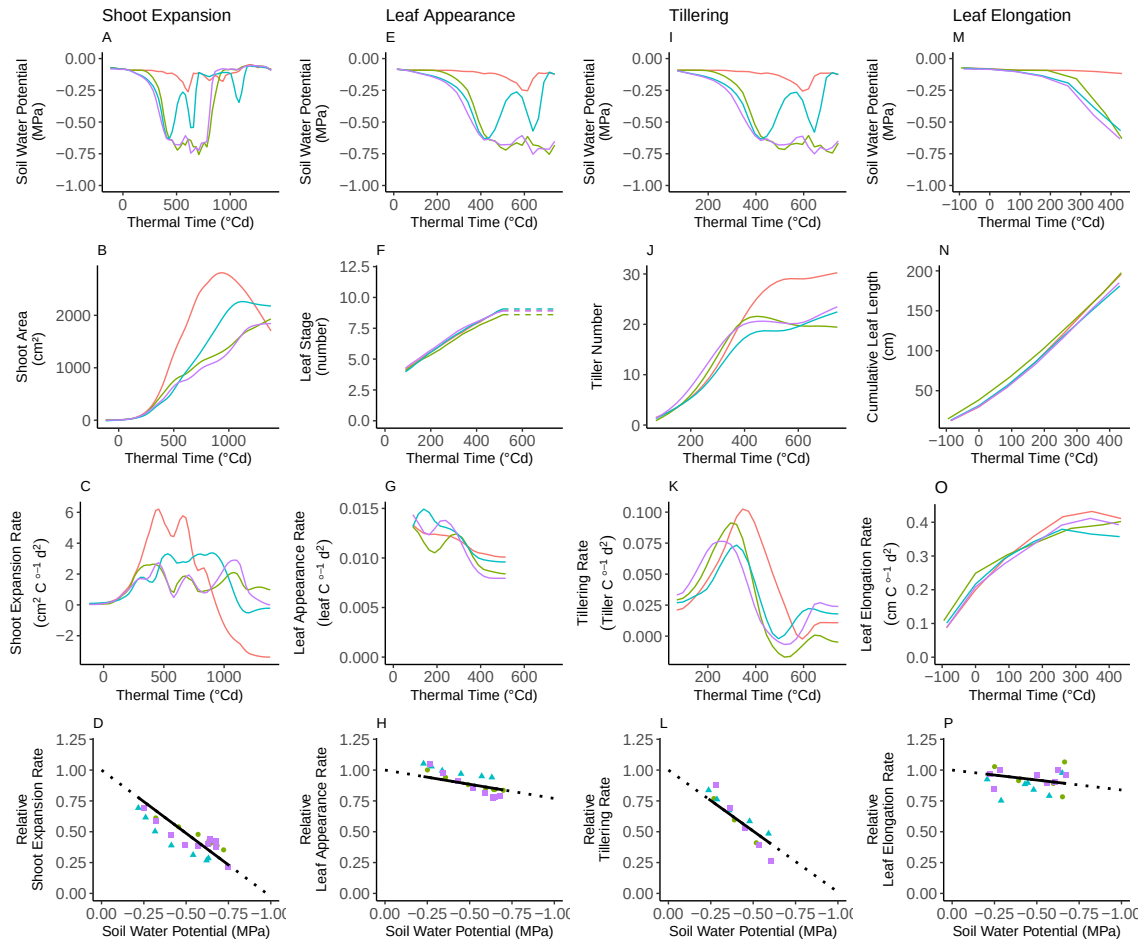


FIGURE 5.3 – Time courses of whole-plant leaf area (shoot area) and sensitivity shoot expansion rate to soil water potential for the bread wheat cultivar Cougar grown under the four watering scenarios. (A) to (D) shoot area versus thermal time after leaf-3 appearance. (E) to (H) shoot expansion rate versus thermal time after leaf-3 appearance. (I) to (L) relative shoot expansion rate versus soil water potential. In (A) to (D) lines are cubic smoothing splines fitted to the experimental data. In (E) to (H) lines are the first derivatives of the fitted splines shown in (A) to (D). In (I) to (L) symbols are the calculated daily values during the period determined as illustrated in Figure 1, and lines are linear regressions forced to unit when soil water potential is null. Red lines, well-watered scenario (S1); blue triangles and lines, early drought scenario (S3); green circles and lines, late drought scenario (S2), and purple squares and lines, long-lasting drought scenario (S4). Fig. 2 and Supplementary Information Fig. S15, S16, S17, S21 show similar plots for shoot area, tiller appearance, main-stem leaf elongation and main-stem leaf appearance, and cumulative internode length for the six genotypes of group A, respectively.

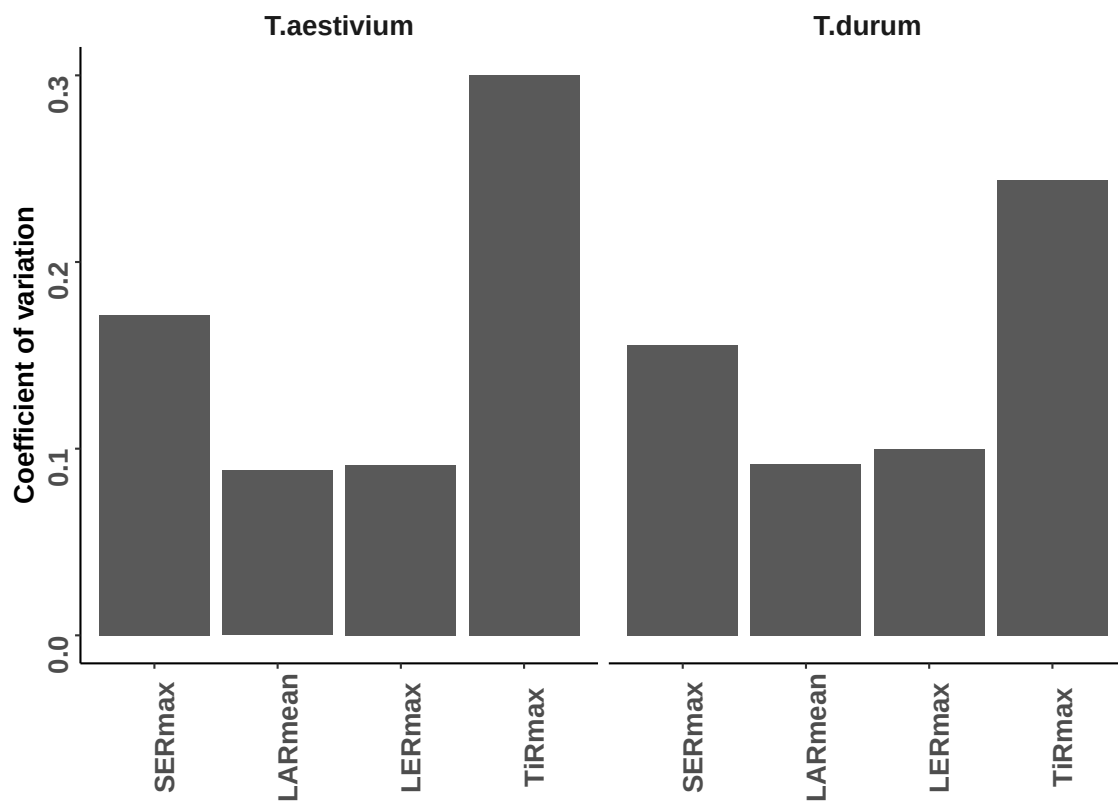


FIGURE 5.4 – Coefficient of variation of maximum shoot expansion rate (SER_{max}) and its three underlying processes, average main stem leaf appearance rate (LAR_{mean}), maximum main stem leaf elongation rate (LER_{max}), and maximum tiller appearance rate (TiR_{max}) for 47 bread wheat and 38 durum wheat cultivars grown under well-watered conditions (scenario 1).

components LER_{max} and LAR_{max} were associated with SER_{max} on the second dimension, explaining less variance. These results were confirmed by an analysis of variance (Table 5.2), which showed that the three components all explaining parts of the variance of SER_{max} , but with a higher total variation that can be attributed to TIR_{max} . Looking at the bivariate relationships between such variables showed significant relationships between SER_{max} and TiR_{max} (Fig. 5.5), either in each species or considering the two species together. This was not the case for LAR_{max} and LER_{max} for which the correlation with SER_{max} was not significant for at least one the two species. As suggested by the principal component analysis, TiR_{max} was largely correlated with the final leaf number (FLN), which in turn was also associated with SER_{max} .

Overall, tillering was the underlying process with the highest relative variability, and the most associated with shoot expansion, indicating a higher relative contribution to the genetic variability of shoot expansion, compared with leaf appearance and individual-leaf elongation.

5.2.3 The plasticity of whole plant leaf area expansion under drought is driven by that of tiller appearance rate

A similar analysis was performed in order to test the relative contributions of the sensitivity to soil water deficit of the three underlying processes to that of whole plant leaf area expansion. In both species, the sensitivity of TiR to soil water deficit had range similar to that of shoot expansion (both SER and TiR were

TABLE 5.2 – Analysis of variance of the maximum value of shoot expansion rate (SER_{max}) as a function of its three underlying components, average main-stem leaf appearance rate (LAR_{mean}), maximum main-stem leaf elongation rate (LER_{max}), and maximum tiller appearance rate (TIR_{max}) and the potential covariable final main stem leaf number (FLN).

Factor	D.f.	Sun of squares	Mean square	F-value	P-value
LAR_{mean}	1	12.30	12.30	10.77	< 0.01
TiR_{max}	1	22.40	22.38	19.59	< 0.001
LER_{max}	1	17.88	17.88	15.68	< 0.001
FLN	1	6.01	6.10	5.34	< 0.05
Residuals	72	82.27	1.14	NA	NA

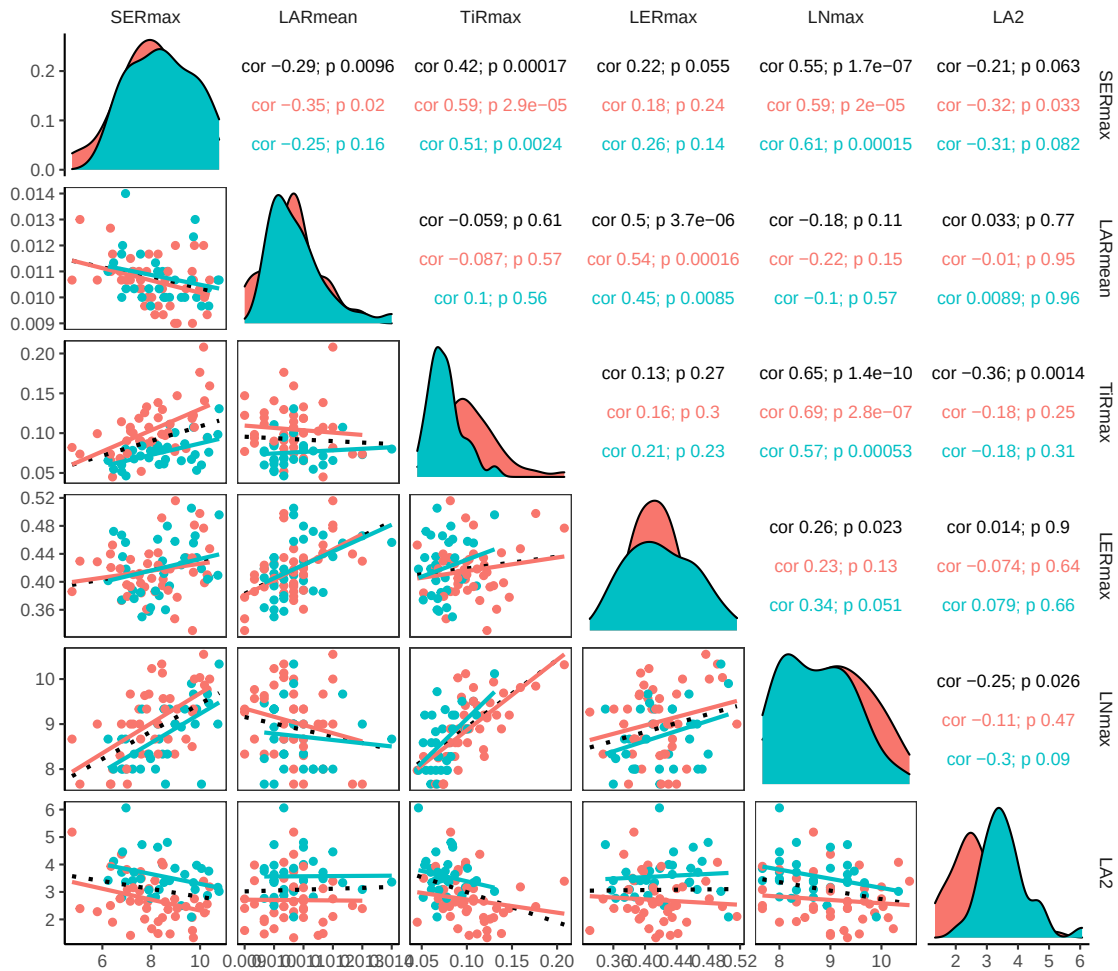


FIGURE 5.5 – Scatter plot matrix of bivariate relationships between maximum shoot expansion rate (SER_{max}), its three underlying processes (average main-stem leaf appearance rate [LAR_{mean}], maximum main-stem leaf elongation rate [LER_{max}], and maximum tiller appearance rate [TIR_{max}]), and two potential covariables (final main stem leaf number [FLN], and surface area of main-stem leaf 2 [LA2]) for 38 genotypes of durum wheat (blue circles and lines) and 47 genotypes of bread wheat (red circles and lines) grown under well-watered conditions. The distribution of each variable for each species is given in the diagonal. The upper half of the scatter plot matrix gives the Pearson's correlation coefficients (r) and associated P-values, calculated for both species together (black font), and separately for durum wheat (blue font) and bread wheat (red font).

null at around -1 MPa; Fig. 5.6). On the contrary, the sensitivity of LAR and LER was close to 0 MPa⁻¹ in both species (Fig. 5.6) and that of LAR was even negative for most genotypes (Supplementary Information Table S12 and S13). The principal component analysis performed on all variables (Supplementary Information Fig. S18) suggested that S_{TiR} was associated with S_{SER} (on dimension 1), while less variance was explained by S_{LAR} and S_{LER} (on dimensions 2 and 3). This was confirmed by an analysis of variance between S_{SER} and the three underlying sensitivities (Table 5.3). While the three underlying sensitivities were significantly associated with S_{SER} , less variance was explained by S_{LAR} and S_{LER} compared to S_{TiR} . In both species, S_{SER} and S_{TiR} were significantly correlated (Fig. 5.7). It was not the case for S_{LAR} and S_{LER} , which were not significantly correlated with S_{SER} for at least one species and when both species were considered together. As previously found in other species (parent et al., 2009 in rice; Welcker et al., 2011 in maize), the S_{SER} was negatively correlated with SER_{max} . To the contrary, it was not expected but for both species S_{LER} and S_{TiR} were significantly correlated, indicating similar sensitivities of the rates of leaf elongation and tiller appearance.

Overall, the above analyses showed that the sensitivity of whole plant leaf area expansion was the emergent property of that of the three underlying processes, with the S_{TiR} being the main contributor, followed by S_{LER} and S_{LAR} .

TABLE 5.3 – *Analysis of variance for the sensitivity of shoot expansion rate to soil water potential as a function of its three underlying components, sensitivity of main-stem leaf appearance rate (SLAR), sensitivity of tiller appearance rate (STiR), and sensitivity of main-stem leaf elongation rate (SLER).*

Factor	D.f.	Sun of squares	Mean square	F-value	P-value
SLAR	1	0.001	0.001	7.641	<0.01
STiR	1	0.004	0.004	32.476	<0.001
SLER	1	0.001	0.001	9.052	<0.01
Residuals	72	0.009	0	NA	NA

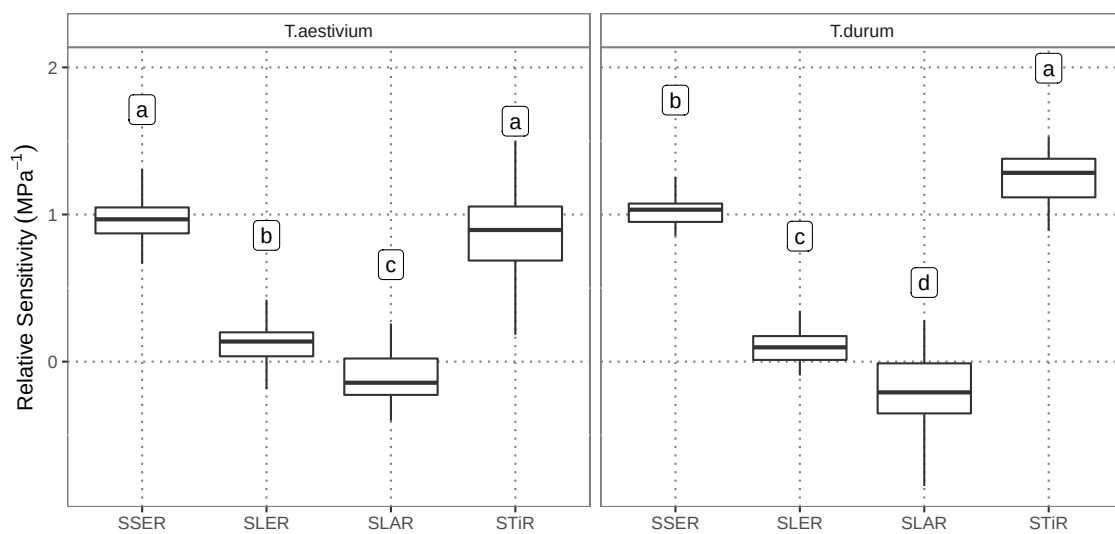


FIGURE 5.6 – Box plot of sensitivities of shoot expansion rate to soil water potential (S_{SER}), and that of its three underlying components, sensitivity of main-stem leaf appearance rate (S_{LAR}), sensitivity of tiller appearance rate (S_{TIR}), and sensitivity of main-stem leaf elongation rate (S_{LER}) for 47 genotypes of bread wheat (A) and 38 genotypes of durum wheat (B). The letters above the whiskers indicate significant difference (Tukey test, $P < 0.05$) between variables. In each box plot, end of vertical lines represent from top to bottom, the 10th, and 90th percentiles, horizontal lines represent from top to bottom, the 25th, 50th, and 75th percentiles of the simulation.

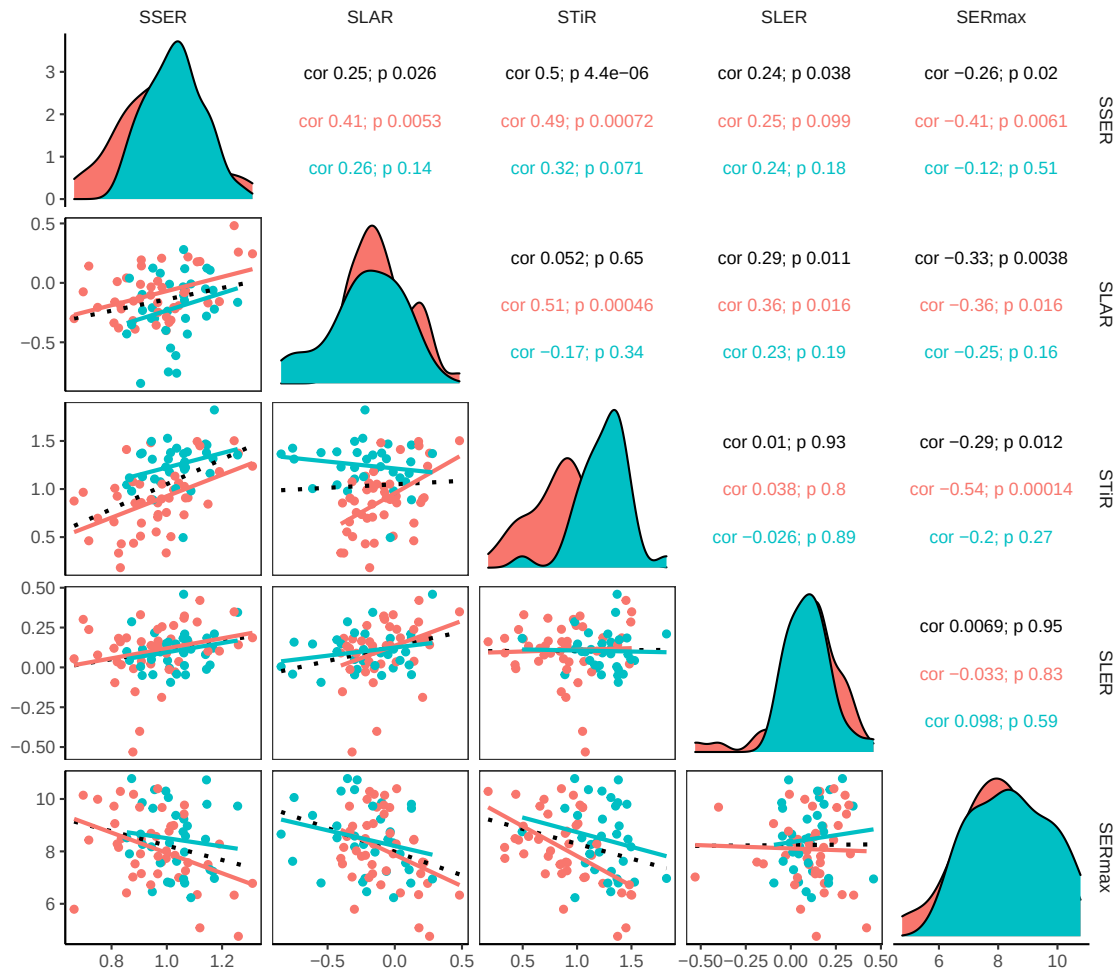


FIGURE 5.7 – Scatter plot matrix of bivariate relationships between sensitivity of shoot expansion rate to soil water potential (S_{SER}), its three underlying processes, sensitivity of main-stem leaf appearance rate (S_{LAR}), sensitivity of tiller appearance rate (S_{TIR}), sensitivity of main stem leaf elongation rate (S_{LER}), and a potential covariable (maximum shoot expansion rate (SER_{max})) for 38 genotypes of durum wheat (blue circles and lines) and 47 genotypes of bread wheat (red circles and lines) grown under well-watered conditions. The distribution of each variable for each species is given in the diagonal. The upper half of the scatter plot matrix gives the Pearson's correlation coefficients (r) and associated P-values, calculated for both species together (black font), and separately for durum wheat (blue font) and bread wheat (red font).

5.3 Discussion

5.3.1 Non-destructive recording of the temporal dynamics of shoot growth processes under drought

In this study, we used a high-throughput plant phenotyping platform to analyze the genetic variability of growth processes and their responses to soil water deficit. We followed the temporal dynamics of whole leaf area expansion, individual main-sem leaf appearance and elongation, and tiller appearance to associate their relative rates to the water deficit experienced at the same time, whether it is different according to the plant and/or the genotype (Jin et al., 2012). However, these results were obtained in pots, under controlled conditions, and their transposability to the field can be questioned. First, the ranking of genotypes made on the sensitivity of the number of spikes per square meter in the field was also partly found on TiR and/or the growth of the whole plant (Supplementary Information Fig. S12). The two genotypes of bread wheat of group A chosen to be the least sensitive in the field were also the least sensitive in the platform. The same was true for durum wheat. This conclusion is supported by previous studies showing that traits related to growth potential and growth sensitivity to water deficit are robust and conserved in the field and in the platform (Tardieu et al., 2015). Although in the platform the number of tillers per plants was much higher than in the field, the relative responses of shoot growth processes to soil water deficit in the platform can be extrapolated to the field, or at least the classification of genotypes found here can be considered as similar in the field and under controlled conditions.

In this study, a very large experimental effort was dedicated to the non-destructive measurements of leaf dimensions and tiller numbers over time (> 10'000 data points each). Indeed, up to now, while there have been some attempts in the literature, there is no automatic procedure to automatically measure in an imaging platform trait such as tiller numbers or individual-leaf length for large plants. As we demonstrated here that S_{TiR} was a crucial trait to explain the whole-plant

sensitivity, new procedure, probably based on deep learning methods should be rapidly developed for analyzing new germplasms or for genetic analyses of such traits.

5.3.2 Structure of genetic variability of whole-plant leaf area expansion under well-watered conditions

To better understand the structure of the genetic variability of shoot expansion we analyzed the respective impacts of the underlying processes and their co-variations. Indeed, the set of constraints and genetic correlations between traits defines the set of possibilities, informing both plant physiologists and breeder about the feasible combinations of traits. In the case of leaf expansion, for example, this question is "can we find a genotype that produce more tillers, more leaves, and larger leaves or is there constraints (e.g., developmental or carbon source-sink ratio) which result in a trade-off between the components of whole-plant growth.

Our working hypothesis, that the genetic variabilities of tiller appearance, leaf appearance and individual-leaf elongation rates could drive all together that of shoot expansion rate was confirmed, as previously found in white clover (Belaygue et al., 1996, considering branching rather than tillering). Each of these three components was associated with the genetic diversity of shoot expansion. The respective impact of these components followed their genetic relative variability (here quantified by their coefficient of variation), with a higher proportion of the genetic diversity of shoot expansion explained by that of TiR , followed by LER and LAR .

Two other processes participated to structure this genetic variability, plant development and early vigor. As found in maize (Parent et al., 2018), final leaf number was the main driver of cycle duration (compared to leaf appearance rate, not shown), and was therefore associated with maximum value of growth processes (SER_{max} , TiR_{max} , LER_{max}) because these processes were integrated over a

longer time. This effect was also visible on maximum rates. In particular, TiR_{max} was positively associated with FLN, as predicted by the Fibonacci series that links leaf and tiller appearance (Bos and Neuteboom, 1998).

The surface area of leaf 2 on the main-stem, a proxy of plant early vigor (Rebetzke and Richards, 1999) presented counter-intuitive correlations, by being negatively associated with SER_{max} and TIR_{max} , probably due to the negative correlation between the size of earlier organs and cycle duration (as observed for maize leaves, Lacube et al., 2020).

To summarize, the genetic variability of shoot expansion was highly structured in both species, with a main contribution of cycle duration via TiR and overall, the relative contributions of each of the three underlying processes.

5.3.3 Plasticity of whole-plant leaf area expansion

We questioned the plasticity of whole-plant leaf area expansion at two levels. The first question was to which sub-variables is the plasticity of leaf growth under water deficit due? The second was whether this calls into question the structure of leaf expansion shown above under well-irrigated conditions. In other words, how water deficit affects the relationships between traits.

We first showed that the sensitivity of leaf growth at the whole-plant level was a consequence of the sensitivities of its three components (all significant in the anova). However, most of the variance in S_{SER} was explained by S_{TIR} . Thus, water deficit did not alter the structure of the genetic variability of shoot expansion, with tiller appearance being the main explanatory process of this genetic diversity. However, under water deficit, the phenotypic space appeared even more constrained. Indeed, a strong correlation appeared between S_{TIR} and S_{LER} , indicating that a genotype that strongly decreases its number of appeared tillers under water deficit, also strongly decreases its individual leaf growth.

In the same way, an important issue for breeding in near-optimal conditions

is to develop a genotype with a high potential of growth and a low-sensitivity to drought while these two traits appear as negatively correlated in many crops (e.g., Welcker et al., 2011 in maize; Parent et al., 2009 in rice). In this study, this constraint was also found in both durum and bread wheat.

5.3.4 Genetic variability of tillering for future drought tolerant wheat cultivars

In environments subject to severe drought at the end of the growth cycle, a breeding strategy may be to favor varieties with limited leaf growth, to maintain a satisfactory soil water status at the end of the cycle, during grain filling (Sinclair, 2017). Our study here shows that focusing on tillering control can be a good strategy. First, we showed that TiR has proportionally larger genetic diversity than LAR or LER. Moreover, excess tillering is a carbon sink, of which at least the structural part will not participate in yield in case of premature senescence (Kebrom et al., 2012). A selection strategy based on this high genetic variability and/or genes (e.g., *TIN* gene; Atsmon and Jacobs, 1977; Hendriks et al., 2016) or identified QTLs (Wang et al., 2016) therefore seems adequate for environments where the probability of having sufficient rainfall at the end of the cycle is very low, thus not requiring any particular tillering plasticity, but instead a low constitutive number of tillers.

However, this type of environment is not the majority and there are generally as many drought scenarios as site x year, so a favorable selection strategy would be to focus on adaptive processes and their plasticity (Rymen and Sugimoto, 2012; Tardieu, 2012). Here, we have shown that tillering was the most plastic growth process compared to other growth components, and largely responsible for whole-plant growth plasticity under water deficit. Although somewhat less plastic, leaf elongation also appears to be a good target for modifying whole-plant leaf area plasticity.

5.3.5 Consequences for crop modelling and ideotype studies

This study can have large impact for process-based modelling and simulations of the genetic variability. Indeed, we observed a highly structured genetic diversity of the components of shoot expansion and their sensitivities to soil water deficit. This can either greatly simplify the formalisms of shoot expansion in process-based model or indicate the crucial constraints between parameters which need to be taken into account in ideotype studies. For example, the sensitivity of individual leaf expansion was correlated with that of tillering. This results that only one genotypic parameter should be considered in ideotype studies or for analyzing the impact of the genetic diversity at large scale (Parent et al., 2018). In the same way, our study indicates that the responses of growth processes to soil water deficit cannot be considered as independent from cycle duration. This can have large consequence in simulations. For example, for predicting the impact of climate change, one must consider the possible impact on such parameters if modifying the earliness of future genotypes. In ideotype studies, relationships between traits should be considered if ideotypes are not to be determined with sets of parameter values which cannot be achieved by breeding. Here, we described the structure of the genetic parameters of growth processes, and depending on the scientific question, it can indicate new formalisms taking into account such correlations or at least can suggest a minimum set of genetic parameters needed for considering the genetic diversity of whole-plant leaf area expansion and its plasticity.

5.4 Conclusion

In this study, we evaluated the relationships and trade-offs between the components of shoot expansion and their sensitivities to water deficit and demonstrated that the genetic diversity of plasticity for shoot area expansion is mostly driven by that of tillering. Overall, this work can have large consequences in (i) ecophysiology, by describing the temporal dynamic patterns of shoot expansion

and its components under drought in a large genetic diversity of bread wheat and durum wheat, (ii) in breeding, by identifying target traits under drought together with its associated genetic variability, and (iii) in modelling, by describing the structure of the genetic parameters of growth processes to be considered in parameterization.

5.5 Material and methods

5.5.1 Genetic material

For this study, a set of 38 durum wheat genotypes (*Triticum turgidum* L. subsp. *durum* (Desf.) Husn.; Supplementary Information Table S1) and 47 bread wheat genotypes (*T. aestivum* L.; Supplementary Information Table S2) that represent a large genetic variability within each species was.

For bread wheat, 34 genotypes were selected from a genetic association panel of 200 lines sampled from 4,506 accessions in a worldwide collection (Balfourier et al., 2019) by minimizing linkage disequilibrium and by imposing agronomic constraints on phenology and plant height (Paux et al., 2022) and 10 European elite cultivars. These genotypes were selected to maximize the diversity of the responses of yield and spike number to water and nitrogen deficits (Supplementary Information Fig. S12A). In addition, four spring wheat RILs from the Multi-Reference Nested Association Mapping population (MR-NAM) developed at the Queensland Alliance for Agriculture and Food Innovation (QAAFI; Christopher et al., 2021) were added for their contrasted transpiration efficiency (SUNTOP_73 and SUNTOP_90; Christopher et al., 2018) and stay-green phenotype under terminal drought (SUNTOP_9 and SUNTOP_41; Fletcher, 2020), together with their three parental lines (Suntop, Drysdale, Dharwar-Dry). Three hybrid bread wheat genotypes (SY_117079, HYB15200646, and SY_117079) were also included in this study together with their three parents (Akamar, Oregain, and Sublim).

For durum wheat, the genotypes were selected from a panel of 257 durum

wheat genotypes composed of 126 elite lines of worldwide origin registered between 1973 and 2015, 69 elite lines developed by the Council for Agricultural Research and Economics (CREA) and the University of Foggia, Italy, 61 lines randomly selected from 480 evolutionary pre-breeding lines derived from an open-pollinated population based on the intercrossing of about 650 accessions from wild subspecies and elite lines (David et al., 2014). The 38 durum wheat genotypes used in this study were selected from the 257 panel to maximize the diversity of the responses of grain yield and grain number per square to water and nitrogen deficit (Supplementary Information Fig. S12B).

Genotypes were separated into two groups. Group A (6 replicates by genotype x water scenario) comprised four elite bread wheat cultivars and two elite durum wheat cultivars with contrasted behavior in the field (Supplementary Information Fig. S12). Two bread wheat cultivars that appeared as “sensitive” (Zitnica and Couga), based their yield response under low-input in field experiments and two “tolerant” (Albatros and Torril), with contrasted response of spike number per square meter. A “sensitive” durum wheat genotype (Nemesis), and a “tolerant” durum wheat (Lahan), based on the maintenance of grain number under drought with contrasted response of grain number per square meter. From the six replicates of genotype x water scenario, three were harvested at anthesis. Ten watering scenarios were tested for group A. Only the four scenarios that were also tested in the group B are analyzed in this study. Group B (3 replicates by genotype x water scenario) comprised the other 43 bread wheat genotypes and 38 durum wheat genotypes (Nemesis was also in both groups).

5.5.2 Description of the PhenoArch plant phenotyping platform and experimental design

The experiment was carried out in the PhenoArch platform, hosted in the Montpellier Plant Phenotyping Platforms (MP3, <https://www6.montpellier.inra.fr/lepse/M3P>). The setup and the functioning of PhenoArch has been described in details in Cabrera-Bosquet et al., 2016). Briefly, PhenoArch is a greenhouse

equipped with conveyor belts carrying carts with one pot each (2,400 pots, average plant density of $12.5 \text{ plant m}^{-2}$). Each day, plants are transported towards a watering station, allowing an accurate weighing of the pots and watering at target soil water potential (SWP), and an imaging station equipped a top and a side RGB cameras. Plants are then moved back to the same positions and orientation, conserved throughout the whole experiment. Micrometeorological conditions at the plant level are monitored every minute at six positions in the glasshouse and average values are recorded every 15 min. The temperature of the meristematic zone of eight plants distributed in the glasshouse was also measured with copper-constantan thermocouples (0.2 mm diameter) vertically inserted between the sheaths of two leaves. Supplemental light was provided either during daytime when outside solar radiation was below 300 W m^{-2} or to extend the photoperiod to 12 h. All information (Micrometeorological conditions, irrigation, plant weight, plant images, and manual plant measurements) were stored in the Phenotyping hybrid Information System (PHIS, Neveu et al., 2018). Air vapor pressure difference (VPD) was calculated at each time step as the difference in water vapor pressure between saturation at air temperature and the current vapor pressure in the air. Thermal time was calculated from hourly average meristem temperature using the beta function proposed by Wang et al. (2017), with minimum, optimum and maximum temperatures of 0, 27.5, and 40 °C, respectively.

The dry weight of soil in each pot was calculated at the beginning of the experiment from the fresh weight of soil and soil water content determined in soil samples (three soil samples were taken every 60 pots). Soil water content was then determined by automatically weighing the pots and removing the weight of plants (estimated daily from the plant imaging, see below) and the pot itself. A water-retention curve previously obtained in several experiments (Alvarez-Prado et al., 2018) by fitting predawn leaf water potential of drying plants measured with a Scholander-type pressure chamber (Soil Moisture Equipment Corp., Santa Barbara, USA) against soil water content with the Van Genuchten curve (Genuchten and Th, 1980) was used to calculation of the mean SWP in each pot at

each weighing time.

5.5.3 Plant growth conditions and drought scenarios in the phenotyping platform

Dry seeds were sown on 28 November 2019 at 2 cm depth in 3 x 5 cm Jiffy pellets (Jiffy Group, Oslo, Norway) and placed in vernalization chamber where air temperature was controlled at 3.2 ± 0.5 °C with a PPFD of $25 \mu\text{mol m}^{-2} \text{s}^{-1}$ and 16 h photoperiod. After 7 weeks in the vernalization chamber, plantlets were placed for 1 week under fluctuating conditions in the greenhouse of the platform (mean temperature 13.9 ± 10 °C). Homogenous plants (scored manually) from each genotype were then transplanted together with their Jiffy pellet into 5 L plastic pots (0.17 m diameter, 0.24 m high) filled with a 30 :70 (v :v) mixture of a clay and organic compost. From that date, plants stayed on the PhenoArch greenhouse under fluctuating conditions of air temperature, VPD and light (Supplementary Information Fig. S12) until final harvest at grain ripeness maturity on 12 May 2020.

The spatial design ordered 12 rows of genotypes of group A and 16 rows of genotypes from group B, with successively 2 rows of group A and 3 of group B (Supplementary Figure S3). Each block of two successive rows with genotypes of group A corresponded to one replicate for group A. Genotypes from group B were fully randomized in the platform.

The whole experiment included 10 watering scenarios (for genotypes of group A) but only four, common to genotypes of group A and B, were analyzed in this study :

- Scenario 1 (S1) : Well-watered conditions (control). SWP was maintained at -0.02 MPa during the whole plant growth cycle ;
- Scenario 2 (S2) : Late and short water deficit, a 3-weeks period of water deficit between the beginning of stem elongation (average of all genotypes) until anthesis of the earliest genotypes. Water was withheld on

- 12 February 2020 until SWP decreased to - 0.54 MPa, after which SWP was maintained at that value. At the end of the 3-weeks period of water deficit (6 March 2020), pots were rewatered over a 5 d period during which SWP was gradually increased to 72, 86, 100, 114, and 124 %, and then maintained at -0.02 MPa until final harvest.
- Scenario 3 (S3) : Early and short water deficit, a 3-weeks period of water deficit starting at the beginning of tillering when the plants had three to four visible leaves depending on the genotype. Water was withheld on 3 February 2020 until SWP decreased to - 0.54 MPa. At the end of the 3-weeks period (28 February 2020), pots were slowly rewatered as in S2.
 - Scenario 4 (S4) : Early and long water deficit, a 31-d period of water deficit starting at the beginning of tillering (3 February 2020) until anthesis of the earliest genotypes (6 March 2020).

5.5.4 Automatic measurements of whole plant shoot area

An analysis pipeline was used to calculate whole plant leaf area (shoot SA) and fresh weight from RGB images (2056 × 2454 dpi) taken from 3 angle views (2 side views and one top view) every 2 to 3 d for each plant in the platform. Briefly, plant pixels from each image were segmented from those of the background and projected plant area were then calculated using calibration curves obtained previously using reference objects to convert pixels into projected plant green area (;Alvarez-Prado et al., 2018). Multiple-linear regression models (previously parametrized from destructive measurements) were then used to calculate whole plant shoot area and plant fresh weight.

5.5.5 Manual plant measurements

The number of leaves on the main-stem (LN) and green tillers per plant (TN) of each plant were counted twice a week from the beginning of tillering to anthesis. The former was calculated as (Haun, 1973) :

$$LN = n + \frac{l}{L} \quad (5.1)$$

where n is the number of mature leaves (with the ligule appeared), l is the exposed length of leaf $n+1$ at the time of measurement, and L is the final length of the blade of leaf $n+1$. The exposed length of a leaf was measured with a ruler as the distance from leaf tip to the upper collar of the sheath tube.

Final (mature) leaf length and width of each newly mature leaf on the main stem were measured with a ruler. The surface area of the second leave on the main stem (LA2), a proxy of plant early vigor (Rebetzke, 1999), was calculated as its length times its wight times an allometric factor or 0.75 (Dornbusch et al., 2011).

5.5.6 Calculation of rates and sensitivities of soil water deficit

Thermal time was calculated from the appearance of leaf 3 of the main stem, which for all genotypes occurred after the transfer of the plants on the PhenoArch conveyor. Thermal time was calculated from hourly average meristem temperature using a nonlinear beta function (Wang and Engel, 1998), with minimum, optimum and maximum temperatures of 0, 27.5, and 40 °C, respectively (Wang et al. 2017; Parent et al., 2012).

We analyzed four processes (Table 5.1), the rate of whole plant leaf area expansion (SER) and the three underlying processes, the rates of tiller appearance (TiR), leaf appearance (LAR), and individual leaf elongation on the main stem (LER). For each process, several variables were compared between genotypes (Table 5.1) : (i) the time course of the corresponding variable as a function of thermal time; (ii) its maximum value; (iii) the time course of the related rate as a function of thermal time; (iv) its maximum (for SER, TiR, and LER) or average value (for LAR); and (v) the relative sensitivity of the rate to soil water deficit (i.e. the slope of the relationship between relative rate and SWP).

SA and TN versus thermal time after leaf-3 appearance were fitted to a cubic smoothing spline function equation using the `smooth.spline()` function of the R package *stats* with *spar* parameter set at 0.6 and 0.4, respectively. LN versus thermal time was fitted to a linear model forced to 3 at 0°Cd. For the control treatment (scenario 1), the spline function or linear model was fitted to the data of all the plants of each genotype ($n = 3$ to 6 replicates for group B and A, respectively), while for the water deficit scenario, the data for each plant were fitted separately. The rate of each process was then calculated as the first derivative of the fitted spline or linear regression. We used the R function `predict()` to calculate SER and TiR. We calculated LAR as the slope of the linear regression. LER was calculated as CCL times LARmean. Finally, for each process and each plant subjected to water deficit, relative rates were calculated by dividing the absolute rates by those of well-watered plants (S1 ; average of all plants of the same genotype) at the same thermal time.

For each process and genotype, we defined a time period during which the rate of the process for the well-watered planted was high enough to accurately calculate SWP sensitivity. This period was defined from the time the rate was at 30% of its maximum value and either the time it had decreased to 20% of its maximum value or the date of the first rewatering, whichever occurred first (Fig. 5.1BD). For each genotype, the sensitivity of each process to soil water deficit was calculated during this period as the slope of the relationship between relative rate versus SWP forced to 1 at SWP is null, using data for the three water deficit scenarios (Fig. 5.1D).

5.5.7 Data analyses

All data analyses were performed with the R statistical software program R (R Core Team 2018).

5.6 References

Alvarez Prado, S., Sanchez, I., Cabrera-Bosquet, L., Grau, A., Welcker, C., Tardieu, F., Hilgert, N., 2019. To clean or not to clean phenotypic datasets for outlier plants in genetic analyses? *Journal of Experimental Botany* 70, 3693–3698. <https://doi.org/10.1093/jxb/erz191>

Atsmon, D., Jacobs, E., 1977. A Newly Bred 'Gigas' Form of Bread Wheat (*Triticum aestivum* L.) : Morphological Features and Thermo-photoperiodic Responses1. *Crop Science* 17, crops1977.0011183X001700010010x. <https://doi.org/10.2135/cropsci1977.0011183X001700010010x>

Bacher, H., Sharaby, Y., Walia, H., Peleg, Z., 2022. Modifying root-to-shoot ratio improves root water influxes in wheat under drought stress. *Journal of Experimental Botany* 73, 1643–1654. <https://doi.org/10.1093/jxb/erab500>

Balfourier, F., Bouchet, S., Robert, S., De Oliveira, R., Rimbert, H., Kitt, J., Choulet, F., Appels, R., Feuillet, C., Keller, B., Praud, S., Baumann, U., Budak, H., Rogers, J., Eversole, K., Alaux, M., Bejar, B., Lafarge, S., Lagendijk, E., Derory, J., Le Gouis, J., Paux, E., 2019. Worldwide phylogeography and history of wheat genetic diversity. *Sci. Adv.* 5, eaav0536. <https://doi.org/10.1126/sciadv.aav0536>

Barnabás, B., Jäger, K., Fehér, A., 2008. The effect of drought and heat stress on reproductive processes in cereals. *Plant, Cell & Environment* 31, 11–38. <https://doi.org/10.1111/j.1365-3040.2007.01727.x>

Baumont, M., Parent, B., Manceau, L., Brown, H.E., Driever, S.M., Muller, B., Martre, P., 2019. Experimental and modeling evidence of carbon limitation of leaf appearance rate for spring and winter wheat. *J Exp Bot* 70, 2449–2462. <https://doi.org/10.1093/jxb/erz012>

Belaygue, C., Wery, J., Cowan, A., Tardieu, F., 1996. Contribution of Leaf Expansion, Rate of Leaf Appearance, and Stolon Branching to Growth of Plant Leaf Area under Water Deficit in White Clover. *Crop Science* 36. <https://doi.org/10.2135/cropsci1996.0011183X003600050028x>

Berry, P.M., Spink, J.H., Foulkes, M.J., Wade, A., 2003. Quantifying the contributions and losses of dry matter from non-surviving shoots in four cultivars of winter wheat. *Field Crops Research* 80, 111–121. [https://doi.org/10.1016/S0378-4290\(02\)00174-0](https://doi.org/10.1016/S0378-4290(02)00174-0)

Bos, H.J., Neuteboom, J.H., 1998. Morphological Analysis of Leaf and Tiller Number Dynamics of Wheat (*Triticum aestivum*L.) : Responses to Temperature and Light Intensity. *Annals of Botany* 81, 131–139. <https://doi.org/10.1006/anbo.1997.0531>

Bradshaw, A.D., 1965. Evolutionary Significance of Phenotypic Plasticity in Plants, in : Caspari, E.W., Thoday, J.M. (Eds.), *Advances in Genetics*. Academic Press, pp. 115–155. [https://doi.org/10.1016/S0065-2660\(08\)60048-6](https://doi.org/10.1016/S0065-2660(08)60048-6)

Cabrera-Bosquet, L., Fournier, C., Brichet, N., Welcker, C., Suard, B., Tardieu, F., 2016. High-throughput estimation of incident light, light interception and radiation-use efficiency of thousands of plants in a phenotyping platform. *New Phytologist*

212, 269–281. <https://doi.org/10.1111/nph.14027>

Christopher, M., Paccapelo, V., Kelly, A., Macdonald, B., Hickey, L., Richard, C., Verbyla, A., Chenu, K., Borrell, A., Amin, A., Christopher, J., 2021. QTL identified for stay-green in a multi-reference nested association mapping population of wheat exhibit context dependent expression and parent-specific alleles. *Field Crops Research* 270, 108181. <https://doi.org/10.1016/j.fcr.2021.108181>

David, J., Holtz, Y., Ranwez, V., Santoni, S., Sarah, G., Ardisson, M., Poux, G., Choulet, F., Genthon, C., Roumet, P., Tavaud-Pirra, M., 2014. Genotyping by sequencing transcriptomes in an evolutionary pre-breeding durum wheat population. *Mol Breeding* 34, 1531–1548. <https://doi.org/10.1007/s11032-014-0179-z>

Faralli, M., Cockram, J., Ober, E., Wall, S., Galle, A., Van Rie, J., Raines, C., Lawson, T., 2019a. Genotypic, Developmental and Environmental Effects on the Rapidity of gs in Wheat : Impacts on Carbon Gain and Water-Use Efficiency. *Frontiers in Plant Science* 10.

Faralli, M., Williams, K.S., Han, J., Corke, F.M.K., Doonan, J.H., Kettlewell, P.S., 2019b. Water-Saving Traits Can Protect Wheat Grain Number Under Progressive Soil Drying at the Meiotic Stage : A Phenotyping Approach. *J Plant Growth Regul* 38, 1562–1573. <https://doi.org/10.1007/s00344-019-09956-3>

Fischer, E., Knutti, R., 2013. Robust projections of combined humidity and temperature extremes. *Nature Climate Change* 3, 126–130. <https://doi.org/10.1038/nclimate1682>

Fletcher, A.L., 2020. Understanding the genetic and physiological basis of transpiration efficiency in Australian wheat (PhD Thesis). The University of Queensland. <https://doi.org/10.14264/b45d330>

Funk, J.L., Jones, C.G., Lerda, M.T., 2007. Leaf- and shoot-level plasticity in response to different nutrient and water availabilities. *Tree Physiology* 27, 1731–1739. <https://doi.org/10.1093/treephys/27.12.1731>

Genuchten, V., Th, M., 1980. A Closed-form Equation for Predicting the Hydraulic Conductivity of Unsaturated Soils 1. *Soil Science Society of America Journal* 44, 892–898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>

Hatfield, J.L., Dold, C., 2019. Water-Use Efficiency : Advances and Challenges in a Changing Climate. *Frontiers in Plant Science* 10.

Hendriks, W., Kirkegaard, J.A., Lilley, J.M., Gregory, P.J., Rebetzke, G.J., 2016. A tillering inhibition gene influences root-shoot carbon partitioning and pattern of water use to improve wheat productivity in rainfed environments. *J. Exp. Bot.* 67, 327–340. <https://doi.org/10.1093/jxb/erv457>

Hilty, J., Muller, B., Pantin, F., Leuzinger, S., 2021. Plant growth : the What, the How, and the Why. *New Phytologist* 232, 25–41. <https://doi.org/10.1111/nph.17610>

Jin, X., Wang, K., Xiao, C., Diao, W., Wang, F., Chen, B., Li, S., 2012. Comparison of two methods for estimation of leaf total chlorophyll content using remote sensing in wheat. *Field Crops Research* 135, 24–29. <https://doi.org/10.1016/j.fcr.2012.06.017>

Kebrom, T.H., Chandler, P.M., Swain, S.M., King, R.W., Richards, R.A., Spielmeier, W., 2012. Inhibition of Tiller Bud Outgrowth in the tin Mutant of Wheat Is Associated with Precocious Internode Development. *Plant Physiol.* 160, 308–318. <https://doi.org/10.1104/pp.112.197954>

Koch, G., Rolland, G., Dauzat, M., Bédiée, A., Baldazzi, V., Bertin, N., Guédon, Y., Granier, C., 2019. Leaf Production and Expansion : A Generalized Response to Drought Stresses from Cells to Whole Leaf Biomass—A Case Study in the Tomato Compound Leaf. *Plants* 8, 409. <https://doi.org/10.3390/plants8100409>

Lacube, S., Fournier, C., Palaffre, C., Millet, E.J., Tardieu, F., Parent, B., 2017. Distinct controls of leaf widening and elongation by light and evaporative demand in maize. *Plant Cell Environ.* 40, 2017–2028. <https://doi.org/10.1111/pce.13005>

Lacube, S., Manceau, L., Welcker, C., Millet, E.J., Gouesnard, B., Palaffre, C., Ribaut, J.-M., Hammer, G., Parent, B., Tardieu, F., 2020. Simulating the effect of flowering time on maize individual leaf area in contrasting environmental scenarios. *Journal of Experimental Botany* 71, 5577–5588. <https://doi.org/10.1093/jxb/eraa278>

Leveau, S., Parent, B., Zaka, S., Martre, P., 2021. Differential sensitivity to temperature and evaporative demand in wheat relatives. *Journal of Experimental Botany*. <https://doi.org/10.1093/jxb/erab431>

Lobell, D.B., Schlenker, W., Costa-Roberts, J., 2011. Climate Trends and Global Crop Production Since 1980. *Science* 333, 616–620. <https://doi.org/10.1126/science.1204531>

Martre, P., He, J., Le Gouis, J., Semenov, M.A., 2015. In silico system analysis of physiological traits determining grain yield and protein concentration for wheat as influenced by climate and crop management. *J. Exp. Bot.* 66, 3581–3598. <https://doi.org/10.1093/jxb/erv049>

Neveu, P., Tireau, A., Hilgert, N., Nègre, V., Mineau-Cesari, J., Brichet, N., Chapuis, R., Sanchez, I., Pommier, C., Charnomordic, B., Tardieu, F., Cabrera-Bosquet, L., 2018. Dealing with multi-source and multi-scale information in plant phenomics : the ontology-driven Phenotyping Hybrid Information System. *New Phytologist*. <https://doi.org/10.1111/nph.15385>

Parent, B., Conejero, G., Tardieu, F., 2009. Spatial and temporal analysis of non-steady elongation of rice leaves. *Plant, Cell & Environment* 32, 1561–1572. <https://doi.org/10.1111/j.1365-3040.2009.02020.x>

Parent, B., Leclere, M., Lacube, S., Semenov, M.A., Welcker, C., Martre, P., Tardieu, F., 2018. Maize yields over Europe may increase in spite of climate change, with an appropriate use of the genetic variability of flowering time. *PNAS* 201720716. <https://doi.org/10.1073/pnas.1720716115>

Parent, B., Tardieu, F., 2014. Can current crop models be used in the phenotyping era for predicting the genetic variability of yield of plants subjected to drought or high temperature ? *J Exp Bot* 65, 6179–6189. <https://doi.org/10.1093/jxb/eru223>

Paux, E., Lafarge, S., Balfourier, F., Derory, J., Charmet, G., Alaux, M., Perchet, G., Bondoux, M., Baret, F., Barillot, R., Ravel, C., Sourdille, P., Le Gouis, J., on behalf of the BREEDWHEAT Consortium, 2022. Breeding for Economically and

Environmentally Sustainable Wheat Varieties : An Integrated Approach from Genomics to Selection. *Biology* 11, 149. <https://doi.org/10.3390/biology11010149>

Pierik, R., Fankhauser, C., Strader, L.C., Sinha, N., 2021. Architecture and plasticity : optimizing plant performance in dynamic environments. *Plant Physiology* 187, 1029–1032. <https://doi.org/10.1093/plphys/kiab402>

Prado, S.A., Cabrera-Bosquet, L., Grau, A., Coupel-Ledru, A., Millet, E.J., Welcker, C., Tardieu, F., 2018. Phenomics allows identification of genomic regions affecting maize stomatal conductance with conditional effects of water deficit and evaporative demand. *Plant, Cell & Environment* 41, 314–326. <https://doi.org/10.1111/pce.13083>

R Core Team, 2018. R : A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

Raup, D.M., Michelson, A., 1965. Theoretical Morphology of the Coiled Shell. *Science* 147, 1294–1295. <https://doi.org/10.1126/science.147.3663.1294>

Rebetzke, G.J., Richards, R.A., 1999. Genetic improvement of early vigour in wheat. *Aust. J. Agric. Res.* 50, 291–302. <https://doi.org/10.1071/a98125>

Reynolds, M., Foulkes, M.J., Slafer, G.A., Berry, P., Parry, M.A.J., Snape, J.W., Angus, W.J., 2009. Raising yield potential in wheat. *Journal of Experimental Botany* 60, 1899–1918. <https://doi.org/10.1093/jxb/erp016>

Rizhsky, L., Liang, H., Mittler, R., 2002. The Combined Effect of Drought Stress and Heat Shock on Gene Expression in Tobacco. *Plant Physiology* 130, 1143–1151. <https://doi.org/10.1104/pp.006858>

Rötter, R.P., Tao, F., Hohn, J.G., Palosuo, T., 2015. Use of crop simulation modelling to aid ideotype design of future cereal cultivars. *J. Exp. Bot.* 66, 3463–3476. <https://doi.org/10.1093/jxb/erv098>

Rymen, B., Sugimoto, K., 2012. Tuning growth to the environmental demands. *Current Opinion in Plant Biology* 15, 683–690. <https://doi.org/10.1016/j.pbi.2012.07.005>

Sanad, M.N.M.E., Campbell, K.G., Gill, K.S., 2016. Developmental program impacts phenological plasticity of spring wheat under drought. *Botanical Studies* 57, 35. <https://doi.org/10.1186/s40529-016-0149-3>

Shipley, B., Lechowicz, M.J., Wright, I., Reich, P.B., 2006. Fundamental Trade-Offs Generating the Worldwide Leaf Economics Spectrum. *Ecology* 87, 535–541. <https://doi.org/10.1890/05-1051>

Sinclair, T.R. (Ed.), 2017. Water-Conservation Traits to Increase Crop Yields in Water-deficit Environments, SpringerBriefs in Environmental Science. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-319-56321-3>

Suneja, Y., Gupta, A.K., Bains, N.S., 2019. Stress Adaptive Plasticity : *Aegilops tauschii* and *Triticum dicoccoides* as Potential Donors of Drought Associated Morpho-Physiological Traits in Wheat. *Frontiers in Plant Science* 10.

Tardieu, F., 2012. Any trait or trait-related allele can confer drought tolerance :

just design the right drought scenario. *J Exp Bot* 63, 25–31. <https://doi.org/10.1093/jxb/err269>

Tardieu, F., Parent, B., 2017. Predictable ‘meta-mechanisms’ emerge from feedbacks between transpiration and plant growth and cannot be simply deduced from short-term mechanisms. *Plant, Cell & Environment* 40, 846–857. <https://doi.org/10.1111/pce.12822>

Tardieu, F., Simonneau, T., Parent, B., 2015. Modelling the coordination of the controls of stomatal aperture, transpiration, leaf growth, and abscisic acid : update and extension of the Tardieu-Davies model. *J. Exp. Bot.* 66, 2227–2237. <https://doi.org/10.1093/jxb/erv039>

Urban, D.W., Sheffield, J., Lobell, D.B., 2015. The impacts of future climate and carbon dioxide changes on the average and variability of US maize yields under two emission scenarios. *Environ. Res. Lett.* 10, 045003. <https://doi.org/10.1088/1748-9326/10/4/045003>

Verbraeken, L., Wuyts, N., Mertens, S., Cannoot, B., Maleux, K., Demuynck, K., De Block, J., Merchie, J., Dhondt, S., Bonaventure, G., Crafts-Brandner, S., Vogel, J., Bruce, W., Inzé, D., Maere, S., Nelissen, H., 2021. Drought affects the rate and duration of organ growth but not inter-organ growth coordination. *Plant Physiology* 186, 1336–1353. <https://doi.org/10.1093/plphys/kiab155>

Wang, E., Martre, P., Zhao, Z., Ewert, F., Maiorano, A., Rötter, R.P., Kimball, B.A., Ottman, M.J., Wall, G.W., White, J.W., Reynolds, M.P., Alderman, P.D., Aggarwal, P.K., Anothai, J., Basso, B., Biernath, C., Cammarano, D., Challinor, A.J., Sanctis, G.D., Doltra, J., Dumont, B., Fereres, E., Garcia-Vila, M., Gayler, S., Hoo-genboom, G., Hunt, L.A., Izaurralde, R.C., Jabloun, M., Jones, C.D., Kersebaum, K.C., Koehler, A.-K., Liu, L., Müller, C., Kumar, S.N., Nendel, C., O’Leary, G., Olesen, J.E., Palosuo, T., Priesack, E., Rezaei, E.E., Ripoche, D., Ruane, A.C., Semenov, M.A., Shcherbak, I., Stöckle, C., Stratonovitch, P., Streck, T., Supit, I., Tao, F., Thorburn, P., Waha, K., Wallach, D., Wang, Z., Wolf, J., Zhu, Y., Asseng, S., 2017. The uncertainty of crop yield projections is reduced by improved temperature response functions. *Nature Plants* 3, 1–13. <https://doi.org/10.1038/nplants.2017.102>

Wang, Z., Liu, Y., Shi, H., Mo, H., Wu, F., Lin, Y., Gao, S., Wang, J., Wei, Y., Liu, C., Zheng, Y., 2016. Identification and validation of novel low-tiller number QTL in common wheat. *Theor Appl Genet* 129, 603–612. <https://doi.org/10.1007/s00122-015-2652-4>

Webber, H., Ewert, F., Olesen, J.E., Müller, C., Fronzek, S., Ruane, A.C., Bourgault, M., Martre, P., Ababaei, B., Bindi, M., Ferrise, R., Finger, R., Fodor, N., Gabaldón-Leal, C., Gaiser, T., Jabloun, M., Kersebaum, K.-C., Lizaso, J.I., Lorite, I.J., Manceau, L., Moriondo, M., Nendel, C., Rodríguez, A., Ruiz-Ramos, M., Semenov, M.A., Siebert, S., Stella, T., Stratonovitch, P., Trombi, G., Wallach, D., 2018. Diverging importance of drought stress for maize and winter wheat in Europe. *Nat Commun* 9, 4249. <https://doi.org/10.1038/s41467-018-06525-2>

Welcker, C., Sadok, W., Dignat, G., Renault, M., Salvi, S., Charcosset, A., Tardieu, F., 2011. A Common Genetic Determinism for Sensitivities to Soil Water Deficit and Evaporative Demand : Meta-Analysis of Quantitative Trait Loci and Introgression Lines of Maize. *Plant Physiology* 157, 718–729. <https://doi.org/>

10.1104/pp.111.176479

Wright, I.J., Reich, P.B., Westoby, M., Ackerly, D.D., Baruch, Z., Bongers, F., Cavender-Bares, J., Chapin, T., Cornelissen, J.H.C., Diemer, M., Flexas, J., Garnier, E., Groom, P.K., Gulias, J., Hikosaka, K., Lamont, B.B., Lee, T., Lee, W., Lusk, C., Midgley, J.J., Navas, M.-L., Niinemets, Ü., Oleksyn, J., Osada, N., Poorter, H., Poot, P., Prior, L., Pyankov, V.I., Roumet, C., Thomas, S.C., Tjoelker, M.G., Veneklaas, E.J., Villar, R., 2004. The worldwide leaf economics spectrum. *Nature* 428, 821–827. <https://doi.org/10.1038/nature02403>

How can wheat growth traits and their sensitivity to water deficit measured in controlled phenotyping platform inform on growth and performance of genotypes in the field?

6.1 Introduction

In the 2000s, the advent of efficient, inexpensive and rapid genomic and genetic methods shifted the bottleneck of genetic / breeding pipelines to phenotyping (Berger et al., 2010; Furbank and Tester, 2011; Yang et al., 2020). Indeed, at that time, there was no efficient method to measure physiological traits on hundreds of genotypes, which therefore required large experimental and financial efforts.

In the 2010s, high-throughput phenotyping platforms (Fiorani and Schurr, 2013; Janni and Pieruschka, 2022) have been developed and thus multiplied, in all their diversity of temporal and spatial scales, for different types of traits, from

metabolites (e.g. HitMe), to shoot (e.g. M3P PhenoArch, Cabrera-Bosquet et al., 2016; M3P PhenoPsis, Granier et al., 2006 and root growth (4PMI; Jeudy et al., 2016 under controlled conditions , or in the semi-controlled field (e.g. PhenoField; Beauchêne et al., 2019; FIP, Kirchgessner et al., 2017). In this landscape, one of the roles of imaging platforms under controlled conditions, such as the M3P PhenoArch platform used in Chapter 5, is to allow the phenotyping of a large number of plants under well characterized and controlled environmental condition, using consistent and reproducible methods between experiments. However, to date, studies using these platforms have mainly focused on the development of new tools and methods but have rarely been used to study specific biological questions related to crop improvement (Prado et al., 2018; Perez et al., 2019). The result is that such platforms, although very expensive and widely developed recently in several laboratories, have not found their undisputed place, as other types of platforms could.

One of the reasons is the mistrust towards platforms under controlled conditions, i.e. plants are grown in pots at a low density and under environmental conditions that are very different from those in the field. For example, in the PhenoArch platform, the density of pots is 12 plants per square meter, which is close to the field density for maize but is far from the 100-400 plants per m² of a wheat field.

In parallel, field phenotyping is experiencing an explosion of new imagery technology (e.g. RGB, multi- and hyper-spectral imagery, LIDAR, NDVI) and analytics methods and pipelines (e.g. deep learning) and platforms (mobile rainout shelters, meteorological and soil sensor networks), and highly-funded international projects are also seeing the future of phenotyping around remote sensing (Ghosal et al., 2018; Machwitz et al., 2021; Fu et al., 2022). Field phenotyping platforms provide crop growth conditions that are closer to agronomic conditions, but they do not allow a total control of environmental conditions such as water deficit scenario and in the field of crops are usually phenotyped with revisit time of several days, while plant responded to environmental conditions with a time

scale of minutes to hours (Tardieu and Parent, 2017). Therefore, field and controlled environment phenotyping platforms should not be considered as alternative tools and methods. We should rather take advantage of their complementarity to better understand and characterize the diversity of physiological and architectural traits associated with plant growth potential and adaptation to environmental constraints, in order to ultimately predict their performance in the diversity of environmental scenarios encountered in the field.

In this Chapter, I therefore analyzed the bread wheat and durum wheat panels constituted in Chapter 4 that maximize the variability of yield sensitivity and its components to nitrogen and water limited conditions; in a high throughput phenotyping platform under controlled conditions (Chapter 5) and in a field platform equipped with phenotyping tools (PhenoMobile) to measure shoot expansion (green area index, NDVI). I analyzed the correlations between dynamic variables related to shoot expansion and grain yield and their sensitivity to water deficit in the platform and in the field, and, more broadly, how dynamic variables of plant growth measured in platform are related to more integrated variables (grain yield and its components) in the field. This study demonstrates that the use of a phenotyping platform under controlled conditions can capture the genetic variability of growth rates measured in the field, as well as part of the variance that explains yield formation.

6.2 Material and Methods

6.2.1 Genetic Material

In this chapter, we analyzed 33 durum wheat and 44 bread wheat cultivars that were phenotyped both under controlled conditions (Chapter 5) and in the field (Chapter 4). These genotypes were selected (see Chapter 4) to maximize the diversity of the response of grain yield and yield components to low-input (water and nitrogen) conditions. Only the genotypes grown both in the platform and in the field were analyzed in this chapter (Supplementary Information Table S9).

6.2.2 Field trials

Two field trials conducted at ARVALIS experimental station at Gréoux-les-bains, France (43°44' N, 5°51' E, 293 m a.s.l.) during the 2017-2018 (trial ARVgre2018) and 2018-2019 (trial ARVgre2019) for bread and durum wheat, respectively. The growth conditions during these trials are summarized in Table 6.1. Crops were sown on 29 October 2017 and 20 November 2018 at a density of 300 seeds m⁻². Plots consisted in 11 rows with an inter-row spacing of 0.175 m and a surface area of 11.9 m². In both trial, two environmental conditions were tested (Table 6.1), a control condition with optimal water and nitrogen supply (named high-input, HI, hereafter) and another condition with low nitrogen and water supply for bread wheat and with low nitrogen for durum wheat (named low-input, LI, hereafter). Each trial used a split-plot design in which the LI and HI treatments were randomized on main plots, cultivar was randomized on the sub-plots and each treatment was replicated two times.

Nitrogen fertilizer applications in the HI treatments were intended to replicate commercial practice. The growth stages and application rates were calculated based on provisional N balance associated with expected grain yields using the N sheet balance method described in (Rémy and Hébert, 1977). For the bread wheat trial, the LI treatment N fertilizer was reduced by 72% compared with the HI treatment. Soil water content at 30-cm and 60-cm below the soil surface was monitored throughout the growing season using tensiometric probes. In the HI treatment with low water supply, irrigation was applied to ensure that water supply did not limit plant growth. In the bread wheat trial, the LI treatment received 15 mm of irrigation (compared with 200 mm in the HI treatment). The durum

TABLE 6.1 – *Environmental conditions in the two field trials analyzed in this study.*

Trial name	Average daily minimum temperature (°C)	Average daily maximal temperature (°C)	Thermal time (°Cd)	Cumulative seasonal solar radiation (MJ m ⁻²)	Cumulative seasonal rainfall (mm)	Irrigation (mm)		Fertilizer (kg N ha ⁻¹)	
						HI	LI	HI	LI
ARVgre2018 (Bread Wheat)	-9.8	35.2	2921	3582	476	70	0	180	50
ARVgre2019 (Durum Wheat)	-8.8	43.9	2749	3786	300	200	15	230	230

TABLE 6.2 – Description of the variables measured or calculated in the field.

Name	Description	Unit
Heading	Heading date	°Cd
SpikeNb	Ear number per square meter	spike m ⁻²
S _{SpikeNb}	Sensitivity of spike number to water deficit	unitless
Yield	Final grain yield	Kg DM ha ⁻¹
S _{Yield}	Sensitivity of grain yield to water deficit	unitless
GrainW	Average grain dry mass at maturity	g DM grain ⁻¹
S _{GrainW}	Sensitivity of grain dry mass to water deficit	unitless
GrainNb	Grain number per square meter at maturity	grain m ⁻²
S _{GrainNb}	Sensitivity of grain number to water deficit	unitless
GrainSpike	Grain number per spike	grain spike ⁻¹
S _{GrainSpike}	Sensitivity of grain number per spike to water deficit	unitless
GAI	Grain area index	m ² m ⁻²
GAI _{max}	Seasonal maximum green area index	m ² m ⁻²
GAIR	Rate of change of green area index	°Cd ⁻¹
GAIR _{max}	Seasonal maximal rate of change of green area index	m ² m ⁻² °Cd
S _{GAI}	Sensitivity of rate of change of green area index to water deficit	Unitless
NDVI	Normalized difference vegetation index	unitless
NDVI _{max}	Seasonal maximum of normalized difference vegetation index	unitless
NDVIR	Seasonal rate of change of normalized difference vegetation index	°Cd ⁻¹
NDVIR _{max}	Seasonal maximum rate of change of normalized difference vegetation index	°Cd ⁻¹
S _{NDVI}	Sensitivity of rate of change of normalized difference vegetation index	unitless
CH	Canopy height (from soil surface to the base of the spike)	m
CH _{max}	Seasonal maximum canopy height	m
CHR	Canopy height expansion rate	m °Cd ⁻¹
CHR _{max}	Seasonal maximum canopy height expansion rate	m °Cd ⁻¹
S _{CH}	Sensitivity of canopy height expansion rate to water deficit	unitless

wheat trial was rainfed. All other crop inputs including weed, disease and pest control, and potassium, phosphate, and sulfur fertilizers, were applied at levels to prevent non-N nutrients or weeds, diseases and pests from limiting grain yield.

Heading date, grain yield, average grain dry mass (*GrainW*, calculated from a sample of 1200 grains) and spike number per square meter (*SpikeNb*) was determined (Table 6.2). Average grain number per spike (*GrainSpike*) was calculated as *GrainNb* divided by *SpikeNb* and grain number per square meter (*GrainNb*) was calculated as *Yield* divided by *GrainNb*.

Normalized difference vegetation index (NDVI), green area index (GAI), and canopy height (CH) were determined with a self-guided rover equipped with RGB and multi-spectral cameras mounted at 0° and 45° zenithal view angles (PHENOMOBILE® V1¹; Madec et al., 2017) using the procedures described by Madec et al. (2017). NDVI values measured at the two view angles were averaged. In both trials, measurements were taken every about 500°Cd and every 100°Cd

1. https://www.researchgate.net/publication/290607111_PHENOMOBILE-V1_A_fully_automated_high_throughput_phenotyping_system

between 1000°Cd and 1500°Cd and grain ripeness maturity. The rates of change of NDVI and GAI were calculated as the first derivative of a cubic smoothing spline fitted to the absolute values versus thermal time after sowing (Absolute value, Fig. 6.1AB; derivative, Fig. 6.1DE). The best *spar* parameter of the R function *smooth.spline()* was first estimated for each genotype / treatment combination, and the median value (0.5 and 0.6 for GAI and NDVI) was used for all the plants. For CH, a segmented linear model was fitted to the absolute values versus thermal time after anthesis (Fig.1C) and the average rate of canopy height expansion (CHmean) was calculated as the slope of the second linear segment.

6.2.3 Experiment in the phenotyping platform PhenoArch

The experiment conducted in the plant phenotyping platform PhenoArch described in Chapter 5 was used in this chapter. All growth traits calculated in Chapter 5 were analyzed, namely the maximum value, maximum or average rate and sensitivity to water deficit of shoot area, individual leaf length, leaf appearance, and tiller appearance (Table 3). The calculation procedures are described in Chapter 5.

In addition, in this chapter we analyzed the elongation of the internodes of the main stem. The length of all internodes on the main stem was measured with a ruler at ripeness maturity. The cumulative length of internodes on the main stem (CIL) was calculated as the sum of individual internode length. The period during which each internode was growing was calculated afterwards based on leaf appearance, by considering that the leaf of phytomer n expand at the same time as the internode of phytomer $n-1$ (Evers et al., 2005). The rate of internode expansion (IER) was calculated during each period as CIL times LARmean. The cumulative internode length (CIL), the maximum internode elongation rate (IER-max), and the sensitivity of internode elongation rate to soil water deficit (IER) were calculated as described in Chapter 5 for leaf expansion traits.

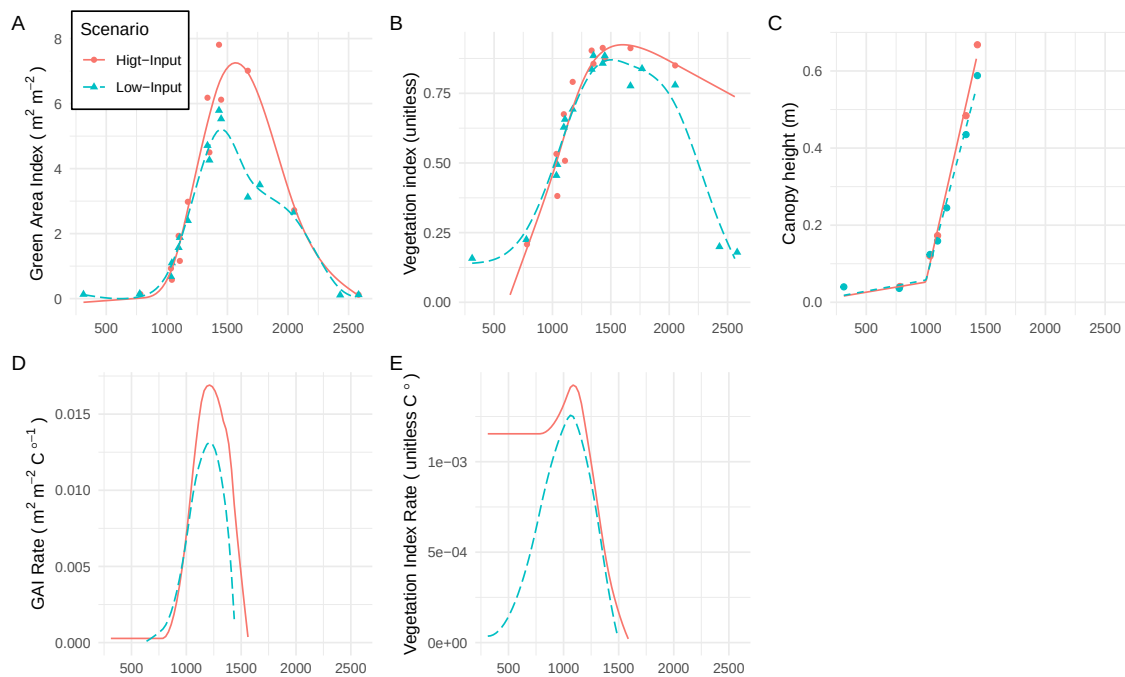


FIGURE 6.1 – Absolute values and rate of change of green area index (A, D), vegetative index (B, E) and canopy height (C) as a function of thermal time after sowing for bread wheat genotype '2109.36'. Grown in the field in Gréoux-les-bains, France, during the 2017-2018 growing season under high and low input conditions. High and low input conditions. Data are means for $n = 2$ independent replicates. Circles are experimental data, lines are cubic smoothing (A, B) and segmented linear (C) spline regressions fitted to the data. In Figure D and E, the lines are the first derivative (rate) of the fitted splines. Solid lines, high input scenario; dashed lines, low input scenario.

6.2.4 Data analysis

For all variables, the sensitivity to low input conditions was calculated as the normalized difference :

$$\text{Sensitivity of variable} = \frac{(\text{high input} - \text{low input})}{\text{high input}} \quad (6.1)$$

Variables measured or calculated in the field and in the platform experiment are described in Table 6.2 and 6.3, respectively. All data analyses were carried out using the statistical software program R (R Core Team, 2018). Principal component analyses (PCAs) were performed using the `PCA()` function of the R package *FactorMineR* (Lê et al., 2008). A hierarchical clustering analysis on principal components (HCPC) was performed using the `pvclust()` function of the R package *pv.clust* (Suzuki and Shimodaira, 2006; Suzuki et al., 2019). One thousand bootstrap samplings were performed to calculate unbiased (AU) *P*-value.

6.3 Results

6.3.1 Comparison of dynamic growth traits in the field and in controlled environment

Together with yield components, we calculated metrics of plant growth that reflect the growth dynamics and its sensitivity to soil limiting conditions, both on the platform and in the field, and for both bread wheat and durum wheat genotypes. Results for the platform experiment are presented in chapter 5.

In the field, the time course of canopy height under both high- and low-input conditions for both bread (Fig. 6.1C) and durum (Supplementary Information Fig. S22) wheat genotypes was well described by a segmented linear model, with a period during which canopy height increased linearly with thermal time that corresponded to period of internode expansion. As illustrated in Fig. 6.1C for the bread wheat cultivar 2109.36C, both CHR and CHmax were reduced under LI

conditions compared with HI conditions. Similar results were obtained for durum wheat (Supplementary Information Fig. S22). The genotypic values of the variables measured in the field are summarized in Supplementary Information Table S9.

In both high- and low-input conditions, GAI (Fig. 6.1A) and NDVI (Fig. 6.1B) showed a period with an almost-linear increase, followed by a period of decrease due to leaf senescence. Fitting spline curves to observed GAI and NDVI data allowed us to calculate their rate of change as a function of thermal time (Fig. 6.1D) and NDVI (Fig. 6.1E). Both types of growth indicators showed that canopy area expansion was reduced under LI conditions compared to HI conditions. In a similar way to that calculated in the platform, genotypes were compared based on (i) the maximum value of the integrated variable; (ii) the maximum rate of change; and (iii) the sensitivity of the rates to limiting conditions (calculated as the normalized difference between LI and HI). Again, similar results were obtained in durum wheat (SI Fig. S1).

6.3.2 Hierarchical clustering discriminated maximal growth traits from growth sensitivity traits, whether in the platform or in the field

To approach and better understand the relationship between dynamic variables related to growth, their sensitivities, together with other variables and sensitivities related to grain yield components, we performed a PCA and a HCPC on all traits measured in the platform and in the field. Results for the two species were largely similar, thus below I only present the results for bread wheat. Results for durum wheat are shown in Supplementary information Figure S23. Similar PCA and HCPC analyses were performed including the field trials presented in Chapter 3, in which variables related to growth were not measured. These results do not affect our conclusions regarding yield components (see below). They are also presented as Supplementary Information Fig. S23 but are not described in the result section.

The number of traits analyzed (40) in the PCA does not allow for direct conclusions (Fig. 6.2AB). However, several indications can be identified at a broader level. First, this analysis did not discriminate field variables from platform variables. Second, it did not discriminate sensitivity variables from rate or yield component variables. Together, these results indicate that correlations should be found between such traits.

Hierarchical classification of these variables allowed for a more detailed approach to the relationships between these variables. (Fig. 6.2C). In particular, the first level of clustering separated a group of variables in which appeared most of the variables related to maximum growth in the field and in the platform (e.g. TIRmax and SERmax in the platform and GAIRmax in the field), and the variables related to the sensitivity of these variables. Note that these results were even clearer for durum wheat and for bread wheat (SI Fig. S23). The approximately unbiased *P*-values were high, especially for a small group (AU = 95, Fig. 6.2C) that included the variables related to tillering (TIRmax, TNmax) and whole plant growth (SERmax, SAmx) measured in the platform, and green area index (GAImax, GAIRmax) measured in the field.

These relationships, observed here on a large scale, could be studied side by side, by biplots (Fig. 6.3) and Pearson's correlations. In particular, the maximum rate of increase of GAI in the field (GAIRmax) was significantly correlated with the maximum number of appeared tiller per plant (TiRmax) and total leaf area (SERmax) measured in platform. It was less clear regarding sensitivity traits. Indeed, while significant relationships were found between sensitivity traits measured in platform (see Chapter 5) or in the field (S_{NDVI} and S_{GAI} , $P < 0.01$; Fig. 6.4), no significant correlation were found between sensitivity traits in the field and in the platform.

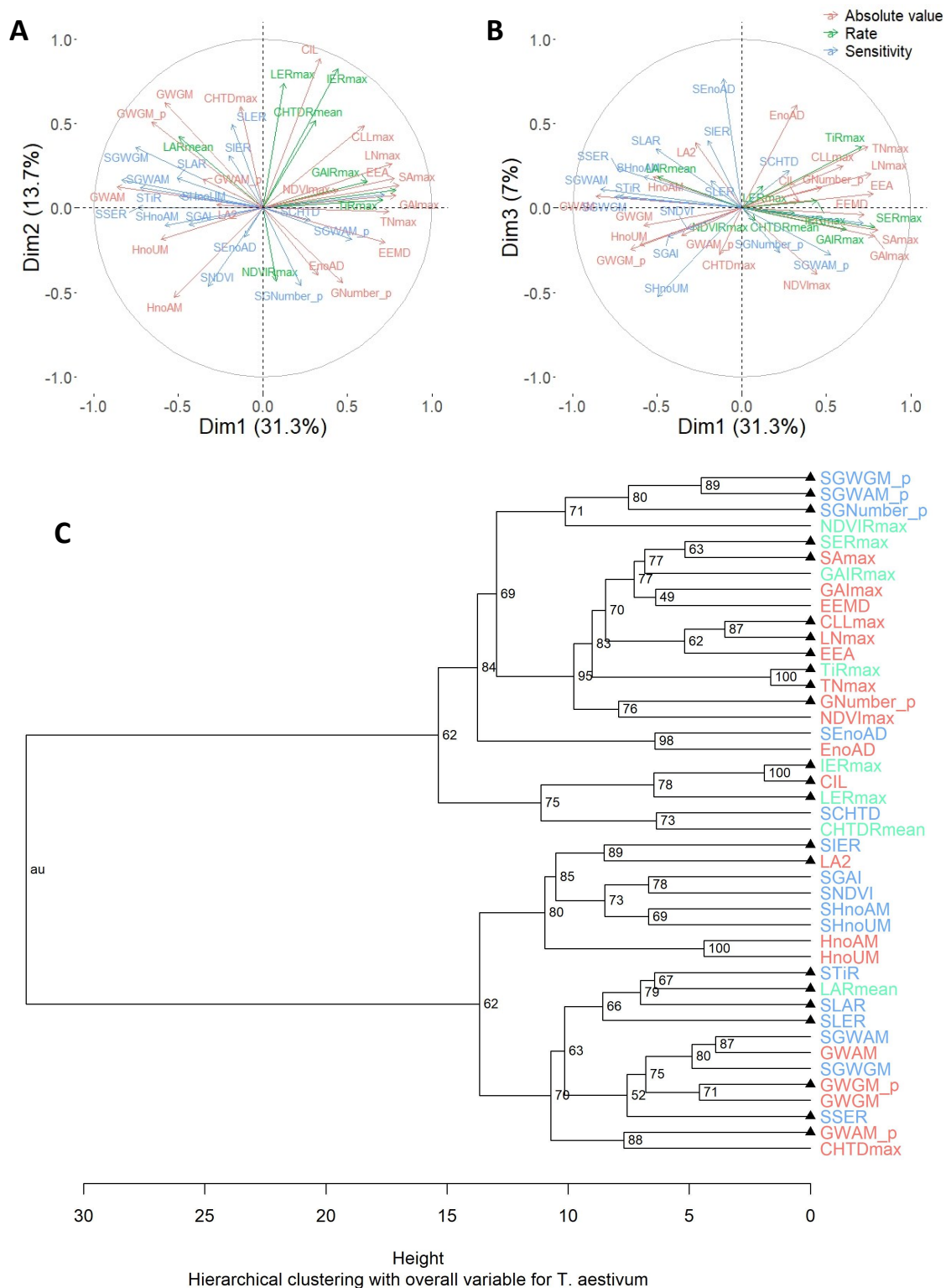


FIGURE 6.2 – Principal component analysis (PCA) and hierarchical clustering on principal components (HCPC) analyses for the 40 traits measured on 48 bread wheat cultivars grown in the field at Gréoux-les-bains, France during the 2017-2018 growing season in high- and low-input conditions in the PhenoArch platform. See Table 6.3 for definitions of variables. Colors display the type of variable : absolute values (raw data, red), rates (green) and sensitivities to either water deficit (in the platform) and low-input conditions in the field (blue). (A) Representation of all variables for (A) PC1 versus PC2 and (B) PC1 versus PC3. (C) Dendrogram of the hierarchical clustering based on all variable. Small triangles indicate which variables were obtained in the platform. The numbers indicate the approximately unbiased (AU) P-value obtained from 1000 bootstrap replications. Colors : as in A



FIGURE 6.3 – Scatter plot matrix of bivariate relationships between variables linked to shoot growth measured in the field or in the platform for all genotypes of bread wheat (red) or durum wheat (green). NDVIR_{max}, maximal rate of vegetation index; GAIR_{max}, maximum rate of change of green area index; CHTDR_{mean}, average rate of canopy height increase; SER_{max}, maximal rate of Shoot expansion rate; TiR_{max}, maximal value of tillering rate; LER_{max}, maximum leaf elongation rate; IER_{max}, Maximum internode elongation rate. See Table 6.3 for further definitions of variables. The right side displays the Pearson correlation coefficients with significance level (blue : T. durum; red : T. aestivum).

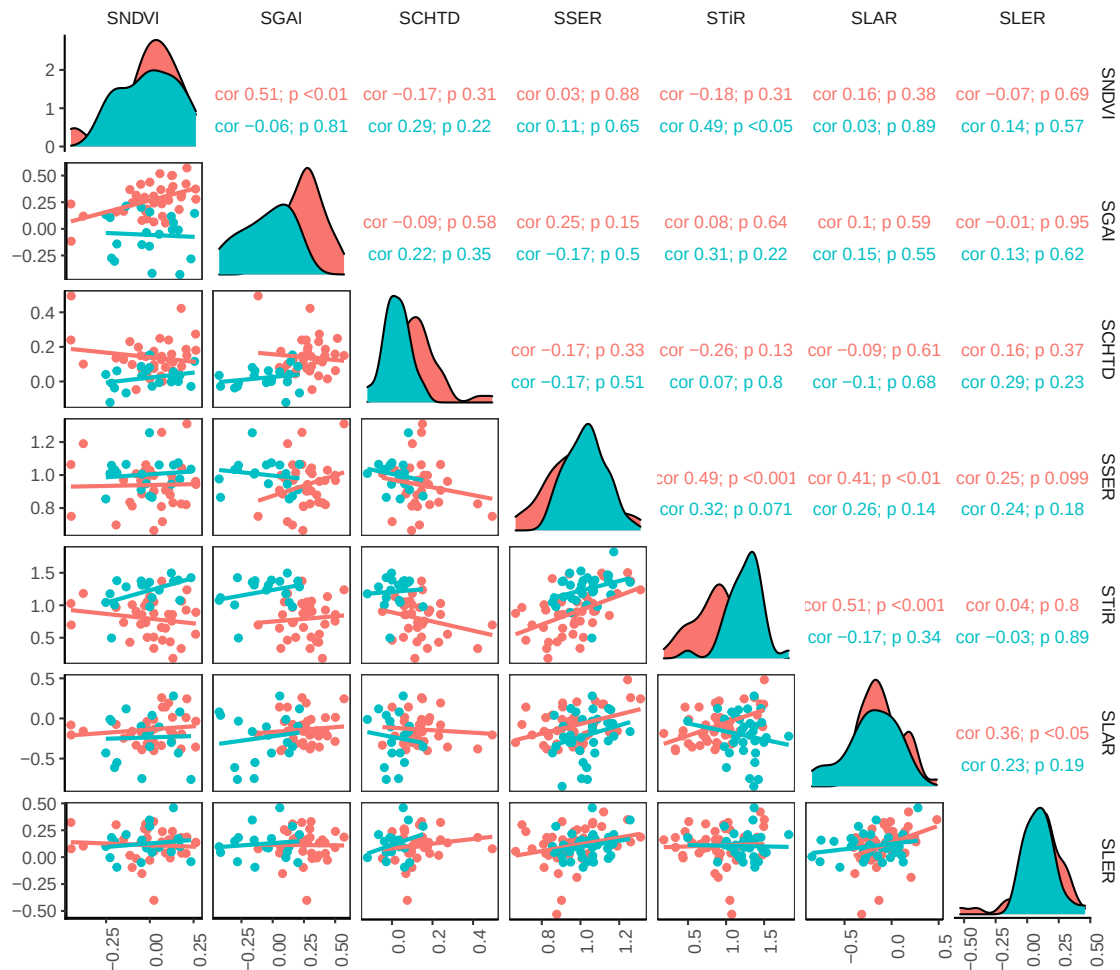


FIGURE 6.4 – Scatter plot matrix of bivariate relationships between variables linked to shoot growth sensitivity in the field or in the platform for all genotypes of bread wheat (red) or durum wheat (blue). SNDVI, sensitivity of vegetation index; SGAI, sensitivity of green area index; SCHTD, sensitivity of canopy height; SSER, sensitivity of Shoot expansion rate; STiR, sensitivity of tillering rate; SLER, sensitivity of leaf elongation rate; SIER, sensitivity of internode elongation rate. See Table 6.3 for further definitions of variables. The right side displays the Pearson correlation coefficients with significance level (blue : *T. durum*; red : *T. aestivum*).

TABLE 6.3 – Description of the variables measured or calculated in the platform.

Name	Description	Description
SA _{max}	Maximum shoot area	cm²
SER _{max}	Maximum rate of shoot expansion	cm ² °Cd
S _{SER}	Sensitivity of shoot expansion rate to water deficit	MPa ⁻¹
LN _{max}	Final leaf number per main stem	leaf main stem ⁻¹
LAR _{mean}	Average main-stem leaf appearance rate	leaf °Cd ⁻¹
S _{LAR}	Sensitivity of main-stem leaf appearance rate to water deficit	MPa ⁻¹
TN _{max}	maximum tiller number per plant	tiller plant ⁻¹
TiR _{max}	Maximum rate of tiller appearance	tiller °Cd ⁻¹
S _{TiR}	Sensitivity of tiller appearance rate to water deficit	MPa ⁻¹
CLL _{max}	Cumulative length of leaf lamina on the main stem	cm main stem ⁻¹
LER _{max}	Maximum leaf expansion rate.	cm °Cd ⁻¹
S _{LER}	Sensitivity of leaf expansion rate to water deficit	MPa ⁻¹
CIL	Cumulative internodes length of the main stem	cm main stem ⁻¹
IER _{max}	Maximum internode expansion rate	cm °Cd ⁻¹
S _{IER}	Sensitivity of internode expansion rate to water deficit	MPa ⁻¹
LA2	Surface area of main-stem leaf 2	cm ²
Heading	Ear emergence date	°Cd
GrainW	Average grain dry mass	g DM grain ⁻¹
GrainNb	Grain number per plant	grain plant ⁻¹
Yield	Grain yield per plant	g DM plant ⁻¹
S _{Yield}	Sensitivity of grain yield to water deficit	number
S _{GrainNb}	Sensitivity of grain number per plant to water deficit	number
S _{GrainW}	Sensitivity of average grain dry mass to water deficit	number

6.3.3 Sensitivities of both growth traits in the platform and in the field correlated to that of grain number in the field

The above paragraph showed that a large cluster included variables related to maximum growth both in the field and in the platform. However, these variables were related to only a few components of grain yield in the field. This was confirmed by the pairwise correlations (Supplementary Information Fig. S24). In contrast, the hierarchical clustering (Fig. 6.2) suggested relationships between grain yield and yield components measured in the field and growth sensitivities, whether measured in the field or in the platform (Fig. 6.2). The large group that included most of the growth sensitivity traits in the field (S_{GAI} , S_{NDVI}) or in the platform (S_{TiR} , S_{SER}), also included most of the variables related to grain yield component sensitivities.

A second level of clustering discriminated a group of variables (AU = 80) with the number of grains (GrainSpike and GrainNb), their sensitivity ($S_{GrainSpike}$ and $S_{GrainNb}$), as well as S_{GAI} and S_{NDVI} . Most of these relationships were significant

in the pairwise correlations (Fig. 6.5). A second group ($AU = 70$) included Yield and S_{Yield} , as well as S_{TiR} and S_{SER} in the platform. Among these variables S_{SER} was correlated with $S_{GrainNb}$ and S_{Yield} in the field (Fig. 6.5, Supplementary Information Fig. S24). However, these sensitivities were not correlated in durum wheat. This can be explained by the fact that yield components were less affected in the ARVgre2019 trial for durum wheat than in ARVgre2018 for bread wheat. The range of variation calculated for the sensitivities of yield components are therefore smaller for durum wheat than for bread wheat.

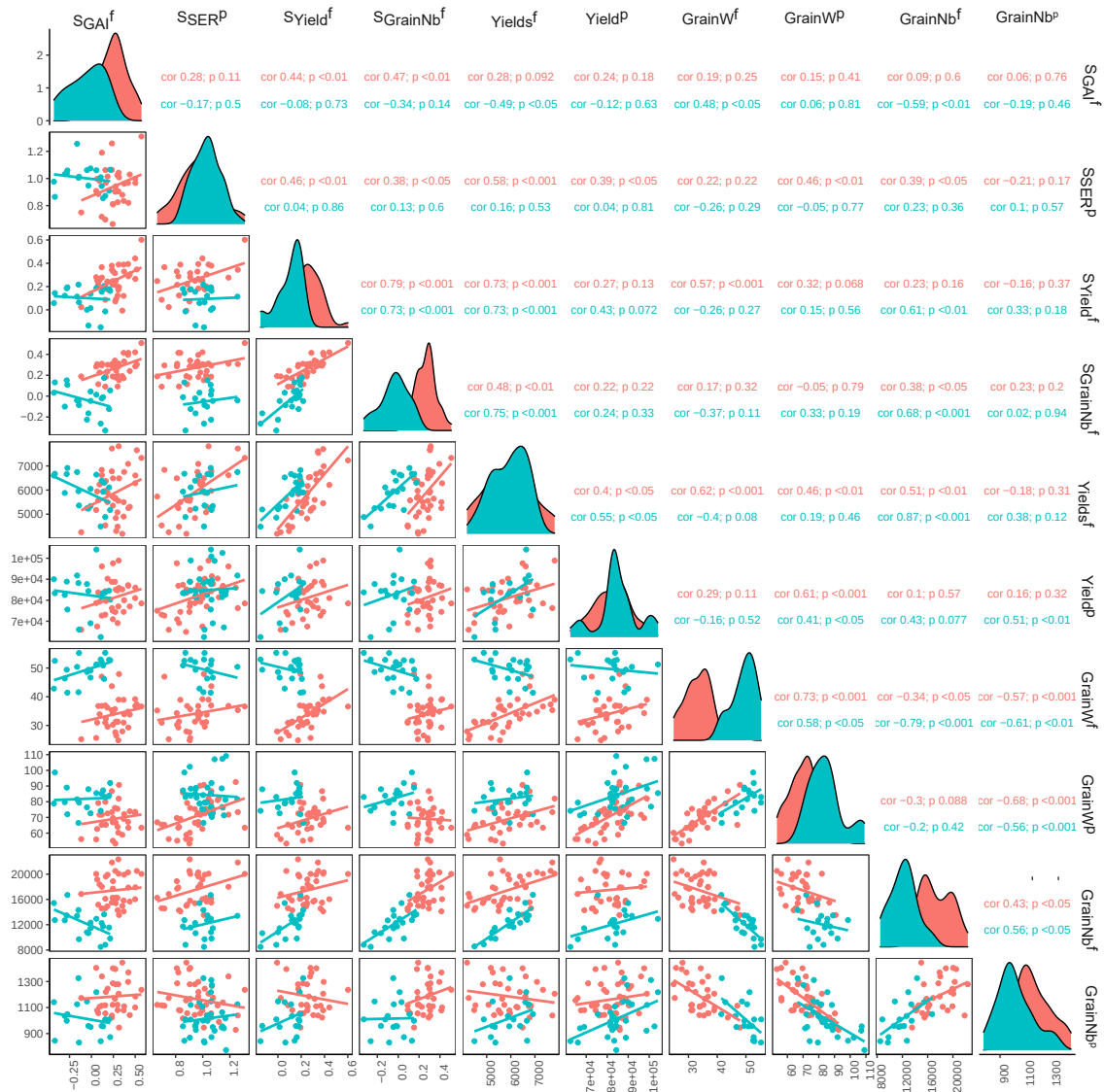


FIGURE 6.5 – Scatter plot matrix of bivariate relationships between variables related to shoot growth sensitivity in the field or in the platform for all genotypes of bread wheat (red) or durum wheat (blue). The right side displays the Pearson correlation coefficients with significance level. SGWAM, sensitivity of yield; SHnoAM, sensitivity of grain number; SSER, sensitivity of shoot expansion rate. See Table 6.3 for further definitions of variables.

6.4 Discussion

6.4.1 Growth traits and their sensitivities to water deficit measured in platform can provide inform on growth potential in the field

Field phenotyping faces several limitations for traits related to plant sensitivity to soil water deficit (Berger et al., 2010) : (i) the random and experimentally risky aspect of water deficit that may or may not occur, even several years in a row under rainfed conditions and in low latitudes; (ii) its very dynamic, non-stable aspect, which complicates the analyses, especially for plants with different phenology, with key stages of development at very different times, under different water status; (iii) The difficulty to quantify this water deficit, which is by essence is very different according to soil depth, and depending on the associated root and soil/root conductance; (iv) The interaction of water deficit with other abiotic stresses, in particular thermal stress, due to the concomitance of drought and high temperature periods and the fact that a decrease in transpiration also affects the temperature of the plant (Parent et al., 2017). For all these reasons, trait-based selection has mainly focused on constitutive traits, easily phenotyped and giving advantages in most water-deficit scenarios. It is therefore reasonable to focus the phenotyping of water-deficit response traits in controlled condition platforms. However, such platforms do not reflect agronomic conditions and therefore the output parameters may not be extrapolated to the field.

In this study, we showed instead that the underlying parameters of growth were strongly correlated in the platform and in the field. These correlations imply that a genetic analysis based on growth parameters should result in genetic markers of interest for field conditions. In addition, parameters related to individual leaf growth and tillering (see Chapter 5) are in the process of being automatically phenotyped in the PhenoArch platform. Knowing that these parameters can be easily linked to genotypic parameters of crop models supports the idea that these highly controlled platforms can be of interest in genetic trait impact studies and for model-assisted breeding. Overall, these results can support the

idea that highly controlled platforms can have a place in academic research and pre-breeding pipelines, and that such place does not depend on the development (or not) of field phenotyping but rather on specific focus on adaptive traits such as drought response traits.

6.4.2 Is the sensitivity of shoot growth in the platform related to that of grain number in the field

Much has been written about the debate on yield control under water deficit, whether its limitation by carbon sinks or sources and/or the most important underlying component, either grain number or grain filling (Sinclair and Jamieson, 2008). I have already discussed this aspect in Chapter 4, concluding that there is, however, the beginnings of a compromise in the literature, targeting grain number as the primary control on yield. Some of these conclusions of Chapter 4 are again found here. Here, in addition to this robustness in contrasted fields (Chapter 4) we also show a very little $G \times E$ interactions on grain dry mass between field and platform (very close in clustering, Fig. 6.2).

Another major result showed the relationship between the sensitivity of growth measured in the platform and the sensitivity of the number of grains in the field. Such a relationship has already been observed in maize for the sensitivity of individual leaf elongation to soil water deficit measured in the PhenoDyn platform (Welcker et al., 2011) and the sensitivity of the number of grain number per square meter in the field, calculated in the same way as in our study (Dignat et al., 2013). In maize, this relationship was interpreted through common determinisms for the sensitivities of different growth processes, and in particular leaf growth and silk growth. Silk growth itself determines grain number in maize (Oury et al., 2016; Turc et al., 2016). Here, in wheat, the putative mechanism involved is necessarily different. However, the relationship between growth and grain number during a pre-flowering sensitivity period is a type of formalism that is widely used in crop models, e.g. Sirius (Martre et al., 2006; He et al., 2012). Here, the fact that this relationship is found on sensitivities (and not absolute values) indicates

that such sensitivities parameters, common to growth and grain number, could be used in new formalisms for grain number calculation, simplifying the phenotyping of hundreds of varieties and contributing to the parsimony of genotypic parameters in crop models.

6.5 References

- Beauchêne, K., Leroy, F., Fournier, A., Huet, C., Bonnefoy, M., Lorgeou, J., de Solan, B., Piquemal, B., Thomas, S., Cohan, J.-P., 2019. Management and Characterization of Abiotic Stress via PhénoField®, a High-Throughput Field Phenotyping Platform. *Front. Plant Sci.* 10, 904. <https://doi.org/10.3389/fpls.2019.00904>
- Berger, B., Parent, B., Tester, M., 2010. High-throughput shoot imaging to study drought responses. *J. Exp. Bot.* 61, 3519–3528. <https://doi.org/10.1093/jxb/erq201>
- Cabrera-Bosquet, L., Fournier, C., Brichet, N., Welcker, C., Suard, B., Tardieu, F., 2016. High-throughput estimation of incident light, light interception and radiation-use efficiency of thousands of plants in a phenotyping platform. *New Phytol.* 212, 269–281. <https://doi.org/10.1111/nph.14027>
- Dignat, G., Welcker, C., Sawkins, M., Ribaut, J.M., Tardieu, F., 2013. The growths of leaves, shoots, roots and reproductive organs partly share their genetic control in maize plants. *Plant Cell Environ.* 36, 1105–1119. <https://doi.org/10.1111/pce.12045>
- Evers, J.B., Vos, J., Fournier, C., Andrieu, B., Chelle, M., Struik, P.C., 2005. Towards a generic architectural model of tillering in Gramineae, as exemplified by spring wheat (*Triticum aestivum*). *New Phytol.* 166, 801–812. <https://doi.org/10.1111/j.1469-8137.2005.01337.x>
- Fiorani, F., Schurr, U., 2013. Future Scenarios for Plant Phenotyping. *Annu. Rev. Plant Biol.* 64, 267–291. <https://doi.org/10.1146/annurev-arplant-050312-120137>
- Fu, P., Montes, C.M., Siebers, M.H., Gomez-Casanovas, N., McGrath, J.M., Ainsworth, E.A., Bernacchi, C.J., 2022. Advances in field-based high-throughput photosynthetic phenotyping. *J. Exp. Bot.* 73, 3157–3172. <https://doi.org/10.1093/jxb/erac077>
- Furbank, R.T., Tester, M., 2011. Phenomics – technologies to relieve the phenotyping bottleneck. *Trends Plant Sci.* 16, 635–644. <https://doi.org/10.1016/j.tplants.2011.09.005>
- Ghosal, S., Blystone, D., Singh, A.K., Ganapathysubramanian, B., Singh, A., Sarkar, S., 2018. An explainable deep machine vision framework for plant stress phenotyping. *Proc. Natl. Acad. Sci.* 115, 4613–4618. <https://doi.org/10.1073/pnas.1716999115>
- Granier, C., Aguirrezabal, L., Chenu, K., Cookson, S.J., Dauzat, M., Hamard, P., Thioux, J.-J., Rolland, G., Bouchier-Combaud, S., Lebaudy, A., Muller, B., Simonneau, T., Tardieu, F., 2006. PHENOPSIS, an automated platform for reproducible phenotyping of plant responses to soil water deficit in *Arabidopsis thaliana*

permitted the identification of an accession with low sensitivity to soil water deficit. *New Phytol.* 169, 623–635. <https://doi.org/10.1111/j.1469-8137.2005.01609.x>

He, J., Le Gouis, J., Stratonovitch, P., Allard, V., Gaju, O., Heumez, E., Orford, S., Griffiths, S., Snape, J.W., Foulkes, M.J., Semenov, M.A., Martre, P., 2012. Simulation of environmental and genotypic variations of final leaf number and anthesis date for wheat. *Eur. J. Agron.* 42, 22–33. <https://doi.org/10.1016/j.eja.2011.11.002>

Janni, M., Pieruschka, R., 2022. Plant phenotyping for a sustainable future. *J. Exp. Bot.* 73, 5085–5088. <https://doi.org/10.1093/jxb/erac286>

Jeudy, C., Adrian, M., Baussard, C., Bernard, C., Bernaud, E., Bourion, V., Buset, H., Cabrera-Bosquet, L., Cointault, F., Han, S., Lamboeuf, M., Moreau, D., Pivato, B., Prudent, M., Trouvelot, S., Truong, H.N., Vernoud, V., Voisin, A.-S., Wipf, D., Salon, C., 2016. RhizoTubes as a new tool for high throughput imaging of plant root development and architecture : test, comparison with pot grown plants and validation. *Plant Methods* 12, 31. <https://doi.org/10.1186/s13007-016-0131-9>

Kirchgessner, N., Liebisch, F., Yu, K., Pfeifer, J., Friedli, M., Hund, A., Walter, A., 2017. The ETH field phenotyping platform FIP : a cable-suspended multi-sensor system. *Funct. Plant Biol.* 44, 154. <https://doi.org/10.1071/FP16165>

Lê, S., Josse, J., Husson, F., 2008. FactoMineR : A Package for Multivariate Analysis. *J. Stat. Softw.* 25, 1–18. <https://doi.org/10.18637/jss.v025.i01>

Machwitz, M., Pieruschka, R., Berger, K., Schlerf, M., Aasen, H., Fahrner, S., Jiménez-Berni, J., Baret, F., Rascher, U., 2021. Bridging the Gap Between Remote Sensing and Plant Phenotyping—Challenges and Opportunities for the Next Generation of Sustainable Agriculture. *Front. Plant Sci.* 12, 749374. <https://doi.org/10.3389/fpls.2021.749374>

Madec, S., Baret, F., de Solan, B., Thomas, S., Dutartre, D., Jezequel, S., Hemmerlé, M., Colombeau, G., Comar, A., 2017. High-Throughput Phenotyping of Plant Height : Comparing Unmanned Aerial Vehicles and Ground LiDAR Estimates. *Front. Plant Sci.* 8.

Martre, P., Jamieson, P.D., Semenov, M.A., Zyskowski, R.F., Porter, J.R., Triboi, E., 2006. Modelling protein content and composition in relation to crop nitrogen dynamics for wheat. *Eur. J. Agron., Modelling Quality Traits and Their Genetic Variability for Wheat* 25, 138–154. <https://doi.org/10.1016/j.eja.2006.04.007>

Oury, V., Caldeira, C.F., Prodhomme, D., Pichon, J.-P., Gibon, Y., Tardieu, F., Turc, O., 2016. Is Change in Ovary Carbon Status a Cause or a Consequence of Maize Ovary Abortion in Water Deficit during Flowering? *Plant Physiol.* 171, 997–1008. <https://doi.org/10.1104/pp.15.01130>

Parent, B., Bonneau, J., Maphosa, L., Kovalchuk, A., Langridge, P., Fleury, D., 2017. Quantifying Wheat Sensitivities to Environmental Constraints to Dissect Genotype x Environment Interactions in the Field. *Plant Physiol.* 174, 1669–1682. <https://doi.org/10.1104/pp.17.00372>

Perez, R.P.A., Fournier, C., Cabrera-Bosquet, L., Artzet, S., Pradal, C., Brichet, N., Chen, T.-W., Chapuis, R., Welcker, C., Tardieu, F., 2019. Changes in the vertical

distribution of leaf area enhanced light interception efficiency in maize over generations of selection. *Plant Cell Environ.* 42, 2105–2119. <https://doi.org/10.1111/pce.13539>

Prado, S.A., Cabrera-Bosquet, L., Grau, A., Coupel-Ledru, A., Millet, E.J., Welcker, C., Tardieu, F., 2018. Phenomics allows identification of genomic regions affecting maize stomatal conductance with conditional effects of water deficit and evaporative demand. *Plant Cell Environ.* 41, 314–326. <https://doi.org/10.1111/pce.13083>

R Core Team, 2018. R : A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

Rémy, J.C., Hébert, J., 1977. Le devenir des engrais dans le sol. *C. r. L'Académie D'Agriculture Fr., français*, biblio 63, 700–710.

Sinclair, T.R., Jamieson, P.D., 2008. Yield and grain number of wheat : A correlation or causal relationship? : Authors' response to "The importance of grain or kernel number in wheat : A reply to Sinclair and Jamieson" by R.A. Fischer. *Field Crops Res.* 105, 22–26. <https://doi.org/10.1016/j.fcr.2007.07.003>

Suzuki, R., Shimodaira, H., 2006. Pvcust : an R package for assessing the uncertainty in hierarchical clustering. *Bioinformatics* 22, 1540–1542. <https://doi.org/10.1093/bioinformatics/btl117>

Suzuki, R., Terada, Y., Shimodaira, H., 2019. pvcust : Hierarchical Clustering with P-Values via Multiscale Bootstrap Resampling.

Tardieu, F., Parent, B., 2017. Predictable 'meta-mechanisms' emerge from feedbacks between transpiration and plant growth and cannot be simply deduced from short-term mechanisms. *Plant Cell Environ.* 40, 846–857. <https://doi.org/10.1111/pce.12822>

Turc, O., Bouteillé, M., Fuad-Hassan, A., Welcker, C., Tardieu, F., 2016. The growth of vegetative and reproductive structures (leaves and silks) respond similarly to hydraulic cues in maize. *New Phytol.* 212, 377–388. <https://doi.org/10.1111/nph.14053>

Welcker, C., Sadok, W., Dignat, G., Renault, M., Salvi, S., Charcosset, A., Tardieu, F., 2011. A Common Genetic Determinism for Sensitivities to Soil Water Deficit and Evaporative Demand : Meta-Analysis of Quantitative Trait Loci and Introgression Lines of Maize. *Plant Physiol.* 157, 718–729. <https://doi.org/10.1104/pp.111.176479>

Yang, W., Feng, H., Zhang, X., Zhang, J., Doonan, J.H., Batchelor, W.D., Xiong, L., Yan, J., 2020. Crop Phenomics and High-Throughput Phenotyping : Past Decades, Current Challenges, and Future Perspectives. *Mol. Plant* 13, 187–214. <https://doi.org/10.1016/j.molp.2020.01.008>

Troisième partie

Conclusion

CHAPITRE 7

Résumé des recherches

Dans le **Chapitre 3**, j'ai étudié la structure de la variabilité génétique des réponses de la croissance foliaire et de la transpiration à la température et au VPD chez 12 espèces et sous-espèces apparentées au blé. Pour cela, j'ai défini des formalismes de réponses et quantifié ces réponses (valeurs de paramètres) en phénotypant cinq variétés par espèce/sous espèce. Mes résultats indiquent que ces traits de réponses sont bien conservés à l'échelle intra-spécifique, en particulier pour les réponses à la température, mais qu'il existe une diversité à l'échelle interspécifique. Cette diversité est largement structurée par les liens phylogénétiques entre les espèces et sous espèces, leur niveau de ploïdie et dans une moindre mesure par la domestication. Enfin, outre le fait de proposer des formalismes et des valeurs de paramètres utiles en modélisation des cultures, mon étude a permis d'identifier des sources potentielles d'amélioration variétale qui résident actuellement dans les espèces apparentées au blé.

Dans le **Chapitre 4**, j'ai tout d'abord étudié les contraintes/reliations entre composantes du rendement (définissant l'espace phénotypique), comment ces re-

lations sont elles-même affectées par des contraintes hydriques et azotées, et enfin, j'ai cherché à sélectionner un sous panel qui respecterait l'espace des possibles observés dans le panel de départ. Plus précisément, j'ai montré ici que l'ensemble des corrélations entre le rendement, ses composantes et leurs sensibilités, définissait un espace phénotypique très contraint. Ces relations étant similaires chez le blé dur et le blé tendre, mes résultats indiquent qu'une hypothèse de contraintes génétiques fortes (proximité, lien entre régions génomiques) pour expliquer les liens entre variables est moins probable que l'hypothèse de contraintes évolutives et agronomiques communes entre les deux espèces. Notre sélection génotypique a respecté cet espace phénotypique, dans le sens où elle a conservé toutes ses composantes, les relations entre variables et la variabilité génétique conduisant à ces relations. Les deux sous panel résultants de cette étape sont ceux utilisés pendant le reste de ma thèse.

Dans le **Chapitre 5**, j'ai cherché à déterminer les compromis entre les composantes de l'expansion foliaire à l'échelle de la plante entière et leurs sensibilités au déficit hydrique. J'ai pu démontrer que la diversité observée de croissance foliaire était largement la résultante de celle du tallage, en tout cas plus que de celle de la vitesse d'émission des feuilles ou l'élongation de feuilles individuelles. De même, j'ai montré que la plasticité de la surface foliaire était principalement déterminée par celle du tallage. Dans l'ensemble, ce travail peut avoir des conséquences importantes en (i) écophysiologie, en décrivant la dynamique de croissance foliaire, ainsi que les parts respectives de ses sous-composantes, leurs plasticité et leurs relations réciproques; (ii) pour la sélection, en identifiant les traits cibles sous déficit hydrique ainsi que la variabilité génétique associée; et (iii) en modélisation, en décrivant des formalisme simples de réponse, ainsi que la structure des paramètres génétiques associés.

Dans le **Chapitre 6**, j'ai questionné l'intérêt de la démarche, en explorant les liens entre ces variables mesurées en plateformes, les variables de croissance mesurées au champ, et jusqu'aux composantes du rendement. Les résultats concernant les paramètres de croissance étaient positifs, au sens qu'ils étaient corrélés.

lés entre la plateforme et le champ. Ces corrélations impliquent par exemple qu'une analyse génétique basée sur des paramètres de croissance mesurés en plateforme devrait aboutir à des marqueurs génétiques d'intérêt pour les conditions au champ. Le fait que ces paramètres puissent être facilement liés aux paramètres génotypiques des modèles de culture renforce l'idée que ces plateformes de phénotypage peuvent être intéressantes pour les études d'impact des caractères génétiques et pour la sélection assistée par modèle. Un autre résultat important est l'existence d'une relation entre la sensibilité de la croissance mesurée dans la plateforme et la sensibilité du nombre de grains au champ. Le fait que cette relation soit trouvée sur les sensibilités (et non sur les valeurs absolues) indique que de tels paramètres de sensibilité, communs à la croissance et au nombre de grains, pourraient être utilisés dans de nouveaux formalismes pour le calcul du nombre de grains, simplifiant le phénotypage de centaines de variétés et contribuant à la diminution des paramètres génotypiques dans les modèles de culture.

CHAPITRE 8

Discussion générale

8.1 Quels approches et concepts traversent cette thèse ?

L'objectif de cette thèse était de contribuer à l'amélioration des modèles de culture, en fournissant à la fois (i) des formalismes de réponses aux contraintes abiotiques pour les processus de développement et de croissance du blé, mais aussi (ii) les valeurs des paramètres de modèles dans des gammes génétiques élargies, leurs liens/contraintes définissant leur espace phénotypique, ainsi que la (iii) pertinence des caractères mesurés en plateforme pour décrire des comportements au champ.

De façon plus concrète, mon étude s'est portée sur :

- les effets de contraintes abiotiques liées au double défi du changement climatique (sécheresse du sol, Chapitres 4, 5, et 6 ; de l'air, Chapitre 3 ; et la température, Chapitre 3) et des politiques publiques pour une agriculture plus durable (azote, Chapitres 4 et 6).

- sur un processus adaptatif, la croissance foliaire, étudiée à différentes échelles temporelles, de la variable dynamique mesurées au pas de temps de la minute (Chapitre 3) à la journée (Chapitre 5), et jusqu'aux valeurs intégrées au champ (Chapitre 6), dans toutes ses composantes spatiales (émission de feuilles, élongation, tallage) et leurs interrelations (Chapitres 3,5 et 6).
- en incluant une diversité génétique élargie (espèces apparentées, Chapitre 3 ou panels de diversité de blé tendre et de blé dur, Chapitres 4, 5 et 6), pour analyser leur impact sur le rendement et ses composantes dans des environnements multi-contraints (Chapitres 4 et 6).

8.1.1 Des formalismes de réponse simple pour des analyses comparatives intra et inter-spécifiques

Le point commun entre les différents chapitres de ma thèse est que j'ai toujours essayé d'analyser les réponses et leur variabilité génétique de manière quantitative. Pour cela j'ai utilisé des formalismes parcimonieux en nombre de paramètres et/ou avec des paramètres aussi indépendants que possibles, et en n'étant surtout pas dogmatique quant aux formalismes utilisés.

Par exemple, dans le Chapitre 3, j'ai choisi d'utiliser un modèle bilinéaire pour décrire les réponses de la croissance de différentes espèces de blé à la température. Pourtant, l'usage au laboratoire était plutôt d'utiliser des formalismes curvilinéaires pour décrire les réponses de processus liés au développement et/ou à la croissance foliaire, tel que ceux décrits dans Parent & Tardieu (2012) ou Wang et al. (2017). Dans ce premier chapitre, la comparaison de plusieurs formalismes donnant l'avantage au formalisme bilinéaire, c'est celui que j'ai adopté.

Toujours dans le Chapitre 3, j'ai décrit les réponses de la croissance foliaire au VPD avec un formalisme bilinéaire. Bien que cet effet du VPD sur l'élongation foliaire soit apparu comme très important, de l'ordre de grandeur de celui observé chez le riz (Parent et al., 2010) et le maïs (Sadok et al., 2007), c'est à notre connaissance, la première étude décrivant ces réponses chez le blé. Ces for-

malismes et valeurs de paramètres devraient donc permettre d'améliorer la prédiction de la croissance foliaire par les modèles écophysiologique de croissance ou les modèle de prédiction, type modèles de culture.

Dans le Chapitre 5, j'ai là encore choisi le formalisme linéaire simple pour décrire et comparer les réponses de plusieurs processus liés à la croissance au déficit hydrique du sol. Choisir un formalisme commun, et aussi simple, permettait de comparer ces réponses sur la base d'une seule valeur de paramètre, et de comparer des processus différents, dans une gamme génétique large. C'est là encore la solution que j'ai choisie pour comparer les réponses de plusieurs processus au champ, sur la base d'un seul paramètre représentant la réponse globale à un environnement complexe affecté par la sécheresse et/ou limité en azote.

8.1.2 L'espace phénotypique comme concept central de la thèse pour étudier les liens entre processus

S'il y avait UN fil rouge à suivre tout le long de cette thèse, ce serait l'utilisation du concept d'espace phénotypique pour décrire les liens entre les processus biologiques et les distances phénotypiques entre espèces et variétés. J'ai décrit à plusieurs reprises dans les introductions des différents chapitres les origines de ce concept et les différents usages qui en sont fait. S'il y avait une seule chose à retenir, c'est que l'utilisation que je fais ici de ce concept diffère légèrement de son utilisation en écologie. En effet, plutôt que de décrire les compromis entre grandes fonctions, en utilisant des valeurs de traits, dépendant de l'environnement (tels que l'épaisseur d'une feuille par exemple), j'ai cherché à utiliser le plus possible des paramètres de modèles, représentant le génotype et non-dépendant de l'environnement (e.g. vitesse maximale d'un processus de croissance, réponses relatives à l'environnement). Ainsi, je pouvais estimer que la distance observée entre deux variétés ou espèces dans l'espace des paramètres représentait des distances phénotypiques indépendantes de l'environnement.

Dans le Chapitre 3, les conclusions tirées de cette analyse de l'espace phéno-

typique étaient multiples. (i) Les comparaisons entre les distances phénotypiques entre génotypes d'une même espèce et au niveau inter-espèces indiquaient une évolution lente des caractères liés aux sensibilités à la température et au VPD. (ii) L'analyse des positions respectives des espèces dans cet espace montrait que ces distances phénotypiques étaient proches des distances phylogénétiques, suggérant un rôle partagé entre évolution naturelle, domestication et sélection et polyploïdisation. Enfin (iii) le fait de regarder en détail les proximités d'espèces et de variétés dans l'espace phénotypique pouvait permettre d'indiquer l'intérêt de telle variété ou espèce pour l'amélioration de certains traits. Par exemple, *Aegilops tauschii* se retrouvant dans l'espace des hexaploïdes suggérait que les réponses spécifiques de ces hexaploïdes pouvait être dues au génome D.

Dans le Chapitre 4, mon utilisation du concept d'espace phénotypique était différente, avec des variables plus proches des traits et dépendants plus de l'environnement. La plasticité des espaces phénotypiques observés permettait ici de tirer des conclusions quant aux interactions $G \times E$ et $G \times E \times M$. Nous avons aussi utilisé ce concept pour vérifier que la diversité génétique d'un sous-panel était bien conservée et représentative de panels de diversité plus large. Cela a été utile pour la suite de mon travail de thèse aux chapitres 5 et 6.

8.2 Limites de l'approche et perspectives

Dans ma thèse, l'échelle d'étude centrale était le processus dynamique mesuré à l'échelle de l'organe (e.g. feuille) jusqu'à la plante entière, au pas de temps de l'heure ou de la journée. Mes résultats ont vocation à être intégrés à des échelles supérieures, soit directement, comme les comparaisons avec des résultats à l'échelle du couvert au Chapitre 6, mais aussi, et surtout, en apportant des formalismes et des paramètres pour des modèles de culture simulant des couverts à large échelles (Chapitres 3 et 5).

Une des limites de cette thèse est peut-être de n'avoir pas fait de lien avec les échelles descendantes, aux niveaux cellulaire ou moléculaire. Les sensibilités aux

différentes variables environnementales analysées ici sont des boîtes noires qui n'impliquent aucune grande fonctions écophysiologique (hydraulique, rapport source-puit en C, ...) ni de mécanismes cellulaires ou moléculaires.

Cependant, les liens entre les échelles décrits dans cette thèse, les protocoles d'analyse à grande échelle devrait permettre à terme de continuer ce travail vers les échelles infras, de déterminer des mécanismes et leurs contrôles génétiques, par exemple, en phénotypant de plus larges populations.

Une autre limite de ce travail est le côté beaucoup moins quantitatif et contrôlé des données et des analyses au champ. Les données récoltées avant et pendant ma thèse, issues des projets ANR Breedwheat et H2020 SolAce ne permettaient pas de différencier des effets propres des limitations hydriques et azotées. Le fait que ces limitations ne soient pas bien mesurées ou caractérisées, interdisaient de construire des courbes de réponse et ainsi pouvoir comparer les comportements au champ et en plateforme sur la même base. Une perspective de ce travail serait donc de quantifier les effets respectifs des limitations hydriques et azotées et d'analyser l'espace phénotypiques de ces réponses afin de décrire les interactions, liens, et contraintes entre les sensibilités à ces limitations a priori très différentes. Des relations fortes, comme des réponses parallèles ou orthogonales suggèreraient des processus communs, avec de fortes possibilités de contrôles génétiques commun. Au contraire, des réponses indépendantes suggèreraient des mécanismes divergent, avec leur propre diversité génétique.

8.3 Conclusion

Dans cette thèse nous avons utilisé une approche basée sur l'analyse des réponses quantitatives aux contraintes environnementales dans une diversité de blé élargies aux espèces apparentées. Le fil conducteur a été l'utilisation d'analyses multivariées et du concept d'espace phénotypique pour explorer les réponses des processus de croissance, de l'organe, à la parcelle. Pris tous ensemble, les résultats de cette thèse devraient avoir des impacts :

- En écophysiologie, en décrivant pour la première fois les réponses de la croissance à la demande évaporative chez le blé, et en questionnant les liens entre les sensibilités des sous-composantes de la croissance (Chapitre 5);
- En Biologie de l'évolution, en donnant des pistes quant à l'évolution de caractères liés aux réponses à la température et au VPD (Chapitre 1);
- En modélisation des cultures, en décrivant les réponses quantitatives, en proposant des formalismes et les gammes de paramètres génotypiques associés dans des gammes de diversité larges (Chapitres 3, 4, 5 et 6);
- En amélioration variétale, en indiquant des sources possibles d'amélioration chez le blé (Chapitre 5) et chez des espèces apparentées (Chapitre 3).

8.4 Références

Parent, B., Suard, B., Serraj, R., Tardieu, F., 2010. Rice leaf growth and water potential are resilient to evaporative demand and soil water deficit once the effects of root system are neutralized. *Plant Cell Environ.* 33, 1256–1267. <https://doi.org/10.1111/j.1365-3040.2010.02145.x>

Parent, B., Tardieu, F., 2012. Temperature responses of developmental processes have not been affected by breeding in different ecological areas for 17 crop species. *New Phytol.* 194, 760–774. <https://doi.org/10.1111/j.1469-8137.2012.04086.x>

Sadok, W., Naudin, P., Boussuge, B., Muller, B., Welcker, C., Tardieu, F., 2007. Leaf growth rate per unit thermal time follows QTL-dependent daily patterns in hundreds of maize lines under naturally fluctuating conditions. *Plant Cell Environ.* 30, 135–146. <https://doi.org/10.1111/j.1365-3040.2006.01611.x>

Wang, E., Martre, P., Zhao, Z., Ewert, F., Maiorano, A., Rötter, R.P., Kimball, B.A., Ottman, M.J., Wall, G.W., White, J.W., Reynolds, M.P., Alderman, P.D., Aggarwal, P.K., Anothai, J., Basso, B., Biernath, C., Cammarano, D., Challinor, A.J., De Sanctis, G., Doltra, J., Dumont, B., Fereres, E., Garcia-Vila, M., Gayler, S., Hoogenboom, G., Hunt, L.A., Izaurralde, R.C., Jabloun, M., Jones, C.D., Kersebaum, K.C., Koehler, A.-K., Liu, L., Müller, C., Kumar, S.N., Nendel, C., O'Leary, G., Olesen, J.E., Palosuo, T., Priesack, E., Rezaei, E.E., Ripoche, D., Ruane, A.C., Semenov, M.A., Shcherbak, I., Stöckle, C., Stratonovitch, P., Streck, T., Supit, I., Tao, F., Thorburn, P., Waha, K., Wallach, D., Wang, Z., Wolf, J., Zhu, Y., Asseng, S., 2017. Author Correction : The uncertainty of crop yield projections is reduced by improved temperature response functions. *Nat. Plants* 3, 833–833. <https://doi.org/10.1038/s41477-017-0032-6>

Quatrième partie

Information supplémentaire

Information supplémentaire du chapitre 3

The following supplementary material is available at JXB online (Leveau et al., 2021 ¹)

TABLE S1 – *List of the 60 studied accessions. (Leveau et al., 2021)*

TABLE S2 – *Alleles of major genes for photoperiod response, cold requirement, and reduced height and predicted phenotypes for the 60 studied accessions. (Leveau et al., 2021)*

TABLE S3 – *Results of the analysis of variance for each of the 17 traits measured in this study. (Leveau et al., 2021)*

1. Leveau, S., Parent, B., Zaka, S., Martre, P., 2021. Differential sensitivity to temperature and evaporative demand in wheat relatives. *Journal of Experimental Botany*. <https://doi.org/10.1093/jxb/erab431>

TABLE S4 – Intra-specific and inter-specific standard deviation for the 17 traits measured in this study. (Leveau et al., 2021)

TABLE S5 – Comparison of the three non-linear models fitted to the leaf elongate rate versus temperature of the leaf growth zone. (Leveau et al., 2021)

TABLE S6 – Estimated cardinal temperature of leaf elongate rate response to temperature for each of the 60 studied accessions. (Leveau et al., 2021)

TABLE S7 – Estimated cardinal temperature of leaf elongate rate response to temperature for each of the 12 groups of wheat related species.(Leveau et al., 2021)

TABLE S8 – Matrix of Spearman’s correlation coefficients between the 17 traits measured in this study. (Leveau et al., 2021)

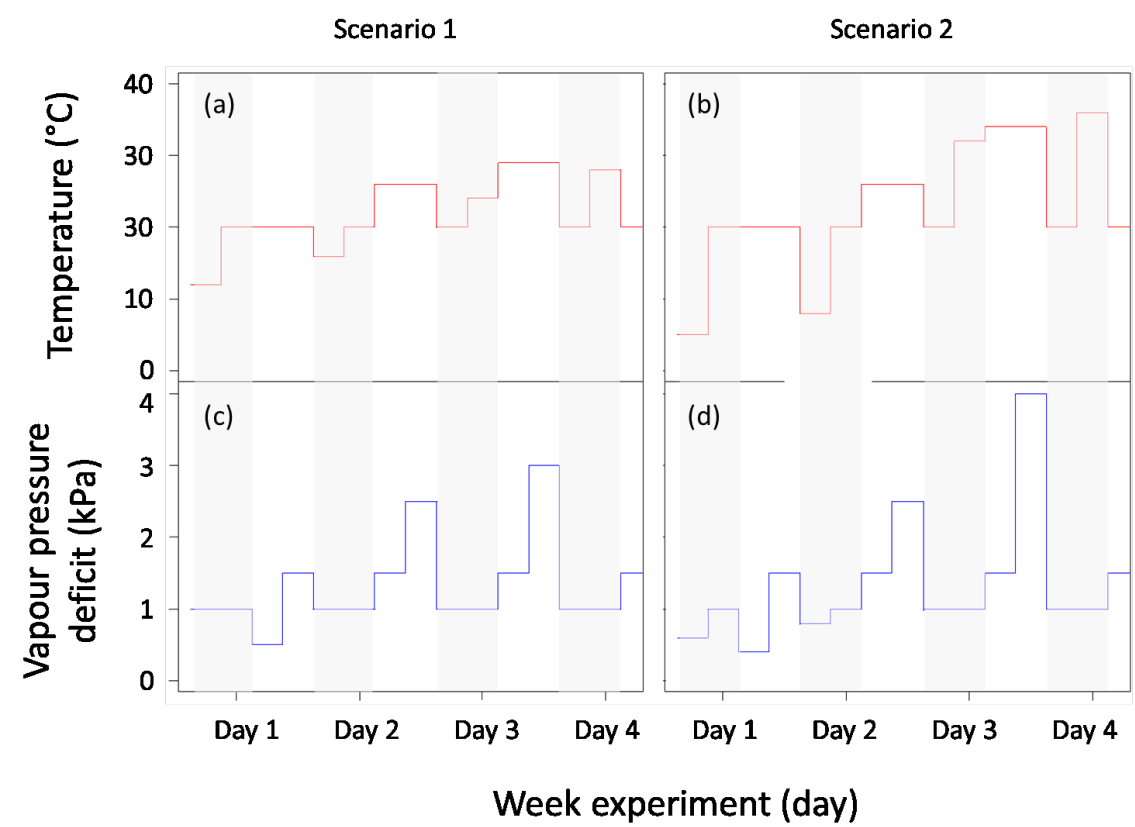


FIGURE S1 – Scenarios of temperature and vapour pressure deficit. Air temperature (a and b) and vapour pressure deficit (VPD, c and d) in scenario 1 (a and c) and scenario 2 (b and d). Vertical grey bandings represent the dark periods.

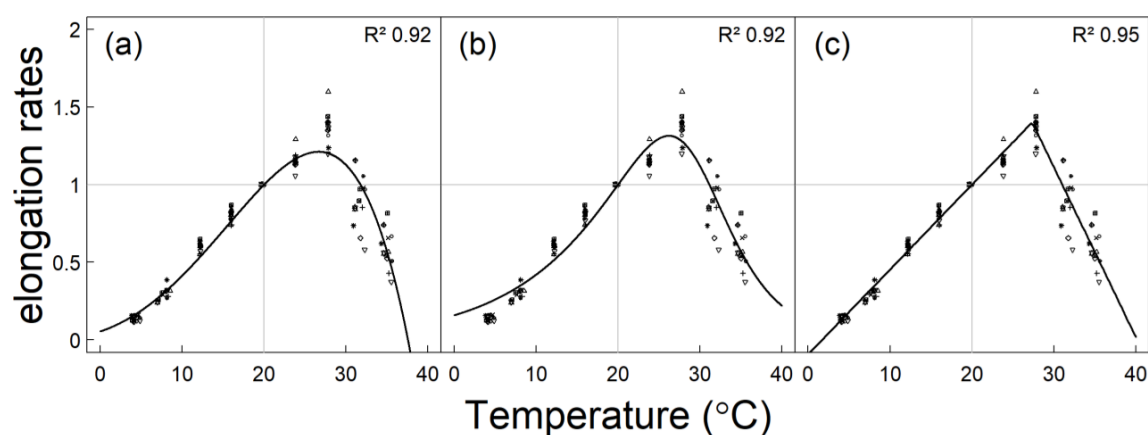


FIGURE S2 – Normalized leaf elongation rate versus temperature of the leaf growth zone fitted with three different non-linear models. Data were fitted with (a) a beta function (Wang and Engel, 1998), (b) an Arrhenius function (Parent and Tardieu, 2012), or (c) a segmented linear function (Eqn 1). LER values measured at the tested temperature were normalized by that at 20°C measured during the same night period in scenario 1 or 2. All models were constrained to unit at the reference temperature. Data are averages of five accessions per species for the 12 groups of wheat related species.

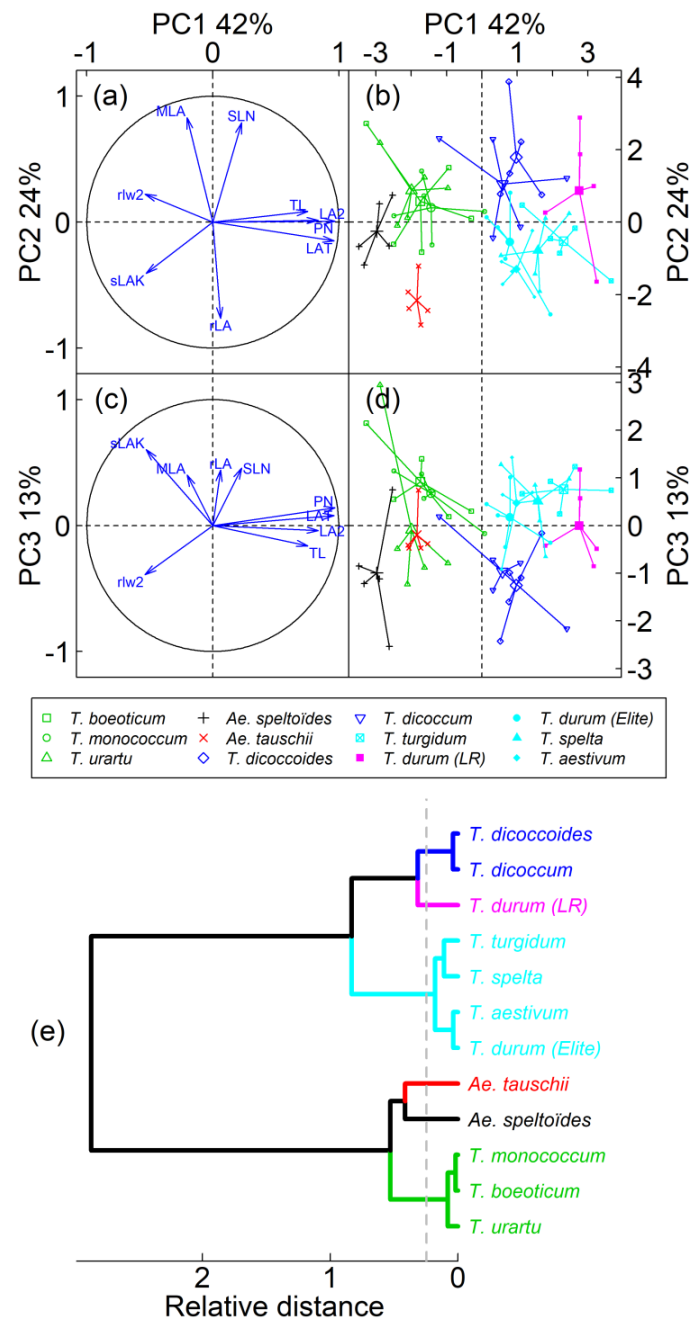


FIGURE S3 – Principal component analysis (PCA) and hierarchical clustering analysis (HCA) for traits related to plant structure for the 12 studied species. (a) and (b) Correlation between plant structure traits on two principal component axes. (c) and (d) Accession values plotted on two principal component axes. Symbol are species and colours correspond to the clusters obtained in (e). Larger symbols are the barycentre of each species. (e) Dendrogram of the HCA performed on the barycentre of each species in the first five axes of the PCA. Dashed vertical line shows the threshold of relative distance used to cluster species.

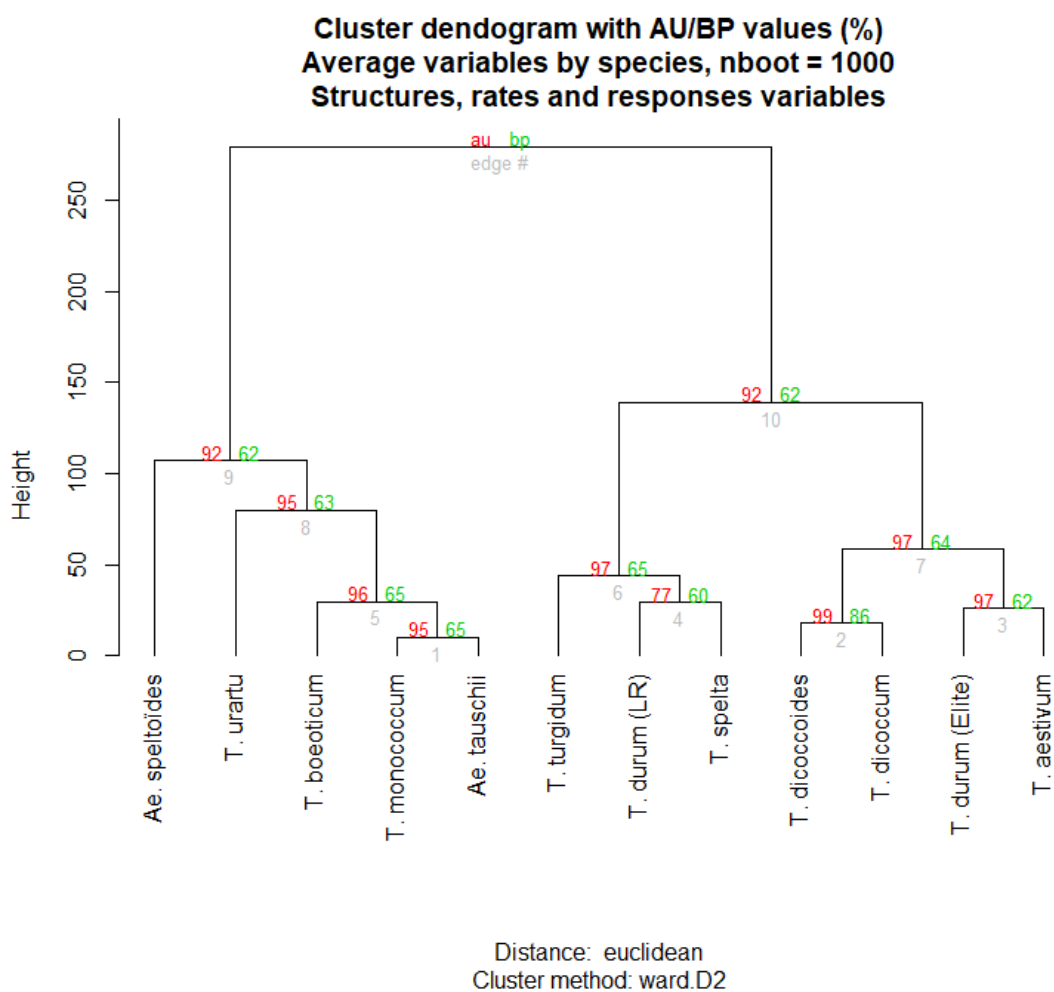


FIGURE S4 – Dendrogram of the hierarchical clustering performed on the average of each species in structure, rate and responses variables. Red values are AU (Approximately Unbiased) P-values (Green values are PB (Bootstrap Probability)). These indicators were calculated with the *pvclust* function from *pv.clust* R package, with 1,000 bootstrap replications

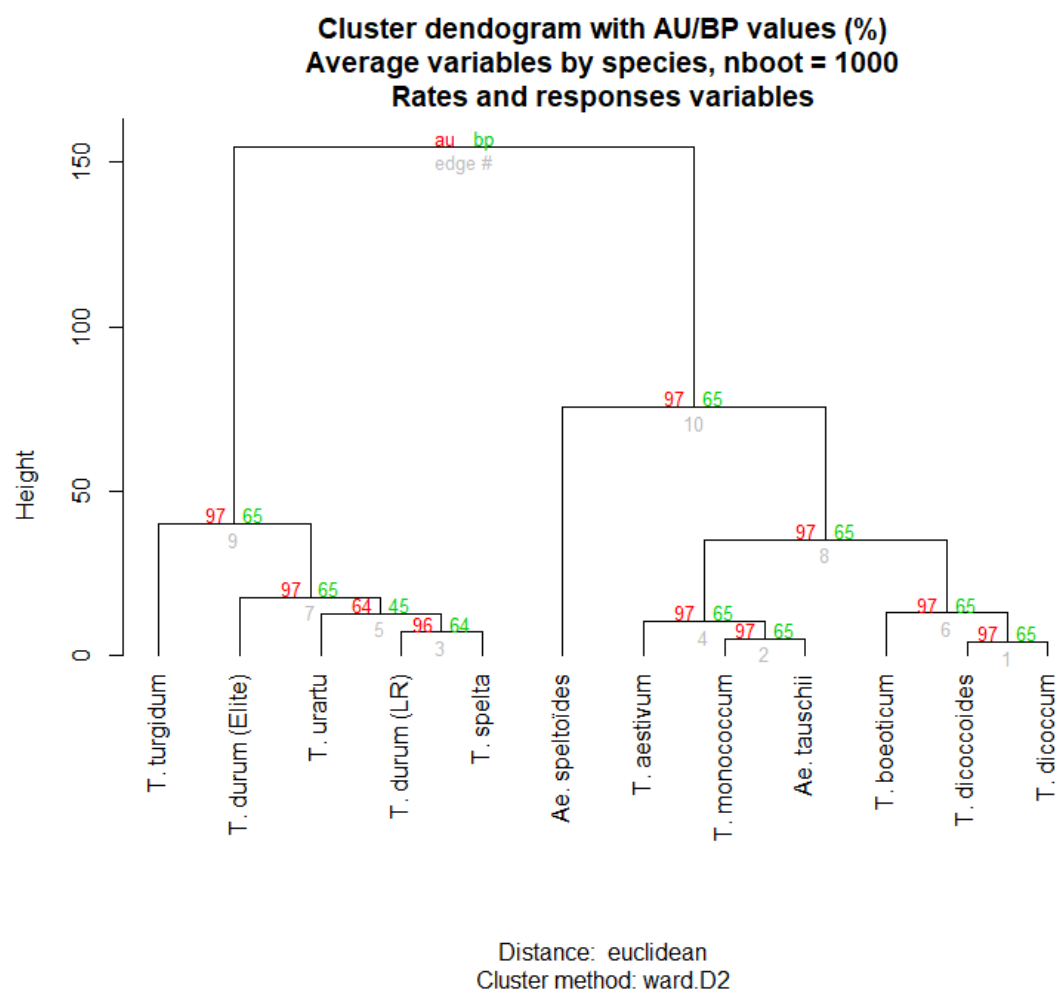


FIGURE S5 – Dendrogram of the Hierarchical clustering performed on the average of each species in rate and responses variables. Red values are AU (Approximately Unbiased) P-values (PB (Bootstrap Probability)). These indicators were calculated with the *pvclust* function from *pvclust* R package, with 1,000 bootstrap replications

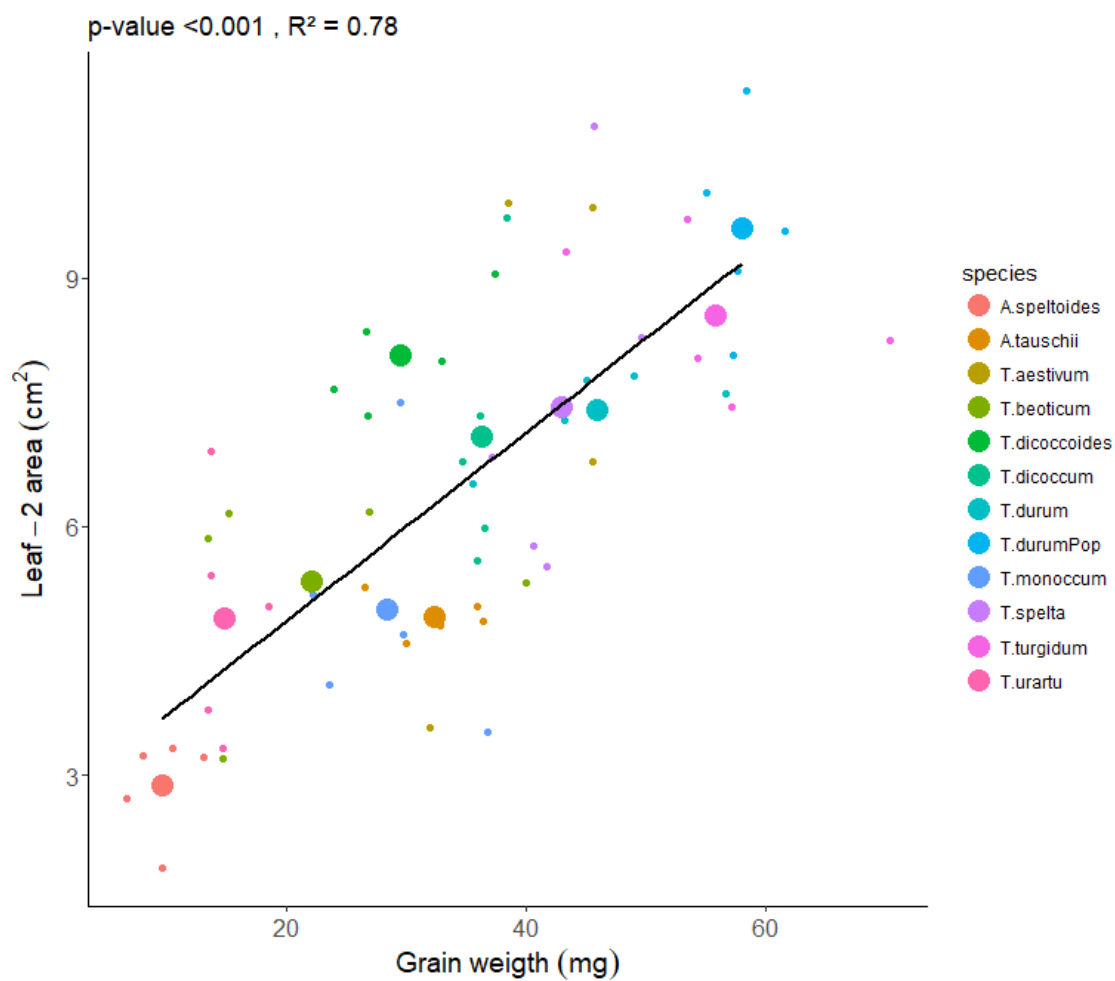


FIGURE S6 – Relationship between area of leaf-2 and individual-grain weight. Small dots : genotype values. Large dots : average of the species. Colors displays the 12 sub-groups analyzed in this study. Black line : linear regression (P -value < 0.001, $R^2 = 0.78$)

Information supplémentaire du chapitre 4

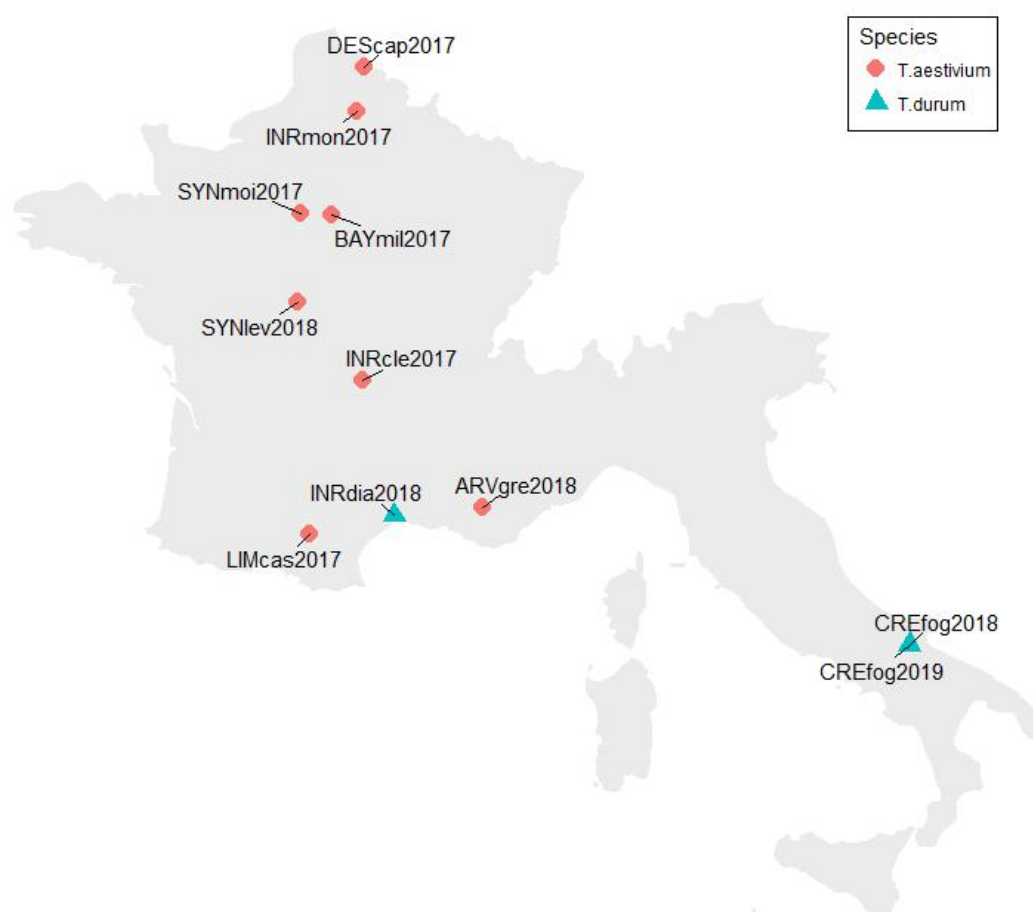


FIGURE S7 – Map of the 10 experimental sites where the field trials analysed in this study were conducted. Red circles, bread wheat trials; blue triangles, durum wheat trials.

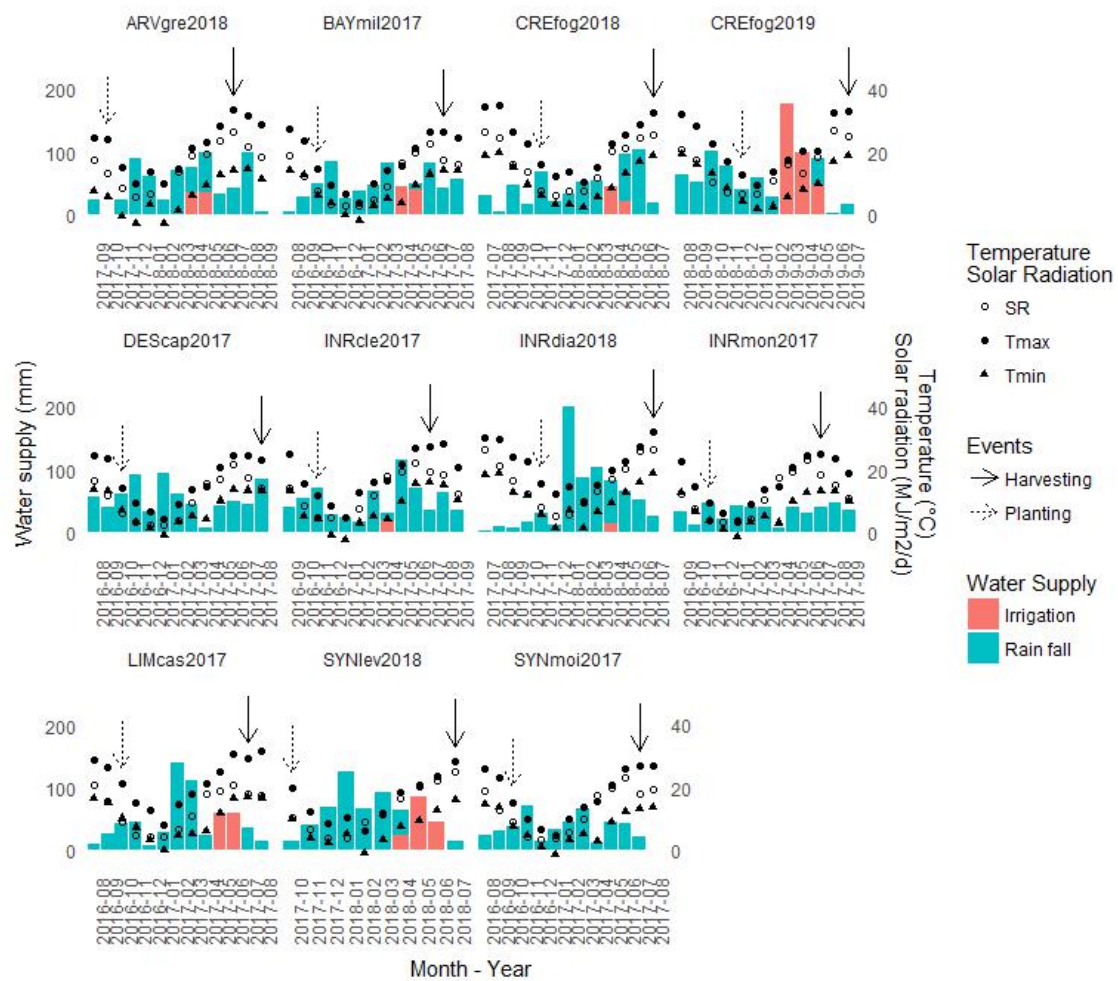


FIGURE S8 – Monthly weather data in the 10 trials analysed in this study. Vertical bars, monthly water-supply (red bars, irrigation; blue bars, precipitations); open circles, monthly average solar radiation (SR); filled circles, monthly average maximum temperature (Tmax); filled triangles, monthly average minimum temperature (Tmin); dashed and solid vertical arrows, sowing and harvest dates, respectively.

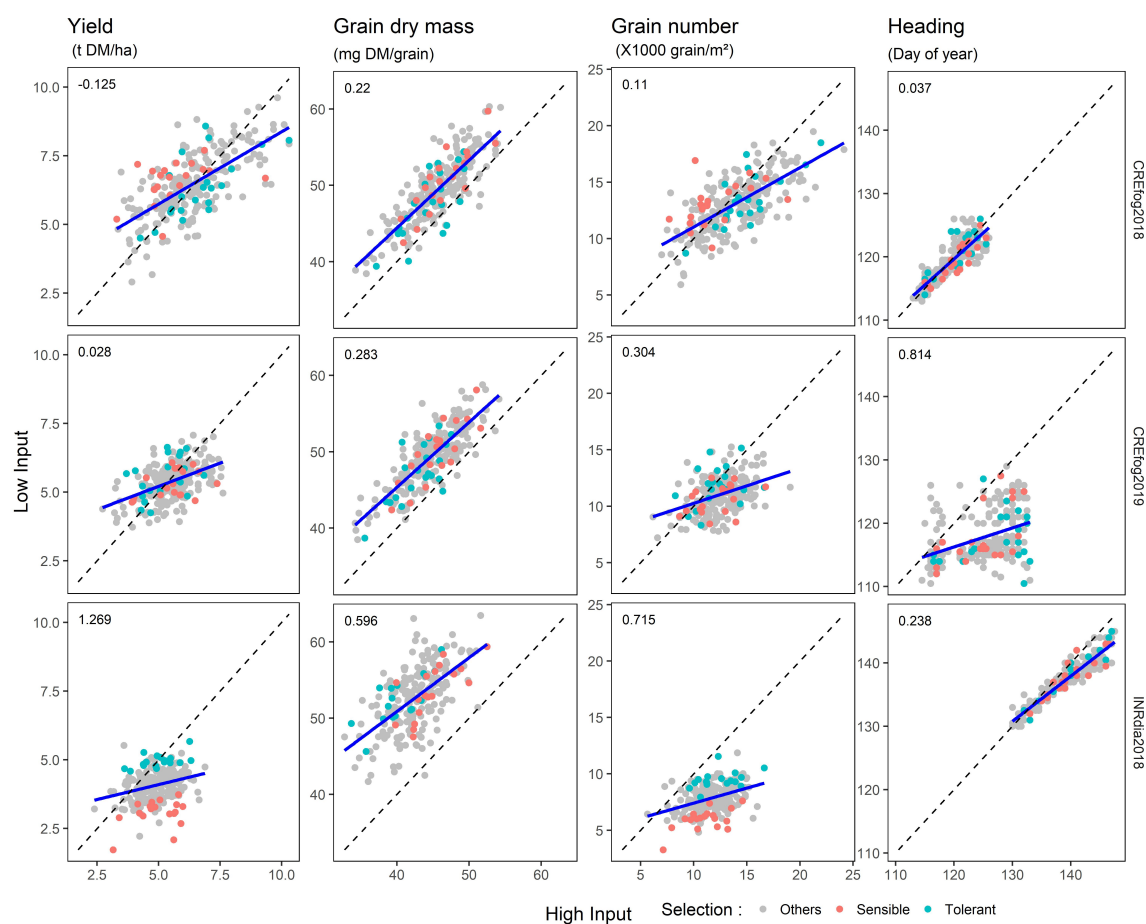


FIGURE S9 – Relationships between grain yield and its components for 232 durum wheat genotypes grown in low-input conditions with their values in high-input conditions. Dashed lines are 1:1 relationships and solid blue lines are linear regressions (Pearson's correlation coefficients and P-values are given in the Supplemnatry Information Table S11). Blue and red circles are the 20 "tolerant" and "sensitive genotypes selected in this study, respectively. In each panel, the value at the top left corner are the stress-index of the trial (see the Material and Method section), calculated for each variable (note that only the values for grain yield were used for genotype selection).

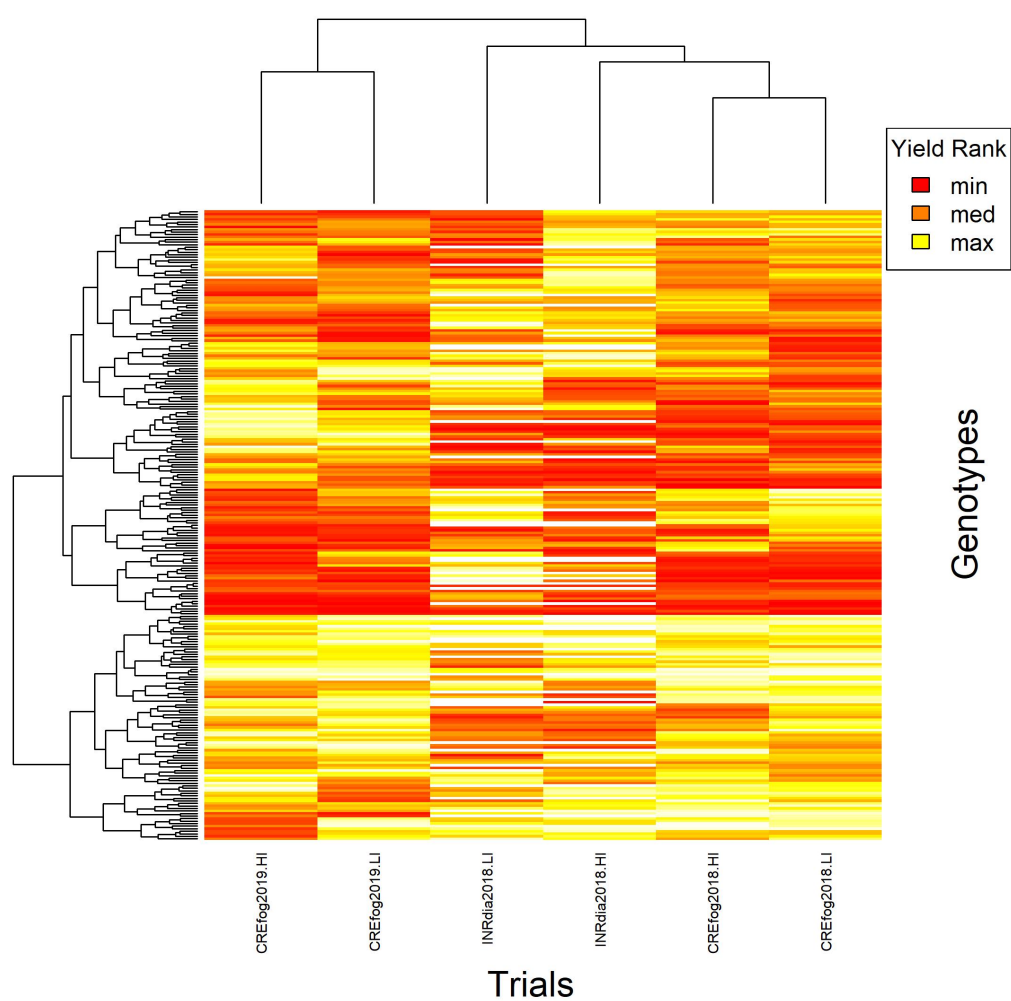


FIGURE S10 – Heat map of grain yield rank for 232 durum wheat genotype grown in the field in six treatment $\times$ trial combinations. The treatments $\times$ trial combinations and genotypes were group using a Hierarchical clustering analysis.

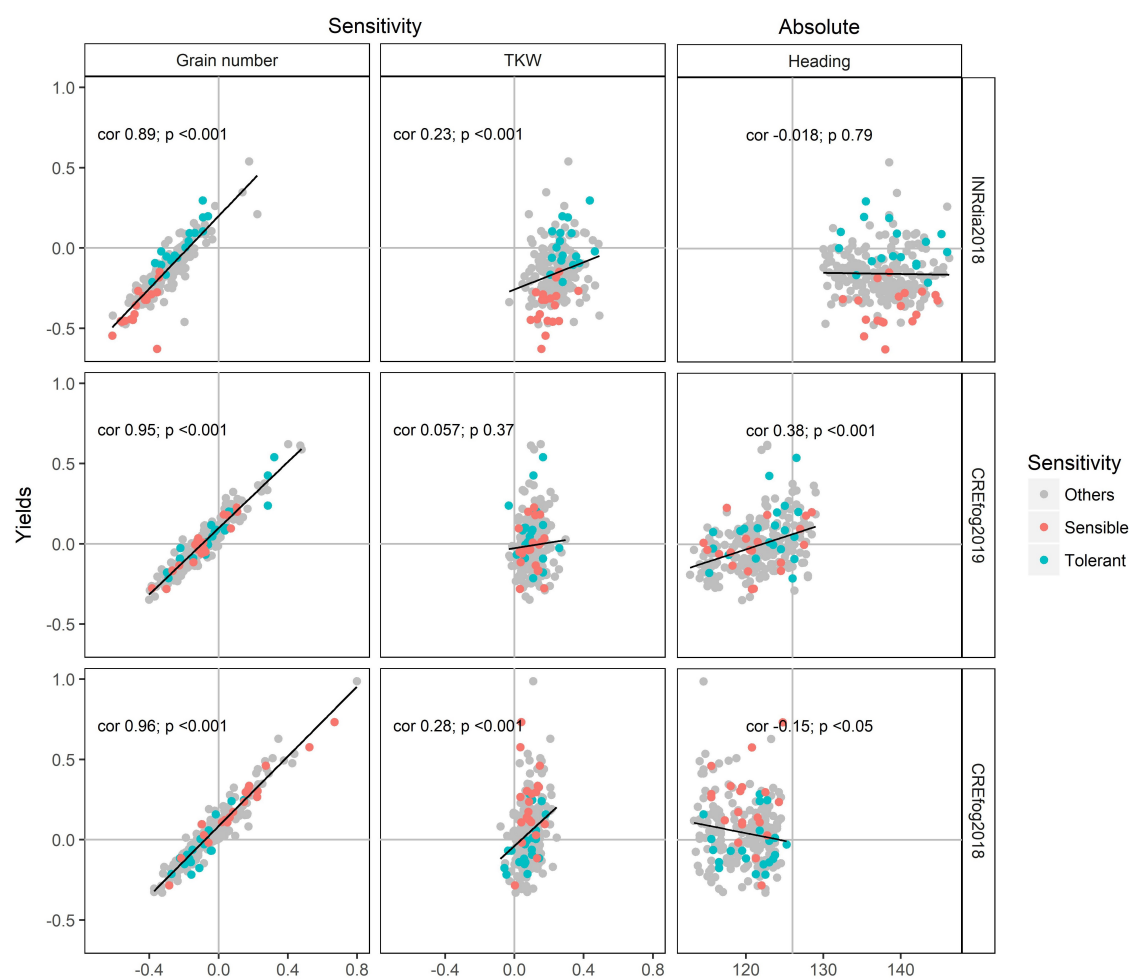


FIGURE S11 – Relationships between the sensitivity of grain yield and the sensitivities of grain number per square meter and individual grain dry mass for 232 durum wheat genotypes grown in field in three trials. Grey dashed line are 1 :1 relationships and solid lines are linear regressions (Pearson correlation coefficients and the associated P-values are shown). Blue and red circles are the 20 “tolerant” and “sensitive” genotypes selected in this study, respectively.

TABLE S9 – List of the 20 “sensitive” and “tolerant” bread-wheat genotypes selected with the method described in thi study and their respective ranks of sensitivities for grain yield and its components.

Genotype	Low-input sensity rank (-)						
	Grain yields	Spike Number per square meter	Grain number per spike	Individual grain dry mass	Grain number per square meter	Heading date	Sensitivity
SANRAFAEL	168.77	118.38	157.5	50.41	180.18	104.78	tolerant
ALBATROS	167.29	51.84	148.74	228.98	157.24	105.23	tolerant
CARSON	161.79	200.77	125.73	239.43	131.08	178.67	tolerant
TCI962395	160.81	124.08	130.01	100.12	149.36	87.77	tolerant
HOKUEI	154.87	227.36	115.01	82.97	166.97	117.69	tolerant
TORRIL	153.68	150.41	131	93.51	158.53	50.11	tolerant
PAGEANT	153.63	66.2	119.12	76.65	140.66	113.94	tolerant
MIRONOVSKAIA 30	148.52	149.39	119.46	195.41	140.12	69.02	tolerant
BEHERT	148.14	63.14	152.14	111.33	152.09	90.13	tolerant
MUSTANG	145.97	114.32	106.02	225.28	121.49	149.01	tolerant
BILINMIYEN96.40	142.35	88.11	124.22	46.23	127.32	111.21	tolerant
MONTJOIE	141.24	116.38	104.72	165.84	159.94	114.83	tolerant
CEREALOR	140.9	122.79	143.01	225.41	124.04	110.47	tolerant
ORMIL	139.33	129.37	138.99	38.09	145.29	127.29	tolerant
EDWIN	139.05	-22.63	136.32	28.58	160.77	84.39	tolerant
ULM	138.64	100.59	106.38	59.99	117.16	113	tolerant
CHARMANY	137.91	77.93	122.22	147.8	135.07	161.72	tolerant
BRUTA	137.78	34.22	93.52	185.24	119.91	104.41	tolerant
2838-39	137.43	74.65	105.63	66.6	122.61	166.85	tolerant
HERKULES	136.53	105.9	131.5	242.74	135.89	175.91	tolerant
MAGISTRAL	136.38	94.29	114.78	6.91	121.38	121.02	tolerant
RIDA	67.44	98.86	84.53	123.32	106.94	139.77	sensible
ARTHUR	66.27	186.41	54.89	190.4	72.04	59.44	sensible
COLORBEN-4	65.18	31.8	41.75	121.46	65.9	108.42	sensible
PAT	64.9	62.06	75.2	8.63	88.07	118.01	sensible
MIRONIVSKA 68	64.43	-8.73	83.96	130.82	61.67	158.97	sensible
DI7202-103	64.07	118.51	34.24	105.32	40.71	62.96	sensible
BRANKA	59.15	9.65	113.6	18.12	104.45	151.86	sensible
308.02.2/WEAVER//F362K2.121	58.97	95.31	85.76	144.15	49.36	103.48	sensible
MCGUIRE	58.17	126.65	44.81	123.24	77.08	55.39	sensible
DACIA	56.41	97.24	47.08	273	43.46	64.83	sensible
SANSA	54.67	108.52	95.76	14.3	51.1	146.86	sensible
KOTARE	54.06	124.89	44.94	68.75	75.62	137.85	sensible
COUGAR	48.52	158.91	46.8	212.46	62.01	118.76	sensible
SALMONE	48.35	43.85	44.88	197.16	48.1	146.82	sensible
ALBIDUM	47.43	101.79	111.84	-2.09	85.49	107.06	sensible
P29	44.91	55.64	133.36	76.65	78.84	96.91	sensible
ZITNICA	34.29	19.35	65.84	132.06	42	49.58	sensible
LUTESCENS EG16/82	25.7	104.18	57.31	123.6	42.95	141.96	sensible
DRUZHBA	25.34	100.05	49.25	-14.14	28.5	99.66	sensible
CH73641	22.91	46.91	74.09	-9.66	40	119.75	sensible

TABLE S10 – List of the 20 “sensitive” and “tolerant” durum wheat genotypes selected with the method described in this study and their respective ranks of sensitivities for grain yield and its components.

Genotype	Low-input sensity rank (-)		Individual grain dry mass	Heading date	sensitivity
	Grain yield	Grain number per square meter			
ACADUR	296.03	226.58	37.41	113.01	tolerant
NORMANNO	282.09	192.62	78.32	36.87	tolerant
ASTERIX	274.82	193.25	167.38	130.90	tolerant
EL4X ₁ 30	274.59	159.75	150.68	205.34	tolerant
CHAM-1 _D P136	270.96	165.51	78.06	232.43	tolerant
RAMIREZ	268.87	182.53	194.02	89.79	tolerant
EL4X ₁ 88	266.52	163.49	196.02	13.45	tolerant
CUSPIDE	265.48	214.55	67.45	104.50	tolerant
SYCISCO	265.27	202.56	75.72	183.83	tolerant
EL4X ₃ 16	263.71	213.58	106.20	50.64	tolerant
ARCOBALENO _D P080	262.20	164.95	212.17	127.72	tolerant
NOBILIS	260.56	182.20	146.20	43.73	tolerant
EL4X ₄ 64	258.34	181.93	124.93	172.61	tolerant
JOYAU	258.11	174.01	148.82	145.66	tolerant
AMMAR-1-DP122	255.92	175.84	51.71	71.33	tolerant
AVENTUR	250.62	185.59	117.08	47.69	tolerant
EL4X ₃ 5	250.19	139.42	170.24	181.73	tolerant
ARNACORIS	249.75	133.01	242.72	112.35	tolerant
EL4X ₄ 12	249.25	109.37	84.30	83.89	tolerant
LGBORIS	246.87	171.93	107.02	153.22	tolerant
SELCUKLU-97	246.66	216.10	226.40	190.57	tolerant
ARDENTE	-16.39	46.82	166.10	105.23	sensible
ACHILLE	-16.44	64.79	49.72	106.71	sensible
GALLARETA _D P041	-20.49	120.38	200.80	29.54	sensible
ORT-1 _D P170	-20.56	30.17	170.98	91.78	sensible
EL4X ₁ 94	-26.26	50.08	203.69	21.48	sensible
PLINIO _D P107	-28.64	47.27	137.83	121.50	sensible
EL4X ₁ 20	-32.08	55.73	87.92	129.53	sensible
SYBANCO	-32.60	174.36	22.00	188.08	sensible
PLUSSUR	-33.41	60.97	79.37	192.58	sensible
MIKI-1 _D P161	-34.95	64.60	121.06	33.57	sensible
CIMMYT23 BISU ₁ /PATKA _{3D} P011	-35.64	23.66	122.72	148.72	sensible
MONDUR	-36.66	84.01	18.64	91.59	sensible
CRESO _D P089	-40.47	48.47	125.02	230.74	sensible
AZEGHAR-2 _D P128	-41.96	84.83	62.83	144.22	sensible
LAHAN	-43.99	10.32	219.14	29.66	sensible
L2574	-62.99	45.03	46.82	84.53	sensible
FURIO _C AMILLO	-65.54	60.84	119.14	145.63	sensible
GIUSTO	-66.14	105.46	152.92	146.36	sensible
STRONGFIELD	-68.39	79.42	34.86	214.61	sensible
CASANOVA	-68.41	19.69	164.43	210.49	sensible

TABLE S11 – Pearson correlation coefficients (*r*) and associated *P*-values for the relationships between grain yield and its components for bread wheat genotypes grow in seven field trials in the low-input conditions with their values in the high-input conditions.

Trial	Grain yield		Spike number per square meter		Grain number per spike		Individual grain dry mass		Grain number per square meter		Heading date	
	<i>r</i>	<i>P</i> -value	<i>r</i>	<i>P</i> -value	<i>r</i>	<i>P</i> -value	<i>r</i>	<i>P</i> -value	<i>r</i>	<i>P</i> -value	<i>r</i>	<i>P</i> -value
SYNlev2018	0.61	< 0.001	0.35	< 0.001	0.39	< 0.001	0.23	< 0.001	0.53	< 0.001	0.98	< 0.001
INRcle2017	0.38	< 0.001	0.4	< 0.001	0.6	< 0.001	0.79	< 0.001	0.61	< 0.001	0.92	< 0.001
DEScap2017	0.7	< 0.001	0.51	< 0.001	0.5	< 0.001	0.1	0.16	0.56	< 0.001	0.95	< 0.001
ARVgre2018	0.55	< 0.001	0.44	< 0.001	0.66	< 0.001	0.84	< 0.001	0.74	< 0.001	0.92	< 0.001
INRmon2017	0.76	< 0.001	0.78	< 0.001	0.87	< 0.001	0.94	< 0.001	0.87	< 0.001	0.98	< 0.001
BAYmil2017	0.65	< 0.001	-	-	-	-	0.93	< 0.001	0.	< 0.001	0.97	< 0.001
SYNmoi2017	0.69	< 0.001	0.11	< 0.001	0.46	< 0.001	0.8	< 0.001	0.75	< 0.001	0.85	< 0.001

Information supplémentaire du chapitre 5

Supplementary Tables

Table S1. Description of the 47 bread wheat genotypes analyzed in this study.

Table S2. Description of the 37 durum wheat genotypes analyzed in this study.

Table S14. Genotypic means of all variables analyzed for the 47 studied bread wheat genotypes.

Table S15. Genotypic means of all variables analyzed for the 38 studied durum wheat genotypes.

Supplementary Figures

Figure S1. Relationship between the weighted sensitivity rank for grain yield and spike number per square meter for the panel of 210 bread wheat genotypes.

Figure S2. Hourly, air temperature, relative humidity and vapor pressure deficit, and photosynthetically photon flux density during the experiment in the high-throughput plant phenotyping platform PhenoArch.

Figure S3. Spatial design of the experiment in the high-throughput plant phenotyping platform PhenoArch.

Figure S4. Time courses of tiller appearance and sensitivity of tiller appearance rate to soil water potential for the four bread wheat and two durum wheat cultivars of group A under the four watering scenarios.

Figure S5. Time courses of cumulative leaf length per plant and sensitivity of leaf elongate rate to soil water potential for the four bread wheat and two durum wheat cultivars of group A under the four watering scenarios.

Figure S6. Time courses of main-stem leaf number and sensitivity of leaf appearance rate to soil water potential for the four bread wheat and two durum wheat cultivars of group A under the four watering scenarios.

Figure S7. Principal component analysis on all variables analyzed in this study for the 38 durum wheat and 47 bread genotypes studied.

Figure S8. Heatmap of the correlation matrix for all variables analyzed in this study calculated for the 38 durum wheat and 47 bread wheat genotypes studied.

Figure S9. Relative sensitivities of shoot expansion rate, main-stem leaf appearance rate, main-stem leaf elongation rate, and tiller appearance rate to soil water potential for the 47 bread wheat and the 38 durum wheat genotypes studied.

Figure S9. Genetic variability in the sensitivity to soil water potential of shoot expansion rate and its three underlying components.

Figure S10. Time courses of cumulative internode length per plant and sensitivity of internode elongation rate to soil water potential for the four bread wheat and two durum wheat genotypes of group A under the four watering scenarios.

TABLE S12 – Genotypic means of all variables analyzed for the 47 studied bread wheat genotypes. S_{Amax} , maximum shoot area; S_{ERmax} , maximum shoot expansion rate; $SSER$, sensitivity of shoot expansion rate to water deficit; FLN , final main-stem leaf number; LAR_{mean} , mean main-stem leaf appearance rate; $SLAR$, sensitivity of main-stem leaf appearance rate to water deficit; TN_{max} , maximum tiller number per plant; TiR_{max} , maximum rate of tiller appearance; $SLER$, sensitivity of tiller appearance rate to water deficit; CLL_{max} , maximum cumulative leaf length per plant; LER_{max} , maximum main-stem leaf elongation rate; S_{LER} , sensitivity of main-stem leaf elongation rate to water deficit; and $LA2$, surface area of main-stem leaf 2; EEA , $XXXX \cdot S_{Amax}$, S_{ERmax} , LN_{max} , LAR_{mean} , TN_{max} , TiR_{max} , CLL_{max} , LER_{max} , $LA2$ and EEA data for the well-water scenario. Data are means for $n = 3$ to 6 replicates.

Accession name	S_{Amax} (cm ² plant ⁻¹)	S_{ERmax} (cm ² °Cd ⁻¹)	$SSER$ (MPa ⁻¹)	FLN (leaf mainstem ⁻¹)	LAR_{mean} (leaf °Cd ⁻¹)	$SLAR$ (Mpa ⁻¹)	TN_{max} (tiller plant ⁻¹)	TiR_{max} (tiller plant ⁻¹ °Cd ⁻¹)	$STiR$ (Mpa ⁻¹)	CLL_{max} (cm plant ⁻¹)	LER_{max} (cm °Cd ⁻¹)	$SLER$ (Mpa ⁻¹)	$LA2$ (cm ² leaf ⁻¹)	EEA (heading date)
AKAMAR	2260	4.76	1.26	8.67	0.011	0.259	27.59	0.082	1.373	202.2	0.386	0.142	5.18	796
ALBATROS	3573	10.39	0.97	10	0.011	0.014	42.07	0.159	0.856	245.1	0.447	0.239	3.18	868
ALBIDUM	4324	10.14	0.7	10.55	0.012	-0.075	62.91	0.208	0.966	299.3	0.477	0.301	3.39	1008
BILINIYEN96.40	2587	7.14	0.97	9.67	0.011	-0.165	32.39	0.107	0.708	230.9	0.393	0.136	2.66	846
BRANKA	3089	7.91	0.91	9.67	0.012	-0.012	34.05	0.107	0.51	228.1	0.44	0.331	3.12	781
BRUTA	3292	8.77	1.01	9.57	0.01	-0.333	32.99	0.111	0.905	231	0.418	-0.013	1.47	823
CEREALOR	2837	7.53	0.89	9	0.01	-0.389	28.84	0.091	0.855	208.7	0.412	-0.153	3.03	814
CH73641	3746	8.49	0.84	9.57	0.009	-0.219	31.47	0.098	0.43	215.2	0.404	0.017	3.98	1089
CHARMANY	4707	10.29	0.83	9.77	0.009	-0.187	35.83	0.094	0.182	252.6	0.41	0.161	2.56	1090
COUGAR	2815	7.15	0.91	9	0.012	0.193	30.24	0.103	0.859	195.9	0.432	0.161	1.42	751
DACIA	3709	9.77	1.06	10	0.012	-0.082	27.95	0.107	1.027	257.6	0.481	0.323	2.75	764
DHARWAR DRY	3410	7.59	0.98	8.33	0.01	0.207	17.72	0.059	0.898	184	0.384	-0.188	3.5	710
DRUZHBA	3851	8.02	1	10.23	0.01	-0.286	29.25	0.09	0.756	245.7	0.405	0.089	1.66	1060
DRYSDALE	1966	5.08	1.12	7.67	0.013	0.18	23.44	0.074	1.45	144.7	0.43	0.42	2.39	630
EDWIN	3982	9.99	0.75	10.03	0.01	-0.206	50.77	0.176	0.696	247.8	0.498	0.078	2.47	1070
HOKUEI	3977	10.18	0.88	9.87	0.01	-0.153	42.01	0.141	0.437	217.4	0.399	-0.008	2.24	894
HTW17000044	2568	7	1.04	8	0.011	-0.192	28.92	0.09	1.063	177.6	0.396	0.037	3.82	681
HYB4200387	2307	7.26	1.07	9	0.011	-0.063	31.92	0.1	0.907	197.5	0.41	0.137	3.16	705
KOTARE	2830	8.15	0.83	8.67	0.009	-0.38	30.53	0.087	0.334	177.9	0.386	0.183	3.41	826
LUTESCENS EG16/82	2648	7.85	1.03	8.33	0.01	-0.218	27.02	0.087	1.134	175.3	0.426	-0.028	2.63	728
MCCUIRE	3261	8.69	0.94	8.67	0.011	-0.034	36.28	0.101	0.892	180.1	0.388	0.138	2.36	729
MIRONOVSKA 68	3862	8.43	0.72	10.33	0.01	0.141	31.93	0.099	0.462	278.2	0.491	0.238	4.08	1288
MIRONOVSKAIA 30	2669	9.42	1.06	8.67	0.01	-0.079	25.91	0.081	0.732	212.1	0.435	0.309	2.47	751
MONTJOIE	3198	8.99	1.03	9	0.012	-0.208	27.93	0.082	0.992	209.5	0.516	0.263	2.11	708
MUSTANG	3932	9.69	0.9	9.27	0.009	-0.134	39.42	0.122	1.051	167.9	0.33	-0.401	1.6	928
MV309.9	3883	5.79	0.67	9	0.011	-0.301	32.11	0.099	0.875	200.5	0.429	0.054	2.28	736
OREGRAIN	2726	8.57	0.96	9.33	0.011	-0.169	35.19	0.119	0.843	202.7	0.409	0.101	2.66	738
ORMIL	3094	8.46	0.91	10.13	0.01	-0.356	30.32	0.097	0.554	217.1	0.418	0.133	2.64	834
P29	2516	7.73	0.81	9.33	0.011	-0.337	37.06	0.122	0.646	229.4	0.395	0.038	2.06	878
PAGEANT	3814	9.73	0.81	9.33	0.011	-0.401	38.57	0.126	0.659	184.2	0.414	0.12	1.33	808
PAT	2677	6.78	1.31	9	0.011	0.245	37.62	0.114	1.238	206.6	0.373	0.186	4.37	723
SUBLIM	2713	6.73	0.94	8.3	0.011	-0.103	22.67	0.068	1.481	179	0.419	0.021	2.76	640
SUNTOP	2994	7.02	0.88	7.67	0.011	-0.318	23.33	0.074	1.076	148	0.379	-0.531	3.09	628
SUNTOP1	2641	6.42	0.85	8	0.012	0.042	15.87	0.045	1.411	167.1	0.453	0.185	1.9	650
SUNTOP73	2675	6.35	1.11	7.67	0.011	0.177	19.92	0.065	1.494	176.9	0.44	0.285	3.93	639
SUNTOP9	2317	6.33	1.24	7.67	0.013	0.481	24.32	0.074	1.502	148.8	0.42	0.349	2.51	612
SUNTOP90	2992	7.57	1.08	8.67	0.011	0.218	29.96	0.09	0.91	208.4	0.398	0.164	3.74	729
SY117079	3279	8.29	0.82	8.33	0.01	0.011	32.15	0.119	0.926	174.4	0.424	0.035	1.92	781
TC1960735	3361	8.96	1.02	8.33	0.009	-0.314	26.14	0.077	0.505	155.3	0.347	0.178	2.74	744
TC1962395	3279	8.29	0.82	8.33	0.01	0.011	32.15	0.119	0.926	174.4	0.424	0.035	1.92	781
TORRIL	2958	7.81	0.97	9.33	0.011	-0.247	40.11	0.133	0.735	171.9	0.361	0.165	2.4	800
ULM	4027	9.06	0.81	9.33	0.009	-0.16	43.71	0.147	1.082	187.6	0.378	-0.096	2.4	1012
308.02.2/WEAVER//F362K2.121	2799	8.28	1.19	9	0.012	-0.158	40.8	0.138	1.188	173.8	0.442	0.102	1.49	715
ZITNICA	2839	9.68	0.98	8.5	0.011	-0.038	37.33	0.119	1.052	174.6	0.434	0.255	2.07	769

TABLE S13 – Genotypic means of all variables analyzed for the 38 studied durum wheat genotypes. *S*Amax, maximum shoot area; *S*ERmax, maximum shoot expansion rate; *S*SEr, sensitivity of shoot expansion rate to water deficit; *FLN*, final main-stem leaf number; *L*ARmean, mean main-stem leaf appearance rate; *S*LAR, sensitivity of main-stem leaf appearance rate to water deficit; *T*Nmax, maximum tiller number per plant; *T*iRmax, maximum rate of tiller appearance; *S*TiR, sensitivity of tiller appearance rate to water deficit; *C*LLmax, maximum cumulative leaf length per plant; *L*ERmax, maximum main-stem leaf elongation rate; *S*LER, sensitivity of main-stem leaf elongation rate to water deficit; and *L*A2, surface area of main-stem leaf 2; *E*EA, XXXX. *S*Amax, *S*ERmax, *L*ARman, *T*Nmax, *T*iRmax, *C*LLmax, *L*ERmax, *L*A2 and *E*EA data for the well-water scenario. Data are means for $n = 3$ to 6 replicates.

Accession name	<i>S</i> Amax (cm ² plant ⁻¹)	<i>S</i> ERmax (cm ² °Cd ⁻¹)	<i>S</i> SEr (Mpa ⁻¹)	<i>L</i> Nmax (leaf mainstem ⁻¹)	<i>L</i> ARmean (leaf °Cd ⁻¹)	<i>S</i> LAR (Mpa ⁻¹)	<i>T</i> Nmax (tiller plant ⁻¹)	<i>T</i> iRmax (tiller plant ⁻¹)	<i>T</i> iRmax (tiller plant ⁻¹ °Cd ⁻¹)	<i>S</i> TiR (Mpa ⁻¹)	<i>C</i> LLmax (cm plant ⁻¹)	<i>L</i> ERmax (cm °Cd ⁻¹)	<i>S</i> LER (Mpa ⁻¹)	<i>L</i> A2 (cm ² leaf ⁻¹)	<i>E</i> EA (Heading date)
ACADUR	3426	7.97	1.06	8.23	0.01	-0.147	26.01	0.078	1.21	1.21	161.7	0.363	0.012	2.77	792
ANTALIS	2898	7.3	1.16	8.33	0.01	0.107	23.58	0.068	1.16	1.16	166	0.376	0.015	3.72	772
ANVERGUR	3499	9.73	1.26	9.67	0.012	-0.164	35.42	0.107	1.35	1.35	225.1	0.456	0.345	3.34	733
ARDEnte	2964	6.8	1.01	8.67	0.01	-0.549	19.54	0.052	1.38	1.38	177.8	0.374	-0.094	2.8	822
ASTERIX	2893	7.24	1.17	8	0.011	-0.063	26.42	0.078	1.32	1.32	165.4	0.388	0.121	3.41	665
AVENTUR	3923	10.3	1	9.67	0.01	-0.4	33.07	0.095	1.23	1.23	210.4	0.404	0.148	2.87	831
AZEGHAR-2 _D P128	2928	8.18	0.94	8	0.011	-0.299	23.11	0.078	1.5	1.5	160.4	0.387	0.05	2.82	754
BALSAMO	3446	10.73	1.14	9	0.011	-0.28	31.56	0.098	1.38	1.38	203.9	0.409	-0.012	3.17	768
BOLIDO _D P034	3412	8.58	1.07	7.67	0.01	-0.299	26.31	0.067	1.22	1.22	148.7	0.36	-0.047	3.42	721
CASANOVA	2546	6.97	1.17	8	0.01	-0.221	16.69	0.046	1.82	1.82	185	0.418	0.21	6.06	633
CHAM _{1D} P136	3115	7.56	0.98	8	0.012	0.081	24.38	0.071	1.08	1.08	162.8	0.385	0.097	3.48	646
COLOSSEO _D P087	3174	8.66	0.91	8.33	0.011	-0.846	28.3	0.082	1.37	1.37	189.7	0.48	0.005	4.81	719
EL4X ₁ 20	3210	6.46	0.95	8	0.011	-0.209	21.41	0.055	1.31	1.31	191.6	0.472	0.085	4.12	638
EL4X ₁ 94	4035	9.86	0.91	9.33	0.01	-0.098	18.02	0.053	1.37	1.37	248.9	0.435	-0.051	3.97	775
EL4X ₃ 16	3580	10.06	1.01	7.67	0.01	-0.749	18.99	0.06	1.31	1.31	181.7	0.419	0.158	3.86	722
EL4X ₃ 5	2508	6.79	1.07	8	0.012	0.119	21.05	0.06	1.28	1.28	183.2	0.467	0.209	3.72	691
EL4X ₄ 28	3402	8.97	1.07	9	0.011	-0.431	31.38	0.101	1.39	1.39	235.3	0.458	0.173	3.86	760
EL4X ₄ 64	3165	8.75	1.06	9.33	0.01	-0.034	36.51	0.111	0.5	0.5	206.9	0.394	0.182	3.41	813
FURIO _C AMILLO	2791	6.24	1.09	8.67	0.011	-0.009	23.36	0.066	0.98	0.98	181.8	0.402	0.114	3.42	705
GIUSTO	3110	8.27	0.91	9	0.011	-0.232	29.28	0.084	1.13	1.13	225.8	0.461	0.113	4.63	732
KOFA	2855	6.83	1	8	0.012	-0.238	22.26	0.066	1.53	1.53	174.9	0.436	0.043	4.46	633
L2574	2885	9.8	1.15	9	0.013	0.124	24.9	0.073	1.46	1.46	190	0.466	0.184	3.09	675
LAHAN	3039	7.91	0.87	8.33	0.01	0.039	23.92	0.069	1.12	1.12	172.4	0.358	0.044	3.07	726
LG BORIS	3540	10.78	0.87	10.33	0.011	-0.352	33.5	0.131	0.98	0.98	233.5	0.496	0.286	3.02	819
LLOYD	4120	9.65	0.95	9.9	0.011	0.124	27.76	0.067	0.89	0.89	258.5	0.505	-0.046	2.09	897
MIK _{1D} P161	2878	6.95	1.06	8.67	0.014	0.28	25.98	0.08	1.37	1.37	171.1	0.456	0.459	3.36	611
MURANO	3677	8.52	0.86	9.33	0.01	-0.43	23.8	0.061	1.04	1.04	217.2	0.412	0.099	3.35	818
NEMESIS	4010	9.23	1.06	8.93	0.01	-0.105	23.15	0.065	1.23	1.23	186.1	0.424	0.091	2.5	817
ORJAUNE	3247	7.62	1.04	9.33	0.01	-0.76	26.13	0.084	1.43	1.43	208.6	0.35	-0.042	4.73	839
PLUSSUR	3520	9.37	1.03	9.33	0.01	-0.612	31.12	0.087	1.01	1.01	243.4	0.434	0.147	3.64	813
QUALIDOU	3351	8.32	1.01	8.67	0.01	-0.012	22.6	0.061	1.18	1.18	183.3	0.38	-0.004	3.09	773
Selcuklu ₉ 7	3934	10.36	0.96	9.47	0.01	-0.299	26.49	0.076	1.11	1.11	243	0.406	0.217	3.76	857
TIZIANA	4736	8.46	1.14	8.33	0.011	-0.051	24.86	0.079	1.47	1.47	197.4	0.43	0.081	4	709

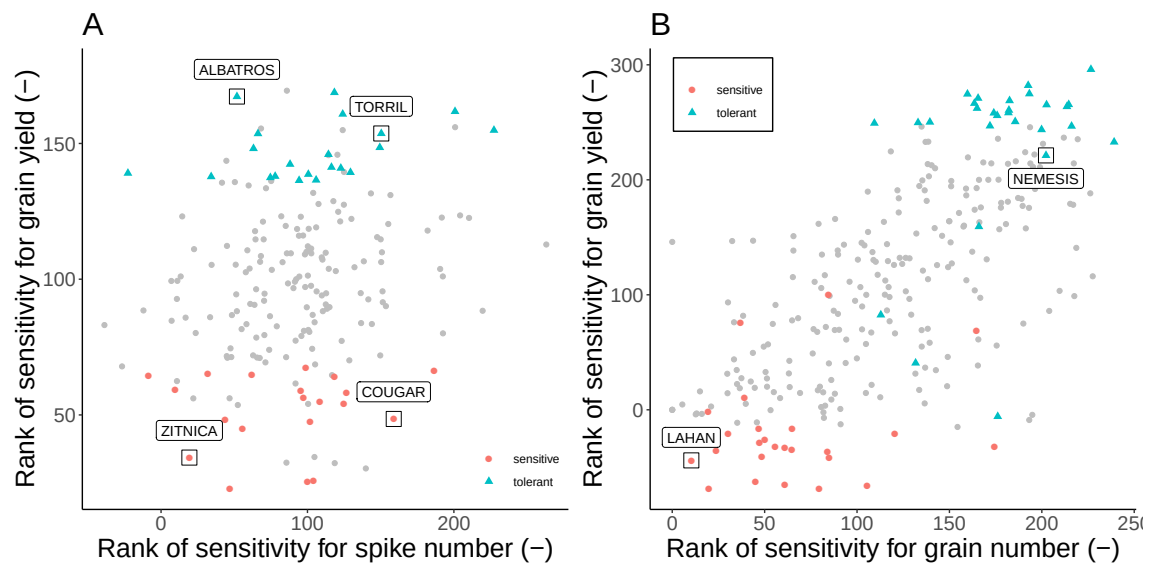


FIGURE S12 – Relationship between the weighted sensitivity rank for grain yield and spike number per square meter for the panel of 273 bread wheat genotypes. Solid line is linear regressions (Pearson's correlation coefficients and the associated P-values are shown). The panel of 273 bread wheat genotypes was grown in the field in seven trials at seven sites covering all wheat grown areas in France. In each trial, genotypes were ranked according to the residuals of the relationship between grain yield or spike number per square meter under low input versus high input. The weighted genotype rank was then calculated as the sum of the ranks in the seven trials weighted by the severity of the water and/or nitrogen deficit in each trial, calculated as the ratio of average grain yield (across the genotypes) or average spike number per square meter under low input to high input. For this study, XX "tolerant" (blue circles) and XX "sensitive" (red circles) genotypes with high and low weighted rank for grain yield, respectively, and with a heading date within XX days of the average of the whole panel, were selected. Four genotypes with contrasted behavior, for which the number of replicates was six (group B), are identified.

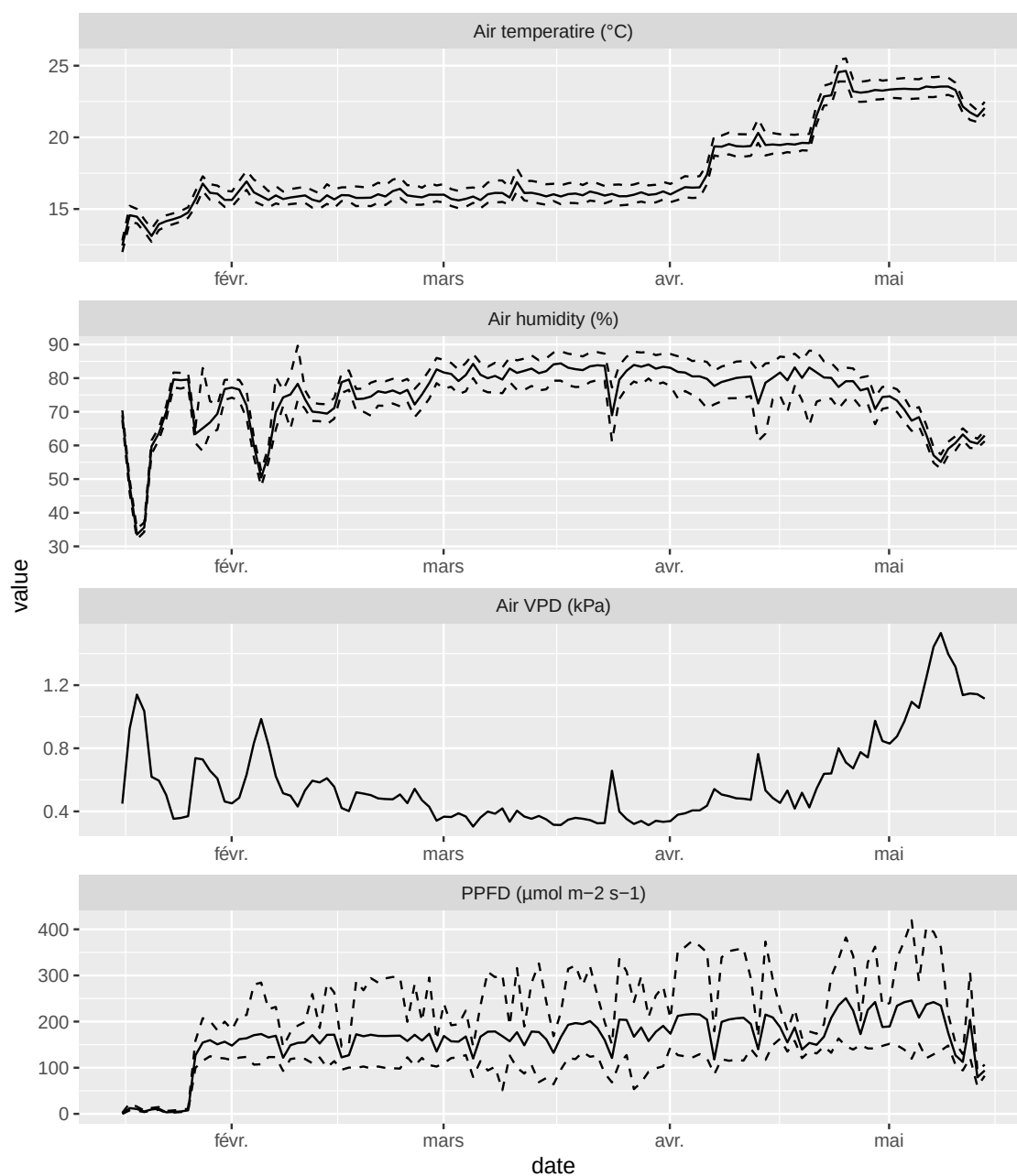


FIGURE S13 – Hourly, air temperature, relative humidity and vapor pressure deficit (VPD), and photosynthetically photon flux density (PPFD) during the experiment in the high-throughput plant phenotyping platform PhenoArch. Air temperature and relative humidity, and PPFD data are the average of 10 sensors distributed throughout the greenhouse. Air VPD was calculated from air temperature and relative humidity data.

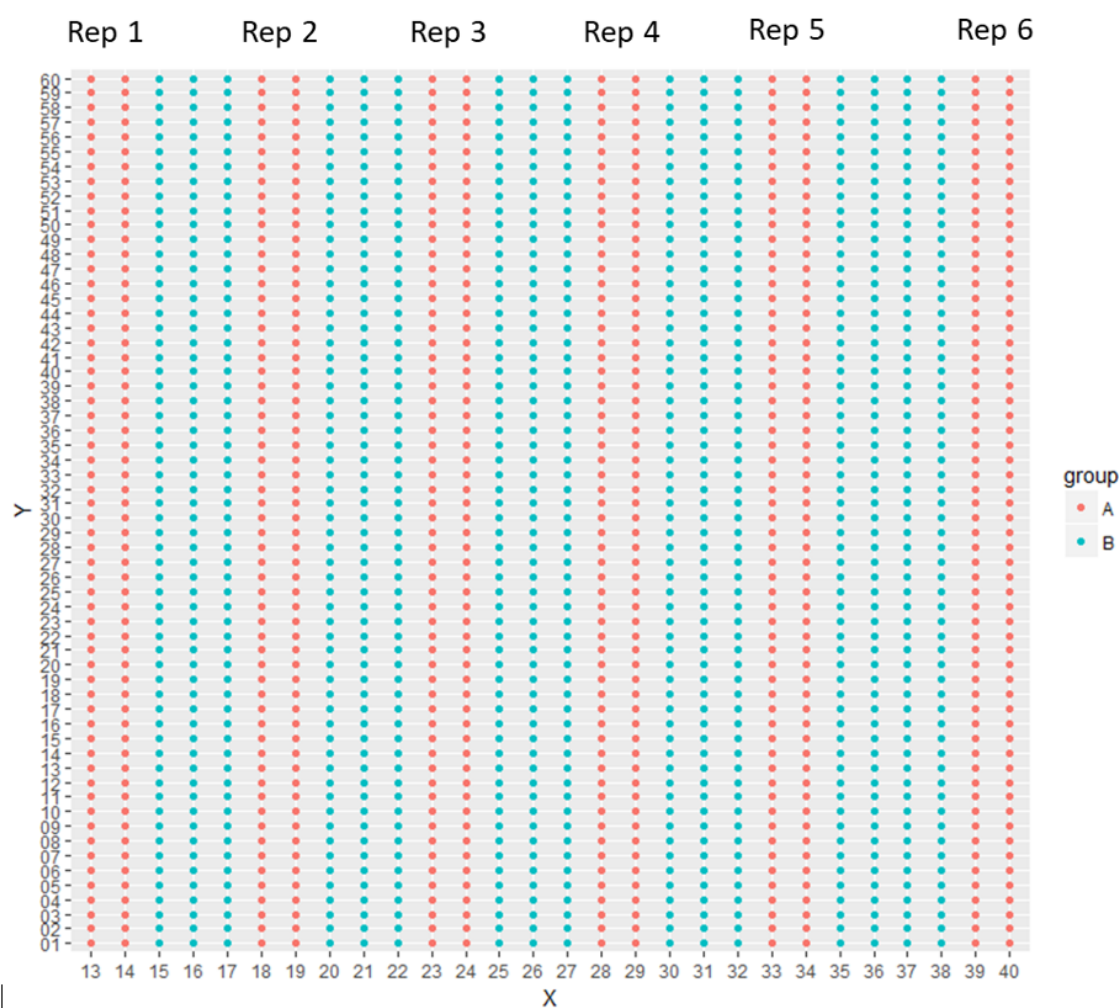


FIGURE S14 – Spatial design of the experiment in the high-throughput plant phenotyping platform PhenoArch. Red and blue circles are genotypes of groups A and B, respectively. Downward triangles, lines, well-watered scenario (S1); upward triangles, early drought scenario (S3); circles, late drought scenario (S2), and squares, long-lasting drought scenario (S4). Each symbol represents a plant.

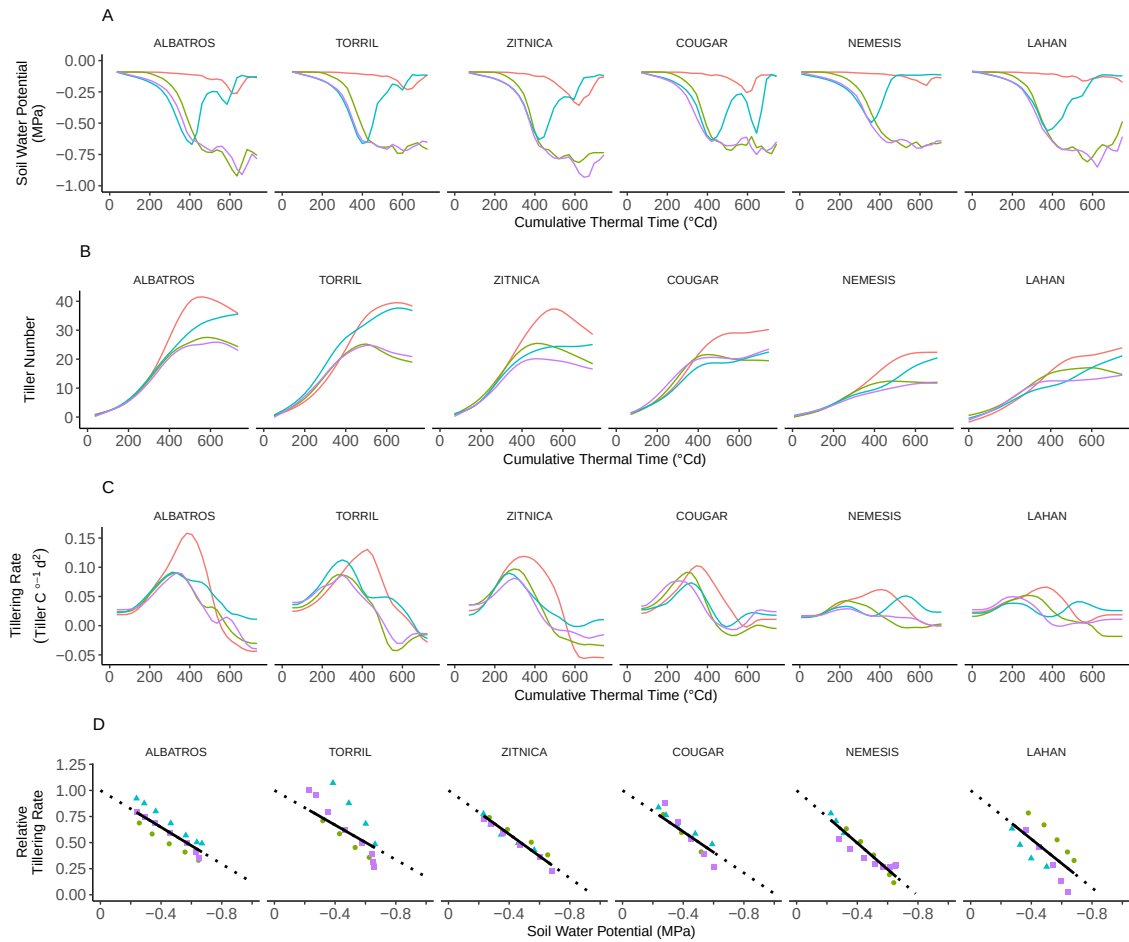


FIGURE S15 – Time courses of tiller appearance and sensitivity of tiller appearance rate to soil water potential for the four bread wheat and two durum wheat cultivars of group A under the four watering scenarios. (A) to (C) soil water potential, appeared tiller number, and tiller appearance rate versus thermal time after leaf-3 appearance, respectively. (D) relative tiller appearance rate versus soil water potential. In (A) and (B) lines are cubic smoothing splines fitted to the experimental data. In (C) lines are the first derivatives of the fitted splines shown in (B). In (D) symbols are the calculated daily values during the period determined as illustrated in Figure 1, and lines are linear regressions forced to unit when soil water potential is null. Red lines, well-watered scenario (S1); blue triangles and lines, early drought scenario (S3); green circles and lines, late drought scenario (S2), and purple square and lines, long-lasting drought scenario (S4).

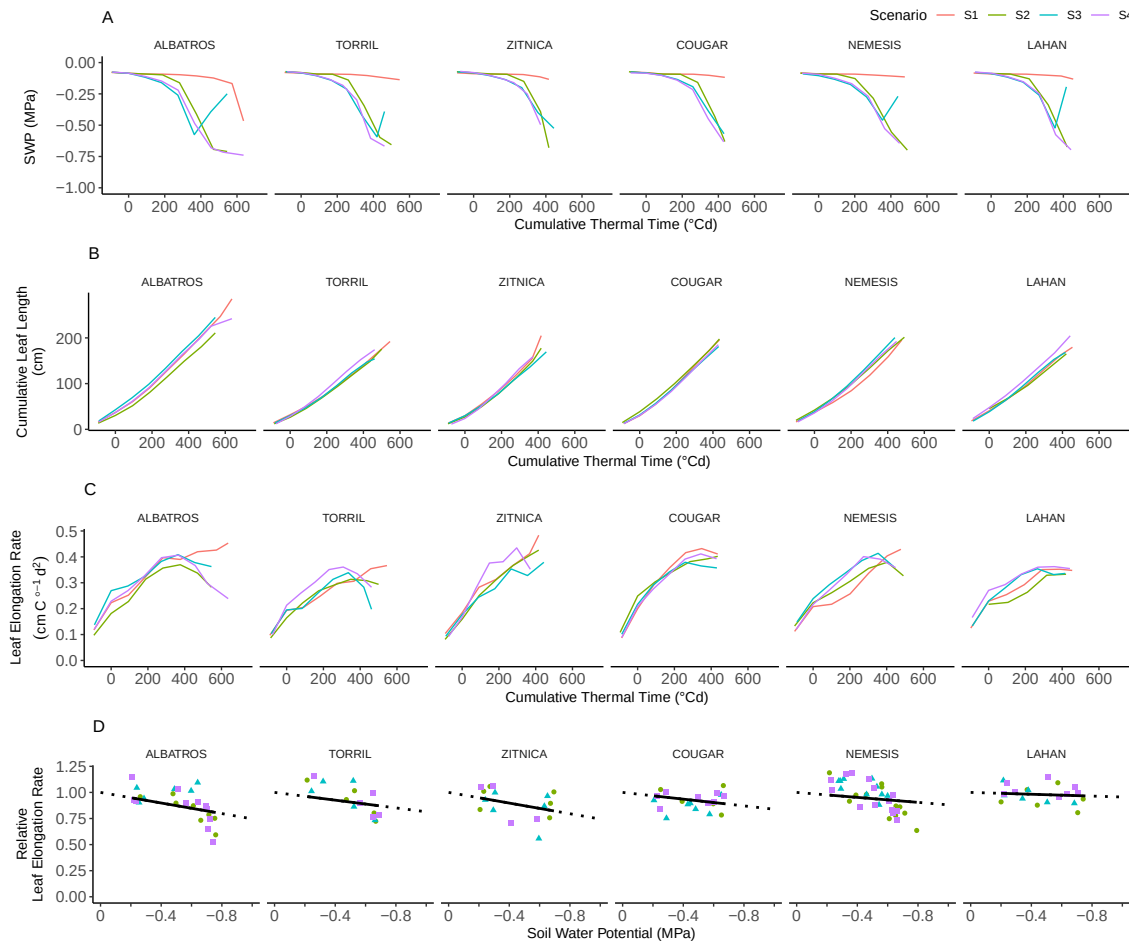


FIGURE S16 – Time courses of cumulative leaf length per plant and sensitivity of leaf elongate rate to soil water potential for the four bread wheat and two durum wheat cultivars of group A under the four watering scenarios. (A) to (C) soil water potential, cumulative leaf length per plant and leaf elongate rate versus thermal time after leaf-3 appearance, respectively. (D) relative leaf elongate rate versus soil water potential. In (A) and (B) lines are cubic smoothing splines fitted to the experimental data. In (C) lines are the first derivatives of the fitted splines shown in (B). In (D) symbols are the calculated daily values during the period determined as illustrated in Figure 1, and lines are linear regressions forced to unit when soil water potential is null. Red lines, well-watered scenario (S1); blue triangles and lines, early drought scenario (S3); green circles and lines, late drought scenario (S2), and purple square and lines, long-lasting drought scenario (S4).

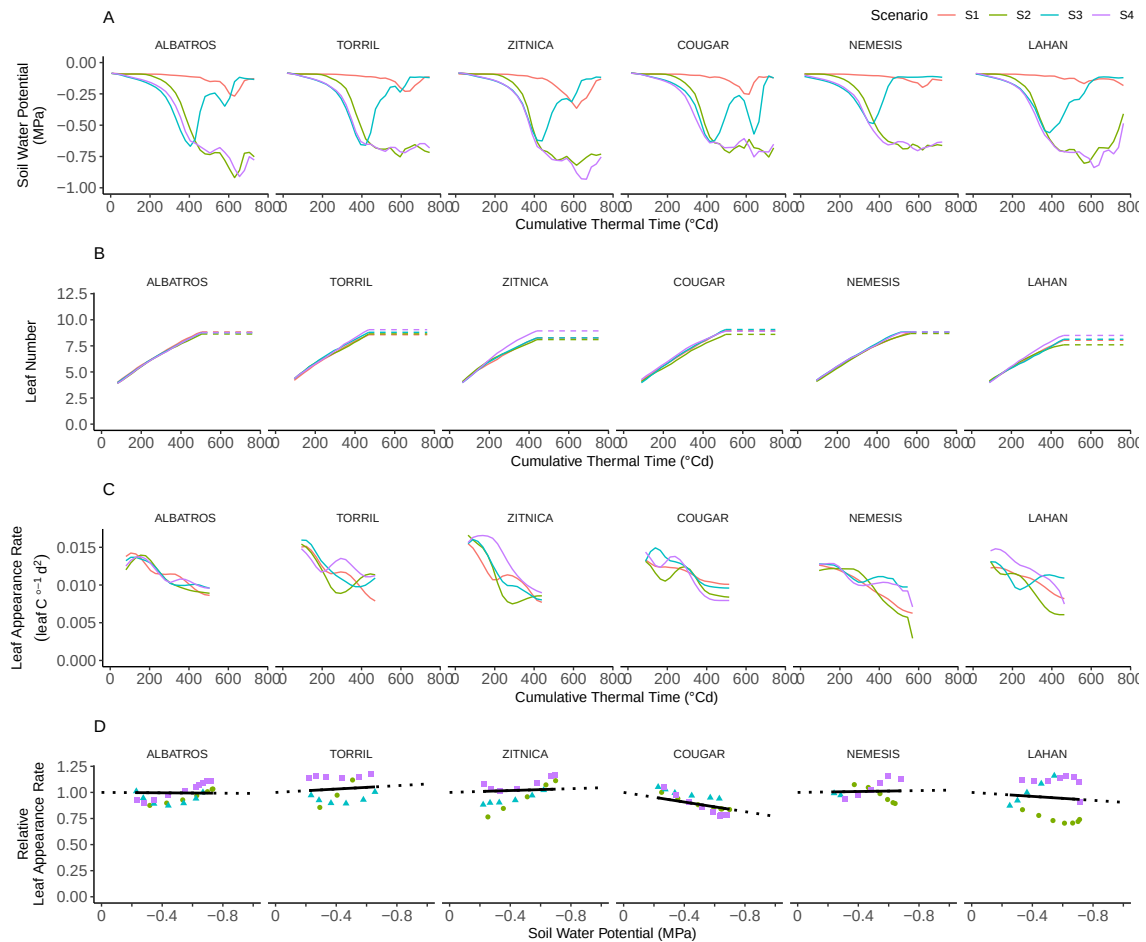


FIGURE S17 – Time courses of main-stem leaf number and sensitivity of leaf appearance rate to soil water potential for the four bread wheat and two durum wheat cultivars of group A under the four watering scenarios. (A) to (C) soil water potential, main-stem leaf number, and leaf appearance rate versus thermal time after leaf-3 appearance, respectively. (D) relative leaf appearance rate versus soil water potential. In (A) and (B) lines are cubic smoothing splines fitted to the experimental data. In (C) lines are the first derivatives of the fitted splines shown in (B). In (D) symbols are the calculated daily values during the period determined as illustrated in Figure 1, and lines are linear regressions forced to unit when soil water potential is null. Red lines, well-watered scenario (S1); blue triangles and lines, early drought scenario (S3); green circles and lines, late drought scenario (S2), and purple square and lines, long-lasting drought scenario (S4).

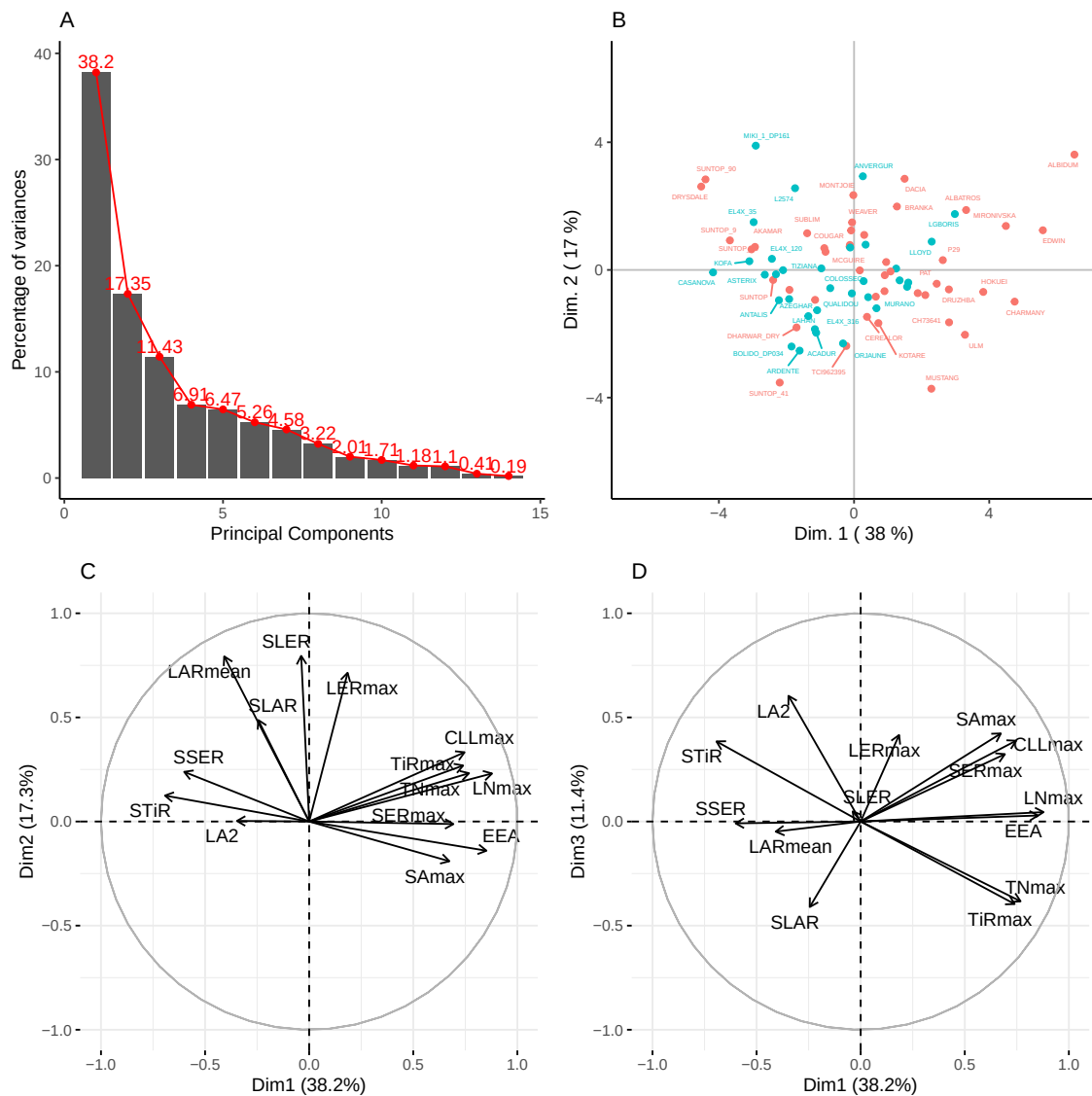


FIGURE S18 – Principal component analysis on all variables analyzed in this study for the 38 durum wheat and 47 bread wheat genotypes studied. (A) Percentage of variance explained by each component. (B) Bi-plots of first versus second components durum wheat and bread wheat genotypes are shown in red and blue, respectively. (C) Bi-plots of second component versus the first one. (D) Bi-plots of third component versus the first one. See Figure 1 for the definition of the variables. SA_{max} , maximum shoot area; SER_{max} , maximum shoot expansion rate; S_{SER} , sensitivity of shoot expansion rate to water deficit; FLN , final main-stem leaf number; LAR_{mean} , mean main-stem leaf appearance rate; S_{LAR} , sensitivity of main-stem leaf appearance rate to water deficit; TN_{max} , maximum tiller number per plant; TiR_{max} , maximum rate of tiller appearance; S_{TiR} , sensitivity of tiller appearance rate to water deficit; CLL_{max} , maximum cumulative leaf length per plant; LER_{max} , maximum main-stem leaf elongation rate; S_{LER} , sensitivity of main-stem leaf elongation rate to water deficit; and $LA2$, surface area of main-stem leaf 2.

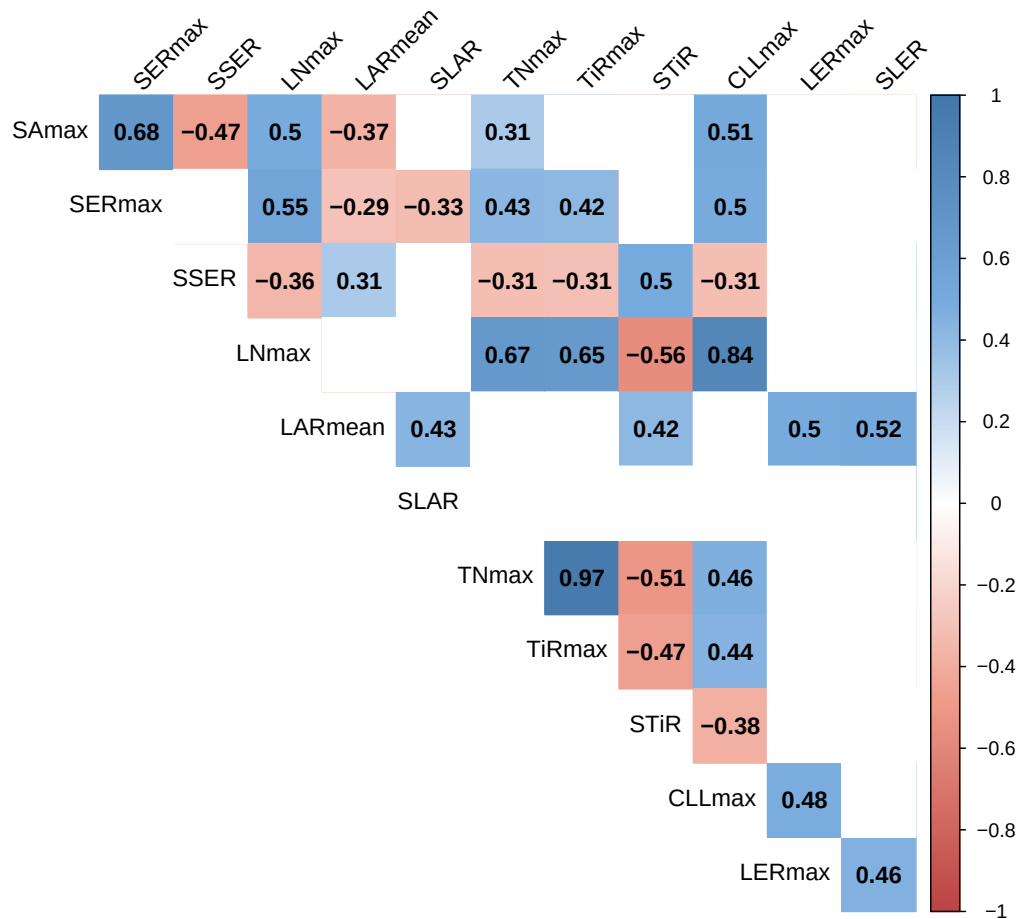


FIGURE S19 – Heatmap of the correlation matrix for all variables analyzed in this study calculated for the 38 durum wheat and 47 bread wheat genotypes studied. Values are Pearson's correlation coefficients. Colors indicate the significance and sign of the correlations (red, negative; blue, positive; and white, non-significant [$P > 0.01$]). SA_{max} , maximum shoot area; SER_{max} , maximum shoot expansion rate; S_{SER} , sensitivity of shoot expansion rate to water deficit; FLN , final main-stem leaf number; LAR_{mean} , mean main-stem leaf appearance rate; S_{LAR} , sensitivity of main-stem leaf appearance rate to water deficit; TN_{max} , maximum tiller number per plant; TiR_{max} , maximum rate of tiller appearance; S_{TiR} , sensitivity of tiller appearance rate to water deficit; CLL_{max} , maximum cumulative leaf length per plant; LER_{max} , maximum main-stem leaf elongation rate; S_{LER} , sensitivity of main-stem leaf elongation rate to water deficit; and LA_2 , surface area of main-stem leaf 2.

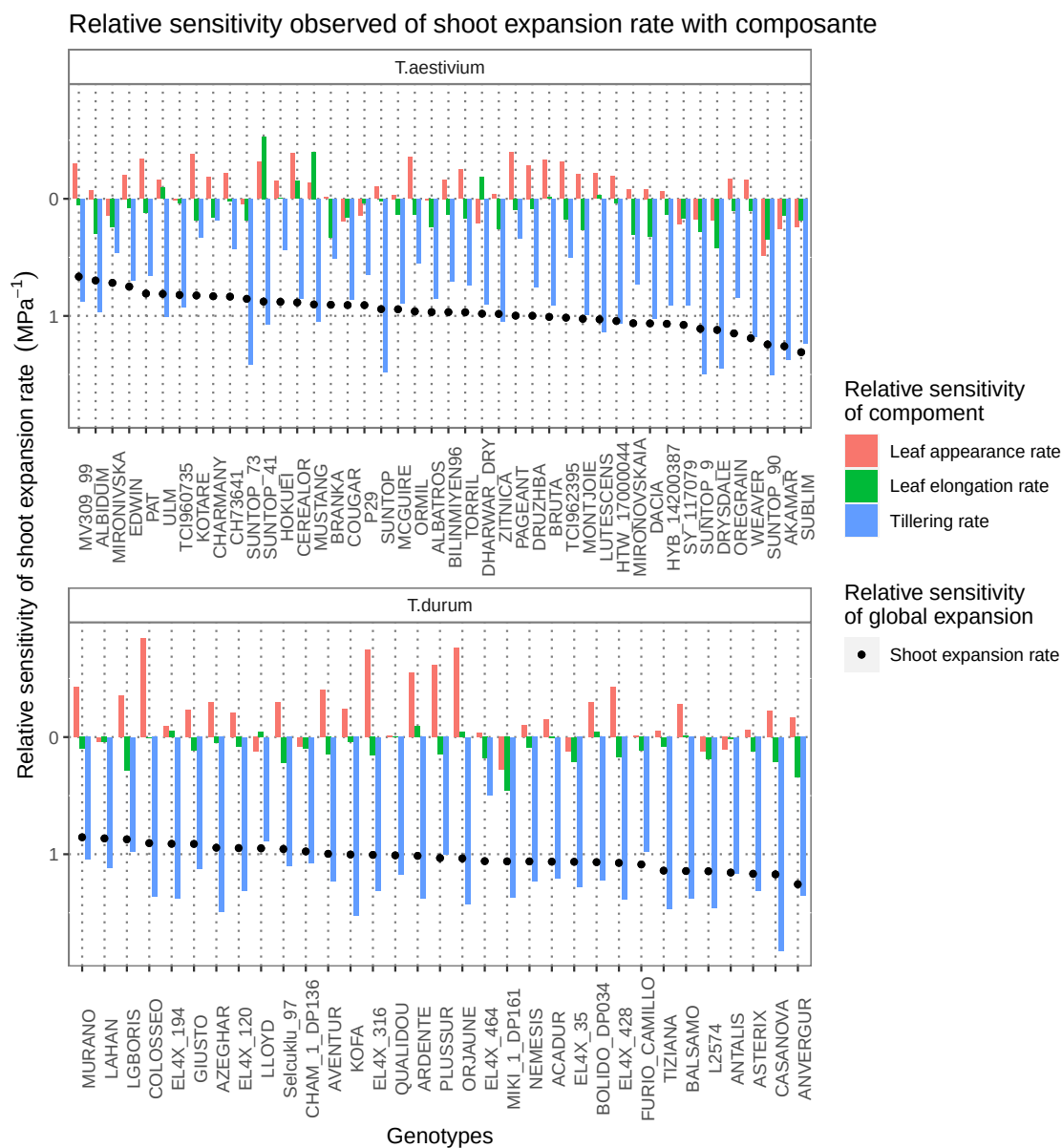


FIGURE S20 – Genetic variability in the sensitivity to soil water potential of shoot expansion rate and its three underlying components. Black circles, relative sensitivities of shoot expansion rate; green bars, relative sensitivities of main-stem leaf appearance rate; red bars, relative sensitivities of main-stem leaf elongation rate; and blue bars, relative sensitivities of tiller appearance rate. Genotypes are rank from left to right by increase value of relative sensitivity of shoot expansion rate to soil water deficit.

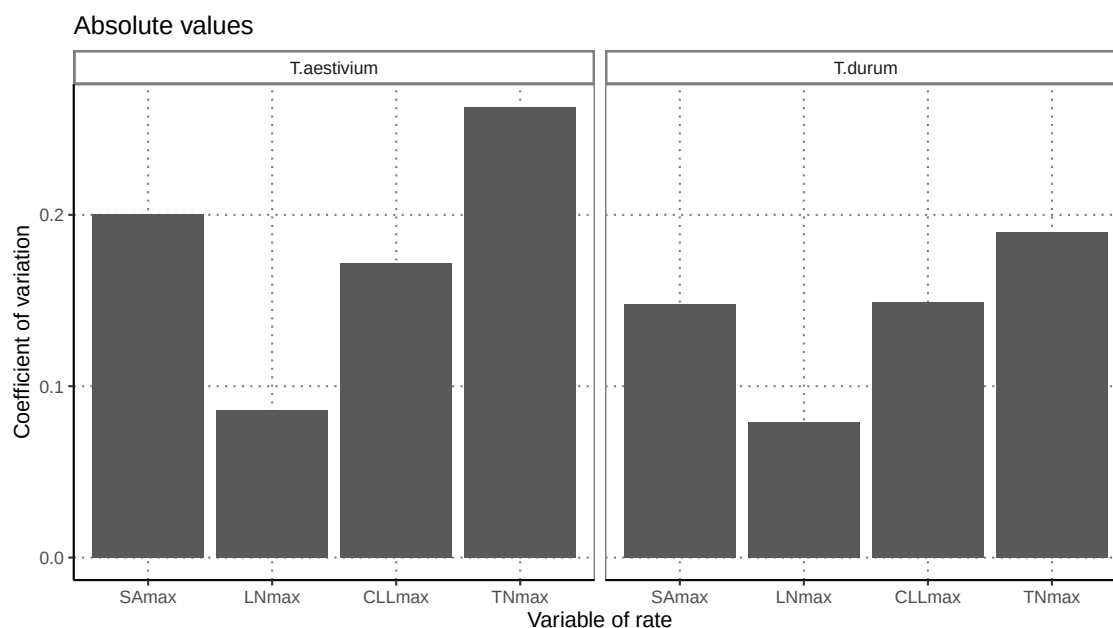


FIGURE S21 – Time courses of cumulative internode length per plant and sensitivity of internode elongation rate to soil water potential for the four bread wheat and two durum wheat genotypes of group A under the four watering scenarios. (A) to (C) soil water potential, of cumulative internode length per plant, and internode elongation rate versus thermal time after leaf-3 appearance, respectively. (D) relative internode elongation rate versus soil water potential. In (A) and (B) lines are cubic smoothing splines fitted to the experimental data. In (C) lines are the first derivatives of the fitted splines shown in (B). In (D) symbols are the calculated daily values during the period determined as illustrated in Figure 1, and lines are linear regressions forced to unit when soil water potential is null. Red lines, well-watered scenario (S1); blue triangles and lines, early drought scenario (S3); green circles and lines, late drought scenario (S2), and purple square and lines, long-lasting drought scenario (S4).

Information supplémentaire du chapitre 6

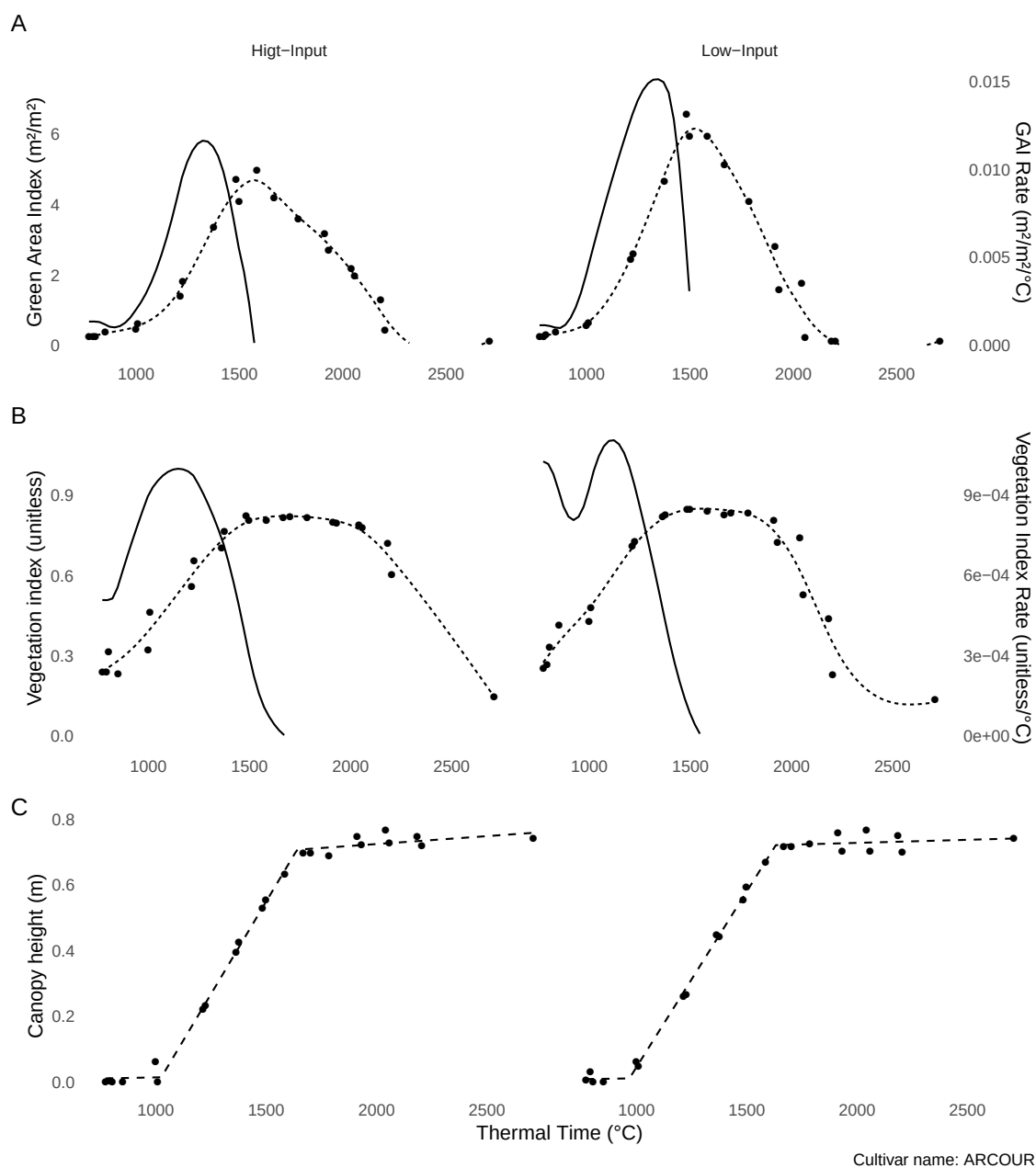


FIGURE S22 – Time courses of Green Area Index, Vegetative Index et Canopy Height and their respective rates for the genotype ARCOUR in the two treatments (high-Input and low-input) of the trial performed in Gréoux-les-bains in 2019 (AVRgre2019). Time is expressed as thermal time (degree day, °C). Dots : observed values. Dashed line : spline curve fitted on observed data (In A and B) or segmented linear models fitted to canopy height (C). Solid line : derivative of the spline curve (A,B).

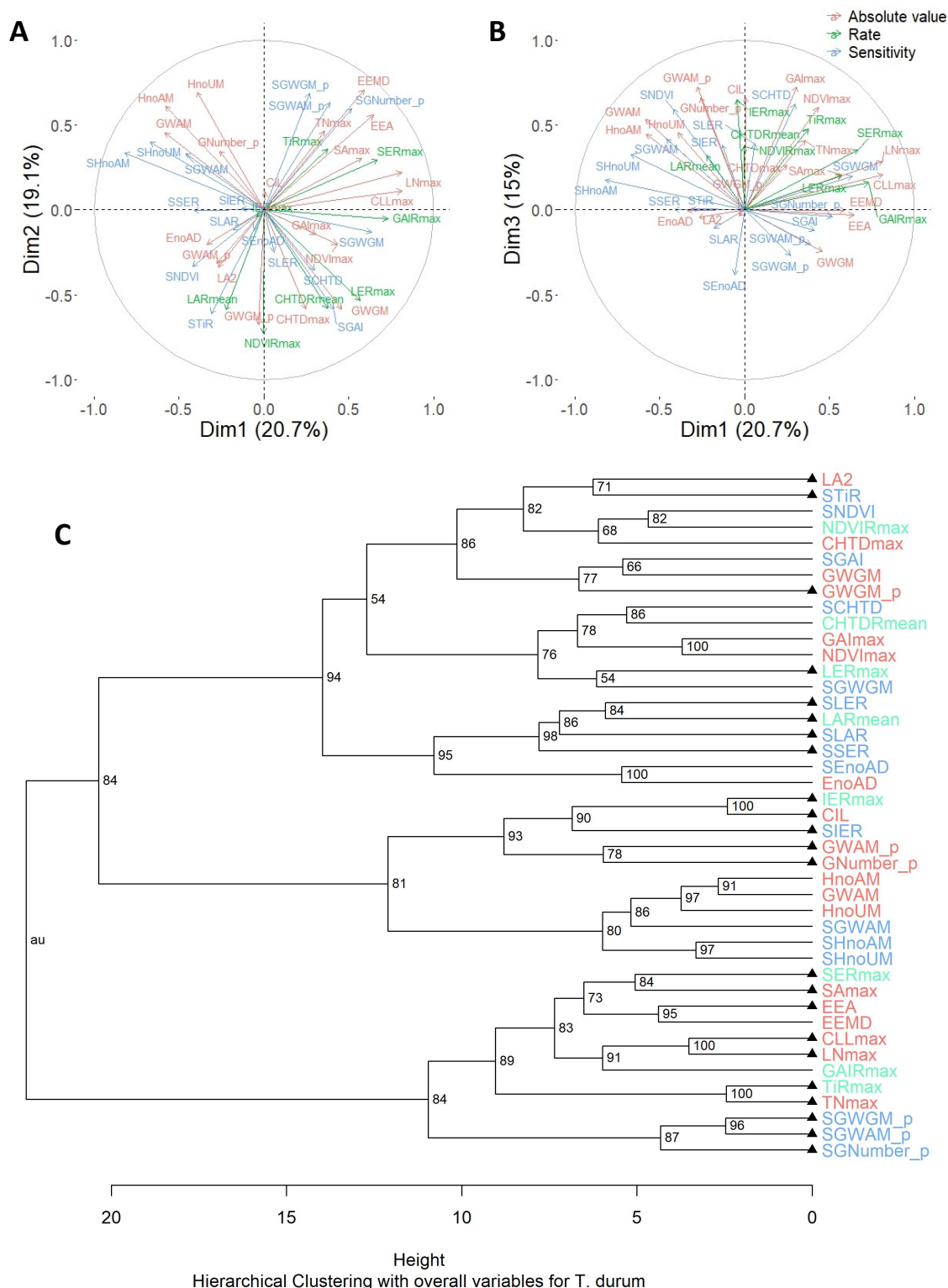


FIGURE S23 – Principal component analysis (PCA) and hierarchical clustering (HC) analysis for the 40 traits measured for the 48 genotypes in the field experiment AVRgre2019 and in the platform experiment. See Table 6.3 for definitions of variables. Colors display the type of variable : absolute values (raw data, red), rates (green) and sensitivities to either water deficit (in the platform) and low-input conditions in the field (blue). (A) Representation of all variables for (A) PC1 versus PC2 and (B) PC1 versus PC3. (C) Dendrogram of the hierarchical clustering based on all variable. Small triangles indicate which variables were obtained in the platform. The numbers indicate the approximately unbiased (AU) P-value obtained from 1000 bootstrap replications. Colors : as in A.

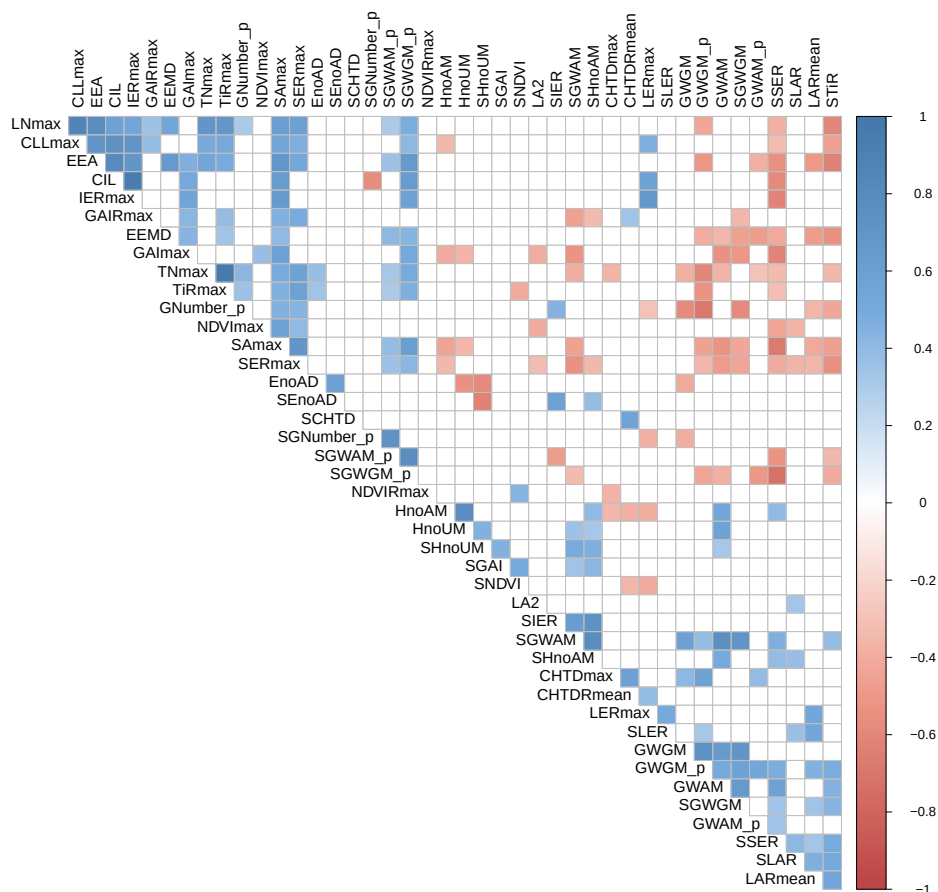


FIGURE S24 – Heatmap of the correlation (Pearson's correlation coefficient) matrix for all variables analyzed in this study for Bread wheat. Variables are described in Table 6.3. Colors indicate the significance and sign of the relationship (red, negative; blue, positive; white, insignificant, P -value > 0.05).

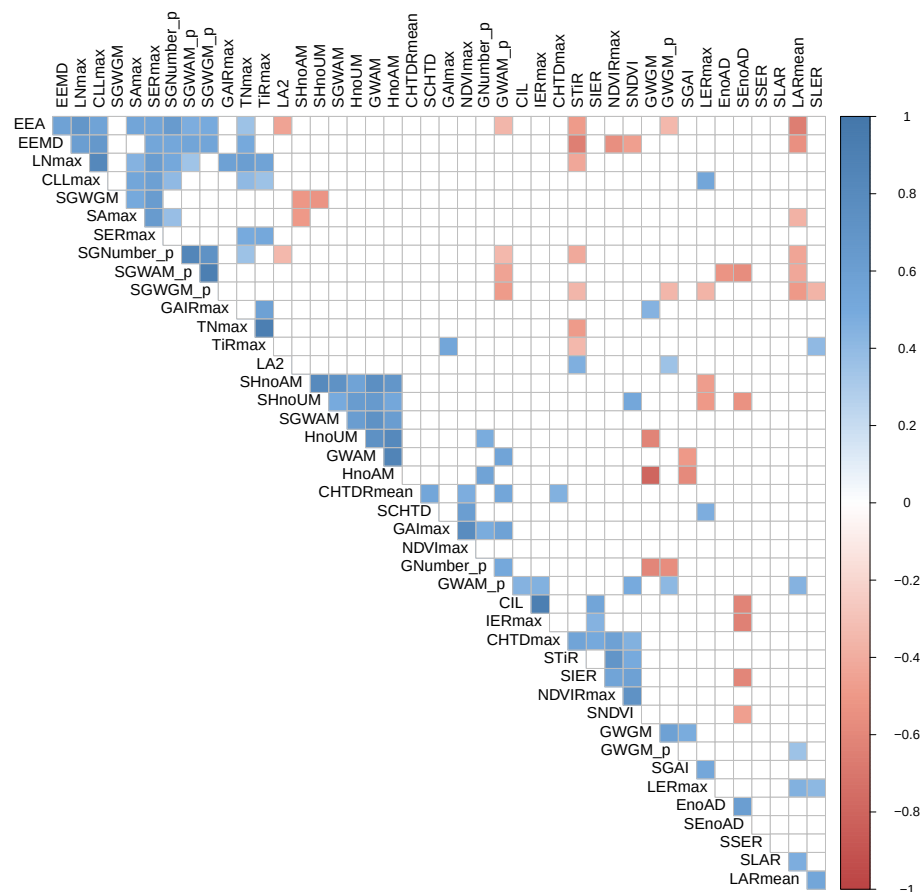


FIGURE S25 – Heatmap of the correlation (Pearson's correlation coefficient) matrix for all variables analyzed in this study for durum wheat. Variables are described in Table 6.3. Colors indicate the significance and sign of the relationship (red, negative; blue, positive; white, insignificant, P -value > 0.05).

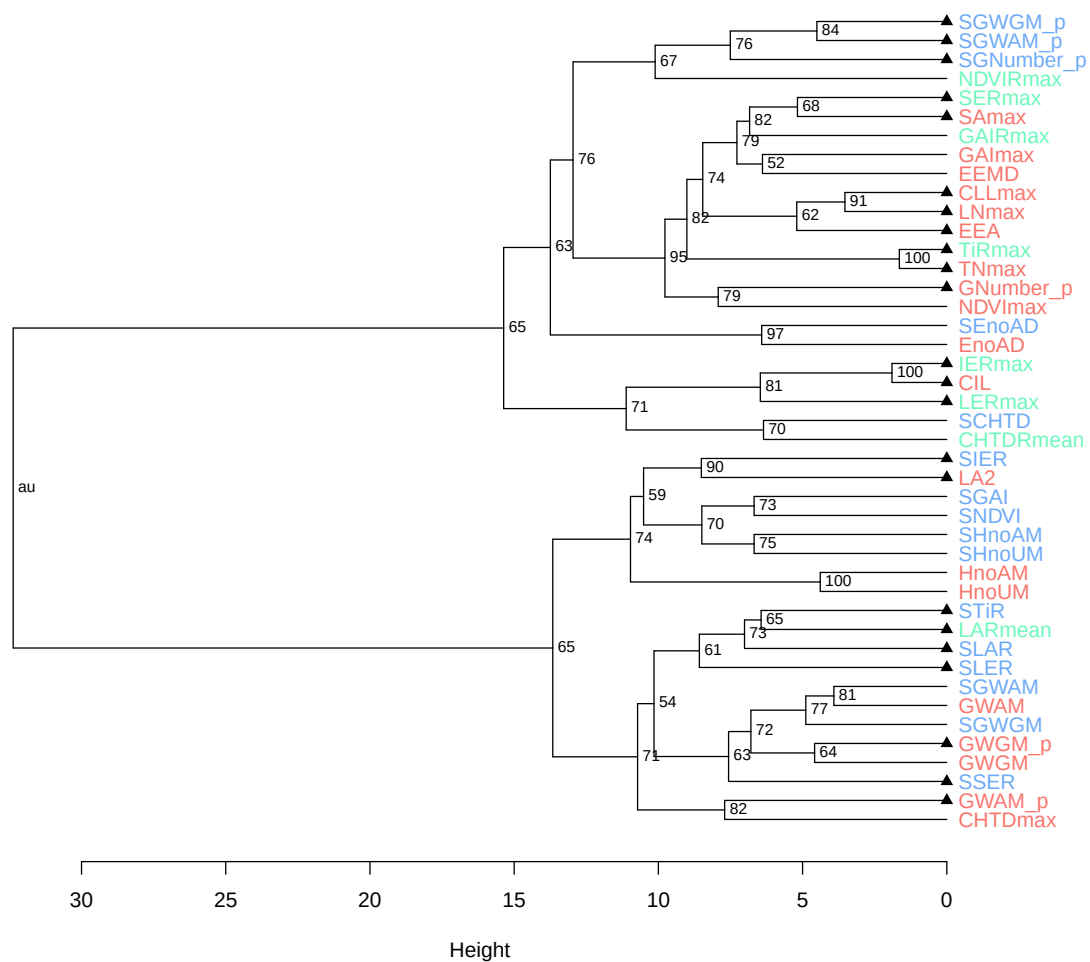


FIGURE S26 – Dendrogram of the hierarchical clustering based on all variable for bread wheat. Small triangles indicate which variables were obtained in the platform. The numbers indicate the approximately unbiased (AU) P-value obtained from 1,000 bootstraps.

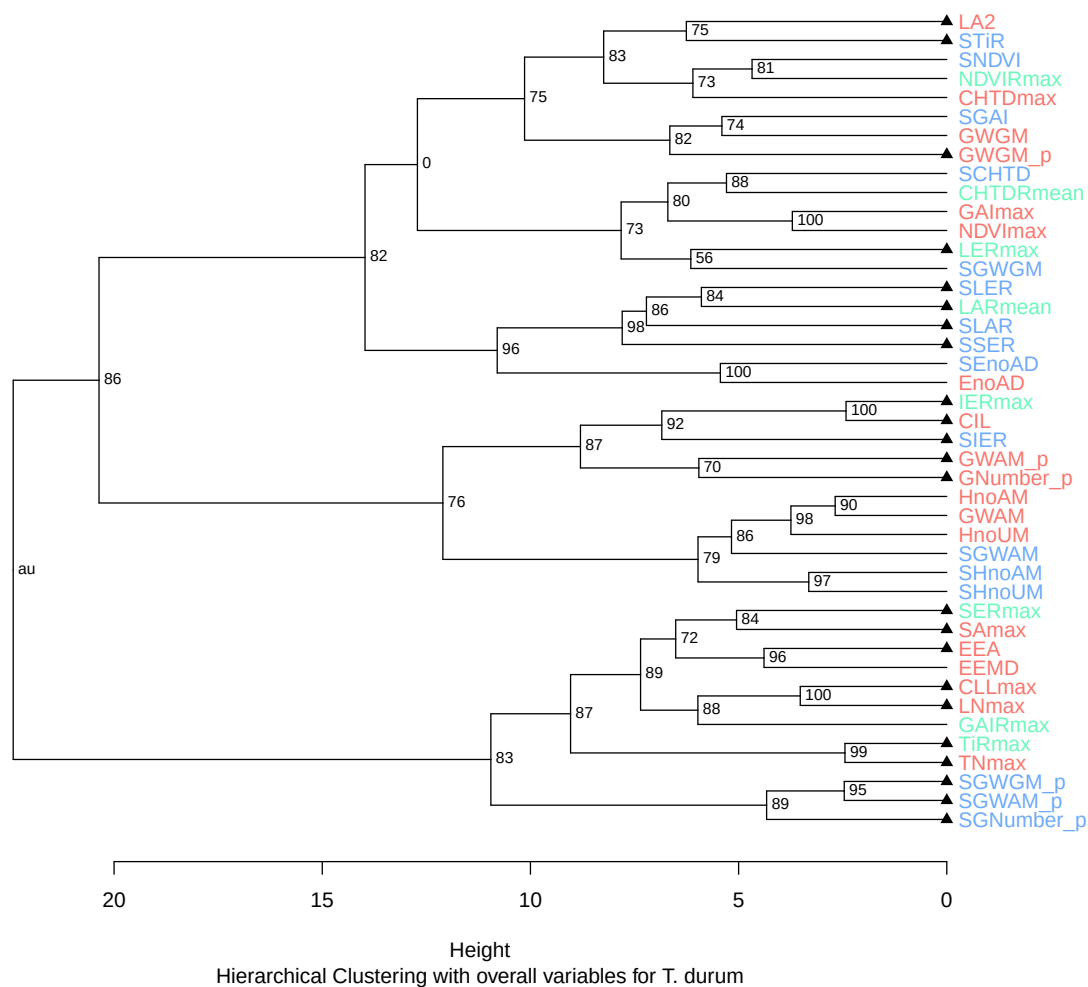


FIGURE S27 – Dendrogram of the hierarchical clustering based on all variable for durum wheat. Small triangles indicate which variables were obtained in the platform. The numbers indicate the approximately unbiased (AU) P-value obtained from 1,000 bootstrap replications